



OSS XML Gateway Reference Guide

For software version DZMS 4.x

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ABOUT THIS GUIDE

DZS Management System (DZMS) is the network management system that provides service-to-subscriber management for DZS™ local loop network elements, including Broadband Access Node (BAN™), Multi-Access Line Concentrator (MALC™), zNID™, Z-Edge 64™, and MXK™, etc. DZMS includes a comprehensive suite of applications that deliver fault, configuration, performance, and security management functions. In addition, DZMS provides for the exposure of its API for automated provisioning by Operations Support Systems (OSS) using web service. The DZS OSS XML Gateway (Optional Software License/Module) provides access to DZMS from service provider OSS systems.

Audience

This guide is written for the system administrator who installs and administers the OSS XML Gateway system and/or the programmer who develops an OSS XML client to provision resources. As a reader of this guide, you should be familiar with basic networking components such as network devices, cards, physical ports, logical interfaces, and permanent virtual circuits (PVCs). To develop an OSS XML client, you need to be familiar with any programming language that supports XML parsing. A basic knowledge of the elements of WSDL file and web services is also required.

Style and notation conventions

The following conventions are used in this document to alert users to information that is instructional, warns of potential damage to system equipment or data, and warns of potential injury or death. Carefully read and follow the instructions included in this document.



Caution: A caution alerts users to conditions or actions that could damage equipment or data.



Note: A note provides important supplemental or amplified information.



Tip: A tip provides additional information that enables users to more readily complete their tasks.



WARNING! A warning alerts users to conditions or actions that could lead to injury or death.



WARNING! A warning with this icon alerts users to conditions or actions that could lead to injury caused by a laser.

Typographical conventions

The following typographical styles are used in this guide to represent specific types of information.

Bold	Used for names of buttons, dialog boxes, icons, menus, profiles when placed in body text, and property pages (or sheets). Also used for commands, options, parameters in body text, and user input in body text.
Fixed	Used in code examples for computer output, file names, path names, and the contents of online files or directories.
Fixed Bold	Used in code examples for text typed by users.
<i>Fixed Bold Italic</i>	Used in code examples for variable text typed by users.
<i>Italic</i>	Used for book titles, chapter titles, file path names, notes in body text requiring special attention, section titles, emphasized terms, and variables.
PLAIN UPPER CASE	Used for environment variables.
Command Syntax	Brackets [] indicate optional syntax. Vertical bar indicates the OR symbol.

Related documentation

OSS XML Gateway User's Guide

OSS XML Gateway Release Notes

OSS Gateway User's Guide

OSS Gateway Reference Guide

MXK-F Hardware Installation Guide

MXK-F Configuration Guide

MXK-F Release Notes

DZMS Release Notes

DZMS Administrator's Guide

DZMS Installation Guide

NetHorizon User's Guide

DZMS HA Platform Installation and Administration Guide

DZMS Web Client Quick Start Guide

NETCONF User's Guide

Refer to the release notes for software installation information and for changes in features and functionality of the product (if any).

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Contacting DZS Quality & Service

All new DZS equipment purchases include one year of HW Warranty and 90 days of Bronze-level Technical Support.

If your product is not within 90 days of the purchase date or you do not have a valid support contract, please contact your local sales representative to get a quote on a support contract.

Customers with a valid support contract or are eligible for 90 days technical support associated with a new product purchase can request technical support by opening a case at:

<https://dzsi.com/support/#TAC>

Customers with a valid support contract have access to technical product documentation, software downloads, knowledge base and consultation on the covered DZS product at the same support portal.

For repair services within the HW Warranty period or under an Extended Warranty support contract, a Return Material Authorization (RMA) must be obtained before sending the equipment for repair. RMA requests can be submitted at:

<https://dzsi.com/support/#RMA>

Technical support

The Technical Assistance Center (TAC) is available with experienced support engineers who can handle questions, assist with service requests, and help troubleshoot systems.

Hours of operation	Monday - Friday, 8 a.m. to 6 p.m, Eastern Time (excluding U.S. holidays)
Telephone (North America)	877-946-6320, prompt #3, #1
Telephone (International)	1-510-777-7133, prompt #3, #1
E-mail	support@dzsi.com
Web - available 24 x 7 to submit and track field issues/ problem reports	https://dzsi.com/support/#TAC Click DZS Problem Reporting System

If you purchased the product from an authorized dealer, distributor, Value Added Reseller (VAR), or third party, contact that supplier for technical assistance and warranty support.

Hardware repair

If the product malfunctions, all repairs must be authorized by DZS with a Return Merchandise Authorization (RMA) and performed by the manufacturer or a DZS-authorized agent. It is the responsibility of users

requiring service to report the need for repair to DZS Quality & Service as follows:

- Complete the RMA Request form (<https://dzsi.com/support/#RMA>) or contact DZS Quality & Service via phone or email:
Hours of operation: Monday - Friday, 8 a.m. to 6 p.m, Eastern Time (excluding U.S. holidays)
E-mail: support@dzsi.com (preferred)
Phone: 877-946-6320 or 510-777-7133, prompt #3, #2
- Provide the part numbers and serial numbers of the products to be repaired.
- All product lines ship with a minimum one year standard warranty (may vary by contract). DZS warrants all repairs for 90 days or the remainder of the standard warranty (whichever is greater).
- DZS will verify the warranty and provide the customer with a repair quote for anything that is not under warranty. DZS requires a purchase order or credit card for out of warranty fees.

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CONFIGURATION REQUESTS

This chapter describes the attributes associated with configuration requests to the OSS XML Gateway. It includes the following sections:

- [Overview, page 14](#)
- [Object hierarchy, page 14](#)
- [Region configuration requests, page 14](#)
- [BAN device configuration requests, page 16](#)
- [MALC/MXK/Raptor device configuration requests, page 24](#)
- [Raptor 100 and Raptor 50 device configuration requests, page 42](#)
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- [N2N/EFM/GBond bonded groups, page 91](#)
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- [OMCI device only objects, page 105](#)
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- [Port access control, page 108](#)
- [SubRegion objects, page 109](#)
- [VdslVectProfile object, page 110](#)
- [DynOmciCpeCfgRule object, page 111](#)
- [DynOmciCpeLldpRem object, page 111](#)
- [Logical configuration requests, page 111](#)
- [lineRedStatusFunction object, page 119](#)

- [Other OSS XML Gateway objects, page 120](#)

Overview

The OSS XML Gateway supports the following configuration requests:

- **get**, **getTemplate**, **list**, and **listTemplate** requests to retrieve data for objects.
- **delete** requests to delete objects.
- **add**, **modify**, **addUsingTemplate** requests to configure objects.

The following tables list the parent and object, indicate whether the object is supported by the OSS XML Gateway, and list the request element attributes that are associated with each object.

Object hierarchy

All OSS XML requests will provide the complete hierarchy information as part of the response. This response information can be used for framing new requests that require parental information.

Region configuration requests

The following table contains the request element attributes associated with Region parents.

Table 1: Region configuration requests

Parent	Object	Supported	Request element attribute
Region	BANDevice MALC19Device MALC23Device MALC319Device S100AE1Device S100AT1Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Raptor100Device Raptor50Device ZEdgeBH2ADevice ZEdgeH2ADevice	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge 64 BH2A Z-Edge 64 H2A	DateAndTime Type active altConfigIP attachedCardId attachedDeviceId attachedPhysicalId attachedShelfId adslSeltUnits bgColor center configSyncAction configSyncExists configSyncFileName configSyncOverflow configSyncPriority configSyncStatus configSyncUserName configSyncUserPassword deviceAddrType deviceIP deviceSubType ellipsoid enableAuthTraps focusX focusY instanceKey ipFilterTransactTimeout mapName monitorEnable name point prefCommName primaryClockSource projectionName radiusAuthIndex revertiveClockSource ringSource secondaryPoint state sysCfgCountryRegion sysContact sysDescription sysLocation sysObjectID sysRebootAdmin sysRebootCardType thermoALowerThold thermoAUpperThold

Table 1: Region configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Region	BANDevice	BAN	thermoBLowerThold
	MALC19Device	MALC 319	thermoBUpperThold
	MALC23Device	MALC 719	unitConversionFactor
	MALC319Device	MALC 723	userAuthMode
	S100AE1Device	Raptor 100	voiceBandWidthCheck
	S100AT1Device	Raptor 50	zmsBlockCli
	Raptor50Device	Raptor 319	zmsConnectionStatus
	Raptor723Device	Raptor 719	zmsExists
	Raptor719Device	Raptor 723	zmsIPAddress
	Raptor319Device	Sechtor 100A E1	zoomFactor
	Raptor100Device	Sechtor 100A T1	
	Raptor50Device	Z-Edge 64 BH2A	
	ZEdgeBH2ADevice	Z-Edge 64 H2A	
	ZEdgeH2ADevice	MXK 819	
	Mx19Device	MXK 823	
	Mx23Device		

BAN device configuration requests

The following table contains the request element attributes associated with BAN Device parents.

Table 2: BAN device configuration requests

Parent	Object	Supported	Request element attribute
BANDevice	AccessList	BAN	accessListIndex name
BANDevice	AdslAlarmProfile	BAN	atucInitFailureTrapEnable atucThresh15MinESs atucThresh15MinLofs atucThresh15MinLols atucThresh15MinLoss atucThresh15MinLprs atucThreshFastRateDown atucThreshFastRateUp atucThreshInterleaveRateDown atucThreshInterleaveRateUp aturThresh15MinESs aturThresh15MinLofs aturThresh15MinLoss aturThresh15MinLprs aturThreshFastRateDown aturThreshFastRateUp aturThreshInterleaveRateDown aturThreshInterleaveRateUp name

Table 2: BAN device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
BANDevice	AdslConfigProfile	BAN	atucChanFastMaxTxRate atucChanFastMinTxRate atucChanInterleaveMaxTxRate atucChanInterleaveMinTxRate atucChanMaxInterleaveDelay atucDownshiftSnrMgn atucMaxSnrMgn atucMinDownshiftTime atucMinSnrMgn atucMinTxThreshAlarm atucMinUpshiftTime atucRateChanRatio atucRateMode atucTargetSnrMgn atucUpshiftSnrMgn aturChanFastMaxTxRate aturChanFastMinTxRate aturChanInterleaveMaxTxRate aturChanInterleaveMinTxRate aturChanMaxInterleaveDelay aturDownshiftSnrMgn aturMaxSnrMgn aturMinDownshiftTime aturMinSnrMgn aturMinTxThreshAlarm aturMinUpshiftTime aturRateChanRatio aturRateMode aturTargetSnrMgn aturUpshiftSnrMgn name
BANDevice	ATMTrafficDesc	BAN	cacDivider name serviceCategory trafficDescrIndex trafficDescrParam1 trafficDescrParam2 trafficDescrParam3 trafficDescrParam4 trafficDescrParam5 trafficDescrType usageControl vclRate

Table 2: BAN device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
BANDevice	ATMTrafficDescTemplate	BAN	Type cacDivider instanceKey name serviceCategory templateCategory templateDescription templateName trafficDescrIndex trafficDescrParam1 trafficDescrParam2 trafficDescrParam3 trafficDescrParam4 trafficDescrParam5 trafficDescrType usageControl vclRate
BANDevice	ATMVcCrossConn	BAN	name vcCrossConnAdminStatus vcCrossConnHighIfIndex vcCrossConnHighVci vcCrossConnHighVpi vcCrossConnIndex vcCrossConnLowIfIndex vcCrossConnLowVci vcCrossConnLowVpi
BANDevice	AtmVpCrossConn	BAN	Type adminStatus highIfIndex highVpi index instanceKey lowIfIndex lowVpi name operStatus
BANDevice	BridgeCfgGroup	BAN	Type groupIndex instanceKey memberNextIndex name

Table 2: BAN device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
BANDevice	BridgeLogical	BAN	name ifIndex vpi vci lineGroupId groupIndex vlanId lowerIfIndex adminStatus learnIp learnUnicast fwdUnicast fwdMulticast fwdDefault brdigeIfStagStripAndInsert
BANDevice	BridgePath	BAN	Type addrType bridgeIfIndex flapControl instanceKey igmpQueryInterval ipOrMacAddr multicastAging name slanId unicastAging vlanId
BANDevice	BulkStatsCollection	BAN	Type collectionId collectionInterval includeChildren instanceKey name operStatus statEnabled statInstance statOid
BANDevice	BulkStatsConfig	BAN	Type collectionEnabled ftpAddress ftpDirPath ftpLogin ftpPassword instanceKey name

Table 2: BAN device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
BANDevice	BulkVcCrossConnTemplate	BAN	Type adminStatus autoCreation instanceKey lineRXTD lineTXTD lineVci lineVpi name templateCategory templateDescription templateName trunkRXTD trunkStartVci trunkTXTD trunkVpi
BANDevice	CardGroup	BAN	cardType cgld name
BANDevice	CommunityName	BAN	communityAccess communityAccessListIndex name
BANDevice	DHCPGroup	BAN	bootFile bootServer clientMatchLength clientMatchOffset clientMatchString defLeaseTime defRouter domain domainName index maxLeaseTime minLeaseTime name priNameServer secNameServer vendorMatchLength vendorMatchOffset vendorMatchString

Table 2: BAN device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
BANDevice	DHCPHost	BAN	IPAddr1 IPAddr2 IPAddr3 IPAddr4 bootFile bootServer clientId defLeaseTime defRouter domain domainName hardwareAddr index maxLeaseTime minLeaseTime name priNameServer secNameServer
BANDevice	DHCPServDefault	BAN	leaseTime maxLeaseTime minLeaseTime name reserveEnd reserveStart
BANDevice	DHCPSubnet	BAN	bootFile bootServer defLeaseTime defRouter domain domainName externalServer index maxLeaseTime minLeaseTime name netmask network priNameServer range1End range1Start range2End range2Start range3End range3Start range4End range4Start secNameServer stickyAddr subnetGroup

Table 2: BAN device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
BANDevice	DslAlarmProfile	BAN	name threshold15MinLofs threshold15MinLols threshold15MinLoss
BANDevice	Hdsl2ConfigProfile	BAN	decoderCoeffA decoderCoeffB frameSyncWord name stuffBits transmitPowerBackoffMode
BANDevice	IpConfig	BAN	defaultTtl forwarding name rdIndex
BANDevice	IpDnsResolverConfig	BAN	domainName firstNameServer name queryOrder rdIndex secondNameServer thirdNameServer
BANDevice	IpFilterSpec	BAN	languageVersion name specDescription specIndex version1 version2
BANDevice	IpRoutingDomain	BAN	name rdIndex
BANDevice	NTPConfig	BAN	Type dayLightSavingTime instanceKey localTimeZone name primaryServerIP secondaryServerIP
BanDevice	Ping	BAN	Type adminStatus dataSize instanceKey name pingIndex probeCount targetAddress timeOut

Table 2: BAN device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
BANDevice	SdslConfigProfile	BAN	connectMode fixBitRate framerType lineRate name ntr powerScale
BANDevice	ShdslConfigProfile	BAN	clockOffset decoderCoeffA decoderCoeffB fixBitRate frameSync lineRate name ntr powerScale repeaterId standard startupMargin transmitPowerBackoffMode wireMode
BANDevice	SubscriberPort	BAN	cpeType deviceRole id lgId maxAllowedLineRate maxCapableLineRate name nextVoiceConnectionIndex providerId subscriberKey
BANDevice	TrapConfig	BAN	ackedSequenceNumber adminStatus allowedCategories communityName destination name resendSequenceNumber sequenceNumber severity
BANDevice	UnnumberedIpAlias	BAN	Type enabled instanceKey ipIfIndex name

Table 2: BAN device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
BANDevice	UnnumberedIpRef	BAN	ipIfIndex ipIndex name

MALC/MXK/Raptor device configuration requests

The following table contains the request element attributes associated with MALC319, MALC 719, MALC 723, MXK 819, MXK 823, Raptor 319, Raptor 719, and Raptor 723 Device parents.

Table 3: MALC/MXK/Raptor device configuration requests

Parent	Object	Supported	Request element attribute
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	AccessList	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	accessListIndex name
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	AdslAlarmProfile	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	atucInitFailureTrapEnable atucThresh15MinESs atucThresh15MinLofs atucThresh15MinLols atucThresh15MinLoss atucThresh15MinLprs atucThreshFastRateDown atucThreshFastRateUp atucThreshInterleaveRateDown atucThreshInterleaveRateUp aturThresh15MinESs aturThresh15MinLofs aturThresh15MinLoss aturThresh15MinLprs aturThreshFastRateDown aturThreshFastRateUp aturThreshInterleaveRateDown aturThreshInterleaveRateUp name

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	AdslConfigProfile	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	atucChanFastMaxTxRate atucChanFastMinTxRate atucChanInterleaveMaxTxRate atucChanInterleaveMinTxRate atucChanMaxInterleaveDelay atucDownshiftSnrMgn atucMaxSnrMgn atucMinDownshiftTime atucMinSnrMgn atucMinTxThreshAlarm atucMinUpshiftTime atucRateChanRatio atucReachExtendedAdsl2 atucRateMode atucTargetSnrMgn atucUpshiftSnrMgn aturChanFastMaxTxRate aturChanFastMinTxRate aturChanInterleaveMaxTxRate aturChanInterleaveMinTxRate aturChanMaxInterleaveDelay aturDownshiftSnrMgn aturMaxSnrMgn aturMinDownshiftTime aturMinSnrMgn aturMinUpshiftTime aturRateChanRatio aturRateMode aturTargetSnrMgn aturUpshiftSnrMgn name
MALC19Device MALC23Device MALC319Device	ATMTrafficContainer	MALC 319 MALC 719 MALC 723	Type instanceKey name oltTrafficContainerCBR oltTrafficContainerCbrCompensate oltTrafficContainerGuaranteedBW oltTrafficContainerIndex oltTrafficContainerMaximumBW

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	ATMTrafficDesc	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	cacDivider cellloss congestion name serviceCategory trafficDescrIndex trafficDescrParam1 trafficDescrParam2 trafficDescrParam3 trafficDescrParam4 trafficDescrParam5 trafficDescrType usageControl vclRate
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	ATMTrafficDescTemplate	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	Type cacDivider instanceKey name serviceCategory templateCategory templateDescription templateName trafficDescrIndex trafficDescrParam1 trafficDescrParam2 trafficDescrParam3 trafficDescrParam4 trafficDescrParam5 trafficDescrType usageControl vclRate
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	ATMVcCrossConn	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	name vcCrossConnAdminStatus vcCrossConnHighIfIndex vcCrossConnHighVci vcCrossConnHighVpi vcCrossConnIndex vcCrossConnLowIfIndex vcCrossConnLowVci vcCrossConnLowVpi

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	AtmVpCrossConn	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	Type adminStatus highIfIndex highVpi index instanceKey lowIfIndex lowVpi name operStatus
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	BridgeCfgGroup	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	Type groupIndex instanceKey memberNextIndex name

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MALC19Device	BridgeLogical	MALC 319	bridgeIfFloodMulticast
MALC23Device		MALC 719	bridgeIfFloodUnknown
MALC319Device		MALC 723	bridgeIfIsPPPoA
Raptor723Device		Raptor 319	bridgeIfMaxVideoStreams
Raptor719Device		Raptor 719	bridgeIfMcastControlList
Raptor319Device		Raptor 723	bridgeIfOutgoingCOSOption
MalcRaptorXPDevice		MALC XP 150	bridgeIfOutgoingCOSValue
Mx19Device		Raptor XP 150	bridgeIfStagCOS
Mx23Device		MXK 819	bridgeIfStagOutgoingCOSOption
Mx3UDevice		MXK 823	bridgeIfStagOutgoingCOSValue
		MXK 319	bridgeIfStagStripAndInsert
			bridgeIfStagTPID
			bridgeIfStagId
			bridgeIfVlanIdCOS
			bridgeStatsBSDAlarmCount
			bridgeStatsBSDDBcastBlocked
			bridgeStatsBSDMcastBlocked
			bridgeStatsBSDUcastBlocked
			customerARP
			customerDHCP
			DhcpLearn
			egressGroupIndex
			filterBroadcast
			fwdUnicast
			fwdMulticast
			fwdDefault
			groupIndex
			ifAdminStatus
			ifIndex
			learnIp
			learnMulticast
			learnUnicast
			lineGroupId
			lowerIfIndex
			maxUnicast
			name
			stripInsert
			TableBasedFilter
			vpi
			vci
			vlanId
			Continued at next page...

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MALC19Device	BridgeLogical (Continued)	MALC 319	The following attributes are only for MALC, MXK and MALCXP, RaptorXP Devices: bridgeStatsBcastRcvd bridgeStatsBcastSent bridgeStatsErrorSent bridgeStatsMcastRcvd bridgeStatsMcastSent bridgeStatsUcastRcvd bridgeStatsUcastSent The following attributes are only for MXK 819, MXK 823, and MXK 319 devices: actChans actHosts errorRx genQueryRx genQueryTx leavesRx leavesTx specQueryRx specQueryTx unknownRx v2ReportsRx v2ReportsTx The following attributes are only for MXK 819, MXK 823, MXK 319, MX 1U (160, 161 and 162) devices: vlanXlateId slanXlateId mvrVlanId The following attributes are only for MXK 819 Device and MXK 823Device: gponOltId gponPortId gponSlotId gponTrafficIdx gtpTemplateName The following attributes are only for MALC19Device, MALC23Device, MALC319Device, MalcRaptorXPDevice, MXK 819Device, and MXK 823Device: rstpEnable Continued at next page...
MALC23Device		MALC 719	
MALC319Device		MALC 723	
Raptor723Device		Raptor 319	
Raptor719Device		Raptor 719	
Raptor319Device		Raptor 723	
MalcRaptorXPDevice		MALC XP 150	
Mx19Device		Raptor XP 150	
Mx23Device		MXK 819	
Mx3UDevice		MXK 823	
Mx1UDevice		MXK 319	
		MX 1U	

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device MalcRaptorXPDevice Mx19Device Mx23Device Mx3UDevice	BridgeLogical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MALC XP 150 Raptor XP 150 MXK 819 MXK 823 MXK 319	The following attributes are only for MALC19Device, MALC23Device, MALC319Device, Raptor723Device, Raptor719Device, Raptor319Device, MXK 819Device, and MXK 823Device: egressGroupTemplate groupTemplate
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	BridgePath	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	Type addrType bridgeIfIndex flapControl instanceKey igmpQueryInterval ipOrMacAddr multicastAging name slanId unicastAging vlanId
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	BulkMulticastMap	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	Type adminStatus autoIncrementVpi autoSelectVci instanceKey lgId multicastMapTemplateId name numChannels otherVclTemplateId rxTrafficDesc startMulticastAddress startVci txTrafficDesc useTDTmpl videoDomain vpi

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	BulkStatsCollection	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	Type collectionId collectionInterval includeChildren instanceKey name operStatus statEnabled statInstance statOid
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	BulkStatsConfig	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	Type collectionEnabled ftpAddress ftpDirPath ftpLogin ftpPassword ftpPassword instanceKey name
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	BulkVcCrossConnTemplate	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	Type adminStatus autoCreation autoIncrementVpi autoSelectVci instanceKey lineRXTD lineTXTD lineVci lineVpi name templateCategory templateDescription templateName trunkRXTD trunkStartVci trunkTXTD trunkVpi
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	BulkVideoCrossConn	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	Type adminStatus instanceKey name otherVciTemplateId vciRange videoCrossConnTemplateId vpi

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	CardGroup	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	cardType cgId name
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	CommunityName	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	communityAccess communityAccessListIndex name
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	DHCPGroup	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	bootFile bootServer clientMatchLength clientMatchOffset clientMatchString defLeaseTime defRouter domain domainName index maxLeaseTime minLeaseTime name priNameServer secNameServer vendorMatchLength vendorMatchOffset vendorMatchString

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	DHCPHost	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	IPAddr1 IPAddr2 IPAddr3 IPAddr4 bootFile bootServer clientId defLeaseTime defRouter domain domainName hardwareAddr index maxLeaseTime minLeaseTime name priNameServer secNameServer
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	DHCPServDefault	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	leaseTime maxLeaseTime minLeaseTime name reserveEnd reserveStart

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	DHCPSubnet	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	bootFile bootServer defLeaseTime defRouter domain domainName externalServer index maxLeaseTime minLeaseTime name netmask network priNameServer range1End range1Start range2End range2Start range3End range3Start range4End range4Start secNameServer stickyAddr subnetGroup
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	DslAlarmProfile	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	name threshold15MinLofs threshold15MinLols threshold15MinLoss
GponPortPhysical	GponAllocIdProfile	MALC319 MALC719	compensated gponPortId guaranteedBw oltLgId onuId trafficClass

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MalcGponCard Mx4GponCard Mx8GponCard	GponOltPhysical	MALC319 MALC719 MXK 819 MXK 823 MXK 319	ackTimeout allocatedAssuredBw allocatedBestEffortBw allocatedCbrBw allocatedCompensatedCbrBw allocatedNonAssuredBw allocatedUbrBw allocCycleLength appendDefaultName autoLearn availAllocIds availDbalAllocIds berSdThreshold berSfThreshold configKeyExchangeInterval dbaMode driftControlInterval driftControlLimit dsBerInterval eqDMeasureCycles fecMode fecRequest gemBlockSize guardBitCount ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId keyExchange lastCacRc lineGroupId loamAlpha lofAlpha losAlpha lpDesc maxAllocIds maxDbalAllocIds maxOnuResponseTime maxRtPropagationDelay minOnuResponseTime minRtPropagationDelay minUsAlloc name opticTemperature opticVoltage opticTxBiasCurrent opticTxPower opticStatus Continued at next page...

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MalcGponCard Mx4GponCard Mx8GponCard	GponOltPhysical	MALC319 MALC719 MXK 819 MXK 823	plsMaxAllocSize powerLevel preassignedEqD rogueOnuDetect scrambler totalAvailableBw totalAvailableCompensatedCbrBw totalGemPorts updateLogicalNames usBerInterval usedAllocIds usedDbalAllocIds

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
GponOltPhysical	GponOnuPhysical (to be continued)	MALC319 MALC719 MXK 819 MXK 823	autoLearn autoUpgrade dsBerInterval dsRemoteBip8Errors errorMessages fecCorrBytes fecCorrCodeWords fecUncorrCodeWords fullSerialNo fullSerialNoHex genericProfile gponOnuPassword ifHCInBroadcastPkts ifHCInMulticastPkts ifHCInOctets ifHCInUcastPkts ifHCOutBroadcastPkts ifHCOutMulticastPkts ifHCOutOctets ifHCOutUcastPkts ifInDiscards ifInErrors ifInUnknownProtos ifOutDiscards ifOutErrors ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId lineGroupId lldpMedList lpDesc meProfile name omciConfigFile ontImageVersionActive ontImageVersionStandby ontIpAddress ontMacAddress ontModel ontPhysicalTraps ontTraps ontVersion onuAdded onuLineStatusTraps opticRssi portDesc Continued at next page...

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
GponOltPhysical	GponOnuPhysical (continued)	MXK 819 MXK 823 MXK 319	powerLevel serialNoVendorId serialNoVendorSpecific serialNoVendorSpecificHex totalRxCodeWords unreceivedBursts updateLogicalNames usBerInterval useRegId usBip8Errors usRemoteBip8Errors usRxPowerMonitoring usRxPowerHighThreshold usRxPowerLowThreshold windowIndications
GponOnuPhysical	GponPortPhysical	MALC319 MALC719 MXK 819 MXK 823	direction encrypted ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId lineGroupId lpDesc multicast name trafficProfileTemplate
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	Hdsl2ConfigProfile	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	decoderCoeffA decoderCoeffB frameSyncWord name stuffBits transmitPowerBackoffMode
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	IpConfig	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	defaultTtl forwarding name rdIndex

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	IpDnsResolverConfig	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	domainName firstNameServer name queryOrder rdIndex secondNameServer thirdNameServer
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	IpRoutingDomain	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	name rdIndex
MALC19Device MALC23Device MALC319Device Mx19Device Mx23Device	McastControlLists	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	controlId ipAddress listType name
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	MTAC	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	interfaceIndex mtacIndex name testId testMode
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	MulticastMap	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	Type ifIndex instanceKey lgId multicastAddress multicastMapIndex name vci videoDomain vpi
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	NTPConfig	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	Type dayLightSavingTime instanceKey localTimeZone name primaryServerIP secondaryServerIP

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	Ping	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	Type adminStatus dataSize instanceKey name pingIndex probeCount targetAddress timeOut
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	TrapConfig	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	ackedSequenceNumber adminStatus allowedCategories communityName destination name resendSequenceNumber sequenceNumber severity
MALC19Device MALC23Device MALC319Device	UlcConfig	MALC 319 MALC 719 MALC 723	Type instanceKey name portIndex portType shelf slot trapEnable
Raptor100Device Raptor50Device Mx19Device Mx23Device	UnnumberedIpRef	Raptor 100 Raptor 50 MXK 819 MXK 823	ipIfIndex ipIndex name
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	VideoCrossConn	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	Type hiLgid instanceKey lowLgid name videoCrossConnAdminStatus videoCrossConnHighIfIndex videoCrossConnHighVci videoCrossConnHighVpi videoCrossConnIndex videoCrossConnLowIfIndex videoCrossConnLowVci videoCrossConnLowVpi

Table 3: MALC/MXK/Raptor device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Raptor100Device Raptor50Device	VideoSystemConfig	Raptor 100 Raptor 50	Type defaultChannelAddress epgServerAddress instanceKey name noChannelAddress ntpServerAddress stbServerAddress videoSystemConfigIndex
MALC19Device MALC23Device MALC319Device Mx19Device Mx23Device	VoipServer	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	Type addr addrType applIndex calleeReqTimer callerReqTimer calleeSpecifyRefresher callerSpecifyRefresher domainName headerMethod instanceKey interDigitTimeOut invite ipTos name protocol register rtcpEnabled rtcpPktInterval sendCallProceedingTone serverIdType serverIndex sessionExpiration sessionMinSE sessionTimer udpPortNumber dtmfMode
MALC19Device MALC23Device MALC319Device Mx19Device Mx23Device	VoipSystemConfig	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	Type domainName instanceKey interDigitTimeOut ipTos name protocol rtcpEnabled rtcpPktInterval sendCallProceedingTone

Raptor 100 and Raptor 50 device configuration requests

The following table contains the request element attributes associated with Raptor100Device and Raptor50Device parents.

Table 4: Raptor 100 and Raptor 50 device configuration requests

Parent	Object	Supported	Request element attribute
Raptor100Device Raptor50Device	AccessList	Raptor 100 Raptor 50	accessListIndex name
Raptor100Device Raptor50Device	AdslAlarmProfile	Raptor 100 Raptor 50	atucInitFailureTrapEnable atucThresh15MinESs atucThresh15MinLofs atucThresh15MinLols atucThresh15MinLoss atucThresh15MinLprs atucThreshFastRateDown atucThreshFastRateUp atucThreshInterleaveRateDown atucThreshInterleaveRateUp aturThresh15MinESs aturThresh15MinLofs aturThresh15MinLoss aturThresh15MinLprs aturThreshFastRateDown aturThreshFastRateUp aturThreshInterleaveRateDown aturThreshInterleaveRateUp name

Table 4: Raptor 100 and Raptor 50 device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Raptor100Device Raptor50Device	AdslConfigProfile	Raptor 100 Raptor 50	atucChanFastMaxTxRate atucChanFastMinTxRate atucChanInterleaveMaxTxRate atucChanInterleaveMinTxRate atucChanMaxInterleaveDelay atucDownshiftSnrMgn atucMaxSnrMgn atucMinDownshiftTime atucMinSnrMgn atucMinTxThreshAlarm atucMinUpshiftTime atucRateChanRatio atucRateMode atucReachExtendedAdsl2 atucTargetSnrMgn atucUpshiftSnrMgn aturChanFastMaxTxRate aturChanFastMinTxRate aturChanInterleaveMaxTxRate aturChanInterleaveMinTxRate aturChanMaxInterleaveDelay aturDownshiftSnrMgn aturMaxSnrMgn aturMinDownshiftTime aturMinSnrMgn aturMinTxThreshAlarm aturMinUpshiftTime aturRateChanRatio aturRateMode aturTargetSnrMgn aturUpshiftSnrMgn name
Raptor100Device Raptor50Device	ATMTrafficDesc	Raptor 100 Raptor 50	cacDivider name serviceCategory trafficDescrIndex trafficDescrParam1 trafficDescrParam2 trafficDescrParam3 trafficDescrParam4 trafficDescrParam5 trafficDescrType usageControl vclRate

Table 4: Raptor 100 and Raptor 50 device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Raptor100Device Raptor50Device	ATMTrafficDescTemplate	Raptor 100 Raptor 50	Type cacDivider instanceKey name serviceCategory templateCategory templateDescription templateName trafficDescrIndex trafficDescrParam1 trafficDescrParam2 trafficDescrParam3 trafficDescrParam4 trafficDescrParam5 trafficDescrType usageControl vclRate
Raptor100Device Raptor50Device	ATMVcCrossConn	Raptor 100 Raptor 50	name vcCrossConnAdminStatus vcCrossConnHighIfIndex vcCrossConnHighVci vcCrossConnHighVpi vcCrossConnIndex vcCrossConnLowIfIndex vcCrossConnLowVci vcCrossConnLowVpi
Raptor100Device Raptor50Device	BridgeLogical	Raptor 100 Raptor 50	name ifIndex vpi vci lineGroupId groupIndex vlanId lowerIfIndex adminStatus learnIp learnUnicast fwdUnicast fwdMulticast fwdDefault brdigelfStagStripAndInsert

Table 4: Raptor 100 and Raptor 50 device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Raptor100Device Raptor50Device	BridgePath	Raptor 100 Raptor 50	Type addrType bridgeIfIndex flagControl instanceKey igmpQueryInterval ipOrMacAddr multicastAging name SlanId unicastAging vlanId
Raptor100Device Raptor50Device	BulkMulticastMap	Raptor 100 Raptor 50	Type adminStatus autoIncrementVpi autoSelectVci instanceKey lgId multicastMapTemplateId name numChannels otherVclTemplateId rxTrafficDesc startMulticastAddress startVci txTrafficDesc useTDTmpl videoDomain vpi
Raptor100Device Raptor50Device	BulkStatsCollection	Raptor 100 Raptor 50	Type collectionId collectionInterval includeChildren instanceKey name operStatus statEnabled statInstance statOid
Raptor100Device Raptor50Device	BulkStatsConfig	Raptor 100 Raptor 50	Type collectionEnabled ftpAddress ftpDirPath ftpLogin ftpPassword instanceKey name

Table 4: Raptor 100 and Raptor 50 device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Raptor100Device Raptor50Device	BulkVcCrossConnTemplate	Raptor 100 Raptor 50	Type adminStatus autoCreation autoIncrementVpi autoSelectVci instanceKey lineRXTD lineTXTD lineVci lineVpi name templateCategory templateDescription templateName trunkRXTD trunkStartVci trunkTXTD trunkVpi
Raptor100Device Raptor50Device	BulkVideoCrossConn	Raptor 100 Raptor 50	Type adminStatus instanceKey name otherVciTemplateId vciRange videoCrossConnTemplateId vpi
Raptor100Device Raptor50Device	CardGroup	Raptor 100 Raptor 50	cardType cgId name
Raptor100Device Raptor50Device	CommunityName	Raptor 100 Raptor 50	communityAccess communityAccessListIndex name

Table 4: Raptor 100 and Raptor 50 device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Raptor100Device Raptor50Device	DHCPGroup	Raptor 100 Raptor 50	bootFile bootServer clientMatchLength clientMatchOffset clientMatchString defLeaseTime defRouter domain domainName index maxLeaseTime minLeaseTime name priNameServer secNameServer vendorMatchLength vendorMatchOffset vendorMatchString
Raptor100Device Raptor50Device	DHCPHost	Raptor 100 Raptor 50	IPAddr1 IPAddr2 IPAddr3 IPAddr4 bootFile bootServer clientId defLeaseTime defRouter domain domainName hardwareAddr index maxLeaseTime minLeaseTime name priNameServer secNameServer
Raptor100Device Raptor50Device	DHCPServDefault	Raptor 100 Raptor 50	leaseTime maxLeaseTime minLeaseTime name reserveEnd reserveStart

Table 4: Raptor 100 and Raptor 50 device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Raptor100Device Raptor50Device	DHCPSubnet	Raptor 100 Raptor 50	bootFile bootServer defLeaseTime defRouter domain domainName externalServer index maxLeaseTime minLeaseTime name netmask network priNameServer range1End range1Start range2End range2Start range3End range3Start range4End range4Start secNameServer stickyAddr subnetGroup
Raptor100Device Raptor50Device	DslAlarmProfile	Raptor 100 Raptor 50	name threshold15MinLofs threshold15MinLols threshold15MinLoss
Raptor100Device Raptor50Device	Hdsl2ConfigProfile	Raptor 100 Raptor 50	decoderCoeffA decoderCoeffB frameSyncWord name stuffBits transmitPowerBackoffMode
Raptor100Device Raptor50Device	IpConfig	Raptor 100 Raptor 50	defaultTtl forwarding name rdIndex
Raptor100Device Raptor50Device	IpDnsResolverConfig	Raptor 100 Raptor 50	domainName firstNameServer name queryOrder rdIndex secondNameServer thirdNameServer

Table 4: Raptor 100 and Raptor 50 device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Raptor100Device Raptor50Device	IpRoutingDomain	Raptor 100 Raptor 50	name rdIndex
Raptor100Device Raptor50Device	NTPConfig	Raptor 100 Raptor 50	Type dayLightSavingTime instanceKey localTimeZone name primaryServerIP secondaryServerIP
Raptor100Device Raptor50Device	MulticastMap	Raptor 100 Raptor 50	Type ifIndex instanceKey lgId multicastAddress multicastMapIndex name vci videoDomain vpi
Raptor100Device Raptor50Device	Ping	Raptor 100 Raptor 50	Type adminStatus dataSize instanceKey name pingIndex probeCount targetAddress timeOut
Raptor100Device Raptor50Device	TrapConfig	Raptor 100 Raptor 50	ackedSequenceNumber adminStatus allowedCategories communityName destination name resendSequenceNumber sequenceNumber severity
Raptor100Device Raptor50Device	UnnumberedIpRef	Raptor 100 Raptor 50	enabled ipIfIndex name

Table 4: Raptor 100 and Raptor 50 device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Raptor100Device Raptor50Device	VideoCrossConn	Raptor 100 Raptor 50	Type hiLgid instanceKey lowLgid name videoCrossConnAdminStatus videoCrossConnHighIfIndex videoCrossConnHighVci videoCrossConnHighVpi videoCrossConnIndex videoCrossConnLowIfIndex videoCrossConnLowVci videoCrossConnLowVpi
Raptor100Device Raptor50Device	VideoSystemConfig	Raptor 100 Raptor 50	Type defaultChannelAddress epgServerAddress instanceKey name noChannelAddress ntpServerAddress stbServerAddress videoSystemConfigIndex

Sechtor 100A device configuration requests

The following table contains the request element attributes associated with Sechtor 100A Device parents.

Table 5: Sechtor 100A device configuration requests

Parent	Object	Supported	Request element attribute
S100AE1Device S100AT1Device	AccessList	Sechtor 100A E1 Sechtor 100A T1	accessListIndex name
S100AE1Device S100AT1Device	ATMTrafficDesc	Sechtor 100A E1 Sechtor 100A T1	cacDivider name serviceCategory trafficDescrIndex trafficDescrParam1 trafficDescrParam2 trafficDescrParam3 trafficDescrParam4 trafficDescrParam5 trafficDescrType usageControl vclRate

Table 5: Sechtor 100A device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
S100AE1Device S100AT1Device	ATMTrafficDescTemplate	Sechtor 100A E1 Sechtor 100A T1	cacDivider instanceKey name serviceCategory templateCategory templateDescription templateName trafficDescrIndex trafficDescrParam1 trafficDescrParam2 trafficDescrParam3 trafficDescrParam4 trafficDescrParam5 trafficDescrType Type usageControl vclRate
S100AE1Device S100AT1Device	BridgeLogical	Sechtor 100A E1 Sechtor 100A T1	name ifIndex vpi vci lineGroupId groupIndex vlanId lowerIfIndex adminStatus learnIp learnUnicast fwdUnicast fwdMulticast fwdDefault brdigeIfStagStripAndInsert
S100AE1Device S100AT1Device	BridgePath	Sechtor 100A E1 Sechtor 100A T1	Type addrType bridgeIfIndex flapControl instanceKey igmpQueryInterval ipOrMacAddr multicastAging name slanId unicastAging vlanId
S100AE1Device S100AT1Device	CardGroup	Sechtor 100A E1 Sechtor 100A T1	cardType cgId name

Table 5: Sechtor 100A device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
S100AE1Device S100AT1Device	CommunityName	Sechtor 100A E1 Sechtor 100A T1	communityAccess communityAccessListIndex name
S100AE1Device S100AT1Device	IpConfig	Sechtor 100A E1 Sechtor 100A T1	defaultTtl forwarding name rdIndex
S100AE1Device S100AT1Device	IpDnsResolverConfig	Sechtor 100A E1 Sechtor 100A T1	domainName firstNameServer name queryOrder rdIndex secondNameServer thirdNameServer
S100AE1Device S100AT1Device	IpRoutingDomain	Sechtor 100A E1 Sechtor 100A T1	name rdIndex
S100AE1Device S100AT1Device	NTPConfig	Sechtor 100A E1 Sechtor 100A T1	Type dayLightSavingTime instanceKey localTimeZone name primaryServerIP secondaryServerIP
S100AE1Device S100AT1Device	Ping	Sechtor 100A E1 Sechtor 100A T1	Type adminStatus dataSize instanceKey name pingIndex probeCount targetAddress timeOut

Table 5: Sector 100A device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
S100AE1Device S100AT1Device	SonetOC3Physical	Sector 100A E1 Sector 100A T1	activationTime circuitIdentifier clockExternalRecovery clockTransmitSource ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId lineCoding lineGroupId lineGroupKey lineType loopbackCfg lpDesc mediumExtLineScrmEnabled mediumExtScrambleEnabled mediumType name pathWidth
S100AE1Device S100AT1Device	SubscriberPort	Sector 100A E1 Sector 100A T1	cpeType deviceRole id lgId maxAllowedLineRate maxCapableLineRate name nextVoiceConnectionIndex providerId subscriberKey

Table 5: Sechtor 100A device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
S100AT1Device	T1Physical	Sechtor 100A T1	activationTime cellScramble channelNumber channelization circuitIdentifier clockSourceEligibility cosetPolynomial csuLineLength fdl groupIndex ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId isLineUsed lineCoding lineGroupId lineGroupKey lineLength lineStatusChangeTrapEnable lineType loopbackConfig lpDesc mode name protocolEmulation sendCode signalMode timeslotAssignment transmitClockSource
S100AE1Device S100AT1Device	TrapConfig	Sechtor 100A E1 Sechtor 100A T1	ackedSequenceNumber adminStatus allowedCategories communityName destination name resendSequenceNumber sequenceNumber severity

Z-Edge 64 device configuration requests

The following table contains the request element attributes associated with Z-Edge 64 Device parents.

Table 6: Z-Edge 64 device configuration requests

Parent	Object	Supported	Request element attribute
ZEdgeBH2ADevice ZEdgeBH2ADevice	AccessList	Z-Edge 64 BH2A ZEdge 64 H2A	accessListIndex name
ZEdgeBH2ADevice ZEdgeH2ADevice	ATMTrafficDesc	Z-Edge 64 BH2A Z-Edge 64 H2A	cacDivider name serviceCategory trafficDescrIndex trafficDescrParam1 trafficDescrParam2 trafficDescrParam3 trafficDescrParam4 trafficDescrParam5 trafficDescrType usageControl vclRate
ZEdgeBH2ADevice ZEdgeH2ADevice	ATMTrafficDescTemplate	Z-Edge 64 BH2A Z-Edge 64 H2A	Type cacDivider instanceKey name serviceCategory templateCategory templateDescription templateName trafficDescrIndex trafficDescrParam1 trafficDescrParam2 trafficDescrParam3 trafficDescrParam4 trafficDescrParam5 trafficDescrType usageControl vclRate
ZEdgeBH2ADevice ZEdgeH2ADevice	BridgeLogical	Z-Edge 64 BH2A Z-Edge 64 H2A	name ifIndex vpi vci lineGroupId groupIndex vlanId lowerIfIndex adminStatus learnIp learnUnicast fwdUnicast fwdMulticast fwdDefault brdigelfStagStripAndInsert

Table 6: Z-Edge 64 device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
ZEdgeBH2ADevice ZEdgeH2ADevice	BridgePath	Z-Edge 64 BH2A Z-Edge 64 H2A	Type addrType bridgeIfIndex flapControl instanceKey igmpQueryInterval ipOrMacAddr multicastAging name slanId unicastAging vlanId
ZEdgeBH2ADevice ZEdgeH2ADevice	CardGroup	Z-Edge 64 BH2A ZEdge 64 H2A	cardType cgId name
ZEdgeBH2ADevice ZEdgeH2ADevice	CommunityName	Z-Edge 64 BH2A ZEdge 64 H2A	communityAccess communityAccessListIndex name
ZEdgeBH2ADevice ZEdgeH2ADevice	DHCPGroup	Z-Edge 64 BH2A Z-Edge 64 H2A	bootFile bootServer clientMatchLength clientMatchOffset clientMatchString defLeaseTime defRouter domain domainName index maxLeaseTime minLeaseTime name priNameServer secNameServer vendorMatchLength vendorMatchOffset vendorMatchString

Table 6: Z-Edge 64 device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
ZEdgeBH2ADevice ZEdgeH2ADevice	DHCPHost	Z-Edge 64 BH2A Z-Edge 64 H2A	IPAddr1 IPAddr2 IPAddr3 IPAddr4 bootFile bootServer clientId defLeaseTime defRouter domain domainName hardwareAddr index maxLeaseTime minLeaseTime name priNameServer secNameServer
ZEdgeBH2ADevice ZEdgeH2ADevice	DHCPServDefault	Z-Edge 64 BH2A Z-Edge 64 H2A	leaseTime maxLeaseTime minLeaseTime name reserveEnd reserveStart
ZEdgeBH2ADevice ZEdgeH2ADevice	DHCPSubnet	Z-Edge 64 BH2A Z-Edge 64 H2A	bootFile bootServer defLeaseTime defRouter domain domainName externalServer index maxLeaseTime minLeaseTime name netmask network priNameServer range1End range1Start range2End range2Start range3End range3Start range4End range4Start secNameServer stickyAddr subnetGroup

Table 6: Z-Edge 64 device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
ZEdgeBH2ADevice ZEdgeH2ADevice	DslAlarmProfile	Z-Edge 64 BH2A Z-Edge 64 H2A	name threshold15MinLofs threshold15MinLols threshold15MinLoss
ZEdgeBH2ADevice ZEdgeH2ADevice	Hdsl2ConfigProfile	Z-Edge 64 BH2A Z-Edge 64 H2A	decoderCoeffA decoderCoeffB frameSyncWord name stuffBits transmitPowerBackoffMode
ZEdgeBH2ADevice ZEdgeH2ADevice	IpConfig	Z-Edge 64 BH2A Z-Edge 64 H2A	defaultTtl forwarding name rdIndex
ZEdgeBH2ADevice ZEdgeH2ADevice	IpDnsResolverConfig	Z-Edge 64 BH2A Z-Edge 64 H2A	domainName firstNameServer name queryOrder rdIndex secondNameServer thirdNameServer
ZEdgeBH2ADevice ZEdgeH2ADevice	IpRoutingDomain	Z-Edge 64 BH2A Z-Edge 64 H2A	name rdIndex
ZEdgeBH2ADevice ZEdgeH2ADevice	NatConfig	Z-Edge 64 BH2A Z-Edge 64 H2A	clearBindings name tcpTimeout udpTimeout
ZEdgeBH2ADevice ZEdgeH2ADevice	PAT	Z-Edge 64 BH2A Z-Edge 64 H2A	Type instanceKey localAddr localPort name patBindIndex portType publicAddr publicPort
ZEdgeBH2ADevice ZEdgeH2ADevice	Ping	Z-Edge 64 BH2A Z-Edge 64 H2A	Type adminStatus dataSize instanceKey name pingIndex probeCount targetAddress timeOut

Table 6: Z-Edge 64 device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
ZEdgeBH2ADevice ZEdgeH2ADevice	PppAuth	Z-Edge 64 BH2A Z-Edge 64 H2A	Type chapName chapSecret id instanceKey ipIfIndex lgId name papPassword papPeerId protocol status subId
ZEdgeBH2ADevice ZEdgeH2ADevice	PppLogical	Z-Edge 64 BH2A Z-Edge 64 H2A	Type callMode frameType hierarchyString ifAdminStatus ifIndex ifType ifTypeExtension instanceKey lineGroupId lineGroupKey lowerIfIndex name numChannels vci vpi

Table 6: Z-Edge 64 device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
ZEdgeBH2ADevice ZEdgeH2ADevice	PppLcp	Z-Edge 64 BH2A Z-Edge 64 H2A	Type accmEnable acfcEnable authId cbDialString cbEnable cbType fcsAltEnable fcsSize ifIndex initialMRU instanceKey lqrPeriod lqrStatus magicNumEnable magicNumber maxConfigRetries maxFailRetries maxPad maxTermRetries mruEnable name numModeEnable pfcEnable qualityProtocol receiveAuthEnable receiveAuthProtocol restartTimer rxACCMMap sdpEnable sendAuthEnable sendAuthIdentity sendAuthProtocol txACCMMap
ZEdgeBH2ADevice ZEdgeH2ADevice	PppNcp	Z-Edge 64 BH2A Z-Edge 64 H2A	Type compProtocol compSlotId instanceKey ipAddrEnable ipIfIndex maxConfigRetries maxFailRetries maxSlotId maxTermRetries name restartTimer

Table 6: Z-Edge 64 device configuration requests (Continued)

Parent	Object	Supported	Request element attribute
ZEdgeBH2ADevice ZEdgeH2ADevice	SdslConfigProfile	Z-Edge 64 BH2A Z-Edge 64 H2A	connectMode fixBitRate framerType lineRate name ntr powerScale
ZEdgeBH2ADevice ZEdgeH2ADevice	ShdslConfigProfile	Z-Edge 64 BH2A Z-Edge 64 H2A	clockOffset decoderCoeffA decoderCoeffB fixBitRate frameSync lineRate name ntr powerScale repeaterId standard startupMargin transmitPowerBackoffMode wireMode
ZEdgeBH2ADevice ZEdgeH2ADevice	SubscriberPort	Z-Edge 64 BH2A Z-Edge 64 H2A	cpeType deviceRole id lgId maxAllowedLineRate maxCapableLineRate name nextVoiceConnectionIndex providerId subscriberKey
ZEdgeBH2ADevice ZEdgeH2ADevice	TrapConfig	Z-Edge 64 BH2A Z-Edge 64 H2A	ackedSequenceNumber adminStatus allowedCategories communityName destination name resendSequenceNumber sequenceNumber severity
ZEdgeBH2ADevice ZEdgeH2ADevice	UnnumberedIpAlias	Z-Edge 64 BH2A Z-Edge 64 H2A	Type enabled instanceKey ipIfIndex name

Card configuration requests

The following table lists the request element attributes associated with specific cards.

Table 7: Card configuration requests

Parent	Object	Supported	Request element attribute
Shelf	AdslCard DS3Card DS3UnchannelizedCard DSLCard E1Atm24PortCard E3Aal52Atm4PortCard E3Atm4PortCard E3Aal5Atm4PortCard E3Atm4PortPolyCard E1Tdm63PortCard EtherCard InfoServCard OC3Aal5MultiModeCard OC3Aal52MultiModeCardOC3 Aal52SingleModeCard OC3Card OC3MultiModeCard OC3Aal5SingleModeCard OC3MultiModeCard OC3PolyCard T1Atm24PortCard T1Tdm64PortCard T3AtmPortPolyCard T3Aal5Atm4PortCard T3Atm4PortCard T3Atm4PortPloyCard VASPCard	BAN	adminStatsEnable adminStatus cardGroupKey cardInterfaceType cardLineType cardLineVoltage cardProvisionState cardReboot cardShelfIndex cardVpiVciRange cardZhoneType identification instanceKey mfgBootRevision mfgCLEICode mfgRevisionCode mfgSerialNumber name processorType slotIndex swAdmin swPreviousVersion swRunningVersion sysCardSpecificSwFileName sysCardSwEnable sysCardSwType sysCardSwUpgradeEnable sysCardUpgradeSwFileName totalMemory wettingCurrent The following attributes apply only to cards that support ATM: atmConfiguration maxAtmConnections maxVci maxVpi vpiState The following attributes apply only to cards that support redundancy isActive isRedundant

Table 7: Card configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Shelf	AdslAC5Malc24PortCard	MALC 319	Type
	AdslAC5BMalc24PortCard	MALC 719	adminStatsEnable
	AdslAC6Malc32PortSpltCard	MALC 723	adminStatus
	AdslAC6MalcCard	Raptor 319	cardGroupKey
	AdslAC6MalcPotsCard	Raptor 719	cardInterfaceType
	AdslAC6MalcSpltCard	Raptor 723	cardLineType
	AdslPots48PortCard		cardProvisionState
	Adsl24PortMalcCard		cardReboot
	Dsl1Ma8PortCard		cardShelfIndex
	E1Ma8PortCard		cardZhoneType
	E1MaF8PortCard		identification
	E1UplinkAtm8PortCard		instanceKey
	E1FUplinkAtm8PortCard		mfgBootRevision
	E3UplinkAtm4PortCard		mfgCLEICode
	E3FUplinkAtm4PortCard		mfgRevisionCode
	E1V52Ma16PortCard		mfgSerialNumber
	E1V52Uni16PortCard		name
	E1Oc12GigEUplinkCard		processorType
	Gshdsl4Wire24PortCard		slotIndex
	GshdslCard		slotsRequired
	GshdslLifelinePotsCard		swAdmin
	MalcDslAtm32PortCard		swPreviousVersion
	MALCOltCard		swRunningVersion
	MtacCard		sysCardSpecificSwFileName
	MtacFcE1Card		sysCardSwEnable
	MtacFcT1Card		sysCardSwType
	MtacE1Card		sysCardSwUpgradeEnable
	MtacT1Card		sysCardUpgradeSwFileName
	OC3UplinkAtmCard		totalMemory
	OC3FUplinkAtmCard		The following attributes apply only to cards that support ATM:
	OC3UplinkAtm2PortCard		
	Pots24PortCard		
	Pots24Port900Card		
	T1Gr303Ma16PortCard		
	T1Gr303Uni16PortCard		atmConfiguration
	T1Ma8PortCard		maxAtmConnections
	T1MaF8PortCard		maxVci
	T1UplinkAtm8PortCard		maxVpi
	T1FUplinkAtm8PortCard		vpiState
	T3UplinkAtm4PortCard		The following attributes apply only to cards that support redundancy
	T3FUplinkAtm4PortCard		
	T1Oc12GigEUplinkCard		
	ULCCard		
	VdslCard		

Table 7: Card configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Shelf	MxUplink2Tg8gCard	MXK 819	Type
	MxUplink8gCard	MXK 823	adminStatsEnable
	Mx20ActiveEthCard		adminStatus
	Mx4GponCard		cardGroupKey
	Mx4NewGponCard		cardInterfaceType
	Mx8GponCard		cardLineType
			cardProvisionState
			cardReboot
			cardShelfIndex
			cardZhoneType
			identification
			instanceKey
			mfgBootRevision
			mfgCLEICode
			mfgRevisionCode
			mfgSerialNumber
			name
			processorType
			slotIndex
			slotsRequired
			swAdmin
			swPreviousVersion
			swRunningVersion
			sysCardSpecificSwFileName
			sysCardSwEnable
			sysCardSwType
			sysCardSwUpgradeEnable
			sysCardUpgradeSwFileName
			totalMemory
			The following attributes apply only to cards that support ATM:
			atmConfiguration
			maxAtmConnections
			maxVci
			maxVpi
			vpiState
			The following attributes apply only to cards that support redundancy
			isActive
			isRedundant

Table 7: Card configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Shelf	AtmConnAvailable	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	allocated available
Shelf	MtacE1Card	MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	adminStatsEnable adminStatus interfaceIndex mtacIndex name testMode
Shelf	MtacT1Card	MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	adminStatsEnable adminStatus interfaceIndex mtacIndex name testMode

Table 7: Card configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Shelf	R50GshdslCard	Raptor50Device	Type adminStatsEnable adminStatus atmConfiguration cardGroupKey cardInserted cardInterfaceType cardLineType cardProvisionState cardShelfIndex cardZhoneType identification instanceKey ipEnabled isActive isRedundant maxAtmConnections maxVci maxVpi mfgBootRevision mfgCLEICode mfgRevisionCode mfgSerialNumber name processorType slotIndex slotsRequired swAdmin swPreviousVersion swRunningVersion sysCardSpecificSwFileName sysCardSwEnable sysCardSwType sysCardSwUpgradeEnable sysCardUpgradeSwFileName totalMemory vpiState

Table 7: Card configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Shelf	Raptor100Device Malc100T1ImaCard Malc100GshdslCard	Raptor 100	adminStatsEnable adminStatus cardInterfaceType cardLineType cardProvisionState cardShelfIndex cardZhoneType identification ipEnabled name slotIndex slotsRequired swAdmin swPreviousVersion swRunningVersion sysCardSpecificSwFileName sysCardSwEnable sysCardSwType sysCardSwUpgradeEnable sysCardUpgradeSwFileName
Shelf	S100AE1Card	Sechtor 100A E1	adminStatsEnable adminStatus cardInterfaceType cardLineType cardProvisionState cardShelfIndex cardZhoneType identification ipEnabled name slotIndex slotsRequired swAdmin swPreviousVersion swRunningVersion sysCardSpecificSwFileName sysCardSwEnable sysCardSwType sysCardSwUpgradeEnable sysCardUpgradeSwFileName

Table 7: Card configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Shelf	S100AT1Card	Sechtor 100A T1	adminStatsEnable adminStatus cardInterfaceType cardLineType cardProvisionState cardShelfIndex cardZhoneType identification ipEnabled name slotIndex slotsRequired swAdmin swPreviousVersion swRunningVersion sysCardSpecificSwFileName sysCardSwEnable sysCardSwType sysCardSwUpgradeEnable sysCardUpgradeSwFileName
Shelf	ZEdgeBH2ACard	Z-Edge 64 BH2A	adminStatsEnable adminStatus cardInterfaceType cardLineType cardProvisionState cardShelfIndex cardZhoneType identification ipEnabled name slotIndex slotsRequired swAdmin swPreviousVersion swRunningVersion sysCardSpecificSwFileName sysCardSwEnable sysCardSwType sysCardSwUpgradeEnable sysCardUpgradeSwFileName

Table 7: Card configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Shelf	ZEdgeH2ACard	Z-Edge 64 H2A	adminStatsEnable adminStatus cardInterfaceType cardLineType cardProvisionState cardShelfIndex cardZhoneType identification ipEnabled name slotIndex slotsRequired swAdmin swPreviousVersion swRunningVersion sysCardSpecificSwFileName sysCardSwEnable sysCardSwType sysCardSwUpgradeEnable sysCardUpgradeSwFileName

Table 7: Card configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MALC19Device MALC23Device MALC319Device	UlcCard	MALC 319 MALC 719 MALC 723	Type adminStatsEnable adminStatus atmConfiguration cardGroupKey cardInserted cardInterfaceType cardLineType cardProvisionState cardReboot cardShelfIndex cardZhoneType identification instanceKey isActive isRedundant maxAtmConnections maxVci maxVpi mfgBootRevision mfgCLEICode mfgRevisionCode mfgSerialNumber name processorType slotIndex slotsRequired swAdmin swPreviousVersion swRunningVersion sysCardSpecificSwFileName sysCardSwEnable sysCardSwType sysCardSwUpgradeEnable sysCardUpgradeSwFileName totalMemory vpiState

Physical configuration requests

The following table shows the request element attributes associated with specific physical objects.

Table 8: Physical configuration requests

Parent	Object	Supported	Request element attribute
AdslCard	Aal5ProxyPhysical	BAN	Type
DS3Card		MALC 319	activationTime
DSLCard		MALC 719	ifAdminStatus
OC3PolyCard		MALC 723	ifIndex
OC3MultiModeCard		Raptor 319	ifType
OC3Aal5SingleModeCard		Raptor 719	ifTypeExtension
OC3Aal5MultiModeCard		Raptor 723	ifTypePortId
OC3Aal52SingleModeCard			instanceKey
OC3Aal52MultiModeCard			isActive
E3Atm4PortPolyCard			lineGroupId
E3Aal5Atm4PortCard			lineGroupKey
T3Atm4PortPolyCard			lpDesc
T3Aal5Atm4PortCard			name
T3Aal52Atm4PortCard			portDesc
T1Ima8PortCard			
E1Ima8PortCard			
T1ImaF8PortCard			
E1ImaF8PortCard			
T3UplinkAtm4PortCard			
T3FUplinkAtm4PortCard			
E3UplinkAtm4PortCard			
E3FUplinkAtm4PortCard			
OC3UplinkAtm2PortCard			
OC3FUplinkAtmCard			
T1UplinkAtm8PortCard			
T1FUplinkAtm8PortCard			
E1UplinkAtm8PortCard			
E1FUplinkAtm8PortCard			

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
AdslCard	AdslPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723	adminUpAlarmEnable activationTime alarmProfName annexMModeEnabled appendDefaultName byPassRelayMaxDuration channelMode configProfName deltCmd dmtConfigMode ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId isAssigned lineGroupId lineStatusChangeTrapEnable lpDesc maxDownstreamToneIndex maxUpstreamToneIndex minDownstreamToneIndex minUpstreamToneIndex name ntrModeEnabled portDesc potsLbfRelayStatus reserveState seltCable seltCmd seltDuration seltGaugeExpected transmissionMode trellisModeEnabled
AdslPhysical	AdslFastPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723	activationTime ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId lineGroupId lineGroupKey lpDesc name

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
AdslPhysical	AdslInterleavedPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 50 Raptor 719 Raptor 723	activationTime ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId lineGroupId lineGroupKey lpDesc name portDesc reserveState

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Card	DSLPhysical (to be continued)	MALC319	activationTime
		MALC719	adminUpAlarmEnable
		MALC723	alarmProfName
		Raptor 319	autoDiscovery
		Raptor 723	bConstellation
		ethX3400	bDataRate
		ethX3300	bondGroupIdx
			bondGroupLgIdx
			configProfName
			configUnitMode
			errThrMonClrInt
			errThrMonEn
			errThrMonInt
			errThrMonNotEn
			ifAdminStatus
			ifIndex
			ifType
			ifTypeExtension
			ifTypePortId
			isActive
			isAssigned
			lpDesc
			lineGroupId
			lineGroupKey
			lineStatusTrapChangeEnable
			lineType
			mntEndTime
			mntMode
			mntStartTime
			name
			networkTime

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
Card	DSLPhysical (continued)	MALC319 MALC719 MALC723 Raptor 319 Raptor 723 ethX3400 ethX3300	portDesc portDescriptionIndex redundancyParam1 reserveState snrMgnCrossingTrapEnable snrMonInt targetWorseCaseSnrMgn targetCurrCondMode targetCurrCondSnrMgn targetWorseCaseMode thrMaxSnrMgnDelta thrMinSnrMgn
Card	MLPPPBondPhysical	MALC-EFM-T 1/E1-24 Any Malc uplink running SW 1.16.3.x	appendDefaultName ifIndex ifType ifTypeExtension ifAdminStatus isActive lpDesc lineGroupId name

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
E1Tdm63PortCard E1V52Ima16PortCard E1V52Uni16PortCard MtacE1Card MtacFcE1Card S100AE1Card	E1Physical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1	activationTime appendDefaultName bondGroupIdx bondGroupLgIdx cellScramble channelNumber channelization circuitIdentifier clockSourceEligibility cosetPolynomial csuLineLength fdl ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId isLineUsed lineCoding lineGroupId lineGroupKey lineLength lineStatusChangeTrapEnable lineType loopbackConfig lpDesc mode name portDesc protocolEmulation sendCode signalMode timeslotAssignment transmitClockSource

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
E1Atm24PortCard	E1ATMPPhysical	BAN	Type
E1Ima8PortCard		MALC 319	activationTime
E1ImaF8PortCard		MALC 719	appendDefaultName
E1UplinkAtm8PortCard		MALC 723	cellScramble
E1FUplinkAtm8PortCard		Raptor 319	channelNumber
E1V52Ima16PortCard		Raptor 719	channelization
E1V52Uni16PortCard		Raptor 723	circuitIdentifier
MALC100T1ImaCard			clockSourceEligibility
			cosetPolynomial
			csuLineLength
			fdl
			ifAdminStatus
			ifIndex
			ifType
			ifTypeExtension
			ifTypePortId
			instanceKey
			isActive
			isLineUsed
			lineCoding
			lineGroupId
			lineGroupKey
			lineLength
			lineStatusChangeTrapEnable
			lineType
			loopbackCfg
			lpDesc
			mode
			name
			portDesc
			protocolEmulation
			sendCode
			signalMode
			timeslotAssignment
			transmitClockSource

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
DS3Card	E3Physical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	Type activationTime appendDefaultName channelization circuitId farEndEquipCode farEndFacilityIdCode farEndFrameIdCode farEndLineIndex farEndLocationIdCode farEndUnitCode ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId instanceKey isActive lineCoding lineGroupId lineGroupKey lineLength lineTrapEnable lineType loopbackCfg lpDesc name remoteLoop sendCode transClockSource

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
E3Aal5Atm4PortCard E3Arm4PortPolyCard	E3UnchannelizedPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor100 Raptor 319 Raptor 719 Raptor 723	activationTime atmFraming channelization circuitId e3Framing farEndEquipCode farEndFacilityIdCode farEndFrameIdCode farEndLineIndex farEndLocationIdCode farEndUnitCode ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId lineCoding lineGroupId lineGroupKey lineLength lineTrapEnable lineType loopbackCfg lpDesc name portDesc remoteLoop scrambleEnabled sendCode transClockSource

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
EtherCard InfoServCard E1UplinkAtm8PortCard E1FUplinkAtm8PortCard E3UplinkAtm4PortCard E3FUplinkAtm4PortCard OC3UplinkAtmCard OC3FUplinkAtmCard OC3UplinkAtm2PortCard T1Ima8PortCard T1ImaF8PortCard T1UplinkAtm8PortCard T1FUplinkAtm8PortCard T3UplinkAtm4PortCard T3FUplinkAtm4PortCard S100AE1Card S100AT1Card ZEdgeBH2ACard ZEdgeH2ACard MxUplink2Tg8gCard MxUplink8gCard	EtherPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge 64 BH2A Z-Edge 64 H2A MXK 819 MXK 823	activationTime aggregationMode appendDefaultName bondGroupIdx bondGroupLgIdx ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId lineGroupId lineGroupKey lpDesc name pauseAdminMode pauseFlowControl portDesc
T1Oc12GigEUplinkCard E1Oc12GigEUplinkCard MxUplink2Tg8gCard MxUplink8gCard Mx20ActiveEthCard	GigEPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	Type activationTime ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId instanceKey isActive lineGroupId lineGroupKey lpDesc name pauseAdminMode portDesc
E1Ima8PortCard E1ImaF8PortCard T1Ima8PortCard T1ImaF8PortCard	ImaPhysicalGroup	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	activationTime ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId lineGroupId lineGroupKey lpDesc name portDesc

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
ZEdgeBH2ACard	IsdnBriPhysical	Z-Edge 64 BH2A	activationTime activationTimer2 ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId lineGroupId lineGroupKey lineLoopBack linePower lineTermClass lpDesc name portDesc
EtherCard InfoservCard E1UplinkAtm8PortCard E1FUplinkAtm8PortCard E3UplinkAtm4PortCard E3FUplinkAtm4PortCard OC3UplinkAtmCard OC3FUplinkAtmCard OC3UplinkAtm2PortCard T1Ima8PortCard T1ImaF8PortCard T1UplinkAtm8PortCard T1FUplinkAtm8PortCard T3UplinkAtm4PortCard T3FUplinkAtm4PortCard S100AE1Card S100AT1Card ZEdgeBH2ACard ZEdgeH2ACard	MAUPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge 64 BH2A Z-Edge 64 H2A	fullDuplex mauClockSource mauDefaultType mauType name negAdminStatus negCapAdvertBits negRemoteFaultAdvert negRestart physicalIfIndex speed

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
OltPhysical	OltOnuPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	Type activationTime ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId instanceKey isActive lineGroupId lineGroupKey lpDesc name oltOnuChurnKey oltOnuPassword oltOnuSerialNum portDesc trapEnable

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MALCOltCard	OltPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	Type activationTime autoLearn bip8 cdrActiveHigh cdrLocation cdrPattern cellScrambling cpeLimit crcRx delimiterPattern delimiterSize hecRxBypass ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId instanceKey isActive lcdAlpha lcdDelta lcdLimit lineGroupId lineGroupKey lpDesc maxPonDistance name oltHec oltLoopback overheadSize portDesc syncBytesClkDivisor txDiscardNonMatchingVpi txSyncBytes utopiaDiscard

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
ZEdgeH2ACard	PotsPhysical	Z-Edge 64 H2A	activationTime ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId impedance lineGroupId lineGroupKey lineType lpDesc maintenanceMode name pcmEncoding portDesc receiveTlp transmitTlp trunkConditioning
InfoServCard T1Ima8PortCard T1ImaF8PortCard E1Ima8PortCard E1ImaF8PortCard MAUPhysical	RS232Physical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	activationTime ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId inFlowType inSpeed isActive lineGroupId lineGroupKey lpDesc name outFlowType outSpeed portType portDesc

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
T1Oc12GigEUpLinkCard E1Oc12GigEUpLinkCard	SonetOc12Physical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	Type activationTime circuitIdentifier clockExternalRecovery clockTransmitSource ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId instanceKey isActive lineCoding lineGroupId lineGroupKey lineType loopbackConfig lpDesc mediumExtLineScrmEnabled mediumExtScrambleEnabled mediumType name pathWidth portDesc
OC3MultiModeCard OC3Aal5SingleModeCard OC3PolyCard	SonetOC3Physical	BAN	activationTime circuitIdentifier clockExternalRecovery clockTransmitSource ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId isActive lineCoding lineGroupId lineGroupKey lineType loopbackCfgr lpDesc mediumExtLineScrmEnabled mediumExtScrambleEnabled mediumType name pathWidth portDesc

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MtacFcT1Card	T1Physical	BAN	activationTime
MtacT1Card		MALC 319	appendDefaultName
S100AT1Card		MALC 719	bondGroupIdx
T1Gr303Ima16PortCard		MALC 723	bondGroupLgIdx
T1Gr303Uni16PortCard		Raptor 319	cellScramble
		Raptor 719	channelization
		Raptor 723	channelNumber
		Sechtor 100A	circuitIdentifier
		T1	clockSourceEligibility
			cosetPolynomial
			csuLineLength
			fdl
			groupIndex
			ifAdminStatus
			ifIndex
			ifTypeExtension
			ifTypePortId
			isLineUsed
			lineCoding
			lineGroupId
			lineGroupKey
			lineLength
			lineStatusChangeTrapEnable
			lineType
			loopbackConfig
			lpDesc
			mode
			name
			sendCode
			signalMode
			timeslotAssignment
			transmitClockSource
			portDesc

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
MALC100T1ImaCard T1Atm24PortCard T1FUplinkAtm8PortCard T1Ima8PortCard T1ImaF8PortCard T1Gr303Ima16PortCard T1ImaF8PortCard T1Tdm64PortCard T1UplinkAtm8PortCard	T1ATMPPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	Type activationTime cellScramble channelNumber channelization circuitIdentifier clockSourceEligibility cosetPolynomial csuLineLength fdl groupIndex ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId imaGroupIdx imaGroupLgIdx instanceKey isActive isLineUsed isUni lineCoding lineGroupId lineGroupKey lineLength lineStatusChangeTrapEnable lineType loopbackConfig lpDesc mode name sendCode signalMode timeslotAssignment transmitClockSource portDesc

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
DS3Card	T3Physical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	Type activationTime channelization circuitId farEndEquipCode farEndFacilityIdCode farEndFrameIdCode farEndLineIndex farEndLocationIdCode farEndUnitCode ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId instanceKey isActive lineCoding lineGroupId lineGroupKey lineLength lineTrapEnable lineType loopbackCfg lpDesc name remoteLoop sendCode transClockSource

Table 8: Physical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
T3Aal5Atm4PortCard T3AtmPortPolyCard	T3UnchannelizedPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor100 Raptor 319 Raptor 719 Raptor 723	activationTime atmFraming channelization circuitId farEndEquipCode farEndFacilityIdCode farEndFrameIdCode farEndLineIndex farEndLocationIdCode farEndUnitCode ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId lineCoding lineGroupId lineGroupKey lineLength lineTrapEnable lineType loopbackCfg lpDesc name remoteLoop scrambleEnabled sendCode transClockSource
VdslCard	VdslPhysical	MALC 319 MALC 719 MALC 723 Raptor100 Raptor 319 Raptor 719 Raptor 723	Type activationTime adminUpAlarmEnable alarmProfName appendDefaultName configProfName ifAdminStatus ifIndex ifType ifTypeExtension ifTypePortId instanceKey isActive isAssigned lineGroupId lineGroupKey lineStatusChangeTrapEnable lpDesc name portDesc reserveState

N2N/EFM/GBond bonded groups

Bond groups are just like physicals with the exception that they can be added manually. Physicals can be assigned to bond groups by assigning ifIndex and lineGroupId of newly created bond group to the bondGroupIdx and bondGroupLgIdx of physical respectively using the following Physical modify operation: E1Physical, EtherPhysical, T1Physical, and DSLPhysical.

Table 9: N2N/EFM/GBond bonded groups

Parent	Object	Support	Request element attribute
Card	EFMBondPhysical	MALC319 MALC719 MALC723 Raptor 319 Raptor 723 ethX3400 ethX3300	appendDefaultName ifIndex ifType ifTypeExtension ifAdminStatus isActive lpDesc lineGroupId name
Card	N2NBondPhysical	MALC319 MALC719 MALC723 Raptor 319 Raptor 723 ethX3400 ethX3300	appendDefaultName ifIndex ifType ifTypeExtension ifAdminStatus isActive lpDesc lineGroupId name
Card	GBondPhysical	MALC319 MALC719 MALC723 Raptor 319 Raptor 723 ethX3400 ethX3300	appendDefaultName ifIndex ifType ifTypeExtension ifAdminStatus isActive lpDesc lineGroupId name

Table 9: N2N/EFM/GBond bonded groups

Parent	Object	Support	Request element attribute
Card	BondGroupAlarmConfig	MALC 319	adminUpAlarm
		MALC 719	bitRateThreshold
		MALC 723	bitRateThresholdValue
		Raptor 319	bitRateThresholdHoldTime
		Raptor 723	ifIndex
		ethX3400	lineGroupId
		ethX3300	statusTrapEnable

GPON traffic profile (GTP) for OMCI

Bandwidth allocation on a T-cont is defined in the GPON traffic profile. Multiple GEM ports can share one T-cont by enabling shared feature in the associated GPON traffic profile. GPON traffic profile IDs have to be specified when creating a GEM port.

Table 10: GPON Traffic Profile

Parent	Object	Support	Request element attribute
Device	GponTrafficProfile	MALC	compensated
		MXK 819	dba
		MXK 823	dbaAssuredUsBw
		MX280	dbaFixedUsCbrBw
		MXP280	dbaFixedUsUbrBw
		MXP280 TP RJ45	dbaMaxUsBw
			extraUsBwType
			guaranteedUpBw
			shared
			trafficClass

OMCI profiles for Smart OMCI

OMCI profiles include ME profile, generic profile, and specific profile.

Table 11: OMCI profiles

Parent	Object	Support	Request element attribute
GponOltPhysical	GponOnuPhysical	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	meProfile genericProfile
Device	GponOnuMeProfile	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	name fileName text profileIndex
Device	GponOnuGenProfile	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	name parentMe
GponOnuPhysical	GponOnuSpecProfile	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	parentMe parentGeneric

Static OMCI objects

By setting the **meProfile** of GponOnuPhysical to a non-internal ME profilename, the MXK uses static OMCI for service provisioning, and need to use the **genericProfile** attribute as well.

Generic profile and Specific profile are needed for Static OMCI provisioning.

For all the modify operations of Generic and Specific profiles, a large text blob of all the variables (including changed and unchanged) will be formed and written to a file and download the same file to device. With this options, OSS modify operation shall form a large text and then perform the modify operation. To simplify the process of forming large texts, each variable of generic and specific profile will be provided with its own Handle_Id and hence each variable can be treated as a regular object.

Objects GponOnuGenProfileVariable and GponOnuSpecProfileVariable have been added to work with individual variables. Only list and modify operations are defined for these objectTypes.

Whenever you run a list query on these two objects, it will return all variables as different response element with having its own Handle_Id. Subsequently, you can use Handle_Id of the variable that you are interested in modifying. When a list is performed for all variables in a profile, all variables defined in the ME profile are returned. If a value has not been assigned, the value will be an empty string. Each variable data contains a Handle_Id, name, value and description attribute. Listing variables from multiple profiles is allowed.

When a variable data value is modified, the appropriate line is updated in the omci profile text. Only "value" may be modified. If "description" is modified, the modification will be ignored. If "name" is modified, the modification will fail. To modify multiple attributes in a single request, multiple request elements need to be added. One request element for each variable to modify. All the modified variables have to be in the same omci profile.

Specyng the name of the attribute is mandatory to successfully modify the value.

Table 12: OMCI profiles

Parent	Object	Support	Request element attribute
GponOltPhysical	GponOnuPhysical	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	meProfile genericProfile
GponOnuGenProfile	GponOnuGenProfileVariable	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	name description value
GponOnuSpecProfile	GponOnuSpecProfileVariable	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	name description value

Unified Service Provisioning: Dynamic OMCI objects

By setting the **meProfile** of **GponOnuPhysical** to an internal ME profile name, the MXK will use the dynamic OMCI for service provisioning on OMCI zNIDs instead of static OMCI, and will not need to use the **genericProfile** attribute anymore.

The internal ME profile is zNID type based. Each DZS supported OMCI zNID has its own internal ME profile. For example, zhone-2520. zhone-default is a universal ME profile. It can be applied to any zNIDs.

By setting the **internalMe** of **DynOmciCpeConnection** to an internal ME profile name, and setting the **rgMode** of **DynOmciCpeConnection** to none, the MXK will use the dynamic OMCI for service provisioning on OMCI zNIDs instead of Residential Gateway.

The following objects are for Dynamic OMCI.

Table 13: Dynamic OMCI objects

Parent	Object	Support	Request element attribute
GponOltPhysical	GponOnuPhysical	MXK 819 MXK 823 MX 19X	meProfile

Table 13: Dynamic OMCI objects

Parent	Object	Support	Request element attribute
Device	DynOmciCpeConnection	MXK 819 MXK 823 MX 19X	autoSelectGem bridgeIfIndex bridgeIfMaxVideoStreams bridgeIfMcastControlList bridgeTemplateName ctmIndex dscpToCosIndex egressGroupIndex egressGroupTemplate floodingGport gponSlotId gponOltId gponOnuId gponPortId gponTrafficIdx groupIndex groupTemplate gtpTemplateName guidedVlanId ifIndex internalMe mvrVlanId name rgMode slanCos slanId slanTpId slanXlateId slanXlateCos slanXlateTpId tpIndex tpType vlanCos vlanId vlanTpId vlanXlateId vlanXlateCos vlanXlateTpId videoGport

Table 13: Dynamic OMCI objects

Parent	Object	Support	Request element attribute
Device	DynOmciCpeIpServer	MXK 819 MXK 823 MX 19X	firewallAccess gateway hostIpOption ipv6HostIpOption ipv6PrefixLength ipv6Gateway ipv6PrimaryDns ipv6SecondaryDns ipv6DnsType nat netMask primaryDns profileName secondaryDns secureFwd serverIndex
Device	DynOmciCpeSipDialPlan	MXK 819 MXK 823 MX 19X	dialPlanFormat dialPlanString serverIndex dialPlanIndex
Device	DynOmciCpeTrafficMgmt	MXK 819 MXK 823 MX 19X	ctmIndex downstreamSir downstreamPir downstreamPriority downstreamWeight profileName upstreamSir upstreamPir upstreamPriority upstreamWeight

Table 13: Dynamic OMCI objects

Parent	Object	Support	Request element attribute
Device	DynOmciCpePwe	MXK 819 MXK 823 MX 19X	caSignalling cdvTolerance channelAssign clockReference dslMode dslFraming dscp encoding expectedPTypePayload expectedPTypeSignalling expectedSrcPayload expectedSrcSignalling fillPolicy jitterBufMax jitterBufDesired lbitRxPolicy lineType lossPktDeclarePolicy lossPktClearPolicy malformDeclarePolicy malformClearPolicy misconnDeclarePolicy misconnClearPolicy mplsTpType ouDeclarePolicy ouClearPolicy payloadSize payloadEncapsDelay profileName ptypePayload ptypeSignalling pweIndex rbitTxSetPolicy rbitTxClearPolicy rbitRxPolicy rtpTimeStampMode serviceType sesThreshold signalling ssrcPayload ssrcSignalling timingMode transport

Table 13: Dynamic OMCI objects

Parent	Object	Support	Request element attribute
Device	DynOmciCpeVideo	MXK 819 MXK 823 MX 19X	accessControlList bandwidthEnforce gemPort igmpVersion igmpFunction immediateLeave maxSimultaneousGroups maxMcastBandwidth profileName robustNess usIgmpRate videoIndex vlan
Device	DynOmciCpeVideoAcl	MXK 819 MXK 823 MX 19X	aclIndex aclIndex aclType destStartIP destEndIP imputedGroupBw profileName srcIP
Device	DynOmciCpeVoipFeatures	MXK 819 MXK 823 MX 19X	announceType cidFeatures callWaitingFeatures cpotFeatures callPresentationFeatures featuresIndex profileName
Device	DynOmciCpeVoipMedia	MXK 819 MXK 823 MX 19X	codecFirstOrder codecSecondOrder codecThirdOrder codecFourthOrder echoCancel faxMode mediaIndex periodFirstOrder periodSecondOrder periodThirdOrder periodFourthOrder profileName suppressionFirstOrder suppressionSecondOrder suppressionThirdOrder suppressionFourthOrder

Table 13: Dynamic OMCI objects

Parent	Object	Support	Request element attribute
Device	DynOmciCpeVoipServer	MXK 819 MXK 823 MX 19X	casEventsPassingMethod criticalDialTimeout dscpMark dtmfEventsPassingMethod maxRetryTime maxRetryAttempts mgcMsgFormat mgcServiceChangeDelay mgcTerminationIdBase mgcVersion oobCasEvents oobDtmfEvents oobToneEvents outboundServer partialDialTimeout piggyBackEvents primaryServer profileName portId releaseTimer rohTimer rtpDscp secondaryServer serverIndex sigProtocol signalingDscp sipDomain sipRegistrar sipRegExpiryTime sipRegStartTime softSwitch
GponOnuPhysical	DynOmciCpeEthSubscriber	MXK 819 MXK 823 MX 19X	adminState alarmSeverity duplex ifIndex internalMe lineStatusAlarm loopback mtu pauseTime portNumber portType powerFeed rate subMode tmProfileIndex videoProfileIndex

Table 13: Dynamic OMCI objects

Parent	Object	Support	Request element attribute
GponOnuPhysical	DynOmciCpeIp	MXK 819 MXK 823 MX 19X	hostIP ifIndex internalMe serverIndex serviceType
GponOnuPhysical	DynOmciCpePweSubscriber	MXK 819 MXK 823 MX 19X	adminState alarmSeverity farEndIP farEndPoint ifIndex internalMe lineLength lineStatusAlarm loopback nearEndPoint portNumber profileIndex
GponOnuPhysical	DynOmciCpeRfSubscriber	MXK 819 MXK 823 MX 19X	adminState alarmSeverity ifIndex internalMe lineStatusAlarm portNumber
GponOnuPhysical	DynOmciCpeVoipSubscriber	MXK 819 MXK 823 MX 19X	adminState dialNumber displayName featuresProfileIndex ifIndex impedance internalMe mediaProfileIndex password portNumber rxGain serverProfileIndex sigCode txGain txPath userName

Unified Service Provisioning: Residential Gateway objects

By setting the **internalMe** of **DynOmciCpeConnection** to an internal ME profile name, and setting the **rgMode** of **DynOmciCpeConnection** to a value other than none (i.e. brouted, bridged, or bridged PPPoE) , the MXK will use the Residential Gateway for service provisioning on OMCI zNIDs instead of Dynamic OMCI.

The following objects are for Unified Service Provisioning Residential Gateway (RG) feature.

Table 14: Residential Gateway objects

Parent	Object	Support	Request element attribute
Device	DynOmciCpeDnsHost	MXK 819 MXK 823 MX 19X	listIndex entryIndex domainName ipAddress
Device	DynOmciCpeDnsHostList	MXK 819 MXK 823 MX 19X	listIndex profileName
Device	DynOmciCpeIfVlan	MXK 819 MXK 823 MX 19X	dhcpServerProfile ifIndex ipServerProfile ipv6Address portFwdListProfile rgMode slanId slanXlateCos slanXlateId slanXlateTpId tpIndex tpType vlanIP vlanId vlanXlateCos vlanXlateId guidedVlanId
Device	DynOmciCpeStaticRouteList	MXK 819 MXK 823 MX 19X	listIndex profileName

Table 14: Residential Gateway objects

Parent	Object	Support	Request element attribute
Device	DynOmciCpeStaticRoute	MXK 819 MXK 823 MX 19X	destinationIp entryIndex gateway listIndex metric netmask
Device	DynOmciCpeSystem	MXK 819 MXK 823 MX 19X	commonIndex ifIndex
Device	DynOmciCpeSystemCommon	MXK 819 MXK 823 MX 19X	acsPassword acsUrl acsUserName adminPassword commonIndex cookieProtection dnsHostListIndex firewall informInterval profileName staticProfileIndex supportPassword tr69Inform userPassword vlanRouting
Device	DynOmciCpeWlan	MXK 819 MXK 823 MX 19X	hideAccessPoint isolateClients maxClients mcastFwd netAuthentication netReauthInterval profileName radiusPort radiusServerIp wepEncryption wepStrength wlanIndex wmmAdvertise wpa2PreAuthentication wpaEncryption wpaGroupRekeyInterval

Table 14: Residential Gateway objects

Parent	Object	Support	Request element attribute
Device	DynOmciCpeWlanAdv	MXK 819 MXK 823 MX 19x	apMode autoChanTimer basicRate beaconInterval bridgeRestrict channel cfgBandwidth dTimInterval dot11nMode dot11nRate dot1nClientOnly dot1nProtection fragmentationThreshold globalMaxClients mcastRate profileName rate54G rtsThreshold txPower wlanIndex wmm wmmApsd wmmNoAck wps wpsAddClientMethod wpsApMode xpressTechnology
Device	DynOmciCpeDhcpServer	MXK 819 MXK 823 MX 19x	endAddress ipv6DhcpServer ipv6DhcpMode ipv6RouterAdvert ipv6LeaseTime ipv6StartAddress ipv6EndAddress leaseTime profileName serverIndex serverListProfile startAddress
Device	DynOmciCpePortFwdList	MXK 819 MXK 823 MX 19x	listIndex profileName

Table 14: Residential Gateway objects

Parent	Object	Support	Request element attribute
Device	DynOmciCpePortFwd	MXK 819 MXK 823 MX 19x	entryIndex fwdType listIndex portEnd portStart privateIp privatePort protocol
Device	DynOmciCpePppoe	MXK 819 MXK 823 MX 19x	authentication ifIndex password retryInterval slanId tpIndex tpType userName vlanId
GponOnuPhysical	DynOmciCpeSystem	MXK 819 MXK 823 MX 19X	commonIndex ifIndex
GponOnuPhysical	DynOmciCpeWlanSubscriber	MXK 819 MXK 823 MX 19X	accessCtrlListProfileIndex adminState devicePin encryptKey ifIndex radiusKey ssid subscriberIndex wdsMacListProfileIndex wlanAdvProfileIndex wlanProfileIndex

OMCI device only objects

The following OMCI objects are configured at the CPE level (CPEs that are attached to ONUs). None of these objects are stored in DZMS database. Hence, all the operations that are initiated from DZMS/OSS/OSS XML on OMCI device only objects will be invoked directly on device and has no impact on the DZMS database.

Since they are not stored in the DZMS database, they do not have the Handle_Id assigned to them. Therefore, all the operations on OMCI device only objects must provide the Handle_Id of parent ONU as its Handle_Id. However, Handle_Type must be mentioned as the corresponding OMCI device only handle.

Except for OmciOntApply, OmciOntReboot, and OmciOntResync, all other objects are indexed by onuIfIndex, ME Id (OmciMeId) and logical port Id (OmciPortId). It is been made mandatory to provide ME Id and logical port Id for all other OMCI modify operations. Providing of onuIfIndex will not be required as ONU automatically retrieves it from the DZMS database by using Handle_Id (of parent ONU physical).

Get operations will always retrieve the info at ONU level hence there is no need to provide OmciMeId and OmciPortId attributes.

Note that Modify operations will fail if there is no ME Id and logical port Id attributes are provided.

Table 15: OMCI device only objects

Parent	Object	Support	Request element attribute
GponOnuPhysical	OmciOntApply	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	apply onuIfIndex
GponOnuPhysical	OmciOntReboot	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	onuIfIndex reboot
GponOnuPhysical	OmciOntResync	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	resync
GponOnuPhysical	OmciPortAdmin	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	onuIfIndex OmciMeId OmciPortId adminState

Table 15: OMCI device only objects

Parent	Object	Support	Request element attribute
GponOnuPhysical	OmciPortStatus	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	autoDetect onuIfIndex OmciMeId OmciPortId statusText
GponOnuPhysical	OmciPortStats	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	onuIfIndex OmciMeId OmciPortId currentStats prevStats

PWE connection

OSS GW supports the add, list and delete operations on the **PweStdConfig** objects of T1/E1 physicals of PWE card of MALC devices. MALC 1.16.2 devices support only SATOP PWE connections in both T1 and E1 mode cards. MALC 1.16.3 devices support SATOP and CESOP connections for T1 mode and support SATOP, CESOP and E1-PRI connections in E1 mode PWE cards.

Following are the criteria OSS client should follow for creating PWE connections:

1. T1 and E1 mode cards support up to 24 bits and 32 bits respectively for **bundleDs0Bundle** attribute.
2. There can be maximum of one SATOP PWE connection per physical.
3. A PWE T1/E1 physical can have maximum of 8 CESOP connections and none of the CESOP connections of same physical can use the bits that are already used by other CESOP connections of the same physical. In other words, bits usage cannot be overlapped.
4. There can be maximum of one E1/PRI PWE connection per physical.
5. SATOP PWE connections cannot have any bit set for bundle. Hence, no value or zero value should be passed for **bundleDs0Bundle** attribute.
6. CESOP PWE connections must have at least one bit set for **bundleDs0Bundle** attribute.
7. E1/PRI connections must have all bits set for **bundleDso0Bundle** attribute. Hence, OSS clients must pass 32 1's as the value for successful creation of E1/PRI PWE connection.

8. **bundleIsdnMode** attribute is only applicable for E1/PRI PWE connections. And it must be set to 2 (lt)/ 3 (nt) for E1/PRI connections. This attribute must not be set or set to 1 (disabled) for SATOP and CESOP connections.

9. If the destination port (outbandLabel) is 2142, there cannot be any other PWE with using destination port as source port of the PWE with destination port 2142 and same destination IP in entire DZMS managed devices.

10. Source port (inbandLabel) can never be 2142 on any PWE entire DZMS managed devices.

11. No 2 PWE connections can use same destination IP (peerIP) and destination port in entire DZMS managed devices.

Table 16: PweStdConfig object and its attributes

Parent	Object	Support	Request element attribute
Physical	PweStdConfig	T1/E1 PWE card of MALC devices	bundleDs0Bundle bundleIfIndex bundleIsdnMode bundleLocalIpAddr bundleTos connectionType ifIndex stdAdminStatus stdDescription stdInboundLabel stdOutboundLabel stdPeerIPAddress tdmJitterBufferDepth tdmPayloadSize tdmPktFillPattern tdmPktReorder tdmPktReplacePolicy

Port access control

The port access control list in DZS products. It is used to control access to internal ports for better usage. Initially it is used just for control TELNET (23) access, but in theory it could be used for other ports as well. On any given port, only the IP addresses complying with source address and network mask will be allowed the access to telnet to the device.

Table 17: Port access control

Parent	Object	Support	Request element attribute
Device	PortAccessControl	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	name accessIndex portNumber srcAddr netMask

SubRegion objects



Note: DZMS supports the addition of sub-regions until 3 nested levels. Root can have any number of regions. Any of these regions can have any number of immediate sub-regions. These sub regions can in turn have sub-regions for 2 more levels. Addition of regions is not supported from OSS GW. Instead, addition of regions must be done using DZMS. OSS supports get, list, delete and modification of name for Region and Sub Region objects. SubRegionOne represents the all immediate child regions of the root level region. SubRegionTwo represents the all immediate child sub-regions of SubRegionOne. SubRegionThree represents the all immediate child sub-regions of SubRegionTwo.

Table 18: SubRegion object

Parent	Object	Support	Request element attribute
Root	Region	Above the device level hence applicable for all devices.	Region_Id name
Region	SubRegionOne	Above the device level hence applicable for all devices.	Region_Id SubRegion_Id_1 SubRegion_Group_1 name
Region	SubRegionTwo	Above the device level hence applicable for all devices.	Region_Id SubRegion_Id_1 SubRegion_Id_2 SubRegion_Group_1 SubRegion_Group_2 name

Table 18: SubRegion object

Parent	Object	Support	Request element attribute
Region	SubRegionThree	Above the device level hence applicable for all devices.	Region_Id SubRegion_Id_1 SubRegion_Id_2 SubRegion_Id_3 SubRegion_Group_1 SubRegion_Group_2 SubRegion_Group_3 name

VdslVectProfile object

Table 19: VdslVectProfile object

Parent	Object	Support	Request element attribute
Device	VdslVectProfile	MXK 819 MXK 823 MX 19X	psdShape, slot, downDisableBandStartTone1, downDisableBandEndTone1, upDisableBandStartTone1, upDisableBandEndTone1, downDisableBandStartTone2, downDisableBandEndTone2, upDisableBandStartTone2, upDisableBandEndTone2, downDisableBandStartTone3, downDisableBandEndTone3, upDisableBandStartTone3, upDisableBandEndTone3, downDisableBandStartTone4, downDisableBandEndTone4, upDisableBandStartTone4, upDisableBandEndTone4

DynOmciCpeCfgRule object

Table 20: DynOmciCpeCfgRule object

Parent	Object	Support	Request element attribute
Device	DynOmciCpeCfgRule	MXK 819 MXK 823 MX 19X	serviceTemplate, adminState, priority, delBeforeApply, matchExpression, targetUni, cfgRuleIndex, profileName

DynOmciCpeLldpRem object

Table 21: DynOmciCpeLldpRem object

Parent	Object	Support	Request element attribute
Device	DynOmciCpeLldpRem	MXK 819 MXK 823 MX 19X	remIndex, ifIndex, tpType, tpIndex, chassisId, portId, sysName, hardwareRev, firmwareRev, softwareRev, modelName, serialNum, retVal, onuIfName

Logical configuration requests

The following table lists the request element attributes associated with specific logicals.

Table 22: Logical configuration requests

Parent	Object	Supported	Request element attribute
AdslPhysical	ATMLogical	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723	aal5Index ifAdminStatus ifConfVccs ifConfVpcs ifCurrMaxVciBits ifCurrMaxVpiBits ifFlmiVci ifFlmiVpi ifIndex ifMaxActiveVciBits ifMaxActiveVpiBits ifMaxVccs ifMaxVpcs ifMyNeighborIfName ifMyNeighborIpAddr ifType name rfc1483Index
DSLPhysical	ATMLogical	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723 Z-Edge 64 BH2A Z-Edge 64 H2A	aal5Index ifAdminStatus ifConfVccs ifConfVpcs ifCurrMaxVciBits ifCurrMaxVpiBits ifFlmiVci ifFlmiVpi ifIndex ifMaxActiveVciBits ifMaxActiveVpiBits ifMaxVccs ifMaxVpcs ifMyNeighborIfName ifMyNeighborIpAddr ifType name rfc1483Index

Table 22: Logical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
E3UnchannelizedPhysical	ATMLogical	BAN	aal5Index ifAdminStatus ifConfVccs ifConfVpcs ifCurrMaxVciBits ifCurrMaxVpiBits ifFlmiVci ifFlmiVpi ifIndex ifMaxActiveVciBits ifMaxActiveVpiBits ifMaxVccs ifMaxVpcs ifMyNeighborIfName ifMyNeighborIpAddr ifType name rfc1483Index
ImaPhysicalGroup	ATMLogical	MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	aal5Index ifAdminStatus ifConfVccs ifConfVpcs ifCurrMaxVciBits ifCurrMaxVpiBits ifFlmiVci ifFlmiVpi ifIndex ifMaxActiveVciBits ifMaxActiveVpiBits ifMaxVccs ifMaxVpcs ifMyNeighborIfName ifMyNeighborIpAddr ifType name rfc1483Index

Table 22: Logical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
SonetOC3Physical	ATMLogical	BAN Sechtor 100A E1 Sechtor 100A T1	aal5Index ifAdminStatus ifConfVccs ifConfVpcs ifCurrMaxVciBits ifCurrMaxVpiBits ifflmiVci ifflmiVpi ifIndex ifMaxActiveVciBits ifMaxActiveVpiBits ifMaxVccs ifMaxVpcs ifMyNeighborIfName ifMyNeighborIpAddr ifType name rfc1483Index
T3UnchannelizedPhysical	ATMLogical	BAN	aal5Index ifAdminStatus ifConfVccs ifConfVpcs ifCurrMaxVciBits ifCurrMaxVpiBits ifflmiVci ifflmiVpi ifIndex ifMaxActiveVciBits ifMaxActiveVpiBits ifMaxVccs ifMaxVpcs ifMyNeighborIfName ifMyNeighborIpAddr ifType name rfc1483Index

Table 22: Logical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
BANDevice MALC 19Device MALC 23Device MALC 319Device Raptor 319Device Raptor 719Device Raptor 723Device Mx19Device Mx23Device	IpLogical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	addr bcastAddr bcastEnabled destAddr dhcp dhcpServerEnable egressFilterId farEndAddr FaultDetectionType fwdEnabled ifAdminStatus ifIndex ifType ingressFilterId ipAddrDynamic lineGroupId lineGroupKey lowerIfIndex maxVideoStreams mcastControlList mcastEnabled mcastFwdEnabled mru name natEnabled netMask pointToPoint rdIndex reasmMaxSize stagCOS stagId stagTPID subnetGroup tosCOS tosOption unnumberedIndex vci vlanCOS vlanId vpi The following attributes are only for MXK 819Device and MXK 823Device: gponOltId gponPortId gponSlotId gponTrafficIdx gtpTemplateName

Table 22: Logical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
BridgeCfgGroup (Also known as Packet Rule Group)	BridgeCfg (Also known as Packet Rule Member)	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 100 Raptor 50 Raptor 719 Raptor 723 MXK 819 MXK 823	Type cfgType cfgValue groupIndex instanceKey memberIndex name packetRule2 packetRule3 packetRule4 packetRule5
EtherPhysical	IpLogical	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge 64 BH2A Z-Edge 64 H2A MXK 819 MXK 823	addr bcastAddr bcastEnabled destAddr dhcp egressFilterId farEndAddr fwdEnabled ifAdminStatus ifIndex ifType ingressFilterId lgId lowerIfIndex mcastEnabled mcastFwdEnabled mru name natEnabled netMask pointToPoint rdIndex reasmMaxSize unnumberedIndex vci vpi The following attributes are only for MXK 819Device and MXK 823Device: gponOltId gponPortId gponSlotId gponTrafficIdx gtpTemplateName

Table 22: Logical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
McastControlLists	McastControlList	MALC 319 MALC 719 MALC 723 MXK 819 MXK 823	controlId ipAddress listType name precedence
ZEdgeBH2ADevice ZEdgeH2ADevice	PppLogical	Z-Edge 64 BH2A Z-Edge 64 H2A	Type callMode frameType hierarchyString ifAdminStatus ifIndex ifType ifTypeExtension instanceKey lineGroupId lineGroupKey lowerIfIndex name numChannels vci vpi
SubVoiceConn	SubVoiceEbsEndPoint	MALC 319 (with ULCS POTS cards) MALC 719 (with ULCS POTS cards) MALC 723 (with ULCS POTS cards)	name lineGroupId
SubVoiceConn	SubVoiceGr303EndPoint	MALC 319 MALC 719 MALC 723	endPointIndex name igCrv igName
SubVoiceConn	SubVoicePotsEndPoint	MALC 319 MALC 719 MALC 723	endPointIndex huntGroupIndexOne huntGroupIndexTwo huntGroupIndexThree lineGroupId name

Table 22: Logical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
SubVoiceConn	SubVoiceVoipEndPoint	MALC 319 MALC 719 MALC 723	Type directoryNumber endPointIndex framesPerPacket g711Fallback g726ByteOrder instanceKey ipIfIndex name passWord preferredCodecuserId
SubVoiceConn	SubVoiceV52EndPoint	MALC 319 MALC 719 MALC 723	endPointIndex igName userPortId userType isdnChannelId name
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	UnnumberedIpRef	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Raptor 100 Raptor 50	ipIfIndex ipIndex name

Table 22: Logical configuration requests (Continued)

Parent	Object	Supported	Request element attribute
BAN MALC19Device MALC23Device MALC319Device S100AE1Device	V52Ig	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1	Type actionInProgress adminStatus autoConfig instanceKey isdnEnvFuncStartAddr linkInServiceTimer linkOutOfServiceTimer localInterfaceId localProvVariant maxOutstandingFrames n200 n201 name nationalPstnRegion operStatus operStatusCausePortAlignmentRequest portAlignmentStatus priLogicalChanId protocolSpec provVariantReq provVariantReqStatus pstnLayer3StartAddr remoteInterfaceId remoteProvVariant startupCheckLinkId startupUnblockUserPorts switchVendor t200 t203
V52Ig	V52UserPort	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1	Type address adminStatus block igName inUse instanceKey isdnDsCPathId isdnFCPathId isdnPCPathId name regOrLeaseUser userPortType

lineRedStatusFunction object

Table 23: lineRedStatusFunction object

Parent	Object	Support	Request element attribute
Device	lineRedStatusFunction	MXK 819 MXK 823 MX 19X MXK- F devices	lineRedStatusFunction

Other OSS XML Gateway objects

The following table shows the objects that have parents in the preceding tables.

Table 24: Other OSS XML Gateway objects

Parent	Object	Supported	Request element attribute
Aal2VclAtmEndPoint	Aal2Elcp	BAN MALC 319 MALC 719 MALC 723 Sector 100A E1 Sector 100A T1 Z-Edge 64 BH2A Z-Edge 64 H2A	Type audioService instanceKey lineGroupId lineGroupKey logicalIfIndex maxLengthFrame maxSDULength name pcmEncoding portId portType preferredApIndex vci vpi

Table 24: Other OSS XML Gateway objects (Continued)

Parent	Object	Supported	Request element attribute
ATMLogical	Aal2VclAtmEndPoint	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge 64 BH2A Z-Edge 64 H2A	aal2VclAudioProfileIdentifier aal2VclMaxCpsSduSize aal2VclMaxLengthFrame aal2VclMaxNumberMultiplexCh annels aal2VclMaxSDULength aal2VclNextCid aal2VclPCMEncoding aal2VclSscsDefaultType aal2VclTrunkType adminStatus atmVccAal2AppId atmVccAal2SigVccI atmVccAal2VccI audioService cas castType cellloss congestion connKind crossConnId faultDetectionType frameMode logicalIfIndex maxCid minCid name oamF5Ping oltTrafficContainerIndex operStatus rasTimer timerCU vccAalType vci vclAdminStatus vclCastType vclConnKind vclCrossConnId vclLastChange vclOperStatus rxTrafficDescrIndex txTrafficDescrIndex vpi

Table 24: Other OSS XML Gateway objects (Continued)

Parent	Object	Supported	Request element attribute
ATMLogical	Aal5VclAtmEndPoint	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723 Z-Edge 64 BH2A Z-Edge 64 H2A	adminStatus castType connKind cpcsRxSduSize cpcsTxSduSize crossConnId encapsType faultDetectionType lastChange lineGroupId logicalIfIndex name oamF5Ping oltTrafficContainerIndex operStatus rxTrafficDescrIndex txTrafficDescrIndex vccAalType vci vpi
AccessList	AccessListEntry	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge 64 BH2A Z-Edge 64 H2A MXK 819 MXK 823	accessListIndex accessListIpAddress name

Table 24: Other OSS XML Gateway objects (Continued)

Parent	Object	Supported	Request element attribute
Aal2VclAtmEndPoint	ATMCid	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge 64 BH2A Z-Edge 64 H2A	frameMode adminStatus audioService cas inUse index lineGroupId logicalIfIndex maxLengthFrame maxSDULength name pcmEncoding preferredApIndex rasTimer sscsType vci vpi
ATMLogical	AtmVpi	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723	Type instanceKey lineGroupId lineGroupKey logicalIfIndex maxVci maxVciPerVp name oamF4Ping switchType vpi
MtacCard	CardExternalRelay	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	name relayId

Table 24: Other OSS XML Gateway objects (Continued)

Parent	Object	Supported	Request element attribute
CardGroup	CardGroupEntry	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge 64 BH2A Z-Edge 64 H2A MXK 819 MXK 823	cgId holdActive name shelfIndex slotIndex weight
IpRoutingDomain	DHCPLease	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723 Z-Edge 64 BH2A Z-Edge 64 H2A MXK 819 MXK 823	clientHostname clientId ddnsFwdName ddnsRevName domain ends flags hardwareAddress hostname ipAddress name starts
ImaPhysicalGroup	ImaLink	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	cardInstanceKey groupIndex ifIndex name
IpLogical	IpAliasAddr	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723 Z-Edge 64 BH2A Z-Edge 64 H2A MXK 819 MXK 823	addr bcastAddr ifIndex name netMask

Table 24: Other OSS XML Gateway objects (Continued)

Parent	Object	Supported	Request element attribute
AdslCard	IpDhcpClientResource	BAN	defaultLease minLease name offerTimeout
ZEdgeBH2ACard	IpDhcpClientResource	Z-Edge 64 BH2A	defaultLease minLease name offerTimeout
ZEdgeH2ACard	IpDhcpClientResource	Z-Edge 64 H2A	defaultLease minLease name offerTimeout
IpRoutingDomain	IpDnsResolverHost	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge 64 BH2A Z-Edge 64 H2A MXK 819 MXK 823	alias1 alias2 alias3 alias4 hostName ipAddress name rdIndex

Table 24: Other OSS XML Gateway objects (Continued)

Parent	Object	Supported	Request element attribute
IpFilterSpec	IpFilterStmt	BAN	action actionArg arbMask arbOffset arbValueBase arbValueHigh arbValueLow dstAddrHigh dstAddrLow dstPortHigh dstPortLow ipProtocol modifier name newStmtIndex specIndex srcAddrHigh srcAddrLow srcPortHigh srcPortLow stmtIndex
IpLogical	IpNetToMedia	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Z-Edge 64 BH2A Z-Edge 64 H2A	ifIndex mappingType name netAddress physAddress
IpRoutingDomain	IpRip2Config	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723 Z-Edge 64 BH2A Z-Edge 64 H2A MXK 819 MXK 823	address advertiseStaticRoutes authKey authType defaultMetric name rdIndex receive send sendPoisonReverse srcAddress

Table 24: Other OSS XML Gateway objects (Continued)

Parent	Object	Supported	Request element attribute
IpRoutingDomain	IpRip2Global	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723 Z-Edge 64 BH2A Z-Edge 64 H2A MXK 819 MXK 823	adminState name rdIndex updateTime
IpRoutingDomain	IpStaticRoute	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge 64 BH2A Z-Edge 64 H2A MXK 819 MXK 823	dest backupMetric enable ifNumber metric name netmask nextHop rdIndex routeType pingFailMax pingInterval
ZEdgeBH2ACard	IsdnAlarmProfile	Z-Edge 64 BH2A	amiViolations ifIndex name unbalancedFrames

Table 24: Other OSS XML Gateway objects (Continued)

Parent	Object	Supported	Request element attribute
ATMLogical	OtherVclAtmEndPoint	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723 Z-Edge 64 BH2A Z-Edge 64 H2A	adminStatus castType connKind crossConnId encapsType faultDetectionType lineGroupId logicalIfIndex name oamF5Ping oltTrafficContainerIndex, operStatus rxTrafficDescrIndex txTrafficDescrIndex vccAalType vci vpi
PotsPhysical	PotsEmConfig	MALC 319 MALC 719 MALC 723	ifIndex interDigitDuration leadSignalingType name operation
PotsPhysical	PotsFxoConfig	MALC 319 MALC 719 MALC 723	dialType digitDuration ifIndex interDigitDuration name numberRings pulseInterDigitDuration pulseRate signalType supervisoryDisconnect
PotsPhysical	PotsFxsConfig	MALC 319 MALC 719 MALC 723	digitDuration ifIndex interDigitDuration name ringBack ringCadence ringFrequency ringTimer signalType

Table 24: Other OSS XML Gateway objects (Continued)

Parent	Object	Supported	Request element attribute
RS232Physical	RS232AsyncConfig	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	autoBaud bits ifIndex name parity stopBits
ATMLogical	VplAtmEndPoint	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 50 Raptor 319 Raptor 719 Raptor 723	Type adminStatus castType connKind crossConnId faultDetectionType instanceKey lastChange lineGroupId lineGroupKey logicalIfIndex name operStatus rxTrafficDescrIndex txTrafficDescrIndex vpi

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PERFORMANCE REQUESTS

This chapter describes the attributes associated with performance requests to the OSS XML Gateway. It includes the following sections:

- [Overview, page 131](#)
- [Session level request attributes, page 131](#)
- [Request element attributes to identify objects, page 132](#)
- [Performance requests by product, page 132](#)
- [Region and device, page 133](#)
- [Card, page 141](#)
- [Physical, page 149](#)
- [Logical, page 147](#)
- [Other objects, page 194](#)

Overview

The OSS XML Gateway supports `getStatus` performance requests.

getStatus retrieves information about an object from the network element (as opposed to the database).

getStatus requests are *synchronous* requests, which are requests in which the client waits for a response to the request before continuing with other processing. For details on how the OSS XML Gateway processes synchronous requests, see the *Overview* chapter.

Session level request attributes

To obtain network element parameter values, the OSS XML Gateway client issues a **getStatus** request. A **getStatus** request requires session-level attributes, such as what operation to perform, operator name and session ID obtained from the **authentication** request, and so on (see [Table 25](#)).

Table 25: Session-level request attributes

Object Attribute	Description	Valid Values (Comments)	Scope
RequestType	Request type.	getStatus	Mandatory
Version	DZMS/OSS XML software version.	1.2 and higher	Mandatory
operName	Operator login name (obtained from authenticate request).	Valid ZMS operator name	Mandatory
SessionID	ID that identifies the session with the OSS XML Gateway (obtained from authenticate request).	Valid ZMS operator password	Mandatory

Request element attributes to identify objects

A **getStatus** request includes a single request element. The request element identifies the object handle and specifies one or more *Retrieves_Attrs*, each defining the data attribute to be obtained for that object. For example, you can form a request element to obtain status for a physical port. In the request element, identify the particular port and specify the status attributes to be retrieved.

[Table 26](#) lists the request element attributes required to identify an object handle and a *Retrieve_Attr*.

Table 26: Request element attributes to identify an object

Object Attribute	Description	Valid Values (Comments)	Scope
Handle_Type	Type of object.	Valid handle type	Mandatory
Handle_Group	The group_id of an object. Each DZMS object can be put into a group, and each group is identified by a unique group identifier. This supports network resource partitioning. (This attribute must be provided for an operator who does not have view-all privileges.)	Valid handle group	Optional
Handle_Id	Unique ID of the object. The DZMS database generates a unique key that identifies the object.	Valid handle ID	Mandatory
Retrieve_Attr	Specifies attribute to be retrieved.	Valid attribute	Mandatory

Performance requests by product

The following sections describe the request element attributes associated with the BAN, MALC, Sechtor 100A and Z-Edge products. Each table lists the parent and object, indicates which SLMS products are supported by the OSS

XML Gateway, and lists the request element attributes that are associated with each object.

Region and device

This section describes OSS XML Gateway objects that are children of either Region or Device.

Table 27: Objects contained within region and device

Parent	Object	Supported	Attributes
Region	BANDevice	BAN	systemActiveCalls SystemConfigurationNumCards systemConfigurationNumShelves systemIncomingCallsBlocked systemIncomingCallsCompleted systemOutgoingCallsBlocked systemOutgoingCallsCompleted systemUpTime zmsBlockCli
Region	MALC19Device MALC23Device MALC319Device	MALC 319 MALC 719 MALC 723	systemActiveCalls SystemConfigurationNumCards systemConfigurationNumShelves systemCurrentClockSource systemSecondaryClockSource systemIncomingCallsBlocked systemIncomingCallsCompleted systemOutgoingCallsBlocked systemOutgoingCallsCompleted systemUpTime zmsBlockCli
Region	S100AE1Device S100AT1Device	Sechtor 100A E1 Sechtor 100A T1	fanOperStatus systemActiveCalls systemConfigurationNumCards systemConfiguratoInNumShelves systemIncomingCallsBlocked systemIncomingCallsCompleted systemOutgoingCallsBlocked systemOutgoingCallsCompleted systemUpTime thermoAOperStatus thermoATemp thermoBOperStatus thermoBTemp zmsBlockCli

Table 27: Objects contained within region and device (Continued)

Parent	Object	Supported	Attributes
Region	ZEdgeBH2ADevice ZEdgeH2ADevice	Z-Edge BH2A Z-Edge H2A	systemActiveCalls systemConfigurationNumCards systemConfigurationNumShelves systemIncomingCallsBlocked systemIncomingCallsCompleted systemOutgoingCallsBlocked systemOutgoingCallsCompleted systemUpTime zmsBlockCli
BANDevice	ATMVcCrossConn	BAN	vcCrossConnH2LLastChange vcCrossConnH2LOperStatus vcCrossConnL2HLastChange vcCrossConnL2HOperStatus
BANDevice	AtmVpCrossConn	BAN	adminStatus h2LLastChange h2LOperStatus highIfIndex highVpi index l2HLastChange l2HOperStatus lowIfIndex lowVpi
BANDevice	BulkStatsCollection	BAN	operStatus
BANDevice	BulkStatsConfig	BAN	operStatus
BANDevice	IpConfig	BAN	forwDatagrams fragCreates fragFails fragOks inAddrErrors inDelivers inDiscards inHdrErrorss inReceives outDiscards outNoRoutes outRequests reasmFails reasmOks reasmReqds reasmTimeout routingDiscards unknownProtos

Table 27: Objects contained within region and device (Continued)

Parent	Object	Supported	Attributes
BANDevice	IpFilterSpec	BAN	counter name statementIndex specDescription specIndex threshold version
BANDevice	IpRoutingDomain	BAN	cidrRouteNumber rdIndex
BANDevice	ShelfFan	BAN	index location speed threshold
BANDevice	ShelfTemperature	BAN	highThreshold index location lowThreshold temperature
BANDevice	SoftwareDownload	BAN	sysCardSwSpecificVers sysCardSwUpgradeVers sysSwStatus sysSwUpgradeStatus
Device	Bridge	MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	ifInErrors ifHCInMulticastPkts ifHCOutMulticastPkts ifHCInBroadcastPkts ifHCOutBroadcastPkts ifLastChange ifHCOutUcastPkts ifHCInUcastPkts ifInUnknownProtos ifOutErrors
Device	IpLogical	MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	ifInErrors ifHCInMulticastPkts ifHCOutMulticastPkts ifHCInBroadcastPkts ifHCOutBroadcastPkts ifLastChange ifHCOutUcastPkts ifHCInUcastPkts ifInUnknownProtos ifOutErrors

Table 27: Objects contained within region and device (Continued)

Parent	Object	Supported	Attributes
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	ATMVcCrossConn	MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	vcCrossConnH2LLastChange vcCrossConnH2LOperStatus vcCrossConnL2HLastChange vcCrossConnL2HOperStatus
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	AtmVpCrossConn	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	adminStatus h2LLastChange h2LOperStatus highIfIndex highVpi index l2HLastChange l2HOperStatus lowIfIndex lowVpi
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor723Device Raptor719Device Raptor319Device	BulkStatsCollection	MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	operStatus
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor723Device Raptor719Device Raptor319Device	BulkStatsConfig	MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	operStatus

Table 27: Objects contained within region and device (Continued)

Parent	Object	Supported	Attributes
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor723Device Raptor719Device Raptor319Device	IpConfig	MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	forwDatagrams fragCreates fragFails fragOks inAddrErrors inDelivers inDiscards inHdrErrorss inReceives outDiscards outNoRoutes outRequests reasmFails reasmOks reasmReqds reasmTimeout routingDiscards unknownProtos
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor723Device Raptor719Device Raptor319Device	IpRoutingDomain	MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	cidrRouteNumber rdIndex
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	ShelfFan	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	index location speed threshold
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	ShelfTemperature	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	highThreshold index location lowThreshold temperature
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor723Device Raptor719Device Raptor319Device	SoftwareDownload	MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	sysCardSwSpecificVers sysCardSwUpgradeVers sysSwStatus sysSwUpgradeStatus

Table 27: Objects contained within region and device (Continued)

Parent	Object	Supported	Attributes
S100AE1Device S100AT1Device	ATMVcCrossConn	Sechtor 100A E1 Sechtor 100A T1	vcCrossConnH2LLastChange vcCrossConnH2LOperStatus vcCrossConnL2HLastChange vcCrossConnL2HOperStatus
S100AE1Device S100AT1Device	IpConfig	Sechtor 100A E1 Sechtor 100A T1	forwDatagrams fragCreates fragFails fragOks inAddrErrors inDelivers inDiscards inHdrErrorss inReceives outDiscards outNoRoutes outRequests reasmFails reasmOks reasmReqds reasmTimeout routingDiscards unknownProtos
S100AE1Device S100AT1Device	IpRoutingDomain	Sechtor 100A E1 Sechtor 100A T1	cidrRouteNumber rdIndex
S100AE1Device S100AT1Device	SoftwareDownload	Sechtor 100A E1 Sechtor 100A T1	sysCardSwSpecificVers sysCardSwUpgradeVers sysSwStatus sysSwUpgradeStatus

Table 27: Objects contained within region and device (Continued)

Parent	Object	Supported	Attributes
S100AE1Device	V52Ig	Sechtor 100A E1	totalIsdnCalls currActiveCalls currCallsBlocked currInboundCalls currIsdnCalls intercIsdnCalls currLapdRcvd currLapdRcvdErrs currLapdSent currOutboundCalls currOutboundCallsBlocked currPstnInboundCalls currPstnOutboundCalls currV52ProtocolErrors intervCallsBlocked intervInboundCalls intervLapdRcvd intervLapdRcvdErrs intervLapdSent intervOutboundCalls intervOutboundCallsBlocked intervPstnInboundCalls intervPstnOutboundCalls, intervV52ProtocolErrors intervalNum maxConfiguredCalls operStatus operStatusCause peerSlot peerStatus portAlignmentStatus provVariantReqStatus remoteInterfaceId remoteProvVariant shelf slot statsInvalidIntervals statsTimeElapsed statsValidIntervals totalCallsBlocked totalInboundCalls totalLapdRcvd totalLapdRcvdErrs totalLapdSent totalOutboundCalls totalOutboundCallsBlocked totalPstnInboundCalls totalPstnOutboundCalls totalV52ProtocolErrors

Table 27: Objects contained within region and device (Continued)

Parent	Object	Supported	Attributes
ZEdgeBH2ADevice ZEdgeH2ADevice	IpConfig	Z-Edge BH2A Z-Edge H2A	forwDatagrams fragCreates fragFails fragOks inAddrErrors inDelivers inDiscards inHdrErrorss inReceives outDiscards outNoRoutes outRequests reasmFails reasmOks reasmReqds reasmTimeout routingDiscards unknownProtos
ZEdgeBH2ADevice ZEdgeH2ADevice	IpRoutingDomain	Z-Edge BH2A Z-Edge H2A	cidrRouteNumber rdIndex
ZEdgeBH2ADevice ZEdgeH2ADevice	PppLcp	Z-Edge BH2A Z-Edge H2A	badAddrs badCtrls badFCS inGoodOctets inLQRs l2pACCM l2rACCComp l2rProtoComp lastRecvLqrPkt localMRU localPeriod outLQRs p2lACCM phyIndex pktTooLongs quality r2lACCComp r2lProtoComp remoteMRU remotePeriod rxFCSSize txFCSSize

Table 27: Objects contained within region and device (Continued)

Parent	Object	Supported	Attributes
ZEdgeBH2ADevice ZEdgeH2ADevice	PppNcp	Z-Edge BH2A Z-Edge H2A	l2rCompProtocol localMaxSlotID operStatus r2lCompProtocol remoteMaxSlotID
ZEdgeBH2ADevice ZEdgeH2ADevice	SoftwareDownload	Z-Edge BH2A	sysCardSwSpecificVers sysCardSwUpgradeVers sysSwStatus sysSwUpgradeStatus

Card

The following table describes the objects contained by the parent Shelf.

Table 28: Card objects

Parent	Object	Supported	Request Element Attribute
Shelf	S100AE1Card S100AT1Card	Sechtor 100A E1 Sechtor 100A T1	availMemory cardMemStatus mfgBootRevision mfgCLEICode mfgSerialNumber operStatus peakMemoryUsage portStatus postResults processorIdle processorType processorUsage swRunningVers swStatus swUpgradeVers sysCardSwSpecificVers sysCardSwUpgradeVers totalMemory upTime

Table 28: Card objects (Continued)

Parent	Object	Supported	Request Element Attribute
Shelf	AdslAC5BMalc24PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	adminStatus availMemory cardAdminResets cardCLEICode cardEepromVersion cardFaultResets cardFeatureBits cardLineType cardMemStatus cardSerialNumber cardState cardSubType cardType cardVersion mfgBootRevision mfgCLEICode mfgRevisionCode mfgSerialNumber operStatus peakMemoryUsage portStatus postResults processorIdle processorType processorUsage swRunningVers swStatus swUpgradeVers sysCardSwSpecificVers sysCardSwUpgradeVers totalMemory upTime

Table 28: Card objects (Continued)

Parent	Object	Supported	Request Element Attribute
Shelf	T1Tdm64PortCard	BAN	adminStatus availMemory cardAdminResets cardCLEICode cardEepromVersion cardFaultResets cardFeatureBits cardLineType cardMemStatus cardSerialNumber cardState cardSubType cardTypecardVersion mfgBootRevision mfgCLEICode mfgRevisionCode mfgSerialNumber operStatus peakMemoryUsage portStatus postResults processorIdle processorType processorUsage swRunningVers swStatus swUpgradeVers sysCardSwSpecificVers sysCardSwUpgradeVers totalMemory upTime

Table 28: Card objects (Continued)

Parent	Object	Supported	Request Element Attribute
Shelf	T3Aal5Atm4PortCard	BAN	availMemory cardMemStatus mfgBootRevisoin mfgCLEICode mfgRevisionCode mfgSerialNumber operStatus peakMemoryUsage portStatus postResults processorIdle processorType processorUsage swRunningVers swStatus swUpgradeVers sysCardSwSpecificVers sysCardSwUpgradeVers totalMemory upTime

Table 28: Card objects (Continued)

Parent	Object	Supported	Request Element Attribute
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	Gshdsl4Wire24PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	adminStatus availMemory cardActiveCalls cardAdminResets cardCLEICode cardEepromVersion cardFaultResets cardFeatureBits cardIncomingCallsBlocked cardIncomingCallsCompleted cardLineType cardMemStatus cardOutgoingCallsBlocked cardOutgoingCallsCompleted cardSerialNumber cardState cardSubType cardType cardVersion mfgBootRevision mfgCLEICode mfgRevisionCode mfgSerialNumber operStatus peakMemoryUsage portStatus postResults processorIdle processorType processorUsage swRunningVers swStatus swUpgradeVers sysCardSwSpecificVers sysCardSwUpgradeVers totalMemory upTime

Logical

These objects describe logical network entities.

The following table describes logical layer objects.

Table 29: Logical objects

Parent	Object	Supported	Request Element Attribute
AdslPhysical	ATMLogical	BAN	ifConfVccs ifConfVpcs ifCurrMaxVciBits ifCurrMaxVpiBits ifDescr ifDs3PlcpAlarmState ifDs3PlcpSEFSs ifDs3PlcpUASs ifFeHECEvents ifFeNCDEvents ifFeOCDEvents ifHECEvents ifIlmiVci ifIlmiVpi ifMaxActiveVciBits ifMaxActiveVpiBits ifMaxVccs ifMaxVpcs ifMtu ifNCDEvents ifOCDEvents ifPhysAddress ifSpeed ifTCAlarmState
DSLPhysical	ATMLogical	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723 Z-Edge BH2A Z-Edge H2A	ifDescrATMPvc ifDs3PlcpAlarmState ifDs3PlcpUASs ifMtu ifOCDEvents ifPhysAddress ifSpeed ifTCAlarmState
E3UnchannelizedPhysical	ATMLogical	BAN	ifDescr ifDs3PlcpAlarmState ifDs3PlcpUASs ifMtu ifOCDEvents ifPhysAddress ifSpeed ifTCAlarmState

Table 29: Logical objects (Continued)

Parent	Object	Supported	Request Element Attribute
ImaPhysicalGroup	ATMLogical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	ifDescr ifDs3PlcpAlarmState ifDs3PlcpUASs ifMtu ifOCDEvents ifPhysAddress ifSpeed ifTCAlarmState
SonetOC3Physical	ATMLogical	BAN Sechtor 100A E1 Sechtor 100A T1	ifDescr ifDs3PlcpAlarmState ifDs3PlcpUASs ifMtu ifOCDEvents ifPhysAddress ifSpeed ifTCAlarmState
T3UnchannelizedPhysical	ATMLogical	BAN	ifDescr ifDs3PlcpAlarmState ifDs3PlcpUASs ifMtu ifOCDEvents ifPhysAddress ifSpeed ifTCAlarmState

Physical

The following table describes physical layer objects.

Table 30: Physical objects

Parent	Object	Supported	Request Element Attribute
AdslCard DS3Card DSLCard OC3PolyCard OC3MultiModeCard OC3Aal5SingleModeCard OC3Aal5MultiModeCard OC3Aal52SingleModeCard OC3Aal52MultiModeCard E3Atm4PortPolyCard E3Aal5Atm4PortCard T3Atm4PortPolyCard T3Aal5Atm4PortCard T3Aal52Atm4PortCard T1Ima8PortCard E1Ima8PortCard T1ImaF8PortCard E1ImaF8PortCard T3UplinkAtm4PortCard T3FUplinkAtm4PortCard E3UplinkAtm4PortCard E3FUplinkAtm4PortCard OC3UplinkAtm2PortCard OC3FUplinkAtmCard T1UplinkAtm8PortCard T1FUplinkAtm8PortCard E1UplinkAtm8PortCard E1FUplinkAtm8PortCard	Aal5ProxyPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	ifDescr ifHCInBroadcastPkts ifHCInMulticastPkts ifHCInOctets ifHCInBits ifHCInUcastPkts ifHCOutBroadcastPkts ifHCOutMulticastPkts ifHCOutOctets ifHCOutBits ifHCOutUcastPkts ifInDiscards ifInErrors ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange ifMtu ifOperStatus ifOutDiscards ifOutErrors ifOutOctets ifOutBits ifOutUcastPkts ifPhysAddress ifSpeed
AdslPhysical	AdslFastPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	ifDescr ifInDiscards ifInErrors ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange ifMtu ifOperStatus ifOutDiscards ifOutErrors ifOutOctets ifOutBits ifOutUcastPkts ifPhysAddress ifSpeed

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
AdslPhysical	AdslInterleavedPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	ifDescr ifInDiscards ifInErrors ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange ifMtu ifOperStatus ifOutDiscards ifOutErrors ifOutOctets ifOutBits ifOutUcastPkts ifPhysAddress ifSpeed

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
AdslCard	AdslPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	atucActualTxMode atucChanBackgroundErrBlks atucChanCorrectedBlks atucChanCrcBlockLength atucChanCurr15MinCorrectedBlks atucChanCurr15MinRxBlks atucChanCurr15MinTimeElapsed atucChanCurr15MinTxBlks atucChanCurr15MinUncorrectBlks atucChanCurr1DayCorrectedBlks atucChanCurr1DayRxBlks atucChanCurr1DayTimeElapse atucChanCurr1DayTxBlks atucChanCurr1DayUncorrectBlks atucChanCurrTxRate atucChanErrFast atucChanErrInterleave atucChanFECCorrectedFast atucChanFECCorrectedInterleave atucChanFastRetrainFailures atucChanFastRetrains atucChanInterleaveDelay atucChanIntervCorrectedBlks atucChanIntervNum atucChanIntervRxBlks atucChanIntervTxBlks atucChanIntervUncorrectBlks atucChanIntervValidData atucChanInvalidIntervals atucChanNonSES atucChanPerfBE atucChanPerfCD atucChanPerfCU atucChanPrev1DayCorrectedBlks atucChanPrev1DayMoniSecs atucChanPrev1DayRxBlks atucChanPrev1DayTxBlks atucChanPrev1DayUncorrectBlks atucChanPrevTxRate atucChanRxBlks atucChanTxBlks atucChanUncorrectBlks atucChanValidIntervals atucCurr15MinEss atucCurr15MinInits atucCurr15MinLofs atucCurr15MinLols atucCurr15MinLoss atucCurr15MinLprs

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
AdslCard	AdslPhysical (Continued)	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	atucCurr15MinTimeElapsed, atucCurr1DayEss, atucCurr1DayInits, atucCurr1DayLofs, atucCurr1DayLols, atucCurr1DayLoss, atucCurr1DayLprs, atucCurr1DayTimeElapsed, atucCurrAtn, atucCurrAttainableRate, atucCurrOutputPwr, atucCurrSnrMgn, atucCurrStatus, atucDisconnects, atucElapseds, atucEss, atucFECs, atucHiBER, atucInits, atucIntervEss, atucIntervInits, atucIntervLofs, atucIntervLols, atucIntervLoss, atucIntervLprs, atucIntervNum, atucIntervValidData, atucInvSerialNumber, atucInvVendorID, atucInvVersionNumber, atucInvalidIntervals, atucLofs, atucLols, atucLoss, atucLprs, atucPerfCurr15MinFecsL, atucPerfCurr1DayFecsL, atucPerfFecsL, atucPerfLossL, atucPerfPrev1DayFecsL, atucPerfSuccessfulConnects, atucPrev1DayEss, atucPrev1DayInits, atucPrev1DayLofs, atucPrev1DayLols, atucPrev1DayLoss, atucPrev1DayLprs, atucPrev1DayMoniSecs, atucSEs, atucUASs, atucValidIntervals, aturChanBackgroundErrBlks, aturChanCorrectedBlks, aturChanCrcBlockLength, aturChanCurr15MinCorrectedBlks, aturChanCurr15MinRxBlks, aturChanCurr15MinTimeElapsed, aturChanCurr15MinTxBlks, aturChanCurr15MinUncorrectBlks, aturChanCurr1DayCorrectedBlks, aturChanCurr1DayRxBlks, aturChanCurr1DayTimeElapsed,

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
AdslCard	AdslPhysical (Continued)	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	aturChanCurr1DayTxBlks, aturChanCurr1DayUncorrectBlks, aturChanCurrTxRate, aturChanErrFast, aturChanErrInterleave, aturChanFECCorrectedFast, aturChanFECCorrectedInterleave, aturChanInterleaveDelay, aturChanIntervCorrectedBlks, aturChanIntervNum, aturChanIntervRxBlks, aturChanIntervTxBlks, aturChanIntervUncorrectBlks, aturChanIntervValidData, aturChanInvalidIntervals, aturChanNonSES, aturChanPerfBE, aturChanPerfCD, aturChanPerfCU, aturChanPrev1DayCorrectedBlks, aturChanPrev1DayMoniSecs, aturChanPrev1DayRxBlks, aturChanPrev1DayTxBlks, aturChanPrev1DayUncorrectBlks, aturChanPrevTxRate, aturChanRxBlks, aturChanTxBlks, aturChanUncorrectBlks, aturChanValidIntervals, aturCurr15MinEss, aturCurr15MinLofs, aturCurr15MinLoss, aturCurr15MinLprs, aturCurr15MinTimeElapsed, aturCurr1DayEss, aturCurr1DayLofs, aturCurr1DayLoss, aturCurr1DayLprs, aturCurr1DayTimeElapsed, aturCurrAtn, aturCurrAttainableRate, aturCurrOutputPwr, aturCurrSnrMgn, aturCurrStatus, aturElapsedds, aturEss, aturFECs, aturHiBER, aturIntervEss, aturIntervLofs,

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
AdslCard	AdslPhysical (Continued)	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	aturIntervLoss, aturIntervLprs, aturIntervNum, aturIntervValidData, aturInvSerialNumber, aturInvVendorID, aturInvVersionNumber, aturInvalidIntervals, aturLofs, aturLoss, aturLprs, aturPerfCurr15MinFecsL, aturPerfCurr1DayFecsL, aturPerfFecsL, aturPerfLossL, aturPerfLprs, aturPerfPrev1DayFecsL, aturPrev1DayEss, aturPrev1DayLofs, aturPrev1DayLoss, aturPrev1DayLprs, aturPrev1DayMoniSecs, aturSESS, aturUASs, aturValidIntervals, deltDevice, deltFailReason, deltTimeCompleted, downAttainLineRate, downLineRate, ifAdminStatus, ifDescr, ifHCInBroadcastPkts, ifHCInMulticastPkts, ifHCInOctets, ifHCInBits ifHCInUcastPkts, ifHCOutBroadcastPkts, ifHCOutMulticastPkts, ifHCOutOctets, ifHCOutBits ifHCOutUcastPkts, ifInDiscards, ifInErrors, ifInOctets, ifInBits ifInUcastPkts, ifInUnknownProtos, ifLastChange, ifMtu, ifOperStatus,

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
AdslCard	AdslPhysical (Continued)	BAN MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	ifOutDiscards, ifOutErrors, ifOutOctets, ifOutBits ifOutUcastPkts, ifPhysAddress ifSpeed lineAlarmStatus lineCoding lineType lineUpTime seltBridgeTapsSupported seltDevice seltLnInbandNoise seltStatus seltTimeCompleted seltTimeLeft upAttainLineRate upLineRate vendorSpecificPointer
DslIma8PortCard	ImaPhysicalGroup	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	ifDescr ifHCInBroadcastPkts ifHCInMulticastPkts ifHCInOctets ifHCInBits ifHCInUcastPkts ifHCOutBroadcastPkts ifHCOutMulticastPkts ifHCOutOctets ifHCOutBits ifHCOutUcastPkts ifInDiscards ifInErrors ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange ifMtu ifOperStatus ifOutDiscards ifOutErrors ifOutOctets ifOutBits ifOutUcastPkts ifPhysAddress ifSpeed

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
E1Tdm63PortCard E1V52Ima16PortCard E1V52Uni16PortCard MtacE1Card MtacFcE1Card S100AE1Card	E1Physical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1	currentCSSs currentDMs currentESs currentIndex currentLCVs currentLESs currentPCVs currentSEFSs currentSESs currentUASs farEndCurrentBESs farEndCurrentCSSs farEndCurrentDMs farEndCurrentESs farEndCurrentIndex farEndCurrentLESs farEndCurrentPCVs farEndCurrentSEFSs farEndCurrentSESs farEndCurrentUASs farEndIntervalBESs farEndIntervalCSSs farEndIntervalDMs farEndIntervalESs farEndIntervalIndex farEndIntervalLESs farEndIntervalNumber farEndIntervalPCVs farEndIntervalSEFSs farEndIntervalSESs farEndIntervalUASs farEndIntervalValidData farEndInvalidIntervals farEndTimeElapsed farEndTotalBESs farEndTotalCSSs farEndTotalDMs farEndTotalESs farEndTotalIndex farEndTotalLESs farEndTotalPCVs farEndTotalSEFSs farEndTotalSESs farEndTotalUASs farEndValidIntervals ifAdminStatus, ifDescr

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
E1Tdm63PortCard E1V52Ima16PortCard E1V52Uni16PortCard MtacE1Card MtacFcE1Card S100AE1Card	E1Physical (Continued)	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1	ifHCInBroadcastPkts ifHCInMulticastPkts ifHCInOctets ifHCInBits ifHCInUcastPkts ifHCOutBroadcastPkts ifHCOutMulticastPkts ifHCOutOctets ifHCOutBits ifHCOutUcastPkts ifInDiscards ifInErrors ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange ifMtu ifOperStatus ifOutDiscards ifOutErrors ifOutOctets ifOutBits ifOutUcastPkts ifPhysAddress ifSpeed intervalBESSs intervalCSSs intervalDMs intervalESSs intervalIndex intervalLCVs intervalLESSs intervalNumber intervalPCVs intervalSEFSs intervalSESSs intervalUASs intervalValidData invalidIntervals lineStatus lineStatusLastChange loopbackStatus timeElapsed totalBESSs totalCSSs totalDMs totalESSs totalIndex totalLCVs totalLESSs totalPCVs totalSEFSs totalSESSs

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
E1Tdm63PortCard E1V52Ima16PortCard E1V52Uni16PortCard MtacE1Card MtacFcE1Card S100AE1Card	E1Physical (Continued)	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1	totalUASs transmitClockSource validIntervals

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
DS3Card	E3Physical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	currentCCVs currentCESs currentCSESs currentIndex currentLCVs currentLESs currentPCVs currentPESs currentPSESs currentSEFSs currentUASs farEndCurrentCCVs farEndCurrentCESs farEndCurrentCSESs farEndCurrentIndex farEndCurrentInvalidIntervals farEndCurrentUASs farEndIntervalCCVs farEndIntervalCESs farEndIntervalCSESs farEndIntervalIndex farEndIntervalNumber farEndIntervalUASs farEndIntervalValidData farEndTimeElapsed farEndTotalCCVs farEndTotalCESs farEndTotalCSESs farEndTotalIndex farEndTotalUASs farEndValidIntervals ifDescr ifHCInBroadcastPkts ifHCInMulticastPkts ifHCInOctets ifHCInBits ifHCInUcastPkts ifHCOutBroadcastPkts ifHCOutMulticastPkts ifHCOutOctets ifHCOutBits ifHCOutUcastPkts ifInDiscards ifInErrors ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
DS3Card	E3Physical (continued)	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	ifMtu ifOperStatus ifOutDiscards ifOutErrors ifOutOctets ifOutBits ifOutUcastPkts ifPhysAddress ifSpeed intervalCCVs intervalCESS intervalCSEs intervalIndex intervalLCVs intervalLESS intervalNumber intervalPCVs intervalPESs intervalPSESs intervalSEFSs intervalUASs intervalValidData invalidIntervals lineStatus lineStatusLastChange loopbackStatus timeElapsed totalCCVs totalCESS totalCSEs totalIndex totalLCVs totalLESS totalPCVs totalPESs totalPSESs totalSEFSs totalUASs validIntervals

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
E3Aal5Atm4PortCard E3Arm4PortPolyCard	E3UnchannelizedPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor100 Raptor 319 Raptor 719 Raptor 723	currentCCVs currentCESs currentCSEs currentIndex currentLCVs currentLESs currentPCVs currentPESs currentPSESs currentSEFSs currentUASs farEndCurrentCCVs farEndCurrentCESs farEndCurrentCSEs farEndCurrentIndex farEndCurrentInvalidIntervals farEndCurrentUASs farEndIntervalCCVs farEndIntervalCESs farEndIntervalCSEs farEndIntervalIndex farEndIntervalNumber farEndIntervalUASs farEndIntervalValidData farEndTimeElapsed farEndTotalCCVs farEndTotalCESs farEndTotalCSEs farEndTotalIndex farEndTotalUASs farEndValidIntervals ifDescr ifHCInBroadcastPkts ifHCInMulticastPkts ifHCInOctets ifHCInBits ifHCInUcastPkts ifHCOutBroadcastPkts ifHCOutMulticastPkts ifHCOutOctets ifHCOutBits ifHCOutUcastPkts ifInDiscards ifInErrors ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
E3Aal5Atm4PortCard E3Arm4PortPolyCard	E3UnchannelizedPhysical (continued)	BAN MALC 319 MALC 719 MALC 723 Raptor100 Raptor 319 Raptor 719 Raptor 723	ifMtu ifOperStatus ifOutDiscards ifOutErrors ifOutOctets ifOutBits ifOutUcastPkts ifPhysAddress ifSpeed intervalCCVs intervalCESSs intervalCSESSs intervalIndex intervalLCVs intervalLESSs intervalNumber intervalPCVs intervalPESs intervalPSESs intervalSEFSs intervalUASs intervalValidData invalidIntervals lineStatus lineStatusLastChange loopbackStatus timeElapsed totalCCVs totalCESSs totalCSESSs totalIndex totalLCVs totalLESSs totalPCVs totalPESs totalPSESs totalSEFSs totalUASs validIntervals

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
E1Atm24PortCard E1Ima8PortCard E1ImaF8PortCard E1UplinkAtm8PortCard E1FUplinkAtm8PortCard E1V52Ima16PortCard E1V52Uni16PortCard MALC100T1ImaCard	E1ATMPPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	currentBESs currentCSSs currentDMs currentESs currentIndex currentLCVs currentLESSs currentPCVs currentSEFSs currentSESs currentUASs farEndCurrentBESs farEndCurrentCSSs farEndCurrentDMs farEndCurrentESs farEndCurrentIndex farEndCurrentLESSs farEndCurrentPCVs farEndCurrentSEFSs farEndCurrentSESs farEndCurrentUASs farEndIntervalBESs farEndIntervalCSSs farEndIntervalDMs farEndIntervalESs farEndIntervalIndex farEndIntervalLESSs farEndIntervalPCVs farEndIntervalSEFSs farEndIntervalSESs farEndIntervalUASs farEndIntervalValidData farEndInvalidIntervals farEndTimeElapsed farEndTotalBESs farEndTotalCSSs farEndTotalDMs farEndTotalESs farEndTotalIndex farEndTotalLESSs farEndTotalPCVs farEndTotalSEFSs farEndTotalSESs farEndTotalUASs farEndValidIntervals ifDescr ifHCInBroadcastPkts ifHCInMulticastPkts ifHCInOctets ifHCInBits

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
E1Atm24PortCard	E1ATMPPhysical (continued)	BAN	ifHCInUcastPkts
E1Ima8PortCard		MALC 319	ifHCOutBroadcastPkts
E1ImaF8PortCard		MALC 719	ifHCOutMulticastPkts
E1UplinkAtm8PortCard		MALC 723	ifHCOutOctets
E1FUplinkAtm8PortCard		Raptor 319	ifHCOutBits
E1V52Ima16PortCard		Raptor 719	ifHCOutUcastPkts
E1V52Uni16PortCard		Raptor 723	ifInDiscards
MALC100T1ImaCard			ifInErrors
			ifInOctets
			ifInBits
			ifInUcastPkts
			ifInUnknownProtos
			ifLastChange
			ifMTU
			ifOperStatus
			ifOutDiscards
			ifOutErrors
			ifOutOctets
			ifOutBits
			ifOutUcastPkts
			ifPhysAddress
			ifSpeed
			intervalBESSs
			intervalCSSs
			intervalDMs
			intervalESSs
			intervalIndex
			intervalLCVs
			intervalLESSs
			intervalNumber
			intervalPCVs
			intervalSEFSs
			intervalSESs
			intervalUASs
			intervalValidData
			invalidIntervals
			lineStatus
			lineStatusLastChange
			loopbackStatus
			timeElapsed
			totalBESSs
			totalCSSs
			totalDMs
			totalESSs
			totalIndex
			totalLCVs
			totalLESSs
			totalPCVs
			totalSEFSs
			totalSESs
			totalUASs
			transmitClockSource
			validIntervals

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
S100AE1Card	EtherPhysical	Sechtor 100A E1	InPauseFrame collCount collFrequencies controlInUnknownOpCodes ifAdminStatus ifDescr ifHCInBroadcastPkts ifHCInMulticastPkts ifHCInOctets ifHCInBits ifHCInUcastPkts ifHCOutBroadcastPkts ifHCOutMulticastPkts ifHCOutOctets ifHCOutBits ifHCOutUcastPkts ifHighSpeed ifInBroadcastPkts ifInDiscards ifInErrors ifInMulticastPkts ifInNUcastPkts ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange ifMtu ifOperStatus ifOutBroadcastPkts ifOutDiscards ifOutErrors ifOutMulticastPkts ifOutNUcastPkts ifOutOctets ifOutBits ifOutUcastPkts ifPhysAddress ifSpeed outPauseFrame pauseOperMode statsAlignmentErrors statsCarrierSenseErrors statsDeferredTransmission statsDuplexStatus statsExcessiveCollisions statsFCSErrors statsSQETestErrors

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
S100AE1Card	EtherPhysical (continued)	Sechtor 100A E1	statsFrameTooLongs statsIndex statsInternalMacReceiveErrors statsInternalMacTransmitErrors statsLateCollisions statsMultipleCollisionFrames statsSingleCollisionFrames statsSymbolErrors

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Card	EtherPhysical Note: For retrieving the statistics at GIGE port level, perform the statistics retrieval operation on EtherPhysical with the corresponding Handle_Id of Gige Port.	MALC 319 MALC 719 MALC 723	ifHCInBroadcastPkts ifHCInMulticastPkts ifHCInOctets ifHCInBits ifHCInUcastPkts ifHCOutBroadcastPkts ifHCOutMulticastPkts ifHCOutOctets ifHCOutBits ifHCOutUcastPkts ifHighSpeed ifInBroadcastPkts ifInDiscards ifInErrors ifInMulticastPkts ifInNUcastPkts ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange ifOutBroadcastPkts ifOutDiscards ifOutErrors ifOutMulticastPkts ifOutNUcastPkts ifOutOctets ifOutBits ifOutUcastPkts ifPhysAddress ifSpeed statsCarrierSenseErrors statsDeferredTransmission statsDuplexStatus statsExcessiveCollisions statsFCSErrors statsFrameTooLongs statsIndex statsInternalMacReceiveErrors statsInternalMacTransmitErrors statsLateCollisions statsMultipleCollisionFrames statsSQETestErrors statsSingleCollisionFrames statsSymbolErrors transceiverType upperBitRateMarginPercentage vendorName vendorOui

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Card	EtherPhysical Note: For retrieving the statistics at GIGE port level, perform the statistics retrieval operation on EtherPhysical with the corresponding Handle_Id of Gige Port.	MXK 819 MXK 823 MXK 319	ifHCInBroadcastPkts ifHCInMulticastPkts ifHCInOctets ifHCInBits ifHCInUcastPkts ifHCOutBroadcastPkts ifHCOutMulticastPkts ifHCOutOctets ifHCOutBits ifHCOutUcastPkts ifHighSpeed ifInBroadcastPkts ifInDiscards ifInErrors ifInMulticastPkts ifInNUcastPkts ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange ifOutBroadcastPkts ifOutDiscards ifOutErrors ifOutMulticastPkts ifOutNUcastPkts ifOutOctets ifOutBits ifOutUcastPkts ifPhysAddress ifSpeed statsCarrierSenseErrors statsDeferredTransmission statsDuplexStatus statsExcessiveCollisions statsFCSErrors statsFrameTooLongs statsIndex statsInternalMacReceiveErrors statsInternalMacTransmitErrors statsLateCollisions statsMultipleCollisionFrames statsSQETestErrors statsSingleCollisionFrames statsSymbolErrors transceiverType upperBitRateMarginPercentage vendorName vendorOui

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Card	EtherPhysical (continued)	MXK 819 MXK 823 MXK 319	zhoneEtherStatsAverageRxThroughput zhoneEtherStatsAverageTxThroughput zhoneEtherStatsBroadcastPkts zhoneEtherStatsCRCAlignErrors zhoneEtherStatsCollisions zhoneEtherStatsDropEvents zhoneEtherStatsDroppedFrames zhoneEtherStatsFragments zhoneEtherStatsIPMCBridgedPkts zhoneEtherStatsIPMCRoutedPkts zhoneEtherStatsIPMCRxDroppedPkts zhoneEtherStatsIPMCTxDroppedPkts zhoneEtherStatsJabbers zhoneEtherStatsMcastRxBytes zhoneEtherStatsMcastRxDroppedBytes zhoneEtherStatsMcastTxBytes zhoneEtherStatsMulticastPkts zhoneEtherStatsOctets zhoneEtherStatsOversizePkts zhoneEtherStatsOversizeRxPkts zhoneEtherStatsOversizeTxPkts zhoneEtherStatsPkts zhoneEtherStatsPkts1024to1518Octets zhoneEtherStatsPkts128to255Octets zhoneEtherStatsPkts1519to2047Octets zhoneEtherStatsPkts2048to4095Octets zhoneEtherStatsPkts256to511Octets zhoneEtherStatsPkts4095to9216Octets zhoneEtherStatsPkts512to1023Octets zhoneEtherStatsPkts64Octets zhoneEtherStatsPkts65to127Octets zhoneEtherStatsRXNoErrors zhoneEtherStatsReceivedPkts1024to1518Octets

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Card	EtherPhysical (continued)	MXK 819 MXK 823 MXK 319	zhoneEtherStatsReceivedPkts128to255Octets zhoneEtherStatsReceivedPkts1519to2047Octets zhoneEtherStatsReceivedPkts2048to4095Octets zhoneEtherStatsReceivedPkts256to511Octets zhoneEtherStatsReceivedPkts4095to9216Octets zhoneEtherStatsReceivedPkts512to1023Octets zhoneEtherStatsReceivedPkts64Octets zhoneEtherStatsReceivedPkts65to127Octets zhoneEtherStatsRxBandwidthOccupancy zhoneEtherStatsRxPkts zhoneEtherStatsTXNoErrors zhoneEtherStatsTransmittedPkts1024to1518Octets zhoneEtherStatsTransmittedPkts128to255Octets zhoneEtherStatsTransmittedPkts1519to2047Octets zhoneEtherStatsTransmittedPkts2048to4095Octets zhoneEtherStatsTransmittedPkts256to511Octets zhoneEtherStatsTransmittedPkts4095to9216Octets zhoneEtherStatsTransmittedPkts512to1023Octets zhoneEtherStatsTransmittedPkts64Octets zhoneEtherStatsTransmittedPkts65to127Octets zhoneEtherStatsTxBandwidthOccupancy zhoneEtherStatsTxPkts zhoneEtherStatsUndersizePkts

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Mx4GponCard Mx8GponCard	GponOltPhysical	MXK 819 MXK 823	zhoneGponOltStatisticsDownstrea mDiscardedFrames zhoneGponOltStatisticsDownstrea mGemFrames zhoneGponOltStatisticsDownstrea mOmciFrames zhoneGponOltStatisticsDownstrea mPloamFrames zhoneGponOltStatisticsDownstrea mPonCpuPkts zhoneGponOltStatisticsDownstrea mPonValidEthernetPkts zhoneGponOltStatisticsDownstrea mTmCrcPkts zhoneGponOltStatisticsDownstrea mTmDroppedCpuPkts zhoneGponOltStatisticsDownstrea mTmMacLookupMiss zhoneGponOltStatisticsDownstrea mTmPktsDroppedGemPidNotEnab led zhoneGponOltStatisticsDownstrea mTmPktsForwardedFromHmToPo n zhoneGponOltStatisticsDownstrea mTmQ0DroppedPkts zhoneGponOltStatisticsDownstrea mTmQ0ValidPkts zhoneGponOltStatisticsDownstrea mTmQ1DroppedPkts zhoneGponOltStatisticsDownstrea mTmQ1ValidPkts zhoneGponOltStatisticsDownstrea mTmQ2DroppedPkts zhoneGponOltStatisticsDownstrea mTmQ2ValidPkts zhoneGponOltStatisticsDownstrea mTmQ3DroppedPkts zhoneGponOltStatisticsDownstrea mTmQ3ValidPkts zhoneGponOltStatisticsDownstrea mTmQ4DroppedPkts zhoneGponOltStatisticsDownstrea mTmQ4ValidPkts zhoneGponOltStatisticsDownstrea mTmQ5DroppedPkts zhoneGponOltStatisticsDownstrea mTmQ5ValidPkts zhoneGponOltStatisticsDownstrea mTmQ6DroppedPkts

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Mx4GponCard Mx8GponCard	GponOltPhysical (Continued)	MXK 819 MXK 823	zhoneGponOltStatisticsDownstrea mTmQ6ValidPkts zhoneGponOltStatisticsDownstrea mTmQ7DroppedPkts zhoneGponOltStatisticsDownstrea mTmQ7ValidPkts zhoneGponOltStatisticsDownstrea mTmValidPkts zhoneGponOltStatisticsDownstrea mTxBytes zhoneGponOltStatisticsDownstrea mValidGemFrames zhoneGponOltStatisticsUpstreamD iscardedFrames zhoneGponOltStatisticsUpstreamD roppedPktsInactivePorts zhoneGponOltStatisticsUpstreamD roppedPloamsFifoFull zhoneGponOltStatisticsUpstreamG emFrames zhoneGponOltStatisticsUpstreamId lePloamFrames zhoneGponOltStatisticsUpstreamO mciFrames zhoneGponOltStatisticsUpstreamPl oamFrames zhoneGponOltStatisticsUpstreamP onErrorPloams zhoneGponOltStatisticsUpstreamP onInvalidPkts zhoneGponOltStatisticsUpstreamP onValidNotIdlePloams zhoneGponOltStatisticsUpstreamP onValidPkts zhoneGponOltStatisticsUpstreamT mDroppedCpuPkts zhoneGponOltStatisticsUpstreamT mDroppedPktsCreError zhoneGponOltStatisticsUpstreamT mDroppedPktsSecurity zhoneGponOltStatisticsUpstreamT mLearnFailures zhoneGponOltStatisticsUpstreamT mQ0DroppedPkts zhoneGponOltStatisticsUpstreamT mQ0ValidPkts zhoneGponOltStatisticsUpstreamT mQ1DroppedPkts

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Mx4GponCard Mx8GponCard	GponOltPhysical (Continued)	MXK 819 MXK 823	zhoneGponOltStatisticsUpstreamT mQ1ValidPkts zhoneGponOltStatisticsUpstreamT mQ2DroppedPkts zhoneGponOltStatisticsUpstreamT mQ2ValidPkts zhoneGponOltStatisticsUpstreamT mQ3DroppedPkts zhoneGponOltStatisticsUpstreamT mQ3ValidPkts zhoneGponOltStatisticsUpstreamT mQ4DroppedPkts zhoneGponOltStatisticsUpstreamT mQ4ValidPkts zhoneGponOltStatisticsUpstreamT mQ5DroppedPkts zhoneGponOltStatisticsUpstreamT mQ5ValidPkts zhoneGponOltStatisticsUpstreamT mQ6DroppedPkts zhoneGponOltStatisticsUpstreamT mQ6ValidPkts zhoneGponOltStatisticsUpstreamT mQ7DroppedPkts zhoneGponOltStatisticsUpstreamT mQ7ValidPkts zhoneGponOltStatisticsUpstreamV alidGemFrames

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Mx4GponCard Mx8GponCard	GponOltPhysical (Continued)	MXK 819 MXK 823	zhoneEtherStatsAverageRxThroughput zhoneEtherStatsAverageTxThroughput zhoneEtherStatsBroadcastPkts zhoneEtherStatsCRCAlignErrors zhoneEtherStatsCollisions zhoneEtherStatsDropEvents zhoneEtherStatsDroppedFrames zhoneEtherStatsFragments zhoneEtherStatsIPMCBridgedPkts zhoneEtherStatsIPMCRoutedPkts zhoneEtherStatsIPMCRxDroppedPkts zhoneEtherStatsIPMCTxDroppedPkts zhoneEtherStatsJabbers zhoneEtherStatsMcastRxBytes zhoneEtherStatsMcastRxDroppedBytes zhoneEtherStatsMcastTxBytes zhoneEtherStatsMulticastPkts zhoneEtherStatsOctets zhoneEtherStatsOversizePkts zhoneEtherStatsOversizeRxPkts zhoneEtherStatsOversizeTxPkts zhoneEtherStatsPkts zhoneEtherStatsPkts1024to1518Octets zhoneEtherStatsPkts128to255Octets zhoneEtherStatsPkts1519to2047Octets zhoneEtherStatsPkts2048to4095Octets zhoneEtherStatsPkts256to511Octets zhoneEtherStatsPkts4095to9216Octets zhoneEtherStatsPkts512to1023Octets zhoneEtherStatsPkts64Octets zhoneEtherStatsPkts65to127Octets zhoneEtherStatsRXNoErrors zhoneEtherStatsReceivedPkts1024to1518Octets zhoneEtherStatsReceivedPkts128to255Octets zhoneEtherStatsReceivedPkts1519to2047Octets

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Mx4GponCard Mx8GponCard	GponOltPhysical (continued)	MXK 819 MXK 823	zhoneEtherStatsReceivedPkts2048to4095Octets zhoneEtherStatsReceivedPkts256to511Octets zhoneEtherStatsReceivedPkts4095to9216Octets zhoneEtherStatsReceivedPkts512to1023Octets zhoneEtherStatsReceivedPkts64Octets zhoneEtherStatsReceivedPkts65to127Octets zhoneEtherStatsRxBandwidthOccupancy zhoneEtherStatsRxPkts zhoneEtherStatsTXNoErrors zhoneEtherStatsTransmittedPkts1024to1518Octets zhoneEtherStatsTransmittedPkts128to255Octets zhoneEtherStatsTransmittedPkts1519to2047Octets zhoneEtherStatsTransmittedPkts2048to4095Octets zhoneEtherStatsTransmittedPkts256to511Octets zhoneEtherStatsTransmittedPkts4095to9216Octets zhoneEtherStatsTransmittedPkts512to1023Octets zhoneEtherStatsTransmittedPkts64Octets zhoneEtherStatsTransmittedPkts65to127Octets zhoneEtherStatsTxBandwidthOccupancy zhoneEtherStatsTxPkts zhoneEtherStatsUndersizePkts

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Card	LinkAggPhysical	MALC 319	ifInErrors
		MALC 719	ifHCOutMulticastPkts
		MALC 723	ifSpeed
		MXK 819	ifHCInMulticastPkts
		MXK 823	ifOutDiscards
		MXK 319	ifHCOutBits
			ifHCOutOctets
			ifHCOutBits
			ifHCInUcastPkts
			ifHCOutUcastPkts
			ifHCOutBroadcastPkts
			ifHCInOctets
			ifHCInBits
			ifInDiscards
			ifHCInBroadcastPkts
			ifInUnknownProtos
			ifOutErrors

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Card	LinkAggPhysical	MXK 819 MXK 823 MXK 319	zhoneEtherStatsAverageRxThroughput zhoneEtherStatsAverageTxThroughput zhoneEtherStatsBroadcastPkts zhoneEtherStatsCRCAlignErrors zhoneEtherStatsCollisions zhoneEtherStatsDropEvents zhoneEtherStatsDroppedFrames zhoneEtherStatsFragments zhoneEtherStatsIPMCBridgedPkts zhoneEtherStatsIPMCRoutedPkts zhoneEtherStatsIPMCRxDroppedPkts zhoneEtherStatsIPMCTxDroppedPkts zhoneEtherStatsJabbers zhoneEtherStatsMcastRxBytes zhoneEtherStatsMcastRxDroppedBytes zhoneEtherStatsMcastTxBytes zhoneEtherStatsMulticastPkts zhoneEtherStatsOctets zhoneEtherStatsOversizePkts zhoneEtherStatsOversizeRxPkts zhoneEtherStatsOversizeTxPkts zhoneEtherStatsPkts zhoneEtherStatsPkts1024to1518Octets zhoneEtherStatsPkts128to255Octets zhoneEtherStatsPkts1519to2047Octets zhoneEtherStatsPkts2048to4095Octets zhoneEtherStatsPkts256to511Octets zhoneEtherStatsPkts4095to9216Octets zhoneEtherStatsPkts512to1023Octets zhoneEtherStatsPkts64Octets zhoneEtherStatsPkts65to127Octets zhoneEtherStatsRXNoErrors zhoneEtherStatsReceivedPkts1024to1518Octets zhoneEtherStatsReceivedPkts128to255Octets

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Card	LinkAggPhysical (continued)	MXK 819 MXK 823 MXK 319	zhoneEtherStatsReceivedPkts1519to2047Octets zhoneEtherStatsReceivedPkts2048to4095Octets zhoneEtherStatsReceivedPkts256to511Octets zhoneEtherStatsReceivedPkts4095to9216Octets zhoneEtherStatsReceivedPkts512to1023Octets zhoneEtherStatsReceivedPkts64Octets zhoneEtherStatsReceivedPkts65to127Octets zhoneEtherStatsRxBandwidthOccupancy zhoneEtherStatsRxPkts zhoneEtherStatsTXNoErrors zhoneEtherStatsTransmittedPkts1024to1518Octets zhoneEtherStatsTransmittedPkts128to255Octets zhoneEtherStatsTransmittedPkts1519to2047Octets zhoneEtherStatsTransmittedPkts2048to4095Octets zhoneEtherStatsTransmittedPkts256to511Octets zhoneEtherStatsTransmittedPkts4095to9216Octets zhoneEtherStatsTransmittedPkts512to1023Octets zhoneEtherStatsTransmittedPkts64Octets zhoneEtherStatsTransmittedPkts65to127Octets zhoneEtherStatsTxBandwidthOccupancy zhoneEtherStatsTxPkts zhoneEtherStatsUndersizePkts

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Card	VdslPhysical	MXK 819	ifHCInBroadcastPkts
		MXK 823	ifHCInMulticastPkts
		MXK 319	ifHCInOctets
			ifHCInUcastPkts
			ifHCOutBroadcastPkts
			ifHCOutMulticastPkts
			ifHCOutOctets
			ifHCOutUcastPkts
			ifInDiscards
			ifInErrors
			ifInUnknownProtos
			ifOutDiscards
			ifOutErrors
			zhoneDslPerfCRCAnomalies
			zhoneDslPerfES
			zhoneDslPerfInits
			zhoneDslPerfLOSWS
			zhoneDslPerfLofs
			zhoneDslPerfLols
			zhoneDslPerfLoss
			zhoneDslPerfSES
			zhoneDslPerfTimeElapsed
			zhoneDslPerfTotalCRCAnomalies
			zhoneDslPerfTotalES
			zhoneDslPerfTotalInits
			zhoneDslPerfTotalLOSWS
			zhoneDslPerfTotalLofs
			zhoneDslPerfTotalLols
			zhoneDslPerfTotalLoss
			zhoneDslPerfTotalSES
			zhoneDslPerfTotalUAS
			zhoneDslPerfUAS

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Card	VdslPhysical (Continued)	MXK 819 MXK 823 MXK 319	zhoneVdslPerfMonDsCorrOctets zhoneVdslPerfMonDsUncorrOctets zhoneVdslPerfMonUsCorrOctets zhoneVdslPerfMonUsUncorrOctets zhoneVdslPhyrVtucCorrectedRetransmittedCodewords zhoneVdslPhyrVtucRetransmittedCodewords zhoneVdslPhyrVtucUncorrectableRetransmittedCodewords zhoneVdslPhyrVturCorrectedRetransmittedCodewords zhoneVdslPhyrVturRetransmittedCodewords zhoneVdslPhyrVturUncorrectableRetransmittedCodewords zhoneEtherStatsAverageRxThroughput zhoneEtherStatsAverageTxThroughput zhoneEtherStatsBroadcastPkts zhoneEtherStatsCRCAlignErrors zhoneEtherStatsCollisions zhoneEtherStatsDropEvents zhoneEtherStatsDroppedFrames zhoneEtherStatsFragments zhoneEtherStatsIPMCBridgedPkts zhoneEtherStatsIPMCRoutedPkts zhoneEtherStatsIPMCRxDroppedPkts zhoneEtherStatsIPMCTxDroppedPkts zhoneEtherStatsJabbers zhoneEtherStatsMcastRxBytes zhoneEtherStatsMcastRxDroppedBytes zhoneEtherStatsMcastTxBytes zhoneEtherStatsMulticastPkts zhoneEtherStatsOctets zhoneEtherStatsOversizePkts zhoneEtherStatsOversizeRxPkts zhoneEtherStatsOversizeTxPkts zhoneEtherStatsPkts zhoneEtherStatsPkts1024to1518Octets zhoneEtherStatsPkts128to255Octets

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Card	VdslPhysical (Continued)	MXK 819 MXK 823 MXK 319	zhoneEtherStatsPkts1519to2047Octets zhoneEtherStatsPkts2048to4095Octets zhoneEtherStatsPkts256to511Octets zhoneEtherStatsPkts4095to9216Octets zhoneEtherStatsPkts512to1023Octets zhoneEtherStatsPkts64Octets zhoneEtherStatsPkts65to127Octets zhoneEtherStatsRXNoErrors zhoneEtherStatsReceivedPkts1024to1518Octets zhoneEtherStatsReceivedPkts128to255Octets zhoneEtherStatsReceivedPkts1519to2047Octets zhoneEtherStatsReceivedPkts2048to4095Octets zhoneEtherStatsReceivedPkts256to511Octets zhoneEtherStatsReceivedPkts4095to9216Octets zhoneEtherStatsReceivedPkts512to1023Octets zhoneEtherStatsReceivedPkts64Octets zhoneEtherStatsReceivedPkts65to127Octets zhoneEtherStatsRxBandwidthOccupancy zhoneEtherStatsRxPkts zhoneEtherStatsTXNoErrors zhoneEtherStatsTransmittedPkts1024to1518Octets zhoneEtherStatsTransmittedPkts128to255Octets zhoneEtherStatsTransmittedPkts1519to2047Octets zhoneEtherStatsTransmittedPkts2048to4095Octets zhoneEtherStatsTransmittedPkts256to511Octets zhoneEtherStatsTransmittedPkts4095to9216Octets

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
Card	VdslPhysical (Continued)	MXK 819 MXK 823 MXK 319	zhoneEtherStatsTransmittedPkts51 2to1023Octets zhoneEtherStatsTransmittedPkts64 Octets zhoneEtherStatsTransmittedPkts65 to127Octets zhoneEtherStatsTxBandwidthOccu pancy zhoneEtherStatsTxPkts zhoneEtherStatsUndersizePkts
S100AE1Card	MAUPhysical	Sechtor 100A E1	mauAutoNegSupported mauDefaultType mauFalseCarriers mauJabberState mauJabberStateEnters mauMediaAvaStateExits mauMediaAvailable mauOperStatus mauTypeListBits negCapRecvBits negCapabilityBits negConfig negRemoteFaultRecv negRemoteSignaling

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
OltPhysical	OltOnuPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	ifAdminStatus ifDescr ifHCInBroadcastPkts ifHCInMulticastPkts ifHCInOctets ifHCInBits ifHCInUcastPkts ifHCOutBroadcastPkts ifHCOutMulticastPkts ifHCOutOctets ifHCOutBits ifHCOutUcastPkts ifInDiscards ifInErrors ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange ifMtu ifOperStatus ifOutDiscards ifOutErrors ifOutOctets ifOutBits ifOutUcastPkts ifPhysAddress ifSpeed zhoneOltOnuStatusPonAlarmERR zhoneOltOnuStatusPonAlarmLOA zhoneOltOnuStatusPonAlarmMEM zhoneOltOnuStatusPonAlarmREI zhoneOltOnuStatusPonAlarmRXI NH zhoneOltOnuStatusPonAlarmSD

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	OltPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 819 MXK 823	ifAdminStatus ifDescr ifHCInBroadcastPkts ifHCInMulticastPkts ifHCInOctets ifHCInBits ifHCInUcastPkts ifHCOutBroadcastPkts ifHCOutMulticastPkts ifHCOutOctets ifHCOutBits ifHCOutUcastPkts ifInDiscards ifInErrors ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange ifMtu ifOperStatus ifOutDiscards ifOutErrors ifOutOctets ifOutBits ifOutUcastPkts ifPhysAddress ifSpeed zhoneOltStatusLoopback zhoneOltStatusPonAlarmCPE zhoneOltStatusPonAlarmLCD zhoneOltStatusPonAlarmLOS zhoneOltStatusPonAlarmLPHY zhoneOltStatusPonAlarmOAML zhoneOltStatusPonAlarmPEE zhoneOltStatusPonAlarmSUF zhoneOltStatusPonAlarmTF zhoneOltStatusState

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
S100AE1Card	SonetOC3Physical	Sechtor 100A E1	feLineCurrCV feLineCurrES feLineCurrSES feLineCurrUAS feLineIntervCV feLineIntervES feLineIntervNum feLineIntervSES feLineIntervUAS feLineIntervValid fePathCurrCV fePathCurrES fePathCurrSES fePathCurrUAS fePathIntervCV fePathIntervES fePathIntervNum fePathIntervSES fePathIntervUAS fePathIntervvalid fifoOverflowCount hecCorrectedCount hecUncorrectedCount ifAdminStatus ifDescr ifHCInBroadcastPkts ifHCInMulticastPkts ifHCInOctets ifHCInBits ifHCInUcastPkts ifHCOutBroadcastPkts ifHCOutMulticastPkts ifHCOutOctets ifHCOutBits ifHCOutUcastPkts ifInDiscards ifInErrors ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange ifMtu ifOperStatus ifOutDiscards ifOutErrors ifOutOctets ifOutBits

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
S100AE1Card	SonetOC3Physical (continued)	Sechtor 100A E1	ifOutUcastPkts ifPhysAddress ifSpeed invalidIntervals lineBipCount lineCurrCV lineCurrES lineCurrSES lineCurrStatus lineCurrUAS lineFebeCount lineIntervCV lineIntervES lineIntervNum lineIntervSES lineIntervUAS lineIntervValid locdCount oofCount pathBipCount pathCurrCV pathCurrES pathCurrSES pathCurrStatus pathCurrUAS pathFebeCount pathIntervCV pathIntervES pathIntervNum pathIntervSES pathIntervUAS pathIntervValid rxCellCount sectionBipCount sectionCurrCV sectionCurrES sectionCurrSEFS sectionCurrSES sectionIntervCV sectionIntervES sectionIntervNum sectionIntervSEFS sectionIntervSES sectionIntervValid sectionStatus timeElapsed txCellCount validIntervals

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
MALC100T1ImaCard T1Atm24PortCard T1FUplinkAtm8PortCard T1Ima8PortCard T1ImaF8PortCard T1Gr303Ima16PortCard T1ImaF8PortCard T1Tdm64PortCard T1UplinkAtm8PortCard	T1ATMPPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	currentBESs currentCSSs currentDMs currentESs currentIndex currentLCVs curentLESSs currentPCVs currentSEFSs currentSESs currentUASs farEndCurrentBESs farEndCurrentCSSs farEndCurrentDMs farEndCurrentESs farEndCurrentIndex farEndCurentLESSs farEndCurrentPCVs farEndCurrentSEFSs farEndCurrentSESs farEndCurrentUASs farEndIntervalBESs farEndIntervalCSSs farEndIntervalDMs farEndIntervalESs farEndIntervalIndex farEndIntervalLESSs farEndIntervalNumber farEndIntervalPCVs farEndIntervalSEFSs farEndIntervalSESs farEndIntervalUASs farEndIntervalValidData farEndInvalidIntervals farEndTimeElapsed farEndTotalBESs farEndTotalCSSs farEndTotalDMs farEndTotalESs farEndTotalIndex farEndTotalLESSs farEndTotalPCVs farEndTotalSEFSs farEndTotalSESs farEndTotalUASs farEndValidIntervals ifDescr ifHCInBroadcastPkts ifHCInMulticastPkts ifHCInOctets ifHCInBits

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
MALC100T1ImaCard T1Atm24PortCard T1FUpLinkAtm8PortCard T1Ima8PortCard T1ImaF8PortCard T1Gr303Ima16PortCard T1ImaF8PortCard T1Tdm64PortCard T1UpLinkAtm8PortCard	T1ATMPPhysical (continued)	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	ifHCInUcastPkts ifHCOutBroadcastPkts ifHCOutMulticastPkts ifHCOutOctets ifHCOutBits ifHCOutUcastPkts ifInDiscards ifInErrors ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange ifMTU ifOperStatus ifOutDiscards ifOutErrors ifOutOctets ifOutBits ifOutUcastPkts ifPhysAddress ifSpeed intervalBESSs intervalCSSs intervalDMs intervalESSs intervalIndex intervalLCVs intervalLESSs intervalNumber intervalPCVs intervalSEFSs intervalSESSs intervalUASs intervalValidData invalidIntervals lineStatus lineStatusLastChange loopbackStatus timeElapsed totalBESSs totalCSSs totalDMs totalESSs totalIndex totalLCVs totalLESSs totalPCVs totalSEFSs totalSESSs totalUASs transmitClockSource validIntervals

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
DS3Card	T3Physical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	currentCCVs currentCESs currentCSESs currentIndex currentLCVs currentLESs currentPCVs currentPESs currentPSESs currentSEFSs currentUASs farEndCurrentCCVs farEndCurrentCESs farEndCurrentCSESs farEndCurrentIndex farEndCurrentInvalidIntervals farEndCurrentUASs farEndIntervalCCVs farEndIntervalCESs farEndIntervalCSESs farEndIntervalIndex farEndIntervalNumber farEndIntervalUASs farEndIntervalValidData farEndTimeElapsed farEndTotalCCVs farEndTotalCESs farEndTotalCSESs farEndTotalIndex farEndTotalUASs farEndValidIntervals ifDescr ifHCInBroadcastPkts ifHCInMulticastPkts ifHCInOctets ifHCInBits ifHCInUcastPkts ifHCOutBroadcastPkts ifHCOutMulticastPkts ifHCOutOctets ifHCOutBits ifHCOutUcastPkts ifInDiscards ifInErrors ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
DS3Card	T3Physical (continued)	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	ifMtu ifOperStatus ifOutDiscards ifOutErrors ifOutOctets ifOutBits ifOutUcastPkts ifPhysAddress ifSpeed intervalCCVs intervalCESS intervalCSEs intervalIndex intervalLCVs intervalLESS intervalNumber intervalPCVs intervalPESs intervalPSESs intervalSEFSs intervalUASs intervalValidData invalidIntervals lineStatus lineStatusLastChange loopbackStatus timeElapsed totalCCVs totalCESS totalCSEs totalIndex totalLCVs totalLESS totalPCVs totalPESs totalPSESs totalSEFSs totalUASs validIntervals

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
T3Aal5Atm4PortCard	T3UnchannelizedPhysical	BAN	currentCCVs currentCESs currentCSESs currentIndex curentLCVs currentLESs currentPCVs currentPESs curentPSESs currentSEFSs currentUASs farEndCurrentCCVs farEndCurrentCESs farEndCurrentCSESs farEndCurrentIndex farEndCurentInvalidIntervals farEndCurrentUASs farEndIntervalCCVs farEndIntervalCESs farEndIntervalCSESs farEndIntervalIndex farEndIntervalNumber farEndIntervalUASs farEndIntervalValidData farEndTimeElapsed farEndTotalCCVs farEndTotalCESs farEndTotalCSESs farEndTotalIndex farEndTotalUASs farEndValidIntervals ifDescr ifInDiscards ifInErrors ifInOctets ifInBits ifInUcastPkts ifInUnknownProtos ifLastChange

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
T3Aal5Atm4PortCard	T3UnchannelizedPhysical (continued)	BAN	ifMtu ifOperStatus ifOutDiscards ifOutErrors ifOutOctets ifOutBits ifOutUcastPkts ifPhysAddress ifSpeed intervalCCVs intervalCESSs intervalCSEsSs intervalIndex intervalLCVs intervalLESSs intervalNumber intervalPCVs intervalPESs intervalPSESs intervalSEFSs intervalUASs intervalValidData invalidIntervals lineStatus lineStatusLastChange loopbackStatus timeElapsed totalCCVs totalCESSs totalCSEsSs totalIndex totalLCVs totalLESSs totalPCVs totalPESs totalPSESs totalSEFSs totalUASs validIntervals

Table 30: Physical objects (Continued)

Parent	Object	Supported	Request Element Attribute
VdslPhysical	VdslPhysicalStatus	MxK	chanCrcBlockLength chanCurrTxFastFec chanCurrTxRate chanCurrTxSlowBurstProtect chanInterleaveDelay currAnt currAttainableRate currLineRate currOutputPwr currSnrMargin currStatus serialNumber vendorID versionNumber zhoneVdslPhysSide Note: It is mandatory to provide Parent_Type, Parent_Id, Handle_Type and zhoneVdslPhysSide and at least one of the above attribute for successful retrieval of VDSL status information. • Parent_Type must be set to VdslPhysical • Parent_Id must be set to the Handle_Id of the corresponding parent physical. • Handle_Type must be set to VdslPhysicalStatus. • zhoneVdslPhysSide must be set to 1 for CO side and 2 for CPE side information retrieval.

Other objects

The following table describes objects that have parents in the preceding tables.

Table 31: Other OSS XML Gateway objects

Parent	Object Handle type	Supported	Request Element Attribute
Aal2VclAtmEndPoint	ATMCid	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge BH2A Z-Edge H2A	aal2SscsI3661BadLength aal2SscsI3661CRC aal2SscsI3661MsgTooLong aal2SscsI3661MsgTooShort aal2SscsI3661RasTimerExpired aal2SscsI3662IllegalUUI aal2SscsI3662ProfileError aal2SscsI3662Type3CRC cellsReceived cellsSent circuitModeData dtmfDialedDigits faxDemoRemo ifDs3PlcpAlarmState ifDs3PlcpSEFSs ifDs3PlcpUASs ifOCDEvents ifTCAlarmState lastChange mfR1DialedDigits mfR2DialedDigits operStatus statsTimeElapsed
AdslPhysical	AdslSelt	MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	segmentLength segmentGauge segmentType ifIndex
AdslPhysical	AdslDelt	MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	snr snrMargin log qln lnAtten sigAtten attainableRate actAtp ifIndex

Table 31: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Request Element Attribute
ATMLLogical	Aal2VclAtmEndPoint	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge BH2A Z-Edge H2A	aal2CpsPerfBadCPSLength aal2CpsPerfBadPayloadLength aal2CpsPerfHEC aal2CpsPerfIllegalCID aal2CpsPerfIllegalUI aal2CpsPerfPayloadTooLong aal2CpsPerfRessError aal2CpsPerfSTFBadSeq aal2CpsPerfSTFParity aal2CpsPerfTransError aal2ElcpIgLapvReceived aal2ElcpIgLapvSent aal2ElcpIgOperStatus aal2ElcpIgOperStatusChangeCount aal2VclCellsReceived aal2VclCellsSent aal2VclCircuitModeData aal2VclDTMFDialedDigits aal2VclFaxDemoRemo aal2VclMfR1DialedDigits aal2VclMfR2DialedDigits aal2VclStatsTimeElapsed ifDs3PlcpAlarmState ifDs3PlcpSEFSs ifDs3PlcpUASs ifOCDEvents ifTCAlarmState lastChange operStatus totalDiscardedCellsRx totalDiscardedCellsTx totalFabricCellsRx totalFabricCellsTx totalFinalCLP0CellsRx totalFinalCLP0CellsTx totalFinalCLP1CellsRx totalFinalCLP1CellsTx totalInitialCellsRx totalInitialCellsTx

Table 31: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Request Element Attribute
ATMLogical	Aal5VclAtmEndPoint	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Z-Edge BH2A Z-Edge H2A	aal5VccCrcErrors aal5VccOverSizedSDUs aal5VccSarTimeOuts aal5VccVci aal5VccVpi ifDs3PlcpAlarmState ifDs3PlcpSEFSs ifDs3PlcpUASs ifOCDEvents ifTCAlarmState lastChange operStatus totalDiscardedCellsRx totalDiscardedCellsTx totalFabricCellsRx totalFabricCellsTx totalFinalCLP0CellsRx totalFinalCLP0CellsTx totalFinalCLP1CellsRx totalFinalCLP1CellsTx totalInitialCellsRx totalInitialCellsTx
ATMLogical	OtherVclAtmEndPoint	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Z-Edge BH2A Z-Edge H2A	lastChange operStatus totalDiscardedCellsRx totalDiscardedCellsTx totalFabricCellsRx totalFabricCellsTx totalFinalCLP0CellsRx totalFinalCLP0CellsTx totalFinalCLP1CellsRx totalFinalCLP1CellsTx totalInitialCellsRx totalInitialCellsTx
ATMLogical	AtmVpCrossConn	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	adminStatus h2LLastChange h2LOperStatus highIfIndex highVpi index l2HLastChange l2HOperStatus lowIfIndex lowVpi

Table 31: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Request Element Attribute
CardGroup	CardGroupEntry	MALC 319 MALC 719 MALC 723 Raptor 100 Raptor 319 Raptor 719 Raptor 723	operStatus
EtherPhysical	IpDhcpClientIf	Z-Edge BH2A Z-Edge H2A	bootFile domainName hostname rebind renew serverName tftp vendorClassId

Table 31: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Request Element Attribute
ImaPhysicalGroup	ImaLink	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	currFeRxNumFailures currFeRxUnusableSecs currFeSevErroredSecs currFeTxNumFailures currFeTxUnusableSecs currFeUnavailSecs currImaViolations currNeRxNumFailures currNeRxUnusableSecs currNeSevErroredSecs currNeTxNumFailures currNeTxUnusableSecs currNeUnavailSecs currOifAjomalies currRxStuffs currTxStuffs feRxFailureStatus feRxNumFailures feRxState feRxUnusableSecs feSevErroredSecs feTxNumFailures feTxState feTxUnusableSecs feUnavailSecs imaViolations intervFeRxNumFailures intervFeRxUnusableSecs intervFeSevErroredSecs intervFeTxNumFailures intervFeTxUnusableSecs intervFeUnavailSecs intervImaViolations intervNeRxNumFailures intervNeRxUnusableSecs intervNeSevErroredSecs intervNeTxNumFailures intervNeTxUnusableSecs intervNeUnavailSecs intervNum intervOifAnomalies intervRxStuffs intervTxStuffs invalidIntervals neRxFailureStatus neRxNumFailures neRxState neRxUnusableSecs neSevErroredSecs neTxNumFailures

Table 31: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Request Element Attribute
ImaPhysicalGroup	ImaLink (continued)	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	neTxState neTxUnusableSecs neUnavailSecs oifAnomalies relDelay rxLid rxStuffs rxTestPattern testProcStatus timeElapsed totalFeRxNumFailures totalFeRxUnusableSecs totalFeSevErroredSecs totalFeTxNumFailures totalFeTxUnusableSecs totalFeUnavailSecs totalImaViolations totalNeRxNumFailures totalNeRxUnusableSecs totalNeSevErroredSecs totalNeTxNumFailures totalNeTxUnusableSecs totalNeUnavailSecs totalOifAnomalies totalRxStuffs totalTxStuffs txLid txStuffs validIntervals
IpLogical	IpDhcpClientDns	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Z-Edge BH2A Z-Edge H2A	ifIndex
IpLogical	IpDhcpClientRouter	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Z-Edge BH2A Z-Edge H2A	ifIndex

Table 31: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Request Element Attribute
IpRoutingDomain	IpCidrRoute	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge BH2A Z-Edge H2A	age cidrDest cidrInfo cidrMask cidrNextHop ifIndex metric1 metric2 metric3 metric4 metric5 nextHopAS protocol rdIndex routeType
IpRoutingDomain	IpRip2Global	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Z-Edge BH2A Z-Edge H2A	queries rdIndex routeChanges
RS232Physical	RS232AsyncConfig	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	framingErrs overrunErrs parityErrs
RS232Physical	RS232InSignal	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	ifIndex sigChanges sigId sigState
RS232Physical	RS232OutSignal	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	ifIndex sigChanges sigId sigState

Table 31: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Request Element Attribute
SubscriberPort	SubDataConn	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	curlIpAddr ipAddrInUse ipIfOperStatus statsStatus
Device	SubscriberPort	MALC 319 MALC 719 MALC 723	cpeType deviceRole id lgId maxAllowedLineRate maxCapableLineRate name nextVoiceConnectionIndex providerId
SubVoiceConn	SubVoiceAal2EndPoint	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1	subEPActiveCalls subEPIncomingCallsBlocked subEPIncomingCallsCompleted subEPOutgoingCallsBlocked subEPOutgoingCallsCompleted subVoiceEndPointIndex
SubscriberPort	SubVoiceConn	MALC 319 MALC 719 MALC 723	id lgId connectionIndex endPointOneIndex endPointTwoIndex connectionType connectionDesc adminStatus featureSetOne huntGroup esaMode
SubVoiceConn	SubVoiceElcpAal2EndPt	Sechtor 100A E1 Sechtor 100A T1	subEPActiveCalls subEPIncomingCallsBlocked subEPIncomingCallsCompleted subEPOutgoingCallsBlocked subEPOutgoingCallsCompleted subVoiceEndPointIndex

Table 31: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Request Element Attribute
SubVoiceConn	SubVoiceGr303EndPoint	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A T1	subEPActiveCalls subEPIncomingCallsBlocked subEPIncomingCallsCompleted subEPOutgoingCallsBlocked subEPOutgoingCallsCompleted subVoiceEndPointIndex
SubVoiceConn	SubVoiceIsdnEndPoint	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1 Sechtor 100A T1	subEPActiveCalls subEPIncomingCallsBlocked subEPIncomingCallsCompleted subEPOutgoingCallsBlocked subEPOutgoingCallsCompleted subVoiceEndPointIndex
SubVoiceConn	SubVoicePotsEndPoint	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1 Sechtor 100A T1	subEPActiveCalls subEPIncomingCallsBlocked subEPIncomingCallsCompleted subEPOutgoingCallsBlocked subEPOutgoingCallsCompleted subVoiceEndPointIndex
SubVoiceConn	SubVoiceV52EndPoint	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1 Sechtor 100A T1	subEPActiveCalls subEPIncomingCallsBlocked subEPIncomingCallsCompleted subEPOutgoingCallsBlocked subEPOutgoingCallsCompleted subVoiceEndPointIndex
SubVoiceConn	SubVoiceVoipEndPoint	MALC 319 MALC 719 MALC 723	subEPActiveCalls subEPIncomingCallsBlocked subEPIncomingCallsCompleted subEPOutgoingCallsBlocked subEPOutgoingCallsCompleted subVoiceEndPointIndex
V52Ig	V52Link	Sechtor 100A E1	blockStatus
V52Ig	V52UserPort	Sechtor 100A E1	blockStatus operStatus
V52Link	V52CChannel	Sechtor 100A E1	currLogicalChannelId status

3

CONFIGURATION REQUESTS REFERENCE

This chapter describes the configuration request attributes supported for the objects listed in the configuration request tables.

A

aal2VclTrunkType

Specifies the signaling method currently used for the AAL2 VCL trunk.

Values 2: loop start
3: ground start
4: loop reverse battery

Default 2: loop start

Access read-write

accessControlList

This attribute points to a access control group list. The default value 0 indicates that no control list is used.

Values Integer32 (0..255)

Default 0

Access read-write

accessCtrlListProfileIndex (DynOmciCpeWlanSubscriber)

This address of cpe-access-ctrl-list profile associated with this CPE.

Values 0 to 2147483647

Default 0

Access read-write

accessIndex

The index of port access entry in table.

Values 1 to 100

Access read-create

accmEnable

Enable or disable Asynchronous Control Character Map (ACCM) negotiation.

Values 1: enabled

2: disabled

Default disabled

Access read-write

acfcEnable

Enable or disable Address and Control Field Compression (ACFC) negotiation.

Values 1: enabled

2: disabled

Default 1: disabled

Access read-write

acelIndex

This represents the CPE Video Access Control Entry Index.

Values Integer32 (1..2147483647)

Access read-create

aclIndex

This represents the CPE Video Access Control List Index.

Values Integer32 (1..2147483647)

Access read-create

aclType

This attribute defines the video stream type.

Values 1: normal, 2: alwayson, 3: periodic

Default 1: normal

Access read-write

ackTimeout

Timeout length of Acknowledge message.

Access read-write

acsPassword (DynOmciCpeSystemCommon)

User password required for access to the ACS.

Values String (0..64)

Default zhone

Access read-write

acsUrl (DynOmciCpeSystemCommon)

URL for the CPE to connect to the ACS.

Values String (0..256)

Access read-write

acsUserName (DynOmciCpeSystemCommon)

User name required for access to the ACS.

Values String (0..256)

Default admin

Access read-write

action

Filter statement action. The bits set in this object specify actions to take on packets matching this statement.

Values 0: reset

Return a TCP reset packet to the packet sender and drop the packet. This cannot be specified with permit.

1: permit

Stop filtering the packet and allow it to be sent on the associated interface. This cannot be specified with deny.

2: deny

Stop filtering the packet and discard it. This cannot be specified with permit.

3: forward

Forward the packet the IP address specified in fltStmtActionArg.

4: reject

Return an ICMP destination unreachable packet (type 3) to the packet sender with code 13 (communication administratively prohibited). This cannot be specified with permit.

5: log

Write the packet to the log stream.

Access read-create

Notes There are some mutually exclusive settings: reset (0) and permit (1), permit (1) and deny (2), permit (1) and reject (4). No bits set implies to continue filtering on the packet.

actionArg

Filter statement action argument. The meaning of this object depends on the value of fltStmtAction: forward (3): An IP address to forward the packet to. The value of this object must be non-zero. All other values of fltStmtAction have no relation to this object.

Values 0 to 2147483647

Default 0

Access read-create

activationTime

The time the port was activated.

Values 0 to 9223372036854775807

Access sync-only

active

Indicates if the card is active or not in a redundant pair of cards.

Values 1 or 2

Default 1

Notes This object is obsolete and should not be modified.

addr

The IP alias address.

Default 0.0.0.0

Access init-only

addr1

The first subscriber address.

Values 0 to 64

Access read-write

addr2

The second line of the mailing address of the customer (optional).

Access read-write

address (V5.2)

V5.2 user port address.

Values 0 to 32767 (If UserPortType is PSTN) 0 to 8175 (If UserPortType is ISDN)

Default 0

Access init-only

Notes The actual address range is further limited by the following rules:

- PSTN:

$$v52IgPstnLayer3StartAddress \leq v52UserPortAddress < v52IgPstnLayer3StartAddress + \text{maximum PSTN users allowed per interface group}$$
- ISDN:

$$v52IgIsdnEnvFuncStartAddress \leq v52UserPortAddress < v52IgIsdnEnvFuncStartAddress + \text{maximum ISDN users allowed per interface group}$$

In addition, the total number of user ports is restricted by the following rule:

Total PSTN user ports + 2 * Total ISDN BRI user ports ≤ Total user ports supported per platform.

addrType

The type of bridge path.

Values 1: macAddr

The **ipOrMacAddr** represents a MAC Address, and **bridgeIfIndex**, **vlanId** and **ipOrMacAddr** are used for the bridge path.

2: ipAddr

The **ipOrMacAddr** represents an IP Address, and bridgeIfIndex, **vlanId** and ipOrMacAddr are used for the bridge path.

3: vlanIdAddr

The bridgeIfIndex and **vlanId** are used for the bridge path.

4: globalAddr

The bridgeIfIndex is used for the bridge path."

Default 1: macAddr

Access init-only

adminPassword (DynOmciCpeSystemCommon)

Log-in password for user account admin.

Values String (0..16)

Default zhong

Access read-write

adminState

An enumeration of the operational state of RIP for a routing domain. The possible values are either "disabled" or "enabled," where the operation of RIP is completely disabled or enabled on all interfaces within a RD.

Values 1: disabled

2: enabled

Default 1: disabled

Access read-write

adminState (DynOmciCpeCfgRule)

If down, this rule will not participate in rule matching.

Values 1. up

2: down

Default 2

Access read-create

adminState (OmciPortAdmin)

This object is used to place an ONU physical port in a DOWN or UP state. This state is a configuration parameter and is preserved.

Values 0: none
 1: up
 2: down
Default 0
Access read-write

adminState (DynOmciCpeEthSubscriber, DynOmciCpePweSubscriber, DynOmciCpeRfSubscriber, DynOmciCpeVoipSubscriber)

This attribute activates or deactivates the functions performed by the Ethernet port for this subscriber or put the port in test mode.

Values 1: True
 2: False
Default True
Access read-write

adminState (DynOmciCpeWlanSubscriber)

This attribute activates or deactivates the functions performed by the Wireless LAN interface for this subscriber or put the interface in test mode.

Values 1: up
 2: down
 3: testing
Default 1: up
Access read-write

adminStatsEnable

Enable the collection and forwarding of historical statistics on the unit or card. The default value is disabled. Note this value should be logically “anded” with the system statistics enable to determine if the collection and forwarding of statistics should actually be enabled.

Values 1: enable
 2: disable
Default 1: enable
Access read-write

adminStatus

Specifies the administrative status of the unit or card.

Values 1: operational

The unit or card enters the operational state if it is able.

2: disable

The unit or card ceases to accept requests from other elements. If the unit or card is operational and receives this command it will complete processing of open requests if it is able.

3: maintenance

The unit or card will enter a special maintenance state if one is available. If one is not available it will enter the disabled state and adminStatus will be set to disabled.

4: warmreset

The unit or card will perform a warm reset of system software that does not reinitialize or reset hardware resources

5: reset

The unit or card will perform an immediate full reset. Following a successful reset, the unit or card should enter the operational state.

Default 1: operational

Access read-write

adminStatus (V5.2)

The administrative status of the V5.2 interface group, user port or link.

Values 1: inservice

2: outofservice

Default 2: outOfService

Access read-write

adminUpAlarm

The enable status of the Admin Up alarm. If this field is true (1) then the Admin Up alarm is enabled. If this field is false (2) then the Admin up alarm is disabled.

Parent: Card

Object: BondGroupAlarmConfig

Values 1: true

2: false

Default 2: false

Access read-write

adminUpAlarmEnable

Indicates whether DSL line status should be generated when the line is brought up.

Parent: DSLPhysical

Values 1: enabled

2: disabled

Default 2: disabled

Access read-write

adslSeltUnits

This object permits the selection of the metric unit to be used when reporting SELT test results. (i.e for AWG results are in feet and for metric/metricJapan results are in meters). This configuration parameter applies to all the SELT tests run in the device.

Values 1: awg

2: metric

3: metricJapan

Default 1: awg

Access Read-Write

aggregationMode

Values 1: on

The link aggregation is forced to be formed without any LACP negotiation. In other words, the switch will neither send the LACP packet nor process any incoming LACP packet. This is similar to the on state for PAgP.

2: off

The link aggregation will not be formed. We do not send or understand the LACP packet. This is similar to the off state for PAgP.

3: passive

The switch does not initiate the channel, but does understand incoming LACP packets. The peer (in active state) initiates negotiation (by sending out an LACP packet) which we receive and reply to, eventually forming the aggregation channel with the peer. This is similar to the auto mode in PAgP.

4: active

We are willing to form an aggregate link, and initiate the negotiation. The link aggregate will be formed if the other end is running in LACP active or passive mode. This is similar to the desirable mode of PAgP.

Default 1: on

Access read-write

alarmProfName

This object identifies a row in the **DslAlarmProfileTable**. It is assumed that all profile names are unique to the system. The default value is **ifIndex** (an ASCII string value) for static profiles and **zhonedefault** for dynamic profiles.

Type is Integer32.

Values 1 to 32

Access read-write

alarmSeverity (DynOmciCpeEthSubscriber, DynOmciCpePweSubscriber, DynOmciCpeRfSubscriber)

This field is for specifying the severity level of all the alarms from this CPE UNI.

Values 1: critical

2: major

3: minor

4: warning

Default 2: major

Access read-write

alias1

Hostname alias for this entry. The hostname can be either fully-qualified or hostname only.

Values 0 to 255

Access read-write

alias2

Another hostname alias for this entry. The hostname can be either fully-qualified or hostname only.

Values 0 to 255

Access read-write

alias3

Another hostname alias for this entry. The hostname can be either fully-qualified or hostname only.

Values 0 to 255

Access read-write

alias4

Another hostname alias for this entry. The hostname can be either fully-qualified or hostname only.

Values 0 to 255

Access read-write

allocCycleLength

Alloc-ID assignment map cycle length.

Values 1 to 10

Default 2

Access read-write

alphaValue

This indicates the 'alpha' value used to specify the number of consecutive invalid ICP cells to be detected before moving to the IMA Hunt state from the IMA Sync state.

Values 1 or 2

Default 2

Access read-create

altConfigIP

A second IP address of this device. Independent of the different IP interfaces configured and what is active, a manager uses this address as how to address the device. This address may OR may not be an IP address of an existing IP interface. Deletion of a matching IP interface does not cause this address to be modified or deleted. A manager may communicate with the device using any of the configured IP interface addresses. However, any failures such as physical line outages or the card status is inactive, may result in failures to the

manager requests (i.e., timeout). This value is defaulted to 0 and in some cases, may never be configured.

Default 0.0.0.0

Access read-write

annexMModeEnabled

This field indicates whether Annex-M mode is enabled on the adsl modem. If this field is set to 'true' Annex-M mode is enabled, if it set to false it is not.

Values 1: true

2: false

Default 2: false

Access read-write

announceType

This attribute specifies the treatment when a subscriber goes off hook but does not attempt a call.

Values INTEGER (1: silence, 2: reordertone, 3: fastbusy, 4: voice, 5: notapplicable)

Default 2: reordertone

Access read-write

apMode (DynOmciCpeWlanAdv)

Wireless bridge modes of operation.

Values 1: accesspoint

2: wirelessbridge

Default 1: accesspoint

Access read-write

appendDefaultName

Indicates if the default name shall be appended to the user provided name or not.

Values 1: true

2: false

Default 2: false

Access read-write

apply

OMCI ONT apply operation.

Values 0 and 1. Value 1 must be provided for successful completion of apply command.

Access read-write

arbMask

This object is mask for arbitrary value comparisons. The non-zero bits in this field determine the size of the arbitrary field.

Values 0 to 4294967295

Default 0

Access read-create

arbOffset

The offset, in octets, from the beginning of the header to the most significant octet for the arbitrary value comparison.

Values 0 to 64

Default 0

Access read-create

arbValueBase

This field identifies the protocol header to which the arbitrary value comparison applies.

Values 1: none
No arbitrary value comparison.

2: ip
Base is IP header.

3: udp
Base is UDP header.

4: tcp
Base is TCP header.

5: icmp
Base is ICMP header.

6: ipOptions
Base is IP options header.

7: tcpOptions
Base is TCP options header.

Default 1: none

Access read-create

arbValueLow

This object is the inclusive lower bound for arbitrary value comparison. See the filterStatementEntry description for details.

Values 0 to 4294967295

Default 0

Access read-create

arbValueHigh

This object is the inclusive upper bound for arbitrary value comparison. See the filterStatementEntry description for details.

Values 0 to 4294967295

Default 0

Access read-create

atmConfiguration

Specifies bandwidth allocation for rt-VBR and UBR traffic and used by Connection Admission Control (CAC). This parameter only applies to ATM trunking interfaces, such as MALC and Raptor Uplink cards, as well as BAN ATM trunking cards.

Values 1: notApplicable

2: cellRelayOnly (BAN only)

The only supported value for ATM trunking (cell relay) card. Specifies the following bandwidth allocation:

- AAL2/cell relay 99%
- AAL5 1%

For ATM-AAL2/AAL5card, 50% of AAL2/cell relay bandwidth is rt-VBR.

3: cellrelayandmanagement (BAN only)

Specifies the following allocation:

- AAL2/cell relay 95%
- AAL5 5%

For ATM-AAL2/AAL5card, 50% of AAL2/cell relay bandwidth is rt-VBR.

4: dataterm (BAN only)

Specifies the following allocation:

- AAL2/cell relay 5%
- AAL5 95%

For ATM-AAL2/AAL5card, 50% of AAL2/cell relay bandwidth is rt-VBR.

5: voiceGateway (BAN only)

Specifies the following allocation:

(only supported for ATM-AAL2/AAL5 card)

- AAL2/cell relay 95%
- AAL5 5%

For ATM-AAL2/AAL5card, 95% of AAL2/cell relay bandwidth is rt-VBR.

6: hybridlowaal5data (BAN only)

Specifies the following allocation:

- AAL2/cell relay 75%
- AAL5 25%

For ATM-AAL2/AAL5card, 95% of AAL2/cell relay bandwidth is rt-VBR.

7: hybriddefault (BAN only)

Specifies the following allocation:

- AAL2/cell relay 60%
- AAL5 40%

For ATM-AAL2/AAL5 card, 75% of AAL2/cell relay bandwidth is rt-VBR.

8: hybridhighaal5data (BAN only)

Specifies the following allocation:

- AAL2/cell relay 25%
- AAL5 75%

For ATM-AAL2/AAL5 card, 50% of AAL2/cell relay bandwidth is rt-VBR.

9: vbnrt95rt5 (MALC, Raptor)

Specifies the following allocation:

- UBR: 1%
- nrt-VBR: 94%
- CBR/rt-VBR: 5%

10: vbnrt80rt15 (MALC, Raptor)

Specifies the following allocation:

- UBR: 5%
- nrt-VBR: 80%
- CBR/rt-VBR: 15%

11: vbnrt65rt30 (MALC, Raptor)

Specifies the following allocation:

- UBR: 5%
- nrt-VBR: 65%
- CBR/rt-VBR: 30%

12: vbnrt50rt45 (MALC, Raptor)

Specifies the following allocation:

- UBR: 5%
- nrt-VBR: 50%
- CBR/rt-VBR: 45%

13: vbnrt35rt60 (MALC, Raptor)

Specifies the following allocation:

- UBR: 5%
- nrt-VBR: 35%
- CBR/rt-VBR: 60%

14: vbnrt20rt75 (MALC, Raptor)

Specifies the following allocation:

- UBR: 5%
- nrt-VBR: 20%
- CBR/rt-VBR: 75%

15: vbnrt5rt95 (MALC, Raptor)

Specifies the following allocation:

- UBR: 1%
- nrt-VBR: 5%
- CBR/rt-VBR: 94%

atmConfiguration

Specifies bandwidth allocation for rt-VBR and UBR traffic and used by Connection Admission Control (CAC). This parameter only applies to ATM trunking interfaces, such as MALC and Raptor Uplink cards, as well as BAN ATM trunking cards.

Values 1: notApplicable

2: cellRelayOnly (BAN only)

The only supported value for ATM trunking (cell relay) card. Specifies the following bandwidth allocation:

- AAL2/cell relay 99%
- AAL5 1%

For ATM-AAL2/AAL5card, 50% of AAL2/cell relay bandwidth is rt-VBR.

3: cellrelayandmanagement (BAN only)

Specifies the following allocation:

- AAL2/cell relay 95%
- AAL5 5%

For ATM-AAL2/AAL5card, 50% of AAL2/cell relay bandwidth is rt-VBR.

4: dataterm (BAN only)

Specifies the following allocation:

- AAL2/cell relay 5%
- AAL5 95%

For ATM-AAL2/AAL5card, 50% of AAL2/cell relay bandwidth is rt-VBR.

5: voiceGateway (BAN only)

Specifies the following allocation:

(only supported for ATM-AAL2/AAL5 card)

- AAL2/cell relay 95%
- AAL5 5%

For ATM-AAL2/AAL5card, 95% of AAL2/cell relay bandwidth is rt-VBR.

6: hybridlowaal5data (BAN only)

Specifies the following allocation:

- AAL2/cell relay 75%
- AAL5 25%

For ATM-AAL2/AAL5card, 95% of AAL2/cell relay bandwidth is rt-VBR.

7: hybriddefault (BAN only)

Specifies the following allocation:

- AAL2/cell relay 60%
- AAL5 40%

For ATM-AAL2/AAL5 card, 75% of AAL2/cell relay bandwidth is rt-VBR.

8: hybridhighaal5data (BAN only)

Specifies the following allocation:

- AAL2/cell relay 25%
- AAL5 75%

For ATM-AAL2/AAL5 card, 50% of AAL2/cell relay bandwidth is rt-VBR.

9: vbnrt95rt5 (MALC, Raptor)

Specifies the following allocation:

- UBR: 1%

- nrt-VBR: 94%
- CBR/rt-VBR: 5%

10: vbnrt80rt15 (MALC, Raptor)

Specifies the following allocation:

- UBR: 5%
- nrt-VBR: 80%
- CBR/rt-VBR: 15%

11: vbnrt65rt30 (MALC, Raptor)

Specifies the following allocation:

- UBR: 5%
- nrt-VBR: 65%
- CBR/rt-VBR: 30%

12: vbnrt50rt45 (MALC, Raptor)

Specifies the following allocation:

- UBR: 5%
- nrt-VBR: 50%
- CBR/rt-VBR: 45%

13: vbnrt35rt60 (MALC, Raptor)

Specifies the following allocation:

- UBR: 5%
- nrt-VBR: 35%
- CBR/rt-VBR: 60%

14: vbnrt20rt75 (MALC, Raptor)

Specifies the following allocation:

- UBR: 5%
- nrt-VBR: 20%
- CBR/rt-VBR: 75%

15: vbnrt5rt95 (MALC, Raptor)

Specifies the following allocation:

- UBR: 1%
- nrt-VBR: 5%
- CBR/rt-VBR: 94%

Access read-write

atmFraming

The ATM framing mode.

Values 1: dsx3AtmFramingOther
 2: dsx3AtmFramingPLCP
 3: dsx3AtmFramingDirectCellMapped
Default 3: dsx3AtmFramingDirectCellMapped
Access read-write

atmVplAdminStatus

Administrative status of the VPL.

Values up
 down
Default down

atmVplReceiveTrafficDescrIndex

Specifies the ATM traffic descriptor which applies to the receive direction of this VPL.

Values The index value of an existing atm-traf-descr.

Notes Currently this value must be set equal to the value used for the **atmVplTransmitTrafficDescrIndex**.

Traffic descriptors used in active VPLs cannot be modified. You must bring down the VPL before you can modify the traffic descriptor.

atmVplTransmitTrafficDescrIndex

Specifies the ATM traffic descriptor which applies to the transmit direction of this VPL.

Values The index value of an existing atm-traf-descr.

Notes Currently this value must be set equal to the value used for the **atmVplReceiveTrafficDescrIndex**.

Traffic descriptors used in active VPLs cannot be modified. You must bring down the VPL before you can modify the traffic descriptor.

See Also **atmVplReceiveTrafficDescrIndex**

atucMinTxThreshAlarm

Minimum CO Tx Sync rate threshold alarm. If set to 0, this field is disregarded. Otherwise, if Tx Sync rate falls below this value, a trap and alarm will be issued.

Values 0 to 4294967295

Default 0

Access Read-Write

atucReachExtendedAdsl2

Determines whether the ATUC should force downstream READSL2 operation.



Note: READSL2 has been designed to provide rate and reach benefits for long loops. For loop lengths less 16,000 ft, rates will decrease if READSL2 is enabled because the downstream end bin is limited to 128 for READSL2 operation.

Supported for ADSL2/ADSL2+ only.

Values 1: enable

2: disable

Default 1: enable

Access read-write

aturMinTxThreshAlarm

Minimum CPE Tx Sync rate threshold alarm. If set to 0, this field is disregarded. Otherwise, if Tx Sync rate falls below this value, a trap and alarm will be issued.

Values 0 to 4294967295

Default 0

Access Read-Write

audioService (ATMCid)

Defines a number of services for transporting audio traffic and signaling over an AAL2 CID.

Values 1: enabled

2: disabled

3: enabledechocanceloff

Audio service enabled with echo cancellation and silence suppression disabled for a particular CID.

Default 1: enabled

Access read-write

audioService (ATMEndPoint)

Defines a number of services for transporting audio traffic and signaling over an AAL2 VCL.

Values 1: enabled

2: disabled

3: enabledechocanceloff

Audio service enabled with echo cancellation and silence suppression disabled for a particular VCL.

Default 1: enabled

Access read-write

authentication (DynOmciCpePppoe)

The PPP authentication protocol to be used for PPP authentication.

Values 1: auto

2: pap

3: chap

4: mschap

Default 1

Access read-write

autoBaud

A control for the RS232 port's ability to automatically sense input speed.

When rs232PortAutoBaud is enabled, a port may autobaud to values different from the set values for speed, parity, and character size. As a result a network management system may temporarily observe values different from what was previously set.

Values 1: enabled

2: disabled

Default 1

Access read-write

autoChanTimer (DynOmciCpeWlanAdv)

When configured for auto mode, this timer value specifies how often (in minutes) to re-analyze the spectrum to select a low interference channel.

Note: auto channel rescan will only occur when there are no actively connected devices.

Values 0..21474836472: wirelessbridge

Default 15

Access read-write

autoCreation

Determines whether the you need to specify both the upstream and downstream PVC.

Values true

false

Default true

autoDetect

Auto detects the rate and duplex settings of Ethernet physical ports.

Values 1: none (0)

2: auto (1)

3: tenFull(2)

4: hundredFull(3)

5: thousandFull(4)

6: tenAuto(5)

7: tenHalf(6)

8: hundredHalf(7)

9: thousandHalf(8)

10: thousandAuto(9)

11: autoFull(10)

12: autoHalf(11)

13: hundredAuto(12)

autoIncrementVpi

Specifies the system should continue to provision cross connects using a particular VPI until all the VCIs are provisioned. The system will then start creating cross connects on the next available VPI.

Values 1: true

2: false

Default 2: false

Access read-write

autoLearn (APON)

Auto-learn enable flag. This sets up password-protection on the connection between OLT and ONU.

Values 1: oltAutolearnEnable

You must set a password for the connection between OLT and ONU.

2: oltAutolearnDisable

The OLT automatically learns the password for the connection between OLT and ONU.

Default 2: oltAutolearnDisable

autoLearn (GPON)

Enable or disable password auto-learn mode.

Values 1: Enabled

2: Disabled

Default 2: Enabled

autoSelectGem (DynOmciCpeConnection)

Setting this value to 1 will automatically pick the next available GEM on given slot/OLT/ONU.

Values 1: Enabled

2: Disabled

Default 2: Disabled

Access read-write

Notes Applies only to trunking interfaces on the BAN, MALC, or Raptor.

autoSelectVci

Specifies the system should automatically use the next available VCI values for the cross connect.

Values 1: true

2: false

Default 2: false

Access read-write

Notes Applies only to trunking interfaces on the BAN, MALC, or Raptor.

B

backupMetric

This object contains the fallback metric value for routes that are a part of a fallback route pair. Please refer to the DZS User's Reference Manual for more information about fallback route pairs. The value of this object must be greater than both of the staticRouteMetrics of fallback route pair.

Values 0 to 2147483647

Default 0

Access read-create

bandwidthEnforce

The default value of this boolean attribute is false, and specifies that attempts to exceed the max multicast bandwidth be counted but honored. The value true specifies that such attempts be counted and denied. The imputed bandwidth value is taken from the dynamic access control list table, both for a new join request and for pre-existing groups.

Values 1: true

2: false

Default 2: false

Access read-write

basicRate (DynOmciCpeWlanAdv)

The rate when the radio is operating in basic 802.11b/g mode.

Values 1: default

2: all

3: rate1n2

4: stdrates

Default 1: default

Access read-write

bcastAddr

The IP broadcast address used for sending datagrams on the (logical) interface associated with the IP address of this interface alias entry.

Default 0.0.0.0

Access read-write

bcastEnabled

Specifies whether broadcast is enabled on this interface.

Values 1: True

2: False

Default 1: True

Access read-write

bConstellation

TCPAM Constellation of the 2BASE-TL PME.

The value of adaptive(0) in the administrative profile means that the constellation is not fixed and should be set to maximize the attainable rate, under the spectral limitations placed by the efmCuPme2BRegion.

If a Clause 45 MDIO Interface to the PME is present, then this object map to the Constellation1 bits in the 2B general parameter register.

Values 0: adaptive

1: tcpam16

2 : tcpam32

3 : tcpam4

4 : tcpam8

5 : tcpam64

6 : tcpam128

Default 0: adaptive

Access read-create

bDataRate

2BASE-TL PME Data Rate. The rate is fixed when the value of this object is $n \times 64\text{Kbps}$, where $n=3..60$ for 16-TCPAM and $n=12..89$ for 32-TCPAM. The value of 0 in the administrative profile means that data rate is not fixed but is adaptive and should be set to the maximum attainable rate during line probing, under the spectral limitations placed by the efmCuPme2BRegion.

If a Clause 45 MDIO Interface to the PME is present, then this object maps to the Min/Max Data Rate1 bits in the 2B PMD parameters register.

This object MUST be maintained in a persistent manner.

Values 0 to 15532

Default 0

Access read-create

beaconInterval (DynOmciCpeWlanAdv)

The interval at which Beacons are generated.

Values 0 to 65535

Default 100

Access read-write

berSdThreshold

Threshold is $10^{**}(-N)$ for signal degrade. Must exceed SF threshold.

Values 4 to 9

Default 5

Access read-write

berSfThreshold

Threshold is $10^{**}(-N)$ for signal fail.

Values 3 to 8

Default 3

Access read-write

bgColor

Values -2147483647 to 2147483647

Default 12632256

Access read-only

bip8

Bit interleave parity (BIP8), a procedure whereby each bit of a resulting BIP8 parity byte is a parity bit done across a specific bit position.

Values 1: oltBip8Off

2: oltBip8RxCorrectionOnly

3: oltBip8TxGenerationOnly

4: oltBip8RxAndTx

Default 3: oltBip8TxGenerationOnly

Access read-write

bitRateThreshold

The enable status of the Bit Rate Threshold alarm. If this field is true (1) then the Bit Rate Threshold alarm is enabled. If this field is false (2) then the Bit Rate Threshold alarm is disabled.

Values 1: true

2: false

Default 2: false

Access read-write

bitRateThresholdValue

Bit rate threshold value, a value which will generate an alarm if the bitRateThreshold is enabled and the bit rate of this ifIndex drops below this value.

Values 0 - 999999

Default 0

Access read-write

bitRateThresholdHoldTime

This attribute indicates the bit rate threshold holdtime in seconds for which the bit rate of the ifIndex must remain below the bitRateThresholdValue before an alarm will be generated if the bitRateThreshold is enabled.

Values 0 - 999999

Default 0

Access read-write

block

A user port can be blocked by the access network for non urgent or urgent configuration or reconfiguration. If the port is routed through a V5.2 interface to a local exchange, then the block request is granted by the local exchange through the V5 interface only, to avoid interference with call setup, calls in progress or with calls being cleared.

Values 1: blocked

The user port can be unblocked by either accelerated port alignment (group unblocking) and by sequential port alignment.

2: shuttingDown

The user port can be unblocked by either accelerated port alignment (group unblocking) and by sequential port alignment except active ports will not be blocked until all of the calls are completed.

3: unblocked

The user port will be unblocked regardless of its previous port block state.

Default 1: blocked

Access read-write

bondGroupIdx

The interface number of the bond group that uniquely identifies the bond group across the system.

Values 0 to 2147483647

Default 0

Access read-write

bondGroupLgIdx

Bond group line Group Id of the physical that the IP Logical is being created on. This will be 0 for floating IP objects.

Values 0 to 2147483647

Access read-write

bootFile

Name of the initial boot file which is to be loaded by the client. The filename should be a filename recognizable to whatever file transfer protocol the client can be expected to use to load the file.

Default empty string

Access read-write

bootServer

IP Address of the server from which the initial boot file (specified in the BootFile option) is to be loaded. If this IP address is specified as 0.0.0.0 then primary interface address of the DHCP Server is used.

Default 0.0.0.0

Access read-write

bridgelfFloodMulticast

If true, flood all multicast traffic received on this bridge to all other ports on this vlan.

Values 1: true
2: false
Default 2: false
Access read-create

bridgelfFloodUnknown

If true, flood frames with unknown unicast MAC address that would otherwise be discarded.

Values 1: true
2: false
Default 2: false
Access read-create

bridgelfIndex

The ifIndex of the bridge interface.

Values 0 to 2147483647
Default 0
Access init-only

bridgelfMaxVideoStreams

Maximum number of multicast (video) streams that we will support.

Values 0 to 210
Default 0
Access read-create

bridgelfMcastControlList

List of mcast-control-ids this bridge-interface-record is associated with. It is an integer stored as string, and controls which channels a user can watch. Note: unlike for IP, an empty list here allows user to watch all channels.

Values A string of maximum of 255 characters.
Access read-create

bridgelfIndex

ifIndex of the port on which this connection is based. For a GPON connection, this is the GEM port ifIndex and it is a DZMS database only attribute.

Values Integer32 (1..2147483647)

Access read-only

bridgelfsPPPoA

If true, this interface accepts PPPoA packets and will bridge them out as PPPoE packets.

Values 1: true

2: false

Default 2: false

Access read-create

bridgelfOutgoingCOSOption

Determines COS setting on packets bridged out this interface.

disable - COS setting is untouched.

all - COS will be set to bridgelfOutgoingCOSValue.

Values 1:outgoingCOSOptionDisable

2: outgoingCOSOptionAll

Default 1:outgoingCOSOptionDisable

Access read-create

bridgelfOutgoingCOSValue

The COS value loaded into outgoing tagged packets. Applicable only if interface sends tagged packets and bridgelfOutgoingCOSOption is outgoingCOSOption_all.

Values 0-7

Default 0

Access read-create

bridgelfStagCOS

The s-tag COS to be used when doing s-tag insertion.

Values 0-7
Default 0
Access read-create

bridgelfStagId

The s-tag value that we are expecting to put on packets that are Rxed on this interface.

Values 0 to 4095
Default 0
Access read-create

bridgelfStagOutgoingCOSOption

Determines s-tag COS settings on packets bridged out this interface:

disable - s-tag COS setting just passed as Rxed.

all - s-tag COS will be set with s_tagOutgoingCOSValue.

Values 1: sTagCOSDisable
 2: sTagCOSAll
Default 1: sTagCOSDisable
Access read-create

bridgelfStagOutgoingCOSValue

The s-tag COS value loaded into outgoing tagged packets. Applicable only if interface sends S-tagged packets and s_tagOutgoingCOSOption is All.

Values 0-7
Default 0
Access read-create

bridgelfStagStripAndInsert

If 1, enable S-tag tagging in this IF. If 2, disable S-tag tagging in this IF

Values 1 to 2
Default 2
Access read-write

bridgelfStagTPID

The value of the tag protocol identifier that is being used for QinQ packets. (0x8100 is default.)

Values 33024
34984
37120
37376
Default 33024 (8100 'h')
Access read-create

bridgelfVlanIdCOS

The Class-Of-Service (COS) value used for VLAN Tag Insertion.

Values 0-7
Default 0
Access read-create

bridgeRestrict (DynOmciCpeWlanAdv)

Wireless Bridge Restriction Modes of operation.

Values 1: enabled
2: enabledscan
3: disabled
Default 3: disabled
Access read-write

bridgeStatsBSDAlarmConnt

This counter reflects the number of times this interface has transitioned to the alarm state due to the Bridge Packet Storm Detection threshold being exceeded for a pre-defined number of seconds.

Values Unsigned32
Access read-only

bridgeStatsBSDBcastBlocked

Number of broadcast packets dropped due to Bridge Packet Storm Detection threshold exceeded.

Values Unsigned32

Access read-only

bridgeStatsBSDMcastBlocked

Number of multicast packets dropped due to Bridge Packet Storm Detection threshold exceeded.

Values Unsigned32

Access read-only

bridgeStatsBSDUcastBlocked

Number of unicast packets dropped due to Bridge Packet Storm Detection threshold exceeded.

Values Unsigned32

Access read-only

bridgeTemplateName

This is a string type variable and accepts up to 60 characters and is used for specifying bridge template name to automatically create the bridge as part of CPE connection creation if needed.

Access read-create

byPassRelayMaxDuration

The maximum duration in seconds that an Adsl Pots Low Pass Filter Bypass Relay will remain active (meaning closed/energized). As a fail-safe mechanism, the relay will automatically return a line back to normal operational mode when this timer has expired. This object is only valid for Adsl Pots Splitter cards containing the Pots LPF Bypass Relays.

Values 1-300

Default 60

Access read-write

bundleDs0Bundle

This value is a bit field indicating the ds0's in the bundle. No bits selected is also a valid state.

Values 0 .. 11111111111111111111111111111111 (32 1's indicating 32 bits)

Default 0

Access read-create

bundleIfIndex

For non-SATOP (CESOP and E1-PRI) PWE connections, DZMS creates the new if-translate entry and then creates the PWE connection on top of if-translate to support the multiple CESOP connections on top of same physical. As far as OSS is concerned, it is not needed to be mentioned in OSS add and delete operations.

Values 0 .. 2147483647

Access read-only

bundleIsdnMode

Type of ISDN termination. Only valid for E1-mode PWE card physicals.

Values 1 (Disabled), 2 (lt), 3 (nt)

Default 1

Access read-create

bundleLocalIpAddr

The local IP address associated with the DS0 bundle.

Default 0.0.0.0

Access read-create

bundleTos

Type-of-Service bitvalue for the pseudowire.

Values 0..255

Default 0

Access read-create

C

cacDivider

Enables oversubscription for an ATM VCL. During CAC calculations, the system divides the SCR (for nrt-VBR or rt-VBR VCLs) bandwidth by the value specified in the **cac-divider**. It then uses that value to determine if the VCL can be created.

For example, to configure a 4:1 oversubscription, set **cac-divider** to 4.

Default 1

calleeReqTimer

When caller supports timer but did not request one this field specifies the callee's action.

Values **1: yes**
Callee (UAS) will request for timer

2: no
Callee (UAS) will NOT request for timer

Default **2: no**

Access read-create

calleeSpecifyRefresher

When UAC did not specify refresher tag, callee specifies the refresher.

Values **1: uac**
Callee specifies the caller (UAC) as the refresher

2: uas
Callee specifies itself (UAS) as the refresher

Default **1: uac**

Access read-create

callerReqTimer

This field specifies the caller's action.

Values **1: yes**
Caller (UAC) will request for timer when making outbound calls

2: no
Caller (UAC) will NOT request for timer when making outbound calls

Default **2: no**

Access read-create

callerSpecifyRefresher

Caller specifies the refresher.

Values **1: uac**
Caller specifies itself (UAC) as the refresher

2: uas
Caller specifies the callee (UAS) as the refresher

3: omit
Caller does NOT explicitly specify the refresher

Default 3: omit
Access read-create

callMode

Indicates where the PPP call is initiated.

Values 1: nocall
 No call is initiated.
 2: incall
 Call is initiated remotely.
 3: outcall
 Call is initiated locally.

Default 1: nocall
Access read-write

callWaitingFeatures

This attribute contains a bit map of call waiting features.

Values BITS (0: callwaiting, 1: cidannouncement)
Default callwaiting +cidannouncement
Access read-write

callPresentationFeatures

This attribute is a bit map of call presentation features.

Values BITS (0:msgwaitsplashring, 1:msgwaitspecialdialtone, 2:msgwaitvisual, 3:callfwd)
Default msgwaitsplashring+msgwaitspecialdialtone+msgwaitvisual+callfwd
Access read-write

cardInterfaceType

This field is used to identify which interface types are supported for a given card. In the case where a card has exactly one possible interface (such as the ethernet card) this field is exactly the same as the **cardResourceCardType** field. For polymorphic trunk cards, this field reflects the interface card type which is installed.

Values

- 1: unknownType**
- 2: interShelf**
- 3: infoServices**
- 5: voiceASP**
- 6: t3Tdm**
- 7: t3Atm**
- 8: atmTrnkCr**
- 9: ethernet**
- 10: hdsl2**
- 11: adsl**
- 13: oc3Atm**
- 15: atmTrnk**
- 16: atmTrnkAal25**
- 17: atmTrnkT1e1**
- 18: tdmT1e1**
- 1001: sechtor100**
- 2001: zedge64**
- 3001: zplex10**
- 5001: malcUplinkNt1**
- 5002: malcPots**
- 5003: malcMtac**
- 5004: malcAdsl**
- 5005: malcAdslPots**
- 5006: malcGshdsl(5006)**
- 5007: malcGshdslPots**
- 5008: malcPots900**
- 5009: malcDs3**
- 5010: malcAdslAc5**
- 5011: malcOc3**
- 5012: malcMtacFc**
- 5013: malcAc5AnxB**
- 5014: malcT1E1Tdm**
- 5015: malcAdslAc6**
- 5016: malcAdslPotsAc6**

5017: malcAdslSplAc6
5018: malcAdslAnxBAc6
5019: malcAdslAnxBSplAc6
5020: malcAdslC
5021: malcAdslPotsC
5022: malcAdslSplC
5023: malcAdslSplUkAc6
5024: malcAdslSplUkC
5025: malcUlc
5026: malcVdsl
5029: malcOc12
5030: malcGshdsl4w
5031: malcOlt
5032: malcT1E1Atm32
5101: malcT1ImaF
5109: malcDs3F
5111: malcOc3F
5114: malcT1E1TdmF
5201: malc100T1Ima (Raptor 100)
5202: malc100Gshdsl (Raptor 100)
8001: zrg500
8002: zrg300

Default 5001: malcUplinkNt1

Access server-modify-only

CardGroupKey

This is an internal read-only variable. It should not be modified.

cardLineType

This field is used to identify which line types are supported for a MALC Uplink or MTAC card. All other cards only support **unknown**. Not used for MALC OC3/STM1 Uplink cards.

Values 1: unknown
 2: e1
 3: t1
 4: e1ima
 5: ds1ima
 6: e3
 7: ds3
 8: t1UniGr303
 9: t1IimaGr303
 10: e1UniV52
 11: e1IimaV52

Default 1: unknown

Access init-only

Notes This field is only set upon row-creation. Modifications to this value after row creation will not be valid.

cardLineVoltage

Specifies the voltage supplied to all lines (ports) of this card, Used on ISDN line cards.

Values 1: not-used
 2: lv-60-volts
 3: lv-68-volts
 4: lv-95-volts,
 5: lv-100-volts
 6: lv-110-volts

Default 1: not-used

Access read-create

cardProvisionState

The state of provisioning for the card.

Values 0 to 2

Default 0

Access read-write

cardReboot

Causes the specified card to be rebooted. When set to true, thiscard will be rebooted.

If the card is not present, setting this attribute causes an INCONSISTENT_VALUE error.

Parent: Card

Values 1: true

2: false

An attempt to set to **false** results in a WRONG_VALUE error.

All gets will return a value of **false**.

Default 2: false

Access read-write

Notes Only supported on BAN, MALC, and Raptor devices (with the exception of the Raptor 50 and the Raptor 100)

cardShelfIndex

An index number defining a card or shelf.

Values 0 to 255

Default 0

Access init-only

cardType

The type of card.

Values 0 to 32

Access read-write

cardVpiVciRange

The maximum vpi and vci values that may be used for this card.

Values 1: not-applicable

2: vpi-15-vci-63

3: vpi-7-vci-127

Default 1: not-applicable

Access read-create

cardZhoneType

The numerical representation of what unit or card type is present.

Values

- 1: unknownType**
- 2: interShelf**
- 3: infoServices**
- 5: voiceASP**
- 6: t3Tdm**
- 7: t3Atm**
- 8: atmTrnkCr**
- 9: ethernet**
- 10: hdsl2**
- 11: adsl**
- 13: oc3Atm**
- 15: atmTrnk**
- 16: atmTrnkAal25**
- 17: atmTrnkT1e1**
- 18: tdmT1e1**
- 1001: sechtor100**
- 2001: zedge64**
- 3001: zplex10**
- 5001: malcUplinkNt1**
- 5002: malcPots**
- 5003: malcMtac**
- 5004: malcAdsl**
- 5005: malcAdslPots**
- 5006: malcGshdsl(5006)**
- 5007: malcGshdslPots**
- 5008: malcPots900**
- 5009: malcDs3**
- 5010: malcAdslAc5**
- 5011: malcOc3**
- 5012: malcMtacFc**
- 5013: malcAc5AnxB**
- 5014: malcT1E1Tdm**
- 5015: malcAdslAc6**
- 5016: malcAdslPotsAc6**

5017: malcAdslSplAc6
5018: malcAdslAnxBAc6
5019: malcAdslAnxBSplAc6
5020: malcAdslC
5021: malcAdslPotsC
5022: malcAdslSplC
5023: malcAdslSplUkAc6
5024: malcAdslSplUkC
5101: malcT1ImaF
5109: malcDs3F
5111: malcOc3F

Default 5001: malcUplinkNt1

Access init-only

cas

Channel Associated Signalling Bits (CAS) are bits dedicated for connection control across a 1544 Kbps or 2048 Kbps interface that carries 64 Kbps channels.

Values 1: Enabled

2: Disabled

Default 1: Enabled

Access read-write

Notes Procedures are based on the state of a maximum of four signalling bits (A, B, C, D) that are allocated per channel per multiframe. The service provided is the transfer of channel associated signalling information, (such as the transfer of ABCD bits.) Typically, the value of the (A,B,C,D) vector does not change for extended periods of time.

casEventsPassingMethod

If **zhoneCpeVoipOobCasEvents** is disabled, dtmf digits are carried along the voice path. If **zhoneCpeVoipOobCasEvents** is enabled, we look at the dtmf-events-passing-method. If **zhoneCpeVoipCasEventsPassingMethod** is rfc4733, dtmf digits are carried in the RTP payload. If **zhoneCpeVoipCasEventsPassingMethod** is sip-info, dtmf digits are carried along the signaling path in the INFO messages.

Values 1: rfc4733

2: sipinfo

Default 1: rfc4733

Access read-write

caSignalling (DynOmciCpePwe)

This attribute selects the signaling format. It applies to structured interfaces only.

Values 1: base

2: e1cas

3: sfcas

4: ds1esfcas

5: j2cas

Default 1: base

Access read-write

castType

The type of topology of a connection (point-to-point, point-to-multipoint). In the case of point-to-multipoint, the orientation of this VPL or VCL in the connection.

On a host:

- p2mpRoot indicates that the host is the root of the p2mp connection.
- p2mpLeaf indicates that the host is a leaf of the p2mp connection.

On a switch interface:

- p2mpRoot indicates that cells received by the switching fabric from the interface are from the root of the p2mp connection.
- p2mpLeaf indicates that cells transmitted to the interface from the switching fabric are to the leaf of the p2mp connection.

Values 1: p2p

2: p2mpRoot

3: p2mpLeaf

cbDialString

The dialing string used for a callback. The format of this string is determined by callback-type.

Values 0 to 0

Default empty string

Access read-write

Notes This object only applies when callback negotiation is enabled using **callback-enable**.

cbEnable

Enable or disable callback negotiation. Provides a method for an implementation to request a dial-up peer to call back.

Values 1: enabled
2: disabled

Default 1: disabled

Access read-write

Notes

cbType

The method of determining the location to callback.

Values 1: **byauth**
Location is determined by authorization information.

2: **bydialstr**
Location is determined by a dialing string.

3: **byidentifier**
Location is determined by a location ID.

4: **bye164**
Location is determined by an E.164 number.

5: **byename**
Location is determined by a distinguished name.

Default 1: **byauth**

Access read-write

Notes These objects only apply when callback negotiation is enabled using **callback-enable**.

cdrActiveHigh

Clock data recovery activity high.

Values 1: true
2: false
Default 2: false
Access read-write

cdrPattern

Re-sync signal pattern for clock data recovery.

Values 0 to 65535
Default 192
Access read-write

cdvTolerance (DynOmciCpePwe)

This attribute represents the duration of user data that must be buffered by the CES inter working entity to offset packet delay variation. It is expressed in 10 microsecond increments. The default value for DS1 CES is 750 micro seconds; for DS3 CES, it is 1000 micro seconds.

Values Integer32 (0..65535)
Default 750
Access read-write

cellloss

The number of bad sequence number of the STF within any given 15 minutes performance data collection period, which causes the SNMP agent to send an aal2PerfCellLossThreshTrap. One trap will be sent per interval per interface. A value of '0' will disable the trap.

Values 0 to 900
Default 0
Access read-create

cellScrambling (olt)

Cell scrambling option for PON OLT connections.

Values 1: oltCellScramblingOff
2: oltCellScramblingRxOnly
3: oltCellScramblingTxOnly
4: oltCellScramblingRxAndTx
Default 4: oltCellScramblingRxAndTx

Access read-write

cellScramble

Describes the enabled status of the DS1 cell scramble mode.

Values **1: true**
Scramble mode is enabled.
2: false
Scramble mode is disabled.

Default **1: true**

Access read-write

center

Values **0 to 32**

Access read-only

CfgBandwidth

Aggregate radio bandwidth setting selection. The value of cfg-bandwidth depends on the the different combinations of zNID models and the band values.

Values **1 = bandwidth20MHz**
2 = bandwidth40MHz
3 = bandwidth20In2GAnd40In5G
4 = bandwidth80MHz

Access read-write

cfgRuleIndex

Next available entry in zhoneCpeAutoCfgRuleTable.

Access read-create

cgId (CardGroup)

The card group identifier.

Values **0 to 2147483647**

Default **0**

Access read-write

cgroupId (CardGroupEntry)

The unique redundancy group to which the card is assigned.

- Values** 0 to 2147483647
- Default** 0
- Access** init-only

channel (DynOmciCpeWlanAdv)

Defines which channel to use, or “auto” for automatic selection of a channel with low interference. 802.11b and 802.11g use channels to limit interference from other devices.

- Values** 1: auto
- 2: c1
- 3: c2
- 4: c3
- 5: c4
- 6: c5
- 7: c6
- 8: c7
- 9: c8
- 10: c9
- 11: c10
- 12: c11
- 13: c12
- 14: c13
- Default** auto
- Access** read-write

channelAssign

If service type is structured, this attribute is a bit field indicating the 64kbps sub channels are used.

Values BITS (0:channel0, 1: channel1, 2: channel2, 3: channel3, 4: channel4, 5: channel5, 6: channel6, 7: channel7, 8: channel8, 9: channel9, 10: channel10, 11: channel11, 12: channel12, 13:channel13, 14:channel14, 15:channel15, 16:channel16, 17:channel17, 18:channel18, 19:channel19, 20:channel20, 21:channel21, 22:channel22, 23:channel23, 24:channel24, 25:channel25, 26:channel26, 27:channel27, 28:channel28, 29:channel29,30:channel30, 31:channel31)

Default 0:channel0

Access read-write

channelId

The channel number used for this connection.

Values 1 to 2147483647

Default 1

Access read-create

channelization

Indicates whether this T1/E1 connection is channelized or unchannelized.

Values **1: disabled**
Disables channelization.

2: enabledDs0
Indicates that this is a DS-1 channelized into DS-0s

3: enabledDs1
Indicates that this is a DS-2 channelized into DS-1s

Default 2: enabledDs0

Access read-write

Notes Setting this value will create or delete entries in the ifTable for the DS-0s that are within the DS-1.

channelNumber

The channel number.

Values 1 to 2147483647

Default 1

Access init-only

chapName

The username for authenticating with the CHAP protocol.

Default empty string

Access read-write

chapSecret

The password for authenticating with the CHAP protocol.

Default empty string

Access read-write

chassisId

The string value used to identify the chassis component associated with the remote system.

Access read-create

cid

The AAL2 Channel Identifier (CID) for the connection.

Values 16 to 255

Default 16

Access read-create

cidFeatures

This attribute contains a bit map of caller ID features.

Values 0: callingnumber, 1: callingname, 2: cidblocking, 3: cidnumber, 4: cidname, 5: anonymouscidblocking

Maximum callingnumber+callingname+cidblocking+cidnumber+cidname+anonymouscidblocking

Access read-write

circuitId

The transmission vendor circuit identifier, used to facilitate troubleshooting.

Values 0 to 255

Access read-only

circuitIdentifier

This variable contains the transmission vendor's circuit identifier, for the purpose of facilitating troubleshooting.

Default ds1

Access read-write

circuitType

Values 0 to 2147483647

Default 0

Access read-write

city

The city name.

Values 0 to 32

Access read-write

cktLevel

Values 0 to 2147483647

Default 0

Access read-write

clearWaitDuration

Time of Inactive seizure signal to declare call cleared. The default value of this object is 400 milliseconds.

Values 200 to 2000

Default 400

Access read-write

clientHostname

Client hostname (if any) the client used to acquire its address. The default value for this object is an empty string.

Access read-write

clientId

DHCP client identifier of the client that matches this host object. If this field is specified then the dhcpHostHardwareAddress value is not used. However, at least one of these fields must be configured per host object. The default value for this object is an empty octet string.

Access read-write

clientMatchLength

Length of the client identifier substring to match. A value of -1 indicates that the entire string starting at the match offset value should be matched.

Values -1 to 255

Default -1

Access read-write

clientMatchOffset

Offset into the client-passed client identifier string where the string match should start.

Values 0 to 255

Default 0

Access read-write

clientMatchString

Client identifier match string that is used to determine which clients should be placed in this group. The default value for this object is an empty octet string.

Access read-write

clockExternalRecovery

This variable indicates if external clock recovery is enabled for this SONET interface.

Values 1: enabled

2: disabled

Default 1: enabled

Minimum 1

Maximum 2

Access read-write

clockOffset

The value of this variable is used to help achieve a low frequency wander and jitter between network and DSL system clock.

Values -100 to 100

Default 0

Access read-create

clockSourceEligibility

Whether the DS1 can be used as a DS1 Line resource provider in negotiation as a possible transmit clock source for all DS1s.

Values 1: notEligible

2: eligible

Default 1: notEligible

Access read-write

clockTransmitSource

This variable describes the SONET transmit clock source.

Values 1: loopTiming

The transmit clock is derived from the recovered receive clock.

2: throughTiming

The transmit clock is derived from the recovered receive clock of another SONET interface.

3: localTiming

The transmit clock is synthesized from a local clock source or from an external clock that is attached to the box containing the interface.

4: external155MHz

The transmit clock is synthesized from an external 155.52 MHz source.

Default 1: loopTiming

Access read-write

Notes The **external155MHz** option is not valid for Sechtor 100.

clockReference

This attribute specifies the frequency of the common timing reference, in multiples of 8 kHz. The default value is 1 (8 kHz).

Values Integer32 (1..65535)

Default 1

Access read-write

codecFirstOrder

This attribute specifies codec selection as defined by RFC 3551.

- Values**
- 1:** pcmu
 - 2:** gsm
 - 3:** g723
 - 4:** dvi4eightKHz
 - 5:** dvi4sixteenKHz
 - 6:** lpc
 - 7:** pcma
 - 8:** g722
 - 9:** l16twochannels
 - 10:** l16onechannel
 - 11:** qcelp
 - 12:** cn
 - 13:**mpa
 - 14:**g728
 - 15:**dvi411
 - 16:** dvi422
 - 17:** g729

Default 1: pcmu

Access read-write

codecSecondOrder

This attribute specifies codec selection as defined by RFC 3551.

Values 1: pcmu
2: gsm
3: g723
4: dvi4eightKHz
5: dvi4sixteenKHz
6: lpc
7: pcma
8: g722
9: l16twochannels
10: l16onechannel
11: qcelp
12: cn
13:mpa
14:g728
15:dvi411
16: dvi422
17: g729
Default 1: pcmu
Access read-write

codecThirdOrder

This attribute specifies codec selection as defined by RFC 3551.

Values	1: pcmu
	2: gsm
	3: g723
	4: dvi4eightKHz
	5: dvi4sixteenKHz
	6: lpc
	7: pcma
	8: g722
	9: l16twochannels
	10: l16onechannel
	11: qcelp
	12: cn
	13:mpa
	14:g728
	15:dvi411
	16: dvi422
	17: g729
Default	1: pcmu
Access	read-write

codecFourthOrder

This attribute specifies codec selection as defined by RFC 3551.

Values 1: pcmu
 2: gsm
 3: g723
 4: dvi4eightKHz
 5: dvi4sixteenKHz
 6: lpc
 7: pcma
 8: g722
 9: l16twochannels
 10: l16onechannel
 11: qcelp
 12: cn
 13:mpa
 14:g728
 15:dvi411
 16: dvi422
 17: g729

Default 1: pcmu

Access read-write

collectionEnabled

Enables or disables bulk statistics for the device.

Values 1: true
 2: false

Default 2: false

Access read-write

collectionId

Unique identifier of the bulk statistic row.

Values 0 to 2147483647

Default 0

Access init-only

collectionInterval

The collection interval for this statistic. Currently, on 15 minute intervals are supported.

commonIndex (DynOmciCpeSystemCommon, DynOmciCpeSystem)

This represents the CPE system Common Index Value.

Values 0..2147483647

Default 0

Access read-write

communityAccess

The access rights for this particular community name. An access of read-write-admin allows access to those tables with a higher level of security such as the community table. Gets to this table will return E_NOSUCHINSTANCE unless the access rights is read-write-admin.

Values 1: noAccess

This community name does not have any access to this device.

2: read

Allows read access to all fields except admin access fields such as this community table.

3: readAndWrite

Allows read and write access to all fields except admin access fields such as this community table.

4: admin

Read and write access to all fields on this device.

Default 2: read

Access read-write

communityAccessListIndex

The AccessList Table index indicating which IP addresses are allowed to use this community name. If the value is 0, no address checking is performed and all SNMP requests are allowed with the access right as specified in this table.

Values 0 to 50

Default 0

Access read-write

compensated for GponAllocIdProfile

For CBR, indicates whether compensated.

Values 1: True

2: False

Default 1: True

Access read-write

compensated for GponTrafficProfile

For CBR, indicates whether compensated or not.

Values 1: enabled

2: disabled

Default 2: disabled

Access read-write

compProtocol

Sets the IP compression option.

Values 1: none

The local node will not attempt to negotiate any IP compression.

2: vj-tcp

The local node will attempt to negotiate Van Jacobson TCP/IP header compression.

Default 1: none

Access read-write

Notes See RFC 1332 for more information on Van Jacobson compression.

Changing this parameter will take effect when the link is next restarted.

compSlotId

Indicates whether the slot identifier field may be compressed.

Values 1: enabled

The slot identifier may be compressed.

2: disabled

The slot identifier must not be compressed. All compressed TCP packets must set the C bit in every change mask, and must include the slot identifier.

Default 2: disabled

Access read-write

configKeyExchangeInterval

Periodicity for exchanging encryption key with ONUs on this OLT.

A value of 0 disables periodic key exchange, i.e only send key once during ranging.

The key-exchange parameter must be enabled for this value to take effect.

Default **3000**

Access read-write

configProfName

The value of this object identifies the row in the respective DSL configuration profile table. For example, if **zhoneDslLineType** is **HDSL2**, then the profile identifies an HDSL2 configuration profile. It is assumed that all profile names are unique to the system. The default value is **ifIndex** (an ASCII string value) for static profiles and **zhonedefault** for dynamic profiles.

Values **1 to 32**

Access read-write

configSyncAction

Used by DZMS to create a list of configuration synchronization records. It must first write the filename expected and then set the action.

Values **1: noAction**

No list of records is created.

2: createList

Allows the device to determine whether the complete database (synchoverflow occurred) needs to be sent or only those records that have been modified.

3: createFullList

Specifies that DZMS would like to receive the complete database.

Default **1: noAction**

Access read-write

configSyncExists

Indicates whether or not there exists records to be sent to the DZMS configuration synchronization manager for changes that have been made by CLI/HTTP to the data base.

Values 1 or 2
Default 2
Access sync-only

configSyncFileName

File name to be used when creating the configuration synchronization records.

Access read-write

configSyncOverflow

Indicates whether or not the device had to discard updated database records due to resource constraints. If TRUE, traps will be sent to the DZMS configuration synchronization manager indicating a download of the entire data base is required.

Values 1 or 2
Default 2
Access sync-only

configSyncPriority

Indicates the highest priority configuration synchronization trap which is outstanding. This flag is set to none at boot time and whenever a file of synchronization changes has been sent to DZMS. If a parameter change has been made to either the same or lower priority, the device does not have to send a change notification to DZMS. Higher-priority fields must immediately cause traps to be sent to DZMS.

Values 1: none
 2: low
 3: medium
 4: high
Default 1: none
Access sync-only

configSyncStatus

Used by configuration synchronization agent when processing a request to create a list of configuration synchronization records.

Values 1: syncComplete
2: syncPending
3: syncError
4: syncInitializing
Default 1: syncComplete
Access read-only

configSyncUserName

User name to be used when sending the configuration synchronization records. When read, the last user name is returned.

Access read-write

configSyncUserPassword

Password to be used when sending the configuration synchronization records. For use in file transfer protocol (FTP) authentication. When read, always returns the NULL string

Access read-write

configUnitMode

Whether the unit is configured as the Central Office (CO) or Customer Premise Equipment (CPE) side of the connection.

Values 1: co
2: cpe
Default 1: co
Access read-write

congestion

The number of congestion events within any given 15 minutes performance data collection period, which causes the SNMP agent to send an aal2PerfCongestionThreshTrap.

One trap will be sent per interval per interface. A value of '0' will disable the trap.

Values 0 to 900
Default 0
Access read-create

connectionDesc

A text string describing this voice call entry. This value is not used by the system and is provided for operator reference only. It is used to define the external or public phone number assigned by the switch.

Values 0 to 32

Access read-write

connectionIndex

Used to identify this connection pair. The number defines a connection set for this subscriber/line group ID pair.

Values 0 to 2147483647

Default 0

Access read-only

connectionType

The connection type between the two endpoints referred to by the subVoiceEndPointProtocol indexes. Where applicable, the first endpoint refers to the line or premise side of the connection and the second endpoint refers to the central office side of the connection.

Values 1: aal2ToGr303

2: aal2ToV52

3: voipToGr303

4: potsToAal2

5: isdnToAal2

6: potsToGr303

13: ebsToGr303

14: potsToVoip

19: ebsToVoip

Default 1: aal2ToGr303

Access read-write

connectionType (For PweStdConfig object)

PWE connection type of the bundle.

Values 1 (SATOP), 2 (CESOP), 3 (E1-PRI)

Default 1

Access read-create

connectMode

Connection mode compatibility.

Values **1: basic-2b1q-mode**

Basic operating mode.

2: globespan-neg-mode

Globespan mode where CO and CPE can have different rates synchronizing on the lower rate.

3: copper-mtn-mode

Operating mode compatible with Copper Mountain products.

4: flowpoint-mode

Operating mode compatible with FlowPoint Corporation products.

Default **4: flowpoint-mode**

Access read-write

connKind

The use of call control.

The type of call control used for an ATM connection at a particular interface. The use is as follows:

pvc(1): Virtual link of a PVC. Should not be used for an PVC/SVC (i.e., Soft PVC) crossconnect.

svcIncoming(2): Virtual link established after a received signaling request to setup an SVC.

svcOutgoing(3): Virtual link established after a transmitted or forwarded signaling request to setup an SVC.

spvcInitiator(4): Virtual link at the PVC side of an SVC/PVC crossconnect, where the switch is the initiator of the Soft PVC setup.

spvcTarget(5): Virtual link at the PVC side of an SVC/PVC crossconnect, where the switch is the target of the Soft PVC setup.

For PVCs, a pvc virtual link is always cross-connected to a pvc virtual link.

For SVCs, an svcIncoming virtual link is always cross-connected to an svcOutgoing virtual link.

For Soft PVCs, an spvcInitiator is either cross-connected to an svcOutgoing or an spvcTarget, and an spvcTarget is either cross-connected to an svcIncoming or an spvcInitiator.

Values 1: pvc
 2: svcIncoming
 3: svcOutgoing
 4: spvcInitiator
 5: spvcTarget

Default 0

Access read-create

contactPhone

The contact phone number

Values 0 to 32

Access read-write

controlld

A McastControlLists object contains one or more McastControlList elements. ControlId represents the index of parent McastControlLists.

Access not-accessible

cookieProtection (DynOmciCpeSystemCommon)

Protects against malicious attackers attempting to exploit TCP handshaking.

Values 1: enabled
 2: disabled

Default 1: enabled

Access read-write

cosetPolynomial

Describes the enabled status of the coset polynomial mode.

Values 1: true
 Coset polynomial mode is enabled.

2: false
 Coset polynomial mode is disabled.

Default 1: byauth

Access read-write

cpcsRxSduSize

Maximum AAL5 CPCS SDU size, in octets, supported on the receive direction of this VCC.

Values 1 to 65535

Default 1520

Access read-write

cpcsTxSduSize

Maximum AAL5 CPCS SDU size, in octets, supported on the transmit direction of this VCC.

Values 1 to 65535

Default 2048

Access read-write

cpeLimit

Cell phase error limit on customer premises equipment.

Values 0 to 16

Default 4

Access read-write

cpeType

Identifies the type of customer premises equipment that is used by the subscriber. Required for row creation.

Values 1: other

2: zedge64T

3: zedge64S

7: zedgeBH2A

8: zedgeH2A

Default 1: other

Access read-write

cpotFeatures

This attribute is a bit map of call processing features.

Values BITS (0:threeway, 1:calltransfer, 2:callhold, 3:callpark, 4:donotdisturb, 5:flashonemergency, 6:emergencyhold, 7:sixway)

Default threeway+calltransfer+callhold+callpark+donotdisturb+flashonemergency+emergencyhold+sixway

Access read-write

crcRx

Cyclic Redundancy Check for reception.

Values 1: true
2: false

Default 1: true

Access read-write

criticalDialTimeout

This attribute defines the critical dial timeout for digit map processing in milliseconds.

Values Integer32 (0..6553)

Default 4000

Access read-write

crossConnId

This is an internal read-only variable. It should not be modified.

csuLineLength

The length of the T-1 line. This field is not used for E-1 platforms.

Values 1: csu00
0 dB for CSU DS1 WAN for line build out

2: csu75
-7.5 dB for line build out

3: csu150
15.0 dB for line build out

4: csu225
-22.5 dB for line build out

Default 1: csu00

Access read-write

ctmIndex

This represents the CPE Traffic Management Index, values are from the range 1 to n.

Values Integer32 (1..2147483647)

Access read-write

currentStats

Blob of current statistics information returned. Format is:

PhysPort,I1,A1,I2,A2,...In,An

where

PhysPort - physical port ID value

Ii = attribute index (0-15, ascii-coded decimal integer) within ME according to G984.4 standard.

Ai = attribute value (counter, ascii-coded decimal integer)

n = number of attributes returned = number requested.

Maximum number of attributes per ME is 16.

Values Octet String (0..256)

Access read-only

customerARP

If true, send ARP requests selectively or by flood.

Values 1: true
0 dB for CSU DS1 WAN for line build out

2: false
-7.5 dB for line build out

Default 2: false

Access read-create

customerDHCP

If TRUE, forwards broadcast DHCP OFFER or DHCP ACK packets even if bridgeFilterBroadcast is TRUE for that interface.

Values 1: true

2: false

Default 2: false

Access read-create

customerKey

Values 0 to 4294967295

Default 0

Access read-write

customerType

The type of customer.

Values 1: Business

2: Residential

3: Other

Default 2: Residential

Access read-write

D

dataIfIndex

This object should be made equal to the ifIndex used to identify the entry in the interfaceTable that defines the IP record for the interface used by the subscriber.

Values 0 to 2147483647

Default 0

Access sync-only

DateAndTime

The date and time of the system, specified in local time. Only a community name having **admin** privileges will be allowed to modify this value. Although any valid community name is allowed to read it.

Access read-write

dayLightSavingTime

Indicates whether Daylight Savings time for NTP is currently in effect.

Values 1: true
2: false
Default 2: false
Access read-write

dba

A flag to indicate if dynamic or static bandwidth allocation is used. The default value of DbEnabled is false.

Values 1: true
2: false
Default 2: false
Access read-write

dbaAssuredUsBw

DBA assured bandwidth will be allocated when traffic demand exists. The min value of Assured bandwidth can be 0 or 256 kilobits per second. Only multiples of 64 kilo bits per second are allowed.

Values 0..104857
Default 0
Access read-write

dbaFixedUsCbrBw

Fixed CBR bandwidth will be always allocated CBR bandwidth. The min value of Fixed Cbr bandwidth can be 0 or 512 kilobits per second.Only multiples of 64 kilo bits per second are allowed.

Values 0..454208
Default 0
Access read-write

dbaFixedUsUbrBw

Fixed UBR bandwidth will be always allocated UBR bandwidth. The min values of Fixed Ubr bandwidth can be 0 or 128kilo bits per second. Only multiples of 64 kilobits per second are allowed.

Values 0..1048576
Default 0
Access read-write

dbaMaxUsBw

The Max Bandwidth value should be larger or equal to the sum of Fixed UBR, Fixed CBR and Assured. The increments should be done in the order of 64 kilobits per second. The default value of maximum bandwidth is 0.

Values 0..1048576

Default 0

Access read-write

dbaMode

Working DBA mode.

Values 1: predictive

2: piggyback

3: wholeReport

Default 1: predictive

Access read-write

ddnsFwdName

Dynamic DNS forward host name. Not currently used. The default value for this object is an empty string.

Access read-write

ddnsRevName

Dynamic DNS reverse host name. Not currently used. The default value for this object is an empty string.

Access read-write

decoderCoeffA

A 21-bit value corresponding to the decoder coefficient A. The Default value is 0x16e

Values 0 to 2097151

Default 366

Access read-create

decoderCoeffB

A 21-bit value corresponding to the decoder coefficient B. The default value is 0x331

Values 0 to 2097151

Default 817

Access read-create

defaultChannelAddress

The multicast IP address of the channel chosen to be the default channel for a particular head-end provider.

Values 4 to 4

Default 0.0.0.0

Access read-write

defaultLease

Default value for requested lease length. This value should not be less than dhcpMinLease. Maximum allowed value is one week (604800 seconds).

Values 100 to 604800

Default 3600

Access read-write

defaultTtl

The default value inserted into the Time-To-Live field of the IP header of datagrams originated at this entity, whenever a TTL value is not supplied by the transport layer protocol.

Values 1 to 255

Default 64

Access read-write

defLeaseTime

The default time in seconds that will be assigned to a lease if the client requesting the lease does not ask for a specific expiration time. A value of -1 indicates that this field should be ignored. Default value is -1 seconds.

Values -1 to 2147483647

Default -1

Access read-write

defRouter

IP Address of the client's default gateway.

Default 0.0.0.0

Access read-write

delayStart

Minimum time to wait from outgoing seizure to out-dialing digits for delay start signaling. The default value of this object is 300 milliseconds.

Values 20 to 2000

Default 300

Access read-write

delBeforeApply

If true, any existing services on the target uni are removed prior to applying the service template.

If the targetUni field is blank and this field is true, the all services on the CPE are removed before the service template is applied.

Values 1: False

2: True

Default 2: True

Access read-create

delimiterPattern

A unique pattern indicating the start of the ATM cell or minislot, which can be used to perform byte synchronization.

Values 0 to 65535

Default 162

Access read-write

delimiterSize

Delimiter pattern size.

Values 2 to 16

Default 8

Access read-write

deltCmd

Loop diagnostics mode forced (LDSF). Setting this variable to 'forced' commands the device to begin loop diagnostic procedures. The device resets the variable to 'inhibit'. Regardless of the state of zhoneAdsl2LineCmndConfLdsf, the device will remain responsive to loop diagnostics initiated by the far-end ATU.

As per ITU-T G.997.1, paragraph 7.3.1.1.8, before the operator can force loop diagnostics mode, the line must be in the L3 (IDLE) state. When the operator forces loop diagnostics mode, the device will begin loop diagnostic procedures. When loop diagnostic procedures have completed (successfully or not), the device will set the state of zhoneAdsl2LineCmndConfLdsf to 'inhibit', return to and remain in L3 (IDLE) state.

Values 0: inhibit

1: force

Default 0: inhibit

Access read-write

description (GponOnuGenProfileVariable/ GponOnuSpecProfileVariable)

The description of what the variable of OMCI profile for.

Values STRING size (0...64 characters)

Default An empty string

Access read-only

dest

The destination network or host IP address of the static route. A default gateway is one with a destination of 0.0.0.0 and netmask 0.0.0.0.

Default 0.0.0.0

Access init-only

destAddr

The destination address for point-to-point IP interfaces. This parameter is ignored for non point-to-point IP interfaces.

Default 0.0.0.0

Access read-write

destEndIP

Destination IP address of the end of the multicast range. Default value is 0.0.0.0.

Values IPAddress

Default 0.0.0.0

Access read-write

destinationIp (DynOmciCpeStaticRoute)

The IP address of the destination network or host.

Values IPAddress

Default 0.0.0.0

Access read-write

destStartIP

Destination IP address of the start of the multicast range. Default value is 0.0.0.0.

Values IPAddress

Default 0.0.0.0

Access read-write

deviceAddrType

Values 0 to 2

Default 0

Access read-write

deviceIP

The IP address of this device. Independent of the different IP interfaces configured and what is active, a manager uses this value to address the device. This value may or may not be an IP address of an existing IP interface. Deletion of a matching IP interface does not cause this address to be modified or deleted. A manager may communicate with the device using any of the configured IP interface addresses. However, any failures such as physical line outages or the card status is inactive, may result in failures to the manager requests (i.e., timeout).

Default 0.0.0.0

Access read-write

devicePin (DynOmciCpeWlanSubscriber)

Device pin is only applicable when WPS security method is enabled.

Values String (1..8)

Access read-write

deviceRole

Values 0 to 2147483647

Default 0

Access read-write

deviceSubType

The subcategory type (such as MALC23Device) of a generic device type (such as MALCDevice).

Values 0 to 32

Access read-only

Notes This is a database-specific object and should not be modified.

dhcp

Specifies whether this interface is a DHCP client, a DHCP server, both, or neither.

Values 1: No DHCP

2: DHCP Client

3: DHCP Server

4: DHCP Both

Default 1: No DHCP

Access read-write

dhcplearn

none :IP address and Mac address are not learned from DHCP packets.

BIT 1 (mac): MAC addresses are learned from DHCP packets.

ip 2 (ip) : IP addresses are learned from DHCP packets.

Values 0: mac
1: ip
Default none
Access read-create

dhcpServerEnable

If this field is set to TRUE, then ipAddrDynamic has to be STATIC and if and when Dhcp server is not enabled then ipAddrdynamic can set to STATIC, PPP or DHCPCLIENT.

Values 1: True
2: False
Default 2: False
Access Init Only

dhcpServerProfile (DynOmciCpelfVlan)

The address of the DHCP server profile associated with this interface. The value 0 indicates no associated cpe-dhcp-server profile.

Values 0..2147483647
Default 0
Access read-write

diffDelayMax

The maximum number of milliseconds of differential delay among the links that will be tolerated on this interface.

Values 0 to 2147483647
Default 25
Access read-create

dialNumber

This attribute contains the dialing number to make the call.

Values STRING (SIZE (0..32))
Access read-write

dialPlanIndex

This represents the SIP dial plan index value.

Values Integer32 (1..2147483647)

Access read-write

dialPlanFormat

DialPlanFormat defines the dial plan format standard that is supported in the ONT for VoIP.

Values 1: h248

2: nsc

3: vendorspecific

Default 1: h248

Access read-write

dialPlanString

SipDialPlan contains a dial plan used by the VoIP service.

Values STRING (SIZE (0..64))

Access read-write

dialType

Specifies the out dialing type of E&M interface.

Values 1: dtmf

Dual tone multi-frequency digit dialing

2: pulse

Pulse dialing

Default 1: dtmf

Access read-write

digitDuration

Outgoing DTMF digit duration in milliseconds.

Values 50 to 500

measured in milliseconds

Default 100

Access read-write

direction

Direction of data for this port.

Values 1: down
 2: up
 3: bidirectional
Default 3: bidirectional
Access read-write

displayName

This ASCII string attribute defines the customer id used for the display attribute in outgoing SIP messages.

Values STRING (SIZE (0..25))
Access read-write

dmtConfigMode

Discrete Multi-Tone parameter that selects whether there is overlap (echoCancel) or no overlap (freqDivMux) of bins.

Values 1: echoCancel
 2: freqDivMux
Default 2: freqDivMux
Access read-write

dnsHostListIndex (DynOmciCpeSystemCommon)

Address of the static-route-list profile associated with this CPE.

Values 0..2147483647
Default 0
Access read-write

domain

Routing domain to which this subnet object applies.

Values 0 to 2147483647
Default 0
Access read-write

domainName

The default domain name that the client should use when resolving hostnames via DNS. The default value for this object is an empty string.

Access read-write

domainName (VoipServer)

This parameter indicates the Universal Resource Identifier (URI) of the system.

Values A string of maximum of 255 characters.

Access read-create

domainName (DynOmciCpeDnsHost)

The domain name assigned to the host IP address.

Values String (0..256).

Access read-write

dot11nMode (DynOmciCpeWlanAdv)

802.11n MIMO EWC modes of operation. 802.11n improves data rates via MIMO (multiple-input, multiple-output) using spatial streams which each have a channel width of 40 MHz or 20 MHz. Usage of 802.11n in the 2.4 and 5GHz modes should depend on interference with other 802.11 or bluetooth systems on the same frequency. Enhanced Wireless Consortium (EWC) provides extra enhancements (adding the ability to define 20 MHz channels).

Values 1: auto

2: disabled

Default 1: auto

Access read-write

dot11nRate (DynOmciCpeWlanAdv)

Supported 802.11n MIMO rates, in Mbps.

Values 1: auto
 2: use54g
 3: rate6dot5m
 4: rate13m
 5: rate19dot5m
 6: rate26m
 7: rate39m
 8: rate58dot5m
 9: rate65m
 10: rate78m
 11: rate104m
 12: rate117m
 13: rate130m

Default 1: auto

Access read-write

dot1nClientOnly (DynOmciCpeWlanAdv)

Enable or disable the restriction of access to 802.11n clients only. When enabled, prevent 802.11b/g clients from connecting.

Values 1: enabled
 2: disabled

Default 2: disabled

Access read-write

dot1nProtection (DynOmciCpeWlanAdv)

802.11n MIMO protection modes.

Values 1: auto
 2: disabled

Default 1: auto

Access read-write

downDisableBandEndTone <1,2,3,4> downDisableBandStartTone<1,2,3,4>

The four downstream fields downDisableBandStartTone(1-4) and downDisableBandEndTone(1-4) configure bands in which VDSL vectoring is

disabled or not applied at the CO side in the downstream direction. The four upstream fields upDisableBandStartTone(1-4) and upDisableBandEndTone(1-4) configure bands in which VDSL vectoring is disabled or not applied at the CPE side for the upstream direction. The values are VDSL tone indexes. The range for each field is 0 - 4096. The default for all vectoring disable tone fields is 0 which disabled the band.

NOTE:

1) The disable band end tone index must be greater than the corresponding disable band start tone index.

e.g. downDisableBandEndTone1 > downDisableBandStartTone1

upDisableBandEndTone1 > upDisableBandStartTone1

2) disable band start tone index for a subsequent band must be greater than the preceding disable band's disable band end tone index.

e.g. downDisableBandStartTone2 > downDisableBandEndTone1

upDisableBandStartTone2 > upDisableBandEndTone1

3) Only tone bands which follow the first two rules will be applied. If any tone band is zero, or is not starting at a higher index, it and all subsequent tone bands are ignored.

Values 0-4096

Default 0

Access read-create

downstreamSir

This attribute specifies the downstream sustained information rate. Unit is kbps.

Values Integer32 (0..1310720)

Default 0

Access read-write

downstreamPir

This attribute specifies the downstream peak information rate. Unit is kbps.

Values Integer32 (0..1310720)

Default 0

Access read-write

downstreamPriority

Downstream Priority, currently the only used value range is 0-7.

Values Integer32 (0..65535)

Default 0

Access read-write

downstreamPvcKey

Values 0 to 2147483647

Default 0

Access read-write

downstreamPvcType

Values 1 to 32

Access read-write

downstreamWeight

Downstream Weight for weighted round robin scheduling, variable size is 1 byte, default 0.

Values Integer32 (0..255)

Default 0

Access read-write

driftControlInterval

Defines the interval of periodic monitoring on drifts of upstream data.

Values Integer

Default 1000

Access read-write

driftControlLimit

Indicates the limit of upstream data drift.

Values Integer

Default 3

Access read-write

ds1Mode

This attribute specifies the mode of a DS1.

- Values**
- 1: shorthaulnopwr**
 - 2: superframe**
 - 3: unframed**
 - 4: g704**
 - 5: jtg704**

Default 1: shorthaulnopwr

Access read-write

ds1Framing

This attribute specifies the DS1 framing structure.

- Values**
- 1: extsuperframe**
 - 2: superframe**
 - 3: unframed**
 - 4: g704**
 - 5: jtg704**

Default 3: unframed

Access read-write

dsBerInterval

Interval for downstream bit error rate.

Values Integer

Default 5000

Access read-write

dsBerLimit

The maximum allowable bit error rate prior to forward error correction for the DSL interface.

Values 1: 1⁻³
 2: 1⁻⁴
 3: 1⁻⁵
 4: 1⁻⁶
Default 3: 1⁻⁵
Access read-write

dscp (DynOmciCpePwe)

This parameter indicates DSCP value for cpe pwe traffic.

Values **Valid code values:**

<0-63>	Differentiated services codepoint value
af11	Mark packets with AF11 dscp (001010)
af12	Mark packets with AF12 dscp (001100)
af13	Mark packets with AF13 dscp (001110)
af21	Mark packets with AF21 dscp (010010)
af22	Mark packets with AF22 dscp (010100)
af23	Mark packets with AF23 dscp (010110)
af31	Mark packets with AF31 dscp (011010)
af32	Mark packets with AF32 dscp (011100)
af33	Mark packets with AF33 dscp (011110)
af41	Mark packets with AF41 dscp (100010)
af42	Mark packets with AF42 dscp (100100)
af43	Mark packets with AF43 dscp (100110)
cs1	Mark packets with CS1(precedence 1) dscp (001000)
cs2	Mark packets with CS2(precedence 2) dscp (010000)
cs3	Mark packets with CS3(precedence 3) dscp (011000)
cs4	Mark packets with CS4(precedence 4) dscp (100000)
cs5	Mark packets with CS5(precedence 5) dscp (101000)
cs6	Mark packets with CS6(precedence 6) dscp (110000)
cs7	Mark packets with CS7(precedence 7) dscp (111000)
default	Mark packets with default dscp (000000)
ef	Mark packets with EF dscp (101110)

Default 0
Access read-write

dscpMark

Diffserv code point to be used for outgoing RTP packets for this profile. The default value is expedited forwarding (EF) = 0x2E.

Values Integer32 (0..255)
Default 46
Access read-write

dsMaxConstant

Restrict VDSL upstream quadrature amplitude modulation (QAM) constellations to use those that do not exceed this maximum.

Values 1: QAM-4
2: QAM-8
3: QAM-16
4: QAM-32
5: QAM-64
6: QAM-128
7: QAM-256
Default 6: QAM-128
Access read-write

dsMaxRate

The target downstream line rate (in Mbps) that the VDSL interface will attempt to achieve. If this rate cannot be achieved, the highest possible rate will be obtained. The actual downstream rate may differ from this value, depending on line conditions.

Values 0 to 2147483647 (in Mbps/1000)
Default 207900
Access read-write

dsMinMargin

The minimum downstream signal-to-noise margin measured at quadrature phase shift keying (QPSK). This is used to compute maximum constellation size for each symbol rate. Applies to VDSL interfaces only.

Values 0 to 255 (in 1/8 of a decibel)
Default 24 (3.0 dB)
Access read-write

dsRsFrame

Specifies the desired final value for downstream Reed-Solomon code word size and interleaving depth. This determines the final downstream Reed-Solomon parameters that are actually used. Applies to VDSL interfaces only.

Values 1: rs-1-1-1
 2: rs-255-239-1
 3: rs-255-239-16
 4: rs-142-126-29
 5: rs-255-239-143
 6: rs-193-177-191
Default 5: rs-255-239-143
Access read-write

dsSymbolRate

Specifies the symbol rates used by the rate adaptive algorithm. Applies to VDSL interfaces only.

Values 1 ds-sym-rate-sel-bit-0
 2: ds-sym-rate-sel-bit-1
 4: ds-sym-rate-sel-bit-2
 8: ds-sym-rate-sel-bit-3
 16: ds-sym-rate-sel-bit-4
 32: ds-sym-rate-sel-bit-5
 64: ds-sym-rate-sel-bit-6
 128: ds-sym-rate-sel-bit-7
Default 128: ds-sym-rate-sel-bit-7
Access read-write

dstAddrHigh

The inclusive upper bound for the destination IP address range. See the filterStatementEntry description for details.

Default 0.0.0.0
Access read-create

dstAddrLow

The inclusive lower bound for the destination IP address range. See the filterStatementEntry description for details.

Default 0.0.0.0
Access read-create

dstPortHigh

The inclusive upper bound for the transport layer destination port range. See the filterStatementEntry description for details.

Values 0 to 65535

Default 0

Access read-create

dstPortLow

The inclusive lower bound for the transport layer destination port range. See the filterStatementEntry description for details.

Values 0 to 65535

Default 0

Access read-create

dTimInterval (DynOmciCpeWlanAdv)

This interval at which Delivery Traffic Indication Messages (DTIM) are generated. A DTIM message notifies a wireless client that a packet waiting for transmission.

Values 1..255

Default 1

Access read-write

dtmfEventsPassingMethod

If zhoneCpeVoipOobDtmfEvents is disabled, dtmf digits are carried along the voice path. If zhoneCpeVoipOobDtmfEvents is enabled, we look at the dtmf-events-passing-method. If zhoneCpeVoipDtmfEventsPassingMethod is rfc4733, dtmf digits are carried in the RTP payload. If zhoneCpeVoipDtmfEventsPassingMethod sip-info, dtmf digits are carried along the signaling path in the INFO messages.

Values 1: rfc4733

2: sipinfo

Default 1: rfc4733

Access read-write

dtmfMode

Indicates whether Dual-Tone Multi-Frequency (DTMF) digits are to be passed as codec voice packets or special separate packets (rfc2833).

Values 1: **inband**
2: **rfc2833**
Default 2: **rfc2833**
Access read-write

duplex

This attribute sets the Ethernet port duplex.

Values 1: **auto**
2: **full**
3: **half**
Default 1: **auto**
Access read-write

E

e3Framing

The status of the E3 framing mode.

Values 1: **e3FrameOther**
2: **e3FrameG832**
3: **e3FrameG751**
Default 2: **e3FrameG832**
Access read-write

echoCancel

This attribute turns on and off echo cancellation.

Values 1: **enabled**, 2: **disabled**
Default 1: **enabled**
Access read-write

egressFilterId

Index of the egress filter spec provisioned for this interface. A value of 0 means there is no egress filter provisioned for this interface. The filter spec must exist before being provisioned for this interface.

Values 0 to 2147483647

Default 0

Access read-write

egressGroupIndex

Index used as partial address of a group of entries in the packetRuleTable, each of which prescribes a filtering condition for the egress interface.

Values 0 to 2147483647

Access read-create

egressGroupTemplate

The packet rule group temple used for specifying a filtering condition for the egress interface.

Values a string up to 60 characters

Default an empty string

Access write-only

ellipsoid

Values 0 to 10

Default 0

Access read-only

email

E-mail address of the customer.

Access read-write

enableAuthTraps

Indicates whether the device is permitted to generate authenticationFailure traps. The value of this object overrides any configuration information; as such, it provides a means to disable all authenticationFailure traps. It is

recommended that this object be stored in non-volatile memory so that it remains constant across re-initializations of the network management system.

Values 1: Enabled
2: Disabled
Default 1: Enabled
Access read-write

encapsType

An instance of this object only exists when the local VCL end-point is also the VCC end-point, and AAL5 is in use. The type of data encapsulation used over the AAL5 SSCS layer. The definitions reference RFC 1483 Multiprotocol Encapsulation over ATM AAL5 and to the ATM Forum LAN Emulation specification.

Values 7: llcEncapsulation
9: other
Default 7: llcEncapsulation
Access read-write.

encoding

This attribute specifies the line coding scheme.

Values 1: b8zs
2: ami
3: hdb3
4: b3zs
Default 3: hdb3
Access read-write.

encrypted

Specifies whether port-id PLOAM messages are encrypted. Must be false if multicast.

Values 1: True
2: False
Default 2: False
Access read-write.

encryptKey (DynOmciCpeWlanSubscriber)

WEP 64-bit encryption key: 5 ASCII characters or 10 hexadecimal digits
 WEP 128-bit encryption key: 13 ASCII characters or 26 hexadecimal digits
 WPA Passphrase: 64 characters.

Values String (1..64)

Access read-write.

endAddress (DynOmciCpeDhcpServer)

Last address in the pool to be assigned by the DHCP server. Default value is 0.0.0.0.

Values IP address

Default 0.0.0.0

Access Read-Write

endPointIndex

The index into the table that contains the sub voice connection end point information. This index is equal to the index defined for the sub voice connection endpoint in the subVoiceConnectionTable.

Values 0 to 2147483647

Default 0

Access init-only

endPointOneIndex

The index that goes into the associated address table for this end point. EndPoint1 should describe the line or premise side of the connection. This table contains any specific protocol information required to define this half of the voice connection. Required for row creation.

Values 1 to 2147483647

Default 1

Access read-write

endPointTwoIndex

The index that goes into the associated address table for this end point. EndPoint2 should describe the central office side of the connection. This table contains any specific protocol information required to define this half of the voice connection. Required for row creation

Values 1 to 2147483647

Default 1

Access read-write

ends

Time in GMT that the lease will expire.

Values 0 to 4294967295

Default 0

Access read-write

entryIndex(DynOmciCpeStaticRoute)

This represents the CPE Static Route Entry Index.

Access read-create

entryIndex(DynOmciCpePortFwd)

This represents the CPE Port Fwd Entry Index.

Values 1 to 2147483647

Access read-create

entryIndex(DynOmciCpeDnsHost)

Address of the cpe-dns-host-list entry profile associated with this CPE.

Values 0..2147483647

Access read-create

epgServerAddress

The IP address of the EPG (Electronic Programming Guide) for a particular video head-end.

Values 4 to 4

Default 0.0.0.0

Access read-write

eqDMeasureCycles

Defines the number of measurement cycles in the Ranging process.

Values Integer

Default 5

Access read-write

expectedPTypePayload

This attribute specifies the RTP payload type for the payload channel in the PSN to TDM direction. The received payload type may be used to detect malformed packets. To disable the check functions, set the value to its default value 0.

Values Integer32 (0..255)

Default 0

Access Read-Write

expectedPTypeSignalling

This attribute specifies the RTP payload type for the signaling channel in the PSN to TDM direction. The received payload type may be used to detect malformed packets. To disable the check functions, set the value to its default value 0.

Values Integer32 (0..255)

Default 0

Access Read-Write

expectedSsrcPayload

This attribute specifies the RTP synchronization source in the PSN to TDM direction. The received SSRC may be used to detect misconnection (stray packets). To disable the check functions, set the value to its default value 0.

Values Integer32 (0..2147483647)

Default 0

Access Read-Write

expectedSsrcSignalling

This attribute specifies the RTP synchronization source for the signaling channel in the PSN to TDM direction. The received SSRC may be used to detect misconnection (stray packets). To disable the check functions, set the value to its default value 0.

Values Integer32 (0..2147483647)

Default 0

Access Read-Write

externalServer

Non-null IP address value that indicates use of the external DHCP server for this subnet instead of an internal IP address.

Values Valid IP Address

Default 0.0.0.0

Access init only

extraUsBwType

Extra Bandwidth = MaxBw - (Assured+FixedUbr+FixedCbr).

By default the value is non-assured. Non-assured has higher priority for getting unused bandwidth than best effort.

Values 1: nonassured

2: besteffort

Default 1: nonassured

Access Read-Write

F

farEndAddr

IP address of the interface on the far end of this interface.

Default 0.0.0.0

Access read-write

farEndEquipCode

The Far End Equipment Identification code that describes the specific piece of equipment. It is sent within the Path Identification Message.

Values 0 to 10

Access read-only

farEndFacilityIdCode

This code identifies a specific Far End DS3 path. It is sent within the Path Identification Message.

Values 0 to 38

Access read-only

farEndFrameIdCode

The Far End Frame Identification code that identifies where the equipment is located within a building at a given location. It is sent within the Path Identification Message.

Values 0 to 10

Access read-only

farEndIP

When the pseudowire service is transported via IP, this attribute specifies the IP or resolved name of the far-end termination point.

Values IPAddress

Default 0.0.0.0

Access read-write

farEndLineIndex

The index value that uniquely identifies the DS-3 interface to which this entry is applicable. The interface identified by a particular value of this index is the same interface and has the same value as the **dsx3LineIndex** object instance.

Values 0 to 2147483647

Default 0

Access read-only

farEndLocationIdCode

The Far End Location Identification code that describes the specific location of the equipment. It is sent within the Path Identification Message.

Values 0 to 11

Access read-only

farEndPoint

When the pseudo-wire service is transported via IP, this attribute specifies the port number of the near-end TCP/UDP service. The default value is 57000+profile index.

Values Unsigned32 (0..65535)

Default 57000

Access read-write

farEndUnitCode

The Far End code that identifies the equipment location within a bay. It is sent within the Path Identification Message.

Values 0 to 6

Access read-only

faultDetectionType

Used to configure VPL fault detection. When fault detection is enabled, an atmVclOperStatusChange trap is used to tell the Network Management System that a VCL has gone down or has come back up.

Values 1: disabled

Fault detection is disabled.

2: oamF5Loopback

The system uses the standard OAM fault management cells as defined in ITU.610 for fault detection. Automatic loopback diagnostic cells are sent to the IAD.

Default 1: disabled

Access read-write

fax

FAX number of the customer.

Access read-write

faxMode

This attribute selects the fax mode.

Values passthrough(1), t38(2)

Default passthrough(1)

Access read-write

fcsAltEnable

Enable or disable Frame Check Sequence (FCS) alternative negotiation. The negotiated FCS values take effect only during Authentication and Network-Layer Protocol phases.

Values 1: enabled
2: disabled
Default 2: disabled
Access read-write

fcsSize

The size of the Frame Check Sequence (FCS) in bits, the local node will attempt to negotiate for use with the remote node.

Values 16
Default 16
Access read-write
Notes Regardless of the value of this object, the local node will comply with any FCS size negotiations initiated by the remote node, per the PPP specification.
Changing this parameter will take effect when the link is next restarted.

fdl

The Facility Data Link (FDL) exchange being used

Values 1: other
2: ansiT1403
The FDL exchange recommended by ANSI.
3: att54016
ESF FDL exchanges
4:fdlNone
Default 4: fdlNone
Access read-write

featuresIndex

This represents the CPE Voip Features index value.

Values Unsigned32 (1..2147483647)
Access read-create

featuresProfileIndex

This attribute points to the associated cpe-voip-feature profile.

Values Unsigned32 (1..2147483647)

Default 1

Access read-create

featureSetOne

First set of features that are enabled for the subscriber. Absence of the feature in the list will disable the feature for the subscriber.

Values 0: hookflash detection

1: onhook signaling

2: always offhook, call is established as soon as incoming call initiation is made.

3: plar

4: Call Wait

5: Call Transfer

6: Conference (3-way local conference bridging)

7: Line side supervision through Reverse Battery

8: Voice Only

9: Hotline

10: Warmline

11: Line side supervision through tone

12: DTMF mode as RFC-2833

13: DTMF mode as inband

14: COD- Enable Call on Disconnect for SIP

Default 0, 1, 4

Access read-write

fecMode

Enables or disables forward error correction.

Values 1: Enabled

2: Disabled

Default 1: Enabled

Access read-write

fecRequest

Indicates whether OLT turns on FEC indication for all Alloc-Id in the Upstream Bandwidth Map.

Values 1: Enabled

2: Disabled

Default 2: Disabled

Access read-write

fileName

Name of the OMCI ME profile file. This attribute is required during addition of ME profile.

Values STRING (SIZE (0..64 characters))

Access read-create

fillPolicy

This attribute defines the payload bit pattern to be applied toward the TDM service if no payload packet is available to play out. The default value 1 specifies that the ONT apply its internal policy, ie vendor-specific.

Values 1:vendorspecific

2: ais

3: allones

4: allzeros

5: repeatprevdata

6: ds1idle

Default 1:vendorspecific

Access read-write

filterBroadcast

If true, discard non-ARP broadcast packets.

Values 1: true

2: false

Default 2: false

Access read-create

firewall (DynOmciCpeSystemCommon)

Enable/disable firewall. Enabling firewall can protect the CPE from unwanted instruction. When firewall is enabled, incoming connections can be selectively allowed through firewall access and port forwarding settings.

Values 1: enabled
2: disabled

Default 1: enabled

Access read-write

firewallAccess (DynOmciCpelpServer)

Lists the protocols allowed on this interface. The firewall option must be enabled before these settings will take effect.

Values 0: http
1: ping
2: snmp
3: snmptrap
4: ssh
5: telnet

Access read-write

firmwareRev

The vendor-specific firmware revision string as advertised by the remote endpoint.

Access read-create

firstNameServer

The IP address of the first/primary name server for this routing domain.

Default 0.0.0.0

Access read-write

fixBitRate

Normally, if the CO and CPE select different line rates at startup, the lower of the two rates will be selected. By using the fix bit rate, the CO can override the default procedure and require startup at the rate specified in **zhoneSdslConfigLineRate**.

Values 1: fix-bit-enable

2: fix-bit-disable

Default 2: fix-bit-disable

Access read-write

Notes This option is available only if **zhoneDslConfigUnitMode** is **co**. This value is ignored if **zhoneDslConfigUnitMode** is **cpe**.

flags

Indicates how the DHCP host acquired its lease.

Values 0: static

The lease is static. Static leases typically should not appear in the lease database.

1: bootp

The lease was allocated using the BOOTP protocol. BOOTP leases are unsupported.

4: abandoned

A client appears to exist (for example, the device responds to a ping command) on the network even though the DHCP server has not allocated the lease address to the client. In this case, the server abandons that lease and tries to reuse it only when it runs out of available addresses.

5: zhoneCPE

The client that was allocated a lease identified itself as a DZS CPE device (such as a Z-Edge).

Default 0: static

Access read-write

flapControl

This object enables or disable Flap Control in the Bridge Filter Table. The value 'disable(2)' turns Flap Control mode off. The values 'enable(3)' or 'fast(4)' turn flap control mode on. 'Enable(3)' applies a hold time of 60 secs before allowing addr/IP to be moved to a new port, while 'fast(4)' applies a hold time of 0 secs. A value of 'default(1)' indicates 'Don't Care' and defaults to a disable or enable set by another entry.

Values 1: default

2: disable

3: enable

4: fast

Default 1: default

Access read-create

floodingGport

This is the downstream only GEM port for broadcast unknown MAC address packets.

Values Integer32 (0..4095)

Default 0

Access read-write

focusX

Values -2147483647 to 2147483647

Default 0

Access read-only

focusY

Values -2147483647 to 2147483647

Default 0

Access read-only

forwarding

The indication of whether this entity is acting as an IP router in respect to the forwarding of datagrams received by, but not addressed to, this entity. IP routers forward datagrams. IP hosts do not (except those source-routed via the host).

Values 1: forwarding

2: notForwarding

Default 1: forwarding

Access read-write

fragmentationThreshold (DynOmciCpeWlanAdv)

The threshold at which the packets are fragmented.

Values 256..2346

Default 2346

Access read-write

frameMode

The transport of data units as specified in the transmission error detection service of Recommendation I.366.1. This setting is inherited by the CIDs attached to the AAL2 VCL.

Values 1: Enabled

2: Disabled

Default 1: Enabled

Access read-write

Notes There is no assurance of data unit delivery, but if delivered, the relative sequence and the bit integrity of data units are assured. This service corresponds to the service provided by the Common Part of AAL type 5, Recommendation I.363.5, except that the option of corrupted data delivery is not available.

frameType

Indicates which type of framing is used.

Values 3: atmllc

4: atmvc

Default 3: atmllc

Access read-write

framerType

This sets the type of framer used. atm-clear-channel: direct cell mapping, atm-dlcc: uses DLCC framing specification, atm-g991-1: uses framing specified in G.991.1

Values 1: atm-clear-channel

2: atm-dlcc

3: atm-g991-1

Default 1: atm-clear-channel

Access read-write

frameSync

This variable enable the user to select a 14 bit frame sync word (FSW) of his/her choice. The possible values are frame sync default which is 0 no change frame sync word (0xffff), or a user selected 14 bit value.

Values 0 to 65535
Default 45
Access read-create

ftpAddress

The FTP server to which the device sends bulk statistics.

Values **Valid IP address**
 If the FTP server is not accessible, bulk statistics are stored on the local flash disk until the server is available.
Access read-write

ftpDirPath

The directory path on the FTP server where the bulk statistics files are sent. The bulk statistics file names use the following format:

Device-IP_timestamp

where timestamp is in the form YYYY.DD.MM.HH.MM in the device local time. For example: 192.168.80.291_2002.11.06.14.37

Access read-write

ftpLogin

The FTP login name used for the FTP transfer of bulk statistics.

Values **Valid FTP login name**
Access read-write

ftpPassword

The FTP password associated with the FTP login name for bulk statistics. The password must be entered using SNMP or DZMS. It is read-only from the CLI.

Values **Valid FTP password**
Access read-write

fullDuplex

DZMS specific value to indicate if the fullDuplex is disabled or not enabled.

Values **1: disabled**
2: enabled
Default **1: disabled**

Access read-write

fullSerialNo

This is a string combines serialNoVendorId and serialNoVendorSpecific.

The serialNoVendorId is a string of 4 ascii characters representing the vendor ID portion of the onu serial number.

The serialNoVendorSpecific is a string of maximum of 10 ascii decimal digits representing the vendor-specific portion of the onu serial number.

Default 0

Access read-write

fullSerialNoHex

This is a string combines serialNoVendorId and serialNoVendorSpecificHex.

The serialNoVendorId is a string of 4 ascii characters representing the vendor ID portion of the onu serial number.

The serialNoVendorSpecificHex is a string of maximum of 9 hexa-decimal digits representing the vendor-specific portion of the onu serial number.

Default 0

Access read-write

fwdDefault

If 1, attempt to forward via VLAN default paths. If 2, do not attempt to forward via VLAN default paths.

Values 1: True

2: False

Default 2: False

Access read-write

fwdEnabled

Specifies whether IP forwarding is enabled on this interface.

Values 1: True

2: False

Default 1: True

Access read-write

fwdMulticast

If 1, attempt to forward via multicast MAC paths. If 2, do not attempt to forward via multicast MAC paths.

Values 1: True

2: False

Default 2: False

Access read-write

fwdType (DynOmciCpePortFwd)

portRange - Range indicates that any traffic on those ports will be sent to the private IP address.

portRemap - Remap indicates that any traffic on those ports will be sent to the private IP address at the private port.

dmz - When DMZ is chosen it is the only rule allowed on that interface. A DMZ rule is effectively the same as a rule with all ports included. Range rules are more secure than setting a DMZ rule, because Range rules allow ports or groups of ports to be opened up.

Values 1: dmz

2: portrange

3: portremap

Default 1: dmz

Access read-write

fwdUnicast

If 1, attempt to forward via unicast MAC paths. If 2, do not attempt to forward via unicast MAC paths.

Values 1: True

2: False

Default 2: False

Access read-write

G

gammaValue

This indicates the 'gamma' value used to specify the number of consecutive valid ICP cells to be detected before moving to the IMA Sync state from the IMA PreSync state.

Values 1 to 5
Default 1
Access read-create

gateway

The default gateway address used for IP host services.

Values IP address
Default 0.0.0.0
Access read-write

gateway (DynOmciCpeStaticRoute)

Next hop IP address.

Values IP address
Default 0.0.0.0
Access read-write

gemBlockSize

Basic unit in bytes used by all ONUs for DBA reports of their queue occupancy (for piggyback and whole-report DBA only).

Values Integer
Default 16
Access read-write

gemPort

This represents the CPE GEM Port Index. This is a mandatory attribute for addition operation.

Values Integer32 (1..2147483647)
Access read-create

genericProfile

This is the name of a Profile that contains OMCI commands which can configure an ONU. It is expected that this Profile contain generic commands that are common to multiple ONUs.

Values A string of 32 characters

Default Empty string

Access read-write

globalFocusX

Values -2147483647 to 2147483647

Default 0

Access read-only

globalFocusY

Values -2147483647 to 2147483647

Default 0

Access read-only

globalMaxClients (DynOmciCpeWlanAdv)

The maximum number of wireless client devices that may be simultaneously connected to the radio. This value should include the sum total of all active SSIDs.

Values 1 to 128

Default 16

Access read-write

globalZoomFactor

Values 0 to 2147483647

Default 0

Access read-only

gponOltId

The ID of the OLT to add a gpon port and bridge/IP to.

The lowerIfIndex and lineGroupId attributes are ignored when the gponOltId, gponSlotId, gponPortId, and gponTrafficIdx are specified.

Access init-only

gponPortId

The ID of the GPON port to create and add a bridge/IP to.

The lowerIfIndex and lineGroupId attributes are ignored when the gponOltId, gponSlotId, gponPortId, and gponTrafficIdx are specified.

Access init-only

gponSlotId

The slot number of the GPON card to add a gpon port and bridge/IP to.

The lowerIfIndex and lineGroupId attributes are ignored when the gponOltId, gponSlotId, gponPortId, and gponTrafficIdx are specified.

Access init-only

gponTrafficIdx

The index of the gpon-traffic-profile to create the gpon port with.

The lowerIfIndex and lineGroupId attributes are ignored when the gponOltId, gponSlotId, gponPortId, and gponTrafficIdx are specified.

Access init-only

gponOnuPassword

Password (10 bytes).

Access read-write

groupIndex

The value which identifies the IMA group (imaGroupIndex) of which this link is a member.

Values 0 to 2147483647

Default 0

Access read-write

groupIndex (bridgelogicals)

Index used as partial address of a group of entries in the bridgeConfigTable, each of which prescribes a filtering condition for the interface.

Default '0'b

Access read-create

groupTemplate

The packet rule group temple used for specifying a filtering condition for the ingress interface.

Values a string up to 60 characters

Default an empty string

Access write-only

gtpTemplateName

The name of the GPON traffic profile template.

Whenever, gponTrafficIdx is given as non-zero, same characters will be used for creation of dynamic GPON port. Whenever gtpTemplateName is set to an existing GTTP template, DZMS server will check for the matching GTP on device. If found, the profileIndex of same GTP will be used. If not found, DZMS server will automatically create GTP on device and use the corresponding profileIndex for dynamic GPON port creation. If both gponTrafficIdx and gtpTemplateName values are present, gtpTemplateName value will be ignored.

If the GPON port given as part of bridge / host creation already exists, no new GTP will be created as GTP is a read-create variable for GponTrafficProfile.

Values a string, up to 60 characters

Default an empty string

Access read-create

guaranteedBw

Bandwidth guaranteed by this alloc-id. This value is specified in kilobits per second. Only multiples of 512 are allowed, and the value may not exceed 2621440 (i.e. 2.5 gigabits per second).

Values Integer

Access read-write

guaranteedUpBw

Bandwidth guaranteed by this alloc-id. This value is specified in kilobits per second. Only multiples of 512 are allowed, and the value may not exceed 2621440 (i.e. 2.5 gigabits per second).

Values 0..1048576

Default 0

Access read-write

guardBitCount

Number of guard bits for upstream transmission.

Values Integer

Default 32

Access read-write

guidedVlanId (DynOmciCpelfVlan, DynOmciCpeConnection)

The Guided VLAN Id for RG Services. A zero value means a guided vlan does not exist. If a guided vlan is used then uni-vlan and uni-slcn values must be zero.

Values 0...4096

Default 0

Access read-create

H

hardwareAddr

Hardware address of the client that matches this host object. If the dhcpHostClientId field is specified then this field is ignored. However, at least one of these fields must be configured per host object.

Default 0:0:0:0:0:0

Access read-write

hardwareAddress

The MAC address of the network interface that was used to acquire the lease.

Access read-write

hardwareRev

The vendor-specific hardware revision string as advertised by the remote endpoint.

Access read-create

headerErrorControl

Header error control on ONU remote device.

Values 1: **onuOff**
 2: **onuRxCorrectionOnly**
 3: **onuTxGenerationOnly**
 4: **onuRxAndTx**

Default 4: **onuRxAndTx**

Access read-write

headerMethod

This convention is a bit map. Each bit represents a SIP method where the Expires header should be included. If a bit has value 1, then the requests corresponding to that SIP method must include an Expires header line. If a bit has value 0, then the Expires header line will not be added.

Combinations of bits can be set when the Expires header line is required in multiple SIP methods.

Values 0: **invite**
 INVITE method.
 1: **register**
 REGISTER Method.

Access read-create

hecRxBypass

Cyclic Redundancy Check bypass for reception.

Values 1: **true**
 2: **false**

Default 2: **false**

Access read-write

hideAccessPoint (DynOmciCpeWlan)

Enable or disable the suppression of the advertising of the access point's SSID. If enabled, clients will need to configure the SSID to associate.

Values 1: enabled

2: disabled

Default 1: enabled

Access read-write

highIfIndex

The index (in the form *shelf-slot-port-subport-interface/atm* or a user-defined string) of the ATM interface for this cross connect. The **high-if-index** is arbitrary, but by convention it indicates the ATM interface with a numerically higher ifIndex value than the other ATM interface identified in the same cross connect.

Notes The **low-if-index** and the **high-if-index** cannot be equal.

high-vpi

The VPI value associated with the interface specified in the **high-if-index**.

holdActive

When set to true, this indicates that the card, when Active, should remain in the Active state regardless of any system events. This effectively disables redundancy for the card group.

Values 1 or 2

Default 2

Access read-create

Notes The holdActive attribute can be simultaneously set for multiple cards in the same redundancy group.

hostName

Host name for this entry. The host name can be either fully-qualified or host name only.

Access read-write

Notes The maximum length for this name is 255 characters.

hostIP

The address used for IP host services, this attribute has default value 0.0.0.0.

Values IPAddress

Default 0.0.0.0

Access read-write

hostIpOption

This attribute selects an IP related option.

Values 1 (dhcp), 2 (static)

Default 1 (dhcp)

Access read-write

huntGroupIndexOne

The hunt group this subscriber is part of. Each subscriber can be part of 3 hunt groups. The hunt group endpoint index is endpt2 index of the subscriber-voice connection which has huntgroup as TRUE.

Values 0 to 2147483647

Access read-write

huntGroupIndexTwo

The hunt group this subscriber is part of. Each subscriber can be part of 3 hunt groups. The hunt group endpoint index is endpt2 index of the subscriber-voice connection which has huntgroup as TRUE.

Values 0 to 2147483647

Access read-write

huntGroupIndexThree

The hunt group this subscriber is part of. Each subscriber can be part of 3 hunt groups. The hunt group endpoint index is endpt2 index of the subscriber-voice connection which has huntgroup as TRUE.

Values 0 to 2147483647

Access read-write

I

id

A unique numeric ID that defines a subscriber. It is also used as the index to the subDataConnectionTable and subVoiceConnectionTable.

Values 0 to 2147483647

Default 0

Access sync-only

identification

Textual description of the unit or card or access device type.

Default CARD

Access init-only

ifAdminStatus

The desired state of the interface. The testing (3) state indicates that no operational packets can be passed. When a managed system initializes, all interfaces start with ifAdminStatus in the down (2) state. As a result of either explicit management action or per configuration information retained by the managed system, ifAdminStatus is then changed to either the up (1) or testing (3) states (or remains in the down (2) state).

Values 1: up

2: down

3: testing

Default 2: down

Access read-write

ifAlias

An alias name for the interface as specified by a network manager, and provides a non-volatile identifier for the interface.

Values 0 to 64

Access read-write

ifConfVccs

The number of VCCs (PVCC,) currently in use at this ATM interface. Includes the number of PVCCs that are configured at the interface.

Values 0 to 65536

Default 0

Access init-only

ifConfVpcs

The number of VPCs (PVPC) currently in use at this ATM interface. Includes the number of PVPCs that are configured at the interface.

At the ATM UNI, the configured number of VPCs can range from 0 to 256 only.

Values 0 to 4096

Default 0

Access init-only

ifCurrMaxVciBits

The maximum number of VCI Bits that may currently be used at this ATM interface. The value is the minimum of atmInterfaceMaxActiveVciBits, and the atmInterfaceMaxActiveVciBits of the interface's UNI/NNI peer. If the interface does not negotiate with its peer to determine the number of VCI Bits that can be used on the interface, then the value of this object must equal atmInterfaceMaxActiveVciBits.

Values 0 to 16

Default 0

Access init-only

ifCurrMaxVpiBits

The maximum number of VPI Bits that may currently be used at this ATM interface. The value is the minimum of atmInterfaceMaxActiveVpiBits, and the atmInterfaceMaxActiveVpiBits of the interface's UNI/NNI peer. If the interface does not negotiate with its peer to determine the number of VPI Bits that can be used on the interface, then the value of this object must equal atmInterfaceMaxActiveVpiBits.

Values 0 to 12

Default 0

Access init-only

ifDescr

A textual string containing information about the interface. This string should include the name of the manufacturer, the product name and the version of the interface hardware/software.

Access sync-only

ifIlmiVci

The VCI value of the VCC supporting the ILMI at this ATM interface. If the values of atmInterfaceIlmiVpi and atmInterfaceIlmiVci are both equal to zero, ILMI is not supported at this ATM interface.

Values 0 to 2147483647

Default 16

Access init-only

ifIlmiVpi

The VPI value of the VCC supporting the ILMI at this ATM interface. If the values of atmInterfaceIlmiVpi and atmInterfaceIlmiVci are both equal to zero, ILMI is not supported at this ATM interface.

Values 0 to 2147483647

Default 0

Access init-only

ifIndex

The interface number of the port or port/subport that uniquely identifies the interface across the system.

Values 0 to 2147483647

Default 0

Access init-only

ifIndex (for DynOmciCpeLldpRem object)

This represents the CPE Ip index value

Access read-create

ifIndex (for PweStdConfig object)

Represents the ifIndex of parent if-translate.

Values 0 .. 2147483647

Access read-only

ifIndex (for DynOmciCpeEthSubscriber, DynOmciCpePweSubscriber, DynOmciCpeRfSubscriber, DynOmciCpeVoipSubscriber, DynOmciCpeIp, DynOmciCpeConnection, DynOmciCpeWlanSubscriber, DynOmciCpeSystem, DynOmciCpeIfVlan object)

IfIndex of parent ONU physical.

Values Unsigned32 (1..2147483647)

Access read-only

ifMaxActiveVciBits

The maximum number of active VCI bits configured for use at this ATM interface.

Values 0 to 16

Default 0

Access init-only

ifMaxActiveVpiBits

The maximum number of active VPI bits configured for use at the ATM interface. At the ATM UNI, the maximum number of active VPI bits configured for use ranges from 0 to 8 only.

Values 0 to 12

Default 0

Access init-only

ifMaxVccs

The maximum number of VCCs supported at this ATM interface.

Values 0 to 65536

Default 0

Access init-only

ifMaxVpcs

The maximum number of VPCs supported at this ATM interface. At the ATM UNI, the maximum number of VPCs ranges from 0 to 256 only.

Values 0 to 4096

Default 0

Access init-only

ifMtu

The size of the largest packet which can be sent/received on the interface, specified in octets. For interfaces that are used for transmitting network datagrams, this is the size of the largest network datagram that can be sent on the interface.

Values 0 to 2147483647

Default 0

Access init-only

ifMyNeighborIfName

Textual name of the interface on the neighbor system on the far end of this interface, and to which this interface connects. If the neighbor system is manageable through SNMP and supports the object ifName, the value of this object must be identical with that of ifName for the ifEntry of the lowest level physical interface for this port. If this interface does not have a textual name, the value of this object is a zero length string. Note that the value of this object may be obtained in different ways, such as, by manual configuration, or through ILMI interaction with the neighbor system.

Access read-write

ifMyNeighborIpAddr

The IP address of the neighbor system connected to the far end of this interface, to which a Network Management Station can send SNMP messages (as IP datagrams sent to UDP port 161) in order to access network management information concerning the operation of that system. The value of this object can be obtained in different ways, such as by manual configuration or through ILMI interaction with the neighbor system.

Default 0.0.0.0

Access read-write

ifNumber

The interface of the static route. This is for directly connected destinations (i.e. use this instead of staticRouteNextHop). Unnumbered point-to-point interfaces require this value to be valid. In general, this value must be valid if staticRouteNextHop is not valid. If staticRouteNextHop is valid, then this value must represent the interface named by the nexthop address. This index comes from ifIndex of RFC2233.

Values 0 to 2147483647

Default 0

Access init-only

ifOperStatus

The operational state of the interface. If ifAdminStatus is down (2) then ifOperStatus should be down (2). If ifAdminStatus is changed to up(1) then ifOperStatus should change to up (1) if the interface is ready to transmit and receive network traffic. It should change to dormant (5) if the interface is waiting for external actions (such as a serial line waiting for an incoming connection). It should remain in the down (2) state if and only if there is a fault that prevents it from going to the up (1) state; it should remain in the notPresent (6) state if the interface has missing (typically, hardware) components.

Values 1: up

2: down

3: testing

No operational packets can be passed.

4: unknown

5: dormant

6: notPresent

7: lowerLayerDown

Default 2: down

Access read-only

ifSpeed

An estimate of the interface's current bandwidth in bits per second. For interfaces which do not vary in bandwidth or for those where no accurate estimation can be made, this object should contain the nominal bandwidth. If the bandwidth of the interface is greater than the maximum value reportable by this object then this object should report its maximum value (4,294,967,295) and ifHighSpeed must be used to report the interface's speed.

For a sub-layer which has no concept of bandwidth, this object should be zero.

Values 0 to 4294967295

Default 0

Access sync-only

ifType

The type of interface.

Values 0 to 2147483647

Default 0

Access init-only

Notes Additional values for ifType are assigned by the Internet Assigned Numbers Authority (IANA), through updating the syntax of the IANAifType textual convention.

ifTypeExtension

This field is used to specify the Zhone-proprietary interface type when necessary. For example, if the interface is a line-group, iftype is set to other and ifTypeExtension is set to other. By default ifTypeExtension is set to none.

Values 0 to 2147483647

Default 0

Access init-only

ifTypePortId

The ifType port identifier.

Values 0 to 2147483647

Default 0

Access init-only

igmpQueryInterval

How often (in seconds) we as a bridge inject IGMP query values into the stream.

Values 0 to 2147483647

Access read-create

igCrv

The Call Reference Value (CRV) for this connection.

Values 1 to 2147483647

Default 1

Access read-create

igmpFunction

This attribute enables an IGMP function - transparent IGMP snooping only, snooping with proxy reporting (SPR) and IGMP proxy. The function must be consistent with the capabilities specified by the other IGMP configuration attributes.

Values 1: transparentsnoothing

2: snoopwithproxy

3: proxy

Default 1: transparentsnoothing

Access read-write

igmpVersion

This attribute specifies the version of the IGMP to be supported.

Values 1: version1

2: version2

3: version3

Default 2: version2

Access read-write

igName

The Interface Group name. It is a system-wide unique symbolic name that is used for call routing.

Values STRING (SIZE(0.. 32 characters))

Access read-create

imaGroupIfIndex

The IMA group ifIndex.

Values 0 to 2147483647
Default 3
Access init-only

imaGroupIndex

A unique value for the IMA Group.

Values 0 to 2147483647
Default 0
Access init-only

imaGroupLgldx

immediateLeave

This attribute enables or disables the immediate leave function.

Values 1:enabled, 2:disabled
Default 2:disabled
Access read-write

impedance

Specifies the terminating impedance of voice analog interfaces. other - none of the following; arbitrary, within limits, for special cases. ohms600Real - 600 Ohms. ohms600Complex - 600 Ohms + 2.16uF ohms900Complex - 900 Ohms + 2.16uF. ohmsComplex1 - complex 1; 220 Ohms + 820 Ohms || 115nF. This impedance is primarily used in Germany. ohmsComplex2 - complex 2; 270 Ohms + 750 Ohms || 150nF. This impedance is primarily used in United Kingdom and Sweden.

Values 1: other
2: ohms600Real
3: ohms600Complex
4: ohms900Complex
5: ohmsComplex1
6: ohmsComplex2
Default 1: other
Access read-write

impedance (DynOmciCpeVoipSubscriber)

This attribute specifies the impedance for the POTS UNI in ohm units.

Values 1: ohm600
 2: ohm900
 3: complex1
 4: complex2
 5: complex3
Default 1: ohm600
Access read-write

imputedGroupBw

Expressed in bytes/sec, the imputed group bandwidth is used to decide whether or not to honor a join request in the presence of a max multicast bandwidth limit. The default value 0 effectively allows this table entry to avoid max bandwidth limitations.

Values Integer32 (0..2147483647)
Default 0
Access read-write

includeChildren

Whether to include the children of an SNMP OID instance collected as part of the bulk statistics feature.

Values true
 false
Default false
Access read-write

index

The channel identifier.

Values 1 to 7
 Reserved for I.363.2 layer management.
 8
 Some CPE manufactures use this CID for CCS and/or Emulated Loop Control protocol.
 10 to 15
 For future revisions of STR-VMOA-LES-01-00.

16 to 223

Bearer channels and ISDN D-channels.

224 to 255

Reserved for vendor-specific usage.

Default 16

Minimum 0

Maximum 223

Access init-only

Notes The maximum number of connections allowed per ATM VCL is 255. Note that there are restrictions in the use of CID and in order to be LES-compliant.

inFlowType

The port's type of input flow control.

Values 1: none

Indicates no flow control at this level.

2: ctsRts

Indicates use of the ctsRts hardware signal.

3: dsrDtr

Indicates use of the dsrDtr hardware signal.

Default 1: none

Access read-write

informInterval (DynOmciCpeSystemCommon)

Periodic interval (in seconds) at which Inform messages will be generated.

Values 0..2147483647

Default 300

Access read-write

ingressFilterId

The index of the ingress filter spec provisioned for this interface. A value of 0 means that there is no ingress filter provisioned for this interface. The filter spec must exist first in the filterSpecTable (comIpFilter.mib) before being provisioned for this interface.

Values 0 to 2147483647

Default 0

Access read-create

instanceKey

A unique identifier for an object and is equal to the Handle_Id of the object. It can be used as a filter_condition attribute (Filter_Condition=instanceKey=Value of Handle_Id) in list operations to retrieve to a particular object.

Note that Handle_Id cannot be used as a Filter_Condition attribute in list operations.

initialMRU

The initial Maximum Receive Unit (MRU) that the local PPP entity will advertise to the remote entity.

Values 128 to 65535

Default 1500

Access read-write

Notes If set to 0 the local PPP entity will not advertise any MRU to the remote entity and the default MRU will be assumed.

Changing this object will have effect when the link is next restarted.

inSpeed

The port's input speed in bits per second. Note that non-standard values, such as 9612, are probably not allowed on most implementations.

Values 0 to 4294967295

Default 0

Access read-write

interDigitTimeOut

This parameter indicates the duration, in seconds, that the device will wait after each digit is entered before assuming the user has finished entering digits.

Values 0 to 2147483647 (in seconds)

Default 10

Access read-create

interDigitDuration

Outgoing DTMF inter-digit duration in milliseconds. The default value of this object is 100 milliseconds.

Values 50 to 500
Measured in milliseconds.

Default 100

Access read-write

interfaceIndex

This field contains the InterfaceIndex of the physical line to be tested. If no line is currently being tested this value is 0. The ability of a physical line type to support metallic test may vary depending on the line cards installed and the external test equipment. This field may not be modified when an mtac test is in progress as indicated by a non-zero value in this field and the test mode set to one of mtacModeLookIn, mtacModeLookOut or mtacModeBridge

Values 0 to 2147483647

Default 0

Access read-write

internalMe (DynOmciCpeConnection, DynOmciCpeEthSubscriber, DynOmciCpelp, DynOmciCpePweSubscriber, DynOmciCpeRfSubscriber, DynOmciCpeVoipSubscriber)

Setting this field to zhone-xxxx will enable the DZMS to automatically set the internal ME for ONU if not set and continue with the actual CPE connection creation. If ONU has already been set with an internal ME, DZMS will ignore this value during CPE connection add.

Values String (0..35)

Access read-write

invite

For user agent clients, this value is inserted into the Expires header. For proxy servers, if a received request contained an Expires header, then the value in the header takes precedence.

Values 0 to 4294967295 (in seconds)

Default 0

Access read-create

ipAddrEnable

Enable or disable IP Address Negotiation.

Values 1: enabled
2: disabled
Default 2: disabled
Access read-write

IPAddr1

The fixed IP address that should be assigned to this client.

Default 0.0.0.0
Access read-write

IPAddr2

The fixed IP address that should be assigned to this client.

Default 0.0.0.0
Access read-write

IPAddr3

The fixed IP address that should be assigned to this client.

Default 0.0.0.0
Access read-write

IPAddr4

The fixed IP address that should be assigned to this client.

Default 0.0.0.0
Access read-write

ipAddress

The IP address of this entry.

Default 0.0.0.0
Access init-only
Notes 127.1 (127.0.0.1) is a valid IP address

ipAddrDynamic

This is the mode by which the IP address is discovered by the system.

Values **1: static**
IP address is statically configured.

2: ppp
Use PPP to get IP address.

3: dhcpclient
IP address discovered from DHCPCLIENT

4: unnumbered
This is an unnumbered interface

Default **1: static**

Access read-create

IpDesc

A descriptive name for the interface. This name is not used by the system, it is only used for reference by an administrator.

Values A string of 32 characters or less.

Default null

Access read-write

ipEnabled

Values 0 to 2147483647

Default 0

Access read-only

ipFilterTransactTimeout

The filter inconsistency timeout in seconds. A filter spec is considered to be in an inconsistent state when the value of the objects fltSpecVersion1 and fltSpecVersion2 are not equal. This timeout indicates the minimum number of seconds a filter may be in an inconsistent state before the filter specification becomes invalid and the default action for a filter is used as the filter. Provided fltGlobalTimeout is long enough, it should ensure that both an old modification is permanently stalled (ensuring exclusive access) as well as enough time to repair a filter.

Values 1 to 2147483647

Default 5

Access read-write

ipIfIndex (Unnumbered IP)

This value specifies where to borrow the local IP address of the unnumbered IP link.

Values 0 to 2147483647

Default 0

Access read-write

ipIfindex

The index value of the **ip-interface-record** associated with this PPP authentication record.

Values 0 to 2147483647

Default 0

Access read-write

Notes If this entry is to be used for incoming authentication, this object must be set to a non-zero value.

ipIndex (Unnumbered IP)

This value is only used to identify this particular entry in the table, it is not externally visible.

Values 0 to 255

Default 0

Access read-write

ipOrMacAddr

If the **addrType** is set to **1 (macAddr)** this represents a MAC address.

If the **addrType** is set to **2 (ipAddr)** this represents an IP address.

Values 0 to 16

Default 0:0:0:0:0:0

Access init-only

ipProtocol

The IP protocol value that is to be matched.

Values 1: any

Any protocol type is a match (wildcard)

- 2: ip**
Raw IP packet
- 3: tcp**
TCP packet
- 4: udp**
UDP packet
- 5: icmp**
ICMP packet

Default 1: any
Access read-create

ipServerProfile (DynOmciCpelfVlan)

The address of the cpe-ip-server profile associated with this host IP address.
The value 0 indicates no associated cpe-ip-server profile.

Values 0..2147483647
Default 0
Access read-write

ipTos (VoipServer)

This parameter indicates the tos value that is set in the ip header for voice of ip traffic.

Values 0 to 2147483647
Default 0
Access read-create

ipv6Address

This attribute specifies the IPV6 address. The default value is ::/128.
Type: IPV6Address.

Values 2-39
Default ::
Access read-write

ipv6DhcpServer

Enables or disables the IPv6 DHCP server on the LAN interface.

Values enabled(1)
disabled(2)
Default enabled(1)
Access read-write

ipv6DhcpMode

Enables or disables the IPv6 DHCP server on the LAN interface.

Values stateful(1),
stateless(2)
Default stateful(1),
Access read-write

ipv6DnsType

This attribute specifies the DNS mode for this IPv6 interface.

Values default(1),
static(2),
proxy(3)
Default default(1)
Access read-write

ipv6EndAddress

This parameter is for assigning IPv6 address by state-full DHCPV6 server. The value indicates the end of the interface ID pool to be assigned by the DHCPV6 server.

Type: IPV6Address

Values 2-39
Default ::254
Access read-write

ipv6Gateway

This attribute specifies the default gateway IPv6 address. The default value is ::128.

Values 2-39
Default ::
Access read-write

ipv6HostIpOption

This attribute represents the method used to assign an ipv6 address on the WAN-side of the CPE.

Values **unconfigured(2)**
autoconfig(3),
dhcp(4),
prefixDelegation(5),
static(6),
eui64(7),
unnumbered(8),
unnumberedDHCPPD(9)

Default **unconfigured(2),**

Access read-write

ipv6LeaseTime

Specifies the lease time in seconds of client assigned addresses. A value of -1 indicates an infinite lease.

Values **-1, 2147483647**

Default **86400**

Access read-write

ipv6PrefixLength

This attribute specifies the IPv6 prefix length.

Values **1- 128**

Default **64**

Access read-write

ipv6PrimaryDns

This attribute specifies the IPV6 primary DNS IP address. The default value is ::/128

Values **2-39**

Default **::**

Access read-write

ipv6RouterAdvert

Enable or disable Router Advertisement service, where router announces periodically its IPv6 address.

Values enabled(1),
disabled(2)
Default enabled(1),
Access read-write

ipv6SecondaryDns

This attribute specifies the IPv6 secondary DNS IP address. The default value is ::/128.

Values 2-39
Default ::
Access read-write

ipv6StartAddress

This parameter is for assigning IPv6 address by state-full DHCPV6 server. The value indicates the beginning of the interface ID pool to be assigned by the DHCPV6 server.

Type: IPV6Address

Values 2-39
Default ::2
Access read-write

isActive

Reflects the current state of a redundant card. Read-only attribute. Should not be modified.

Values 1: unavailable
2: inactive
3: standby
4: active

isAssigned

Whether the port is in use.

Values 1 or 2
Default 2
Access read-write

isdnChannelId

The ISDN B- or D-channel that is used when subVoiceV52UserType is not set to pots.

Values 0 to 32
Default 0
Access read-create

isdnDsCPathId

If the user port is configured for ISDN, this parameter is equal to the associated C-path ID that carries Ds-type data. A value of 0 (zero) indicates that the user port is not configured to carry Ds-type data.

Values 0 to 48, inclusive
Default 0
Access read-write

isdnEnvFuncStartAddr

The start address for ISDN users. When ISDN users are added, they must have an address greater or equal to this one. This value must match the value on the switch.

Values 0 to 8175, inclusive
Default 0
Access read-write

isdnFCPathId`

If the user port is configured for ISDN, this parameter is equal to the associated C-path ID that carries f-type data. A value of 0 (zero) indicates that the user port is not configured to carry f-type data.

Values 0 to 48, inclusive.
Access read-write

isdnPCPathId

If the user port is configured for ISDN, this parameter is equal to the associated C-path ID that carries p-type data. A value of 0 (zero) indicates that the user port is not configured to carry p-type data.

Values 0 to 48, inclusive.

Access read-write

isdnType

The ISDN channel type for this connection.

Values 1: isdnDChannel

2: isdnBChannel

Default 1: isdnDChannel

Access read-create

isLineUsed

Whether the line is used.

Values 1 or 2

Default 2

Access read-write

isolateClients (DynOmciCpeWlan)

Isolate clients within the wireless network from communicating directly with each other.

Values 1: enabled

2: disabled

Default 2: disabled

Access read-write

isRedundant

Specifies whether a card is running in redundant mode or not. Should not be modified. Read-only attribute.

Values 1: True, in redundant mode
2: False: non redundant

isUni

J

jitterBufMax

This attribute specifies the desired maximum depth of the playout buffer in the PSN to TDM direction. The value is expressed as a multiple of the 125 micro second frame rate. The value 0 selects the ONT's internal policy.

Values Integer32 (0..65535)

Default 0

Access read-write

jitterBufDesired

This attribute specifies the desired nominal fill depth of the playout buffer in the PSN to TDM direction. The value is expressed as a multiple of the 125 micro second frame rate. The value 0 selects the ONT's internal policy.

Values Integer32 (0..65535)

Default 0

Access read-write

K

keyExchange

Specifies whether Single Key exchange should be performed during ONU activation process.

Values 1: Enabled

2: Disabled

Default 2: Disabled

Access read-write

L

languageVersion

The language version of the filter. The language version further details the meaning and use of the objects in filterStatmentTable. The definitions of the filter languages is beyond the scope of this description.

Values 0 to 4294967295

Default 0

Access read-create

IbitRxPolicy

This attribute defines the action toward the TDM interface when far end TDM failure is indicated on packets received from the PSN.

Values 1:ais, 2:repeatlastpkt, 3: idlechannel

Default 1:ais

Access read-write

IcdAlpha

Number of cells in delineation error before the LCD alarm is raised.

Values 1 to 15

Default 1

Access read-write

IcdDelta

Number of cells received NOT in delineation error required to clear LCD Alarm.

Values 1 to 15

Default 2

Access read-write

IcdLimit

Loss of cell delineation limit.

Values 4 to 15

Default 8

Access read-write

leadSignalingType

Specifies the E&M Lead Signaling type.

Values 1: typeI
2: typeII
3: typeIII
4: typeIV
5: typeV

Default 1: typeI

Access read-write

learnIp

If 1, snoop unicast packets to build ARP table. If 2, do not snoop unicast packets to build the ARP table.

Values 1 to 2

Default 2

Access read-write

learnMulticast

If true, snoop multicast packets for IGMP joins/leaves.

Values 1: true
2: false

Default 2: false

Access read-create

learnUnicast

If 1, snoop unicast packets to build MAC paths. If 2, do not snoop unicast packets to build the MAC paths.

Values 1 to 2

Default 2

Access read-write

leaseTime

The global default time in seconds that will be assigned to a lease if the client requesting the lease does not ask for a specific expiration time. Default value is 43200 seconds.

Values 0 to 2147483647

Default 43200

Access read-write

leaseTime (DynOmciCpeDhcpServer)

Lease time in seconds of client assigned addresses. A value of -1 indicates an infinite lease.

Values -1 to 2147483647

Default -1

Access read-write

lgld

The ID number of the line group containing the physical line associated with this interface. There may be a single line or a line pair in a redundancy group.

Values 0 to 2147483647

Default 0

Access read-write

Notes The value 0 is only allowed when the IP interface has not yet been bound to a physical port.

lgid (PPPoA)

The unique identifier of the line group which uses this authentication data.

Values 0 to 2147483647

Default 0

Access read-write

lineCoding

The type of Zero Code Suppression used on this interface, which affects a number of its characteristics.

Values 1: jbz

Jammed Bit Zero Suppression, (AT&T) At least one pulse every 8-bit period is literally implemented by forcing a pulse in bit 8 of each channel. Only seven bits per channel, or 1.344 Mbps, is available for data.

2: b8zs

Bipolar 8 Zero Substitution. The use of a specified pattern of normal bits and bipolar violations which are used to replace a sequence of eight zero bits. B8ZS is used in the European hierarchy at the T-1 rate.

3: hdb3

High Density Bipolar order 3.

4: zbtsi

Zero Byte Time Slot Interchange.

5: ami

Alternate Mark Inversion. No zero code suppression is used, and the line encoding does not solve the problem directly. In this application, the higher layer must provide data which meets or exceeds the pulse density requirements, such as inverting HDLC data.

6: other

other

7: b6zs

Bipolar 6 Zero Substitution. A specified pattern of normal bits and bipolar violations which are used to replace a sequence of six zero bits. B6ZS is used in the North American hierarchy at the T-2 rate.

Default 3: High Density Bipolar order 3

Access read-write

lineCoding (SonetOC3Physical)

This variable describes the line coding for this interface. The B3ZS and CMI are used for electrical SONET/SDH signals (STS-1 and STS-3). The Non-Return to Zero (NRZ) and the Return to Zero are used for optical SONET/SDH signals.

Sechtor 100 only supports 'sonetMediumNRZ'.

Values 1: sonetMediumOther

2: sonetMediumB3ZS

3: sonetMediumCMI

4: sonetMediumNRZ

5: sonetMediumRZ

Access Read-Write

lineRedStatusFunction

Current redundancy function as returned from LineRR.

Values	inactive
	registered but neither active nor standby
	standby
	this is the protection line in a pair
	active
	either standalone or the working line in a protection pair
	unavailable
	not registered with LineRR
Default	N/A
Access	Read-only

lineGroupId

Line Group Id of the physical that the IP Logical is being created on. This will be 0 for floating IP objects.

Values	0 to 2147483647
Default	0
Access	Init Only

lineGroupIndex

The line group entry id. This is a unique number that is obtained by the nextLineGroupId field. The range for lgId is 1..2147483647 and the range for nextLineGroupId is 0..2147483647 where the value 0 indicates that no unassigned entries are available.

Values	0 to 2147483647
Default	0
Access	init-only

lineGroupId (bridgelogicals)

Line Group ID that uniquely identifies the interface on the device.

lineGroupKey

Line group key of the physical that the IP Logical is being created on. This will be 0 for floating IP objects.

Values 0 to 2147483647

Access Init Only

lineLength

The length of the DS-1 or DS-3 WAN in feet. This object provides information for line build out circuitry.

Values 1: dsx0

The Line Build Out (LBO) is 0 feet.

2: dsx133

133 foot LBO

3: dsx266

266 foot LBO

4: dsx399

399 foot LBO

5: dsx533

533 foot LBO

6: dsx655

655 foot LBO

Default 1: dsx0

Access read-write

Notes This field is not used for E-1 platforms.

lineLength (DynOmciCpePweSubscriber)

This attribute specifies the length of the twisted pair cable from a DS1 physical UNI to the DSX-1 cross-connect point.

Values Unsigned32 (0..65535)

Default 0

Access read-write

linePower (IsdnBriPhysical)

State of the ISDN line power.

Values 1: off

No power on line.

2: sealing

Minimum current on line to preclude corrosion.

3: powering

Full current on line.

Default 1: off

Access read-write

lineRate (ShdslConfigProfile)

The line rate selection.

Values

- 1: line-rate-72kbps
- 2: line-rate-80kbps
- 3: line-rate-136kbps
- 4: line-rate-144kbps
- 5: line-rate-200kbps
- 6: line-rate-208kbps
- 7: line-rate-264kbps
- 8: line-rate-272kbps
- 9: line-rate-328kbps
- 10: line-rate-336kbps
- 11: line-rate-392kbps
- 12: line-rate-400kbps
- 13: line-rate-456kbps
- 14: line-rate-464kbps
- 15: line-rate-520kbps
- 16: line-rate-528kbps
- 17: line-rate-584kbps
- 18: line-rate-592kbps
- 19: line-rate-648kbps
- 20: line-rate-656kbps
- 21: line-rate-712kbps
- 22: line-rate-720kbps
- 23: line-rate-776kbps
- 24: line-rate-784kbps
- 25: line-rate-840kbps
- 26: line-rate-848kbps
- 27: line-rate-904kbps
- 28: line-rate-912kbps
- 29: line-rate-968kbps
- 30: line-rate-976kbps
- 31: line-rate-1032kbps
- 32: line-rate-1040kbps
- 33: line-rate-1096kbps
- 34: line-rate-1104kbps

35: line-rate-1160kbps
36: line-rate-1168kbps
37: line-rate-1224kbps
38: line-rate-1232kbps
39: line-rate-1288kbps
40: line-rate-1296kbps
41: line-rate-1352kbps
42: line-rate-1360kbps
43: line-rate-1416kbps
44: line-rate-1424kbps
45: line-rate-1480kbps
46: line-rate-1488kbps
47: line-rate-1544kbps
48: line-rate-1552kbps
49: line-rate-1608kbps
50: line-rate-1616kbps
51: line-rate-1672kbps
52: line-rate-1680kbps
53: line-rate-1736kbps
54: line-rate-1744kbps
55: line-rate-1800kbps
56: line-rate-1808kbps
57: line-rate-1864kbps
58: line-rate-1872kbps
59: line-rate-1928kbps
60: line-rate-1936kbps
61: line-rate-1992kbps
62: line-rate-2000kbps
63: line-rate-2056kbps
64: line-rate-2064kbps
65: line-rate-2120kbps
66: line-rate-2128kbps
67: line-rate-2184kbps
68: line-rate-2192kbps

69: line-rate-2248kbps
70: line-rate-2256kbps
71: line-rate-2312kbps
72: line-rate-2320kbps
73: line-rate-2368kbps
74: line-rate-192kbps
75: line-rate-384kbps
76: line-rate-416kbps
77: line-rate-4w-384kbps
78: line-rate-4w-512kbps
79: line-rate-4w-640kbps
80: line-rate-4w-768kbps
81: line-rate-4w-896kbps
82: line-rate-4w-1024kbps
83: line-rate-4w-1152kbps
84: line-rate-4w-1280kbps
85: line-rate-4w-1408kbps
86: line-rate-4w-1536kbps
87: line-rate-4w-1664kbps
88: line-rate-4w-1792kbps
89: line-rate-4w-1920kbps
90: line-rate-4w-2048kbps
91: line-rate-4w-2176kbps
92: line-rate-4w-2304kbps
93: line-rate-4w-2432kbps
94: line-rate-4w-2560kbps
95: line-rate-4w-2688kbps
96: line-rate-4w-2816kbps
97: line-rate-4w-2944kbps
98: line-rate-4w-3072kbps
99: line-rate-4w-3200kbps
100: line-rate-4w-3328kbps
101: line-rate-4w-3456kbps
102: line-rate-4w-3584kbps

	103: line-rate-4w-3712kbps
	104: line-rate-4w-3840kbps
	105: line-rate-4w-3968kbps
	106: line-rate-4w-4096kbps
	107: line-rate-4w-4224kbps
	108: line-rate-4w-4352kbps
	109: line-rate-4w-4480kbps
	110: line-rate-4w-4608kbps
Default	73: line-rate-2368kbps (2-wire SHDSL cards)
	110: line-rate-4w-4608bps (4-wire SHDSL cards)
Access	read-write

lineRXTD

The line side receiving ATM Traffic Descriptor template handle id.

lineStatusAlarm (DynOmciCpeEthSubscriber, DynOmciCpeRfSubscriber)

This field is for enabling/disabling alarm reporting from this CPE UNI.
Type is Integer32.

Values	1: enabled
	2: disabled
Default	2: disabled
Access	read-write

lineStatusAlarm (DynOmciCpePweSubscriber)

This field is for enabling/disabling alarm reporting from this CPE UNI.
Type is Integer32.

Values	1: enabled
	2: disabled
Default	1: enabled
Access	read-write

lineStatusTrapChangeEnable

Indicates whether DSL line status change traps should be generated for this interface.

For ADSL, when auto is set, a trap is generated on a line status change which is not due to a dying gasp.

Values 1: enabled

2: disabled

3: auto

Default 1: enabled

Access read-write

lineTXTD

The line side transmitting ATM Traffic Descriptor template handle id.

lineTrapEnable

Whether dsx3LineStatusChange traps should be generated for this interface.

Values 1: enabled

2: enabled

Default 1: enabled

Access read-write

lineType

The type of DS3 C-bit or E3 application on this interface. The type of interface affects the interpretation of the usage and error statistics. The rate of DS3 is 44.736 Mbps and E3 is 34.368 Mbps.

Values 1: dsx3other

2: dsx3M23

3: dsx3SYNTRAN

4: dsx3CbitParity

5: dsx3ClearChannel

6: e3other

7: e3Framed

8: e3Plcp

Default 1: dsx3other

Access read-write

Notes C-bits are not used except for sending/receiving AIS.

lineType (for DynOmciCpePwe)

This attribute specifies the line type (T1, E1 or other) used.

Values 1: ds1
2: e1
3: other
Default 2: e1
Access read-write

lineType (PotsPhysical)

The type of DS3 C-bit or E3 application on this interface. The type of interface affects the interpretation of the usage and error statistics. The rate of DS3 is 44.736 Mbps and E3 is 34.368 Mbps.

Values 1: fxs
2: fxo
3: em
Default 1: fxs
Access read-only

lineVci

The line side ATM VCLs VCI

lineVpi

The line side ATM VCLs VPI.

linkInServiceTimer

Specifies the length of time that a V5.2 link must be in a framed state before the interface declares that the link is in service. This value should be less than the equivalent value that is provisioned in the LE.

Values 50 to 10000, inclusive.
Default 200
Access read-write

linkOutOfServiceTimer

Specifies the length of time that a V5.2 link must be in a failed state before the interface declares that the link has failed. This value should always exceed the equivalent value that is provisioned in the LE.

Values 50..10000

Default 2500

Access read-write

listIndex(DynOmciCpeStaticRouteList)

Address of the static-route-list profile associated with this CPE.

Values 0..2147483647

Default 0

Access read-write

listIndex(DynOmciCpePortFwdList, DynOmciCpePortFwd)

This represents the CPE Port Fwd List Index value.

Values 1..2147483647

Access read-create

listIndex (DynOmciCpeDnsHostList, DynOmciCpeDnsHost)

Address of the cpe-dns-host-list profile associated with this CPE.

Values 0..2147483647

Default 0

Access read-create

listType

Defines the following video stream types:

normal - join and leave when desired. Used for video.

always-on - always joined. Used for EBS, not video.

periodic - will join and leave after task complete. Not for video. Used to download the TV guide.

Values Integer
1: normal
2: always-on
3: periodic
Default 1
Access read-create

lldpMedList

A list of cpe-lldp-med-policy profiles.

Values 0-2147483647
Default 0
Access read-create

loamAlpha

Number of loss-of-OAM occurrences to trigger alarm.

Values Integer
Default 3
Access read-write

localAddr (PAT)

The IP Address of a host machine on the private subnet that we want to make active through the usage of reverse port forwarding feature.

Access read-write

localInterfaceId

The interface ID of the IG. Must be unique across the system. This value must match the value on the switch.

Values 1 to 16777215, inclusive.
Default 100
Access read-write

localPort (PAT)

The internal port number to use on the private host that is being forwarded to.

Values 1 to 49151

Access read-write

localProvVariant

The prov(isioning) variant describes a type of provisioning. This value must match the value on the switch.

Values 0 to 127, inclusive

Default 0

Access read-write

localTimeZone

The time zone the device is operating in. Used to provision NTP.

Values 1: internationalDateLineWest

GMT-12:00 hours

2: nome

GMT-11:00

3: hawaii

GMT-10:00

4: yukon

GMT-9:00

5: pacific

GMT-8:00

6: mountain

GMT-7:00

7: central

GMT-6:00

8: eastern

GMT-5:00

9: atlantic

GMT-4:00

10: newfoundland

GMT-3:30

11: brasilia

GMT-3:00

12: azores

GMT-2:00

13: westAfrica

GMT-1:00

- 14: gmt**
GMT/UTC
- 15: centralEuropean**
GMT+1:00
- 16: easternEurope**
GMT+2:00
- 17: baghdad**
GMT+3:00
- 18: tehran**
GMT+3:30
- 19: uae**
GMT+4:00
- 20: afghanistan**
GMT+4:30
- 21: islamabad**
GMT+5:00
- 22: india**
GMT+5:30
- 23: kathmandu**
GMT+5:45
- 24: dhaka**
GMT+6:00
- 25: rangoon**
GMT+6:30
- 26: bangkok**
GMT+7:00
- 27: chinaCoast**
GMT+8:00
- 28: japan**
GMT+9:00
- 29: australia**
GMT+9:30
- 30: guam**
GMT+10:00
- 31: magadan**
GMT+11:00
- 32: newZealand**
GMT+12:00
- 33: rawakiIslands**
GMT+13:00

34: lineIslands
GMT+14:00

Default 14: gmt

Access read-write

lofAlpha

Number of loss-of-frame occurrences to trigger alarm.

Values Integer

Default 4

Access init-only

logicalIndex

Unique value, greater than zero, for each interface. It is recommended that values be assigned contiguously starting from 1. The value for each interface sub-layer must remain constant at least from one re-initialization of the entity's network management system to the next re-initialization.

Values 0 to 2147483647

Access init-only

loginId

The subscriber or user logon ID. The value of the logon may be managed in different ways depending on the service provider. In some cases the service will not require a logon or password. The default value is a NULL string.

Values 0 to 32

Access read-write

loopback

This enables or disables Ethernet loopback.

Values 1: Enabled

2: Disabled

Default 2: Disabled

Access read-write

loopback (DynOmciCpePweSubscriber)

This attribute specifies and reports the loopback configuration of the physical interface.

Values 1: noloop
 2: payload
 3: line
 4: ponside
 5: cesside
Default 1: noloop
Access read-write

loopbackCfg

The loopback configuration.

Values 1 to 6
Default 1
Access read-write

loopbackConfig

This variable represents the desired loopback configuration of the DS1 interface. Agents supporting read/write access should return inconsistentValue in response to a requested loopback state that the interface does not support.

Values 1: noLoop
 Not in the loopback state. A device that is not capable of performing a loopback on the interface shall always return this as its value.

2: lineLoop
 The received signal at this interface does not go through the device (minimum penetration) but is looped back out.

3: localLoop
 The transmitted signal at this interface is looped back and received by the same interface. What is transmitted onto the line is product dependent.

4: payloadLoop
 The received signal at this interface is looped through the device. Typically the received signal is looped back for retransmission after it has passed through the device's framing function.

Default 1: noLoop
Access read-write

Notes Agents supporting read/write access should return inconsistentValue in response to a requested loopback state that the interface does not support.

loopbackMode

Loopback mode for ONU remote device.

Values 1: onuNone
2: onuInward
3: onuLine
Default 3: onuLine
Access read-write

losAlpha

Number of loss-of-signal occurrences to trigger alarm.

Values Integer
Default 4
Access read-write

lossPktClearPolicy

The alarm clear policy attribute defines the anomaly rate that causes the corresponding alarm to be cleared. It is an integer percentage between 0..99. If no more than this density of anomalies occurs during the alarm clear soak interval, the alarm is cleared. The default value 255 selects the ONT's internal policy.

Values Integer32 (0..255)
Default 255
Access read-write

lossPktDeclarePolicy

The alarm declaration policy attribute defines the anomaly rate that causes the corresponding alarm to be declared. It is an integer percentage between 1..100. If this density of anomalies occurs during the alarm onset soak interval, the alarm is declared. The default value 0 selects the ONT's internal policy.

Values Integer32 (0..100)
Default 0
Access read-write

lowerIfIndex

The index (in the form *shelf-slot-port-subport-interface/atm* or a user-defined string) of the ATM interface for this cross connect. The **low-if-index** is arbitrary, but by convention it indicates the ATM interface with a numerically lower ifIndex value than the other ATM interface identified in the same cross connect.

Notes The **low-if-index** and the **high-if-index** cannot be equal.

lowerVpi

The VPI value associated with the interface specified in the **low-if-index**.

IpDesc

A descriptive name for the interface. This name is not used by the system, it is only used for reference by an administrator.

Access read-write

lqrPeriod

The Link Quality Reporting period that the local PPP entity will attempt to negotiate with the remote entity, in units of hundredths of a second.

Values 1000 to 99999

Default 0

Access read-write

Notes Changing this parameter will take effect when the link is next restarted.

lqrStatus

Enable or disable Link Quality Report (LQR) status.

Values 1: disabled

LQR negotiation is not performed.

2: enabled

The local node will attempt to perform LQR negotiation with the remote node.

Default 1: disabled

Access read-write

Notes Changing this parameter will take effect when the link is next restarted.

See RFC 1333 for more information on Link Quality Reporting.

M

magicNumber

The number used in magic number negotiation. When set to zero, a random number will be used for the magic number. This allows us to better prevent having two devices with the same magic number.

Default 0

Access read-write

magicNumEnable

Enable or disable magic number negotiation.

Values 1: false

Magic number negotiation is not performed.

2: true

The local node will attempt to perform magic number negotiation with the remote node.

Default 2: true

Access read-write

Notes Changing this parameter will take effect when the link is next restarted.

maintenanceMode

Indicates the maintenance modes of the Analog voice interface.

Values 1: off

2: ifDigitalLoopback

3: ifAnalogLoopback

Default 1: off

Access read-write

mapName

The name of the map.

Values 1 to 32

Default namap.ivl

Access read-only

mappingType

The type of mapping.

Values **1: other**

Cannot be determined.

2: invalid

has the effect of invalidating the corresponding entry in the ipNetToMediaTable. That is, it effectively disassociates the interface identified with said entry from the mapping identified with said entry. It is an implementation- specific matter as to whether the agent removes an invalidated entry from the table. Accordingly, management stations must be prepared to receive tabular information from agents that corresponds to entries not currently in use.

3: dynamic

mapping is supplied by the ARP server.

4: static

mapping is configured by the user.

Default **4: static**

Access read-write

Notes Proper interpretation of such entries requires examination of the relevant ipNetToMediaType object.

malformClearPolicy

The alarm clear policy attribute defines the anomaly rate that causes the corresponding alarm to be cleared. It is an integer percentage between 0..99. If no more than this density of anomalies occurs during the alarm clear soak interval, the alarm is cleared. The default value 255 selects the ONT's internal policy.

Values **Integer32 (0..255)**

Default **255**

Access read-write

malformDeclarePolicy

The alarm declaration policy attribute defines the anomaly rate that causes the corresponding alarm to be declared. It is an integer percentage between 1..100. If this density of anomalies occurs during the alarm onset soak interval, the alarm is declared. The default value 0 selects the ONT's internal policy.

Values **Integer32 (0..100)**

Default **0**

Access read-write

matchExpression

Logical expression telling which CPEs should be configured according to this rule.

Elements are:

keyword followed by value. Multiple elements can be connected with AND.

Keywords are:

model - specified model ID

slot - specifies slot number or numbers

olt - specifies GPON OLT number or numbers

Example:

model 2426 AND slot 6 AND olt 1-4,7

In this example, apply to model 2426 discovered on olts 1, 2, 3, 4, or 7 on slot 6..

Access read-create

mauClockSource

Clock source for timing.

Values 1: unused
unused

2: automatic
Master/Slave is decided during autonegotiation

3: master
Clocked from local source

4: slave
Loop timing from the received data stream

Access read-write

mauType

Physical Interface Type

Values 1: mauOther
 5: mau10BaseT
 10: mau10BaseTHD
 11: mau10BaseTFD
 15: mau100BaseTXHD
 16: mau100BaseTXFD
Default 1: mauOther
Access read-write

maxAddrAllowed

This value limits the maximum number of IP addresses that can be assigned by the DHCP server to the subscriber's IAD. Allowing multiple addresses to be assigned allows the subscriber to have multiple devices attached to the IAD data port without having to provide DHCP or routing services. The default value is 1. Required for row creation.

Values 1 to 2147483647
Default 1
Access read-write

maxAllowedLineRate

The maximum line rate that the subscriber is provisioned to receive from the service provider. This value is the total line rate or bandwidth of the subscriber's connection including voice, data and video. This number may be less than or equal to the subMaxCapableLineRate. Required for row creation.

Values 1 to 2147483647
Default 1536000
Access read-write

maxAtmConnections

Indicates the maximum number of Atm VPL/VCL connections on a given card.

maxCapableLineRate

The maximum line rate that can be supported by the subscriber line. This rate is determined by several factors including the distance the subscriber is from the CO, the copper facilities to the subscriber and the type of IAD installed. This value is determined by the service provider. Required for row creation.

Values 1 to 2147483647

Default 1536000

Access read-write

maxClients (DynOmciCpeWlan)

The maximum number of wireless client devices that may be simultaneously connected to the wireless network.

Values 1..50

Default 16

Access read-write

maxConfigRetries

Number of Link Control Protocol (LCP) configuration retries without response before the link is brought down.

Values 1 to 999

Default 10

Access read-write

maxDelayDuration

Maximum time to wait from outgoing seizure to out-dialing digits. If the far end does not allow to send digits after the duration specified in the object, a reorder tone will be sent to the calling party. The default value of this object is 2000 milliseconds.

Values 100 to 5000

Default 2000

Access read-write

maxDownstreamToneIndex

Index of end tone (last active tone) in the downstream direction. For POTS this must be in the range 32-255. For ISDN this must be in the range 60-255. Default assumes POTS.

1023 - Default for all Annexes and ADSL2/ADSL2Plus

Values 6-1023

Default 255

Access read-write

maxFailRetries

Number of times a Configuration Negative Acknowledgment is sent before a Configuration Reject is sent

Values 1 to 999

Default 10

Access read-write

maxLeaseTime

The global maximum time in seconds that will be assigned to a lease regardless of the value specified by a client. Default value is 86400 seconds.

Values 0 to 2147483647

Default 86400

Access read-write

maxLengthFrame

Maximum length, in octets, for frame mode data.

Values 1 to 65535

Default 65535

Access read-write

maxMcastBandwidth

This attribute specifies the maximum imputed dynamic bandwidth, in bytes per second, that may be delivered to the client port at anyone time. The default value 0 specifies that no administrative limit is to be imposed.

Values Integer32 (0..2147483647)

Default 0

Access read-write

maxOnuResponseTime

Combined with max rt propagation delay, determines maximum logical PON distance.

Values Integer

Default 5

Access read-write

maxOutstandingFrames

The maximum number of outstanding information frames at layer two.

Values 1 to 7, inclusive

Default 7

Access read-write

maxPad

The number of padded bytes at the end of data. This is used for Self-Describing Padding, a method for placing unambiguous padding octets at the end of the PPP Information field in order to cause messages to fall on a natural boundary.

Values 1 to 255

Default 4

Access read-write

Notes See Self-Describing Padding in RFC 1570.

maxPonDistance

Maximum PON distance in microseconds, used in the ranging process.

Values 0 to 255

Default 200

Access read-write

maxRtPropagationDelay

Maximum round trip propagation delay. Combined with max ONU response time and preassigned delay determines maximum logical PON distance. Default 200 usec for 20 dm.

Values Integer

Default 200

Access read-write

maxSDULength

Maximum size SSSAR-SDU, in octets, that is allowed to be reassembled.

Values 1 to 1536

Default 1536

Access read-write

maxSimultaneousGroups

This attribute specifies the maximum number of dynamic multicast groups that may be replicated to the client port at any one time. The default value 0 specifies that no administrative limit is to be imposed.

Values Integer32 (0.65535)

Default 0

Access read-write

maxSlotId

The maximum slot ID for Van Jacobson TCP/IP header compression.

Values 3 to 16

Default 3

Access read-write

Notes The **ip-comp-protocol** parameter must be set to **vj-tcp** to enable Van Jacobson TCP/IP header compression. See RFC 1332 for more information on Van Jacobson compression.

maxTermRetries

Number of Link Control Protocol (LCP) termination retries without response before the link is brought down.

Values 1 to 999

Default 2

Access read-write

Notes

maxUnicast

Maximum number of unicast addresses for I/F.

Values 0 to 2147483647

Access read-create

maxUpstreamToneIndex

Index of end tone (last active tone) in the upstream direction. For POTS this must be in the range 6-30. For ISDN this must be in the range 33-60. Default assumes POTS.

63 - Default for all Annexes and ADSL2/ADSL2Plus

Values 1-63**Default** 30**Access** read-write

maxVci

Indicates the maximum VCI value on a given card.

Values 0

Ethernet card, InfoServCard, IntegralCard, MtacCard, MtacE1Card, MtacT1Card, PlaceholderCard, Pots24Port900Card, Pots24PortCard, VASPCard

32 to 63

MalcDs1Atm32PortCard

32 to 127

VdslCard

32 to 255

MalcOltCard, Gshdsl4Wire24PortCard, ZRG300Card, ZRG500Card, ZRG700Card

32 to 1023

Adsl24PortMalcCard, Adsl24PortMalcCard, AdslAC5Malc24PortCard, AdslPots48PortCard, DSLCard, E1Atm24PortCard, GshdslCard, MALCOltCard, T1Atm24PortCard, VdslCard

0 to 4095

ZEdgeBH2ACard, ZEdgeH2ACard, S100AE1Card, S100AT1Card, Card, Ds1Ima8PortCard, E1Ima8PortCard, E1ImaF8PortCard, E1UplinkAtm8PortCard, E1Oc12GigEUplinkCard, E3UplinkAtm4PortCard, E3FUplinkAtm4PortCard, OC3Card, OC3UplinkAtmCard, OC3FUplinkAtmCard, OC3UplinkAtm2PortCard, T1Ima8PortCard, T1ImaF8PortCard, T1UplinkAtm8PortCard, T1FUplinkAtm8PortCard, T3UplinkAtm4PortCard, T3FUplinkAtm4PortCard

32 to 4095

T1Oc12GigEUplinkCard

32 to 8191

DS3Card, DS3UnchannelizedCard, Ds1Ima8PortCard, E1Ima8PortCard, E1ImaF8PortCard, E1UplinkAtm8PortCard, E3Aal52Atm4PortCard, E3Aal5Atm4PortCard, E3Atm4PortCard, E3Atm4PortPolyCard, OC3Aal52MultiModeCard, OC3Aal52SingleModeCard, OC3Aal5MultiModeCard, OC3Aal5SingleModeCard, OC3Card, OC3MultiModeCard, OC3PolyCard, T3Aal52Atm4PortCard, T3Aal5Atm4PortCard, T3Atm4PortCard, T3Atm4PortPolyCard

Default 0 (Ethernet card, InfoServCard, IntegralCard, MtacCard, MtacE1Card, MtacT1Card, Pots24Port900Card, Pots24PortCard, VASPCard)**255** ZRG300Card, ZRG500Card, ZRG700Card

1023 (Adsl24PortMalcCard, Adsl24PortMalcCard, AdslAC5Malc24PortCard, AdslPots48PortCard, DSLCard, E1Atm24PortCard, GshdslCard, T1Atm24PortCard), MALCOltCard, VdslCard

4095 (For S100AE1Card, S100AT1Card, ZEdgeBH2ACard, ZEdgeH2ACard, Ds1Ima8PortCard, E1Ima8PortCard, E1ImaF8PortCard, E1UplinkAtm8PortCard, E3UplinkAtm4PortCard, E3FUplinkAtm4PortCard, C3UplinkAtmCard, OC3FUplinkAtmCard, OC3UplinkAtm2PortCard, T1Ima8PortCard, T1ImaF8PortCard, T1Oc12GigEUplinkCard, T1UplinkAtm8PortCard, T1FUplinkAtm8PortCard, 3UplinkAtm4PortCard, T3FUplinkAtm4PortCard

8191 (Card, DS3Card, DS3UnchannelizedCard, E3Aal52Atm4PortCard, E3Aal5Atm4PortCard, E3Atm4PortCard, E3Atm4PortPolyCard, OC3Aal52MultiModeCard, OC3Aal52SingleModeCard, OC3Aal5MultiModeCard, OC3Aal5SingleModeCard, OC3Card, OC3MultiModeCard, OC3PolyCard, T3Aal52Atm4PortCard, T3Aal5Atm4PortCard, T3Atm4PortCard, T3Atm4PortPolyCard)

maxVciPerVp

If vp-switched, but VG connection is vc-switched, this specifies max number of vci on gateway.

Values 0 to 2147483647

Access read-write

maxVideoStreams

This is the maximum number of unique multicast (video) streams that this interface has subscribed to.

Values 0-210

Default 0

Access read-create

maxWinkDuration

Maximum outgoing wink duration (between the wink-pulse leading edge and trailing edge) for wink start signaling. The default value of this object is 200 milliseconds.

Values 100 to 3000

Default 200

Access read-write

maxWinkWaitDuration

Maximum time to wait from the outgoing seizure of the line to the time of sending out a wink-pulse. The default value of this object is 200 milliseconds.

Values 100 to 5000

Default 200

Access read-write

maxRate (DynOmciCpeWlanAdv)

The rate for the multicast traffic.

Values Integer32 (0..65535)

Default 0

Access read-write

maxRetryTime

This attribute specifies the maximum retry time for MGC transactions, in seconds. The default value 0 specifies vendor-specific implementation.

Values Integer32 (0..65535)

Default 0

Access read-write

maxRetryAttempts

This attribute specifies the maximum number of times a message is retransmitted to the MGC. The default value 0 specifies vendor specific implementation.

Values Integer32 (0..65535)

Default 0

Access read-write

mcastControlList

List of Mcast-control-ids this IP interface is associated with.

Values Octet String (0-255)

Access Init Only

mcastEnabled

Specifies whether multicast is enabled on this interface.

Values 1: True

2: False

Default 1: True

Access read-write

mcastFwdEnabled

Specifies whether multicast forwarding is enabled on this interface.

Values 1: True

2: False

Default 1: True

Access read-write

mcastFwd (DynOmciCpeWlan)

Wireless Multicast Forwarding enables the ability to send wireless packets to be intercepted by all nodes in the transmission range of sender.

Values 1: Enabled

2: Disabled

Default 2: Disabled

Access read-write

mcastRate (DynOmciCpeWlanAdv)

The rate for the multicast traffic.

Values 1: auto
 2: rate1m
 3: rate2m
 4: rate5dot5m
 5: rate6m
 6: rate9m
 7: rate11m
 8: rate12m
 9: rate18m
 10: rate24m
 11: rate36m
 12: rate48m
 13: rate54m
Default 1: auto
Access read-write

mediaIndex

This represents the CPE Voip media index value.

Values Unsigned32 (1..2147483647)
Access read-create

mediaProfileIndex

This attribute points to the associated cpe-voip-media profile.

Values Unsigned32 (1..2147483647)
Default 1
Access read-write

mediumType

This variable identifies whether a SONET or a SDH signal is used across this interface. SONET is used in North America, and SDH is used in Europe.

Values 1: sonet
 2: sdh
Default 1: sonet
Access read-write

meProfile (GponOnuPhysical)

This is the name of a Profile that contains OMCI commands which can configure an ONU. It is expected that this Profile contain OMCI commands that configure the Managed Entities (ME) on an ONU.

Values A string of 32 characters

Default Empty string

Access read-write

metric

A dimensionless quantity chosen at the discretion of the provisioner for the purpose of comparing static routes. The meaning of this value when static routes are redistributed to other protocols is protocol-dependent, and the resulting value after redistribution is determined by the route-maps in use for that protocol. Lower numeric values for this number indicate more preferred routes.

Values 0 to 4294967295

Default 1

Access read-write

metric(DynOmciCpeStaticRoute)

Number of hops to reach the destination. A value of 0 indicates this metric is not used.

Access read-write

mfgBootRevision

Revision level of any permanent (PROM) or semi-permanent (flash) code on the unit or card. This code is generally used as a bootstrap loader.

Values 0 to 255

Default 0

Access init-only

mfgCLEIcode

CLEI code for the unit or card if defined, otherwise the value will be set to zero (0) by the card.

Values 0 to 255

Default 0

Access init-only

mfgRevisionCode

Manufacturing hardware revision code for the unit or card.

Values 0 to 255

Default 0

Access init-only

mfgSerialNumber

The manufacturing serial number for this unit or card.

Values 0 to 255

Default 0

Access init-only

mgcMsgFormat

This attribute defines the message format.

Values 1 (long), 2 (short), 3 (binary)

Default 1

Access read-write

mgcServiceChangeDelay

The attribute specifies the service status delay time for changes in line service status. This attribute is specified in seconds. The default value 0 specifies no delay.

Values Integer32 (0..65535)

Default 0

Access read-write

mgcTerminationIdBase

The attribute specifies the base string for the H.248 physical termination id(s) for this ONT. This string is intended to uniquely identify an ONT. Vendor specific termination identifiers (i.e. port ids) are optionally added to this string to uniquely identify a termination on a specific ONT.

Values Integer32 (0..65535)

Default 0

Access read-write

mgcVersion

This integer attribute reports the version of the Megaco protocol in use.

Values Integer32 (0..255)

Default 0

Access read-write

minDownstreamToneIndex

Index of start tone (first active tone) in the downstream direction. For POTS this must be in the range 32-255. For ISDN this must be in the range 60-255. Default assumes POTS.

6 - Default for Annex A if G.Span and/or G.Span Plus are also defined, Default for Annex C/C.x, G.Span and G.Span Plus.

32 - Default for ADSL2/ADSL2Plus

64 - Default for Annex B

Values 6-1023

Default 32

Access read-write

minLease

Minimum accepted lease length. This value should not be greater than dhcpDefaultLease. Maximum allowed value is 1 week (604,800 seconds).

Values 1 to 604800

Default 300

Access read-write

minLeaseTime

The minimum expiration time in seconds that will be assigned to a lease regardless of the value specified by a client. A value of -1 indicates that this field should be ignored. Default value is -1 seconds.

Values -1 to 2147483647

Default -1

Access read-write

minNumRxLinks

The minimum number of receive links required to be Active for the IMA group to be in the Operational state.

Values 1 to 32

Default 1

Access read-create

minNumTxLinks

The minimum number of transmit links required to be Active for the IMA group to be in the Operational state.

Values 1 to 32

Default 1

Access read-create

minOnuResponseTime

GPON_stack uses this value in order to limit the size of the quiet zone.

Values Integer

Default 10

Access read-write

minRtPropagationDelay

If closest ONU location is known, GPON_stack uses this value in order to limit the size of the quiet zone.

Values Integer

Default 0

Access read-write

minUpstreamToneIndex

Index of start tone (first active tone) in the upstream direction. For POTS this must be in the range 6-30. For ISDN this must be in the range 33-60. Default assumes POTS.

1 - Default for Annex B

6 - Default for Annex A, Annex C/C.x, G.Span/ADSL+, G.Span Plus, ADSL2 / ADSL2Plus

Values 1-63
Default 6
Access read-write

minUsAlloc

Size of minimal upstream allocation per frame.

Values Integer
Default 16
Access read-write

misconnClearPolicy

The alarm declaration policy attribute defines the anomaly rate that causes the corresponding alarm to be declared. It is an integer percentage between 1..100. If this density of anomalies occurs during the alarm onset soak interval, the alarm is declared. The default value 0 selects the ONT's internal policy.

Values Integer32 (0..100)
Default 0
Access read-write

misconnDeclarePolicy

The alarm clear policy attribute defines the anomaly rate that causes the corresponding alarm to be cleared. It is an integer percentage between 0..99. If no more than this density of anomalies occurs during the alarm clear soak interval, the alarm is cleared. The default value 255 selects the ONT's internal policy.

Values Integer32 (0..255)
Default 255
Access read-write

mode

The mode of the DS1 connection. Default setting is **csu** for T1 platforms and **other** for E1 systems.

Values 1: other
 2: dsx
 3: csu
Default 1: other (for E1 systems)

3: csu (for T1 systems)

Access read-write

modelName

The vendor-specific model name as advertised by the remote endpoint.

Access read-create

modifier

Filter statement modifier. The bits set in this object logically negate the results of the comparisons made on their respective fields.

- Values**
- 0: notIpSrc**
 - 1: notSrcPort**
 - 2: notDstIp**
 - 3: notPortDst**
 - 4: notProtocol**
 - 5: notArbitrary**
 - 6: notStatemet**

Default No bits set (the default) specifies to use all outcomes as is.

Access read-create

monitorEnable

Values 1 or 2

Default 1

Access read-write

mplsTpType

This attribute specifies the type of ANI-side termination for the MPLS Pseudowire.

- Values**
- 1: ethernet**
 - 2: gem**

Default 1: ethernet

Access read-write

mru

Size of the largest datagram fragment that can be carried by the underlying transport protocol.

Values 0 to 4294967294

Default 1500

Access read-write

mruEnable

Enable or disable Maximum Receive Unit (MRU) negotiation.

Values 1: enabled

2: disabled

Default 1: enabled

Access read-write

mtacIndex

This field is the index field for the metallic test channel which will be connected on the external port. This device supports a maximum of two channels, one or both channels may be supported by a device depending on which lines cards are installed.

Values 1 or 2

Default 1

Access read-only

mtu

This attribute denotes the maximum frame size allowed across this interface.

Values Unsigned (0..65535)

Default 1518

Access read-write

multicast

Specifies whether this is a multicast or unicast port. If unicast, only one onu may be given, and the unicast-onu-id field is used. If multicast is true, then the multicast-onu-ids is used to name the onus.

Values 1: True
2: False
Default 2: False
Access read-write

multicastAging

Multicast Aging period (in tenths of seconds) for the specified VLAN. After the specified aging period is met, the VLAN address is deleted from the routing table.

Values 0 to 2147483647
The maximum value equals 2,485.51 days
Default 0
This value disables the aging period.
Access read-write

multicastSourceAddress

Video multicast source address.

Values 4 to 4
Default 0.0.0.0
Access read-write

multicastMapTemplateId

The multicast MAP ID.

Values 0 to 2147483647
Default 0
Access read-write

multicastSourceIndex

Video multicast source index value.

Values 1 to 65535
Default 1
Access read-write

multicastSourceNetMask

Video multicast source netmask value.

Values 4 to 4
Default 0.0.0.0
Access read-write

mvrVlanId

The Multicast Vlan Registration (MVR) vlan ID. Multicast video packets with this vlan can be translated out this bridge. This attribute is only applicable for downlink bridges.

Values 0 to 4095
Default 0
Access read-create

N

n200

The maximum number of retransmissions for a frame allowed at layer two.

Values 1 to 10, inclusive.
Default 3
Access read-only

n201

The maximum number of octets in the information field of an I frame.

Values 0 to 256, inclusive.
Default 256
Access read-write

name

The name associated with this object.

Values 0 to 64
Access read-write

Notes The default value varies by the object to which this attribute is associated.

name (Port access control)

The name of the PortAccessControl object.

Values STRING (SIZE (0.. 32 characters))

Access read-write

name (V5.2)

The name of the V5.2 end point. This is a DZMS only field which means this value will not be retained across device re-additions in DZMS.

The system-wide unique symbolic name that is used for call routing.

Values STRING (SIZE (0.. 32 characters))

Access read-write

name (OMCI ME profile)

The name of the OMCI ME profile. This attribute is required during addition of ME profile.

Values STRING (SIZE (0.. 64 characters))

Access read-create

name (OMCI generic profile)

The name of the OMCI generic profile. This attribute is required during addition of generic profile.

Values STRING (SIZE (0.. 64 characters))

Access read-create

name (GponOnuGenProfileVariable/ GponOnuSpecProfileVariable)

The name of the variable from OMCI generic or specific profile.

Values STRING (SIZE (0.. 64 characters))

Default An empty string

Access read-only

name (Region/ SubRegionOne/ SubRegionTwo/ SubRegionThree)

The name of the variable from Region, SubRegionOne, SubRegionTwo, and SubRegionThree.

Values STRING (SIZE (0.. 64 characters))

Access read-write

nat (DynOmciCpelpServer)

When it is enabled, NAT/NAPT function is performed to translate between the public IP address and the private addresses. It is only supported on a WAN interface.

Values 1: nat
2: napt
3: disabled
Default 1: nat
Access read-write

natEnabled

Specifies whether NAT is enabled on this interface.

Values 1: True
2: False
Default 2: False
Access read-write

nationalPstnRegion

The regional and country-specific setting for V5.2 PSTN service. The PSTN region dictates regional signaling differences like ring cadence.

Values	1: etsi
	2: germany
	3: uk
	4: italy
	5: finland
	6: austria
	7: belgium
	8: china
	9: france
	10: hong-kong
	11: japan
	12: korea
	13: netherlands
	14: new-zealand
	15: singapore
	16: spain
	17: sweden
	18: switzerland
	19: mexico
	20: argentina
	21: peru
	22: puerto-rico
	23: brazil
	24: malaysia
	25: taiwan
	26: united-arab-emirates
	27: ireland
	28: australia
	29: estonia
Default	1: etsi
Access	read-write

nearEndPort

When the pseudo-wire service is transported via IP, this attribute specifies the port number of the near-end TCP/UDP service. The default value is 57000+profile index.

Values Unsigned32 (0..65535)

Default 57000

Access read-write

negAdminStatus

Setting this object to enabled(1) will cause the interface which has the auto-negotiation signaling ability to be enabled. If the value of this object is disabled(2) then the interface will act as it would if it had no auto-negotiation signaling. Under these conditions, an IEEE 802.3 MAU will immediately be forced to the state indicated by the value of the object zhoneMauDefaultType.

When ifMauAutoNegAdminStatus transitions from enabled to disabled, the operational type of the MAU (as reported by zhoneMauType) transitions to the value specified by the zhoneMauDefaultType object, rather than continuing to operate at the value earlier determined by the auto-negotiation function.

Values 1: enabled

2: disabled

Default 1: enabled

Access read-write

negCapAdvertBits

A value that uniquely identifies the set of capabilities advertised by the local auto-negotiation entity. Capabilities in this object that are not available in ifMauAutoNegCapabilityBits cannot be enabled. Note that the local auto-negotiation entity may advertise some capabilities beyond the scope of this MIB. This is indicated by returning the bit value bOther in addition to any bit values for capabilities that are listed above.

Values	0: bOther 1: b10baseT 2: b10baseTFD 3: b100baseT4 4: b100bseTX 5: b100baseTXFD 6: b100baseT2 7: b100baseT2FD 8: bFdxPause 9: bFdxAPause 10: bFdxSPause 11: bFdxBPause 12: b1000baseX 13: b1000baseXFD 14: b1000baseT 15: b1000baseTFD
Default	0: bOther
Access	read-write

negRemoteFaultAdvert

A value that identifies any local fault indications that this MAU has detected and will advertise at the next auto-negotiation interaction for 1000Mbps MAUs.

Values	1: noError 2: offline 3: linkFailure 4: autoNegError
Default	1: noError
Access	read-write

negRestart

Negotiation signaling restart.

Values	1: restart This will force auto-negotiation to begin link renegotiation.
---------------	--

2: norestart

Auto-negotiation signaling is disabled; a write to this object has no effect.

Default 2: norestart

Access read-write

netAddress

The IP address corresponding to the media-dependent “physical” address.

Default 0.0.0.0

Access init-only

netAuthentication (DynOmciCpeWlan)

Configure the network authentication method.

Values 1: open

2: shared

3: dot1x

4: wpa

5: wpapsk

6: wpa2

7: wpa2psk

8: mixedwpa2wpa

9: mixedwpa2wpapsk

Default 1: open

Access read-write

netMask

The subnet mask associated with the IP address of this interface alias entry.
The value of the mask is an IP address with all the network bits set to 1 and all the hosts bits set to 0.

Default 0.0.0.0

Access read-write

netMask (for DynOmciCpelpServer object)

The subnet mask for IP host services.

Values IP address

Default 255.255.255.0

Access read-write

netmask (Port access control)

Network mask of the portAccessSrcAddr.

Access read-create

netmask (DynOmciCpeStaticRoute)

Destination subnet mask. A value of 0.0.0.0 indicates no destination subnet mask is specified.

Default 0.0.0.0

Access read-write

netReauthInterval (DynOmciCpeWlan)

Configure the WPA2 network re-authentication time, in seconds.

Values 0..2147483647

Default 36000

Access read-write

network

Network address of this subnet object.

Default 0.0.0.0

Access read-write

networkRefClock

Network reference clock usage.

Values 1: true

2: false

Default 2: false

Access read-write

networkTime

Network Time and Power.

Values 0: none
not supported

- 1: localOsc**
use local osc for timing
- 2: refClk8khz**
use Backplane 8KHZ clock for timing
- 3: localOscNotReady**
use local osc for timing but not ready
- 4: refClk8khzNotReady**
use Backplane 8KHZ clock for timing but not ready

Default 0: none

Access read-create

neTxClkMode

Transmit clocking mode used by the near-end IMA group.

Values 1:

2:

Default 1

Access read-create

newStmtIndex

The new statement index for the filter statement. Reading this object will return the same value as the 'fltStmtIndex' portion of its index. Writing to this object will cause the corresponding filter statement to be relocated to the position identified by the value written here. If no statement exists at the current index, 'no such instance' will be returned. If a statement already exists at the new index then 'inconsistent value' is returned. For example, to move the second statement of filter #4 to the third position (to make room for a new statement #2), the following SNMP set-request would be issued: fltStmtIndexNew.4.2 = 3. There is no default value for this object as it is derived from the fltStmtIndex.

Values 0 to 2147483647

Default 0

Access read-write

nextHop

The IP address of the nexthop router of the static route. This may be 0.0.0.0 for static routes that use an interface number rather than a nexthop address, as in the case of a destination directly reachable via an interface requiring no intermediary system to act as a gateway.

Default 0.0.0.0

Access init-only

nextVoiceConnectionIndex

The next available connection index which can be used for this subscriber and lgid pair. This is used for the creation of a voice connection entry.

Values 0 to 2147483647

Default 0

Access read-only

noChannelAddress

The video multicast address of the channel appropriate to display when a user selects a channel that is not available or not in his channel profile.

Values 4 to 4

Default 0.0.0.0

Access read-write

ntpServerAddress

The IP address of the NTP Server for a particular video head-end.

Values 4 to 4

Default 0.0.0.0

Access read-write

ntr (SdslConfigProfile)

Determines whether network timing recovery is enabled or disabled. The default value is disabled

Values 1: ntr-disable

2: ntr-enable

Default 1

Access read-write

ntr (ShdslConfigProfile)

Determines the clocking on the SHDSL line.

Values 1: ntr-local-osc

The line uses the system's internal clock. Also known as local timing.

2: ntr-refck-8khz

The line uses an 8khz clock derived from a timing source such as a T1 interface.

Default 1

Access read-write

Notes Circuit Emulation Service (CES) transports TDM circuits over ATM networks. For CES-based services: change **ntr** default setting of **ntr-local-osc** to **ntr-refck-8khz** in the **shdsl-config** profile.

SHDSL lines configured with the **shdsl-ntr** parameter set to **ntr-refck-8khz** will not sync if a system clock source is not enabled in a **system-clock-profile**. To enable clocking:

```
zSH> update system-clock-profile 1-17-1-0/ds1
Please provide the following: [q]uit.
system-clock-eligibility: -> {false}: true
system-clock-weight: -----> {5}:
.....
Save changes? [s]ave, [c]hange or [q]uit: s
Record updated.
```

numberRings

The number of rings detected before closing loop. The default value of this object is 1 ring.

Values 1 to 10

Default 1

Access read-write

numChannels

The maximum number of channels allowed on this interface.

Values 1

Default 1

Access read-write

Notes Only one channel is supported.

numChannels (IP video)

The number of multicast channels carried over the IP video stream.

Values 0 to 2147483647

Default 1

Access read-write

numModeEnable

Enable or disable Numbered Mode negotiation.

Values 1: enabled

2: disabled

Default 1: disabled

Access read-write

Notes From RFC 1663: “The Link Control Protocol (LCP) Numbered-Mode Configuration Option negotiates the use of Numbered-Mode on the link. By default or ultimate disagreement, Unnumbered-Mode is used.”

numSlots

The number of slots in a chassis.

Values 1 to 18

Default 1

Access read-only

O

oamF5Ping

This is used to send a OAM F5 loopback cell for fault detection. The value will automatically revert to disabled after the set is complete.

Values 1: Disabled (OAM F5 loopback cell is not being sent)

2: EndToEnd (This option uses standard OAM fault management cells as defined in ITU.610 for fault detection. Setting this will send a OAM F5 end-to-end loopback cell to be sent)

3: Segment (This is not supported at this time)

Default 1: Disabled

Access init-only

offerTimeout

Interval to get additional DHCP offers.

Values 0 to 100

Default 5

Access read-write

oltHec

Header Error Control field.

Values 1: oltHecOff
2: oltHecRxCorrectionOnly
3: oltHecTxGenerationOnly
4: oltHecRxAndTx

Default 2: oltHecRxCorrectionOnly

Access read-write

oltLoopback

OLT loopback configuration.

Values 1: loopbackNone
2: loopbackInward
3: loopbackLine

Default loopbackNone

Access read-write

oltOnuChurnKey

Churn key enable flag.

Values 1: onuChurnkeyEnable
2: onuChurnkeyDisable

Default 2: onuChurnkeyDisable

Access read-write

oltOnuPassword

ONU remote-device password.

Values 0 to 10

Access read-write

oltOnuSerialNum

ONU remote-device serial number.

Values 0 to 2147483647

Access read-write

oltTrafficContainerCBR

Specifies whether the traffic in the OLT traffic container is constant bit rate (CBR).

Values 1: true

2: false

Default 1: true

Access read-write

oltTrafficContainerCbrCompensate

Specifies the system should perform ranging compensation. When a ranging window is opened, the system appends to the end of the ranging window all the grants missed due to the ranging window by CBR flows with the compensation flag set.

Values 1: true

2: false

Default 2: false

Access read-write

oltTrafficContainerGuaranteedBW

The sum of the **oltTrafficContainerGuaranteedBW** and **oltTrafficContainerMaximumBW** parameter values for all the ports should be less than or equal to the upstream bit rate (155 Mbps) of the PON OLT card.

Values 0 to 2147483647 (CPS)

Default 0

Access read-write

oltTrafficContainerIndex

Index number of a PON traffic container associated with a VCL definition.

Values 0 to 2147483647

Default 0

Access init only

oltTrafficContainerMaximumBW

The sum of the **oltTrafficContainerGuaranteedBW** and **oltTrafficContainerMaximumBW** parameter values for all the ports should be less than or equal to the upstream bit rate (155 Mbps) of the PON OLT card.

Values 0 to 2147483647 (CPS)

Default 0

Access read-write

Notes The **oltTrafficContainerGuaranteedBW** and **oltTrafficContainerMaximumBW** attribute values must be a multiple of 63 and should be equal.

omciConfigFile

Name of OMCI configuration file for this ONT.

Access read-write

omciMeld

Represents the ME Id of the CPE attached to ONU. This is the ID of the managed entity in the ITU G984.4 standard for the OMCI statistics we want to read.

Values NUMBER (0 .. 2147483647)

Access Not-Accessible

Notes Since access type is not-accessible, an SNMP walk (not providing ME Id and Logical Port Id will retrieve all the available ME Id and logical port ids) shall be performed to know the available ME and logical port ids.

omciPortId

Represents the logical port Id of CPE attached to ONU. This specifies which entity on the ONT we want the statistics for. This is a logical number (1,2,...) which is translated according to the ME profile into an entity ID which is included in the command to the ONT.

Values NUMBER (0 .. 2147483647)

Access Not-Accessible

Notes Since access type is not-accessible, an SNMP walk (not providing ME Id and Logical Port Id will retrieve all the available ME Id and logical port ids) shall be performed to know the available ME and logical port ids.

onuAdded

Indicates to Broadlight that onu is activated. May not be used.

Access read-write

onuld

The onu number for this olt to which we are allocating bandwidth. Cannot be greater than number of onus for the olt. This field cannot be changed once the alloc-id has been created.

Values 1: cbr

2: ubr

Default 2: ubr

Access read-write

onulIndex

Represents the ifIndex of parent GPON ONU physical.

Values Number (0 .. 2147483647)

Access read-only

onulName

ONU interface name.

Access read-create

ontIpAddress

The onu IP address for this olt.

Values valid IP address

Access read-write

ontMacAddress

The onu mac address for this olt.

Values valid MAC address

Access read-only

oobToneEvents

Enables or disables handling of tones via RTP tone events per RFC 4733.

Values 1 (enabled), 2 (disabled)

Default 2 (disabled)

Access read-write

oobDtmfEvents

Enables or disables handling of DTMF via RTP DTMF events per RFC 4733.

Values 1 (enabled), 2 (disabled)

Default 2 (disabled)

Access read-write

oobCasEvents

Enables or disables handling of CAS via RTP CAS events per RFC 4733.

Values 1 (enabled), 2 (disabled)

Default 2 (disabled)

Access read-write

operation

Specifies the operation of the E&M signal.

Values 1: **twoWires**
2-wire operation.

2: **fourWires**
4-wire operation.

Default 1: **twoWires**

Access read-write

operStatus (V5.2)

Interface operational status.

Values 1: **operable**

2: **inoperable**

3: **standby**

4: **degradedService**

5: **inoperableIsInProgress**

6: **inoperableOosInProgress**

Access read-only

Notes The following table describes the relationship between the AdminStatus and OperationalStatus attributes:

AdminStatus	OperationalStatus	Description
outOfService	operable	not valid
outOfService	inoperable	admin out of service
outOfService	standby	not valid
inService	operable	normal operations
inService	inoperable	fault
inService	standby	normal operations

operStatusCause (V5.2)

The cause for Interface operational status.

Values 1:other
2: localDisable
3: remoteOutOfService
4: protocolFailureValues

Access read-only

otherVclTemplateId

The VCL template used for the IP video connection.

Values 0 to 2147483647

Default 0

Access read-write

ouClearPolicy

The alarm clear policy attribute defines the anomaly rate that causes the corresponding alarm to be cleared. It is an integer percentage between 0..99. If no more than this density of anomalies occurs during the alarm clear soak interval, the alarm is cleared. The default value 255 selects the ONT's internal policy.

Values Integer32 (0..255)

Default 255

Access read-write

ouDeclarePolicy

The alarm declaration policy attribute defines the anomaly rate that causes the corresponding alarm to be declared. It is an integer percentage between 1..100. If this density of anomalies occurs during the alarm onset soak interval, the alarm is declared. The default value 0 selects the ONT's internal policy.

Values Integer32 (0..100)

Default 0

Access read-write

outboundServer

This attribute contains the name (IP address or resolved name) of the outbound proxy server for SIP signaling messages. Type is IP address.

Default 0.0.0.0

Access read-write

outFlowType

The port's type of output flow control.

Values 1: none

Indicates no flow control at this level.

2: ctsRts

Indicates use of the ctsRts hardware signal.

3: dsrDtr

Indicates use of the dsrDtu hardware signal.

Default 1: none

Access read-write

outSpeed

The port's output speed in bits per second. Note that non-standard values, such as 9612, are probably not allowed on most implementations.

Values 0 to 4294967295

Default 0

Access read-write

overheadSize (OLT)

Overhead size definition field for OLT card.

Values 1: oltPonOverhead3Bytes
 2: oltPonOverhead6Bytes
 3: oltPonOverhead12Bytes
Default 1: oltPonOverhead3Bytes
Access read-write

overheadSize (ONU)

Overhead size definition field for ONU remote devices.

Values 1: onuOverheadSize3Bytes
 2: onuOverheadSize6Bytes
 3: onuOverheadSize12Bytes
Default 1: onuOverheadSize3Bytes
Access read-write

P

packetRule2/packetRule3/packetRule4/packetRule5

A variable-length field whose meaning depends on packetRuleType.

Values SnmpAdminString (Range 0..255)
Default empty string
Access read-write

papPassword

The password used for authenticating with the PAP protocol.

Default empty string
Access read-write

Notes The **pap-peer-id** and **pap-password** values must be entered as strings with single quotes. After the **ppp-auth** record is saved, the **pap-peer-id** and **pap-password** values will be converted to hexadecimal values. For example, the string 'test' will be returned as 74:65:73:74.

papEnable

Enables or disables Password Authentication Protocol (PAP).

Values 1: enabled
2: disabled

Default 2: disabled

Access read-write

Notes

papPeerId

The username for authenticating with the PAP protocol.

Default empty string

Access read-write

Notes The **pap-peer-id** value must be entered as a string with single quotes. After the **ppp-auth** record is saved, the **pap-peer-id** value will be converted to hexadecimal values. For example, the string 'test' will be returned as 74:65:73:74.

parentGeneric

Name of the parent generic profile. The generic profile is created from the ME profile file. This attribute is required during addition of specific profile. It is a read-create attribute for object GponOnuSpecProfile.

Values STRING (SIZE (0..64 characters))

Access read-create

parentMe (OMCI generic profile)

Name of the parent ME profile. The ME profile is the parent of the generic profile.

This attribute is required during addition of generic profile. It is a read-create attribute for object GponOnuGenProfile.

Values STRING (SIZE (0..64 characters))

Access read-create

parentMe (OMCI specific profile)

Name of the parent ME profile. The ME profile is the parent of the generic profile.

This attribute is not required during addition of specific profile. It is a read-only attribute for object GponOnuSpecProfile.

Values STRING (SIZE (0..64 characters))

Access read-only

parity

The port's sense of a character parity bit.

Values 1: none

2: odd

3: even

4: mark

5: space

Default 1: none

Access read-write

partialDialTimeout

This attribute defines the partial dial timeout for digit map processing in milliseconds.

Values Integer32 (0..65535)

Default 16000

Access read-write

password (ONU)

ONU password.

Values 0 to 32

Access read-write

password (DynOmciCpeVoipSubscriber)

This attribute contains the SIP password used for authentication.

Values STRING (SIZE (0..25))

Access read-write

password (DynOmciCpePppoe)

The login password to be used for PPP authentication.

Values STRING (SIZE (1..25))

Access read-write

patBindIndex

Index for the PAT table.

Values 1 to 10

Default 1

Access read-write

pathWidth

A value that indicates the type of the SONET/SDH Path. For SONET, the assigned types are the STS-Nc SPEs, where N = 1, 3, 12, 24, and 48. STS-1 is equal to 51.84 Mbps. For SDH, the assigned types are the STM-Nc VCs, where N = 1, 4, and 16.

Values 1: sts1

2: sts3cSTM1

3: sts12cSTM4

4: sts24c

5: sts48cSTM16

Default 2: sts3cSTM1

Access read-create

Notes For Sechtor 100 systems, the only valid setting is **sts3cSTM1**.

pauseAdminMode

Configure the administrative pause mode for the GigaBit Ethernet interface. This setting is used if auto-negotiation is not enabled and the interface is full-duplex.

If auto-negotiation is enabled, the pause mode for this interface is determined by auto-negotiation, and the value of this attribute determines the mode to which the interface will automatically revert if/when auto-negotiation is later disabled.

Note that the value of this object is ignored when the interface is not operating in full-duplex mode.

Values 1: disabled

2: enabledXmit

3: enabledRcv

4: enabledXmitAndRcv

Default 1: disabled

Access read-write

Notes This attribute is ignored if the interface is not operating in full-duplex mode.

pauseFlowControl

When autonegotiation is enabled (zhoneMauAutoNegSupported is true), this Pause Flow Control field is used to program the advertised pause capabilities.

When autonegotiation is disabled (manual negotiation), this Pause Flow Control field is used to program the local Mac as instructed.

Values **1: disabled**
2: asymmetricTx
3: asymmetricRx
4: symmetric

Access read-write

pauseTime

This attribute allows the interface to ask the subscriber terminal to temporarily suspend sending data. Units are in pause quanta (1 pause quantum is 512 bit times of the particular implementation).

Values **Unsigned (0..65535)**

Default **0**

Access read-write

payloadsize

Number of payload bytes per packet. Valid only if service type = unstructured or unstructured octet-aligned. Valid choices depend on the TDM service, but must include the following. Other choices are at the vendor's discretion.

DS1192

DS1200, required only if unstructured octet aligned service is supported

E1256

DS31024

E31024

Values **Integer32 (0..65535)**

Default **250**

Access read-write

payloadEncapsDelay

This represents number of 125 microsecond frames to be encapsulated in each pseudowire packet. Valid only if service type = structured. The minimum set of choices for various TDM services is listed below, and is affected by the possible presence of in-band signalling. Other choices are at the vendor's discretion.

NxDS0, no signalling, N = 1 64 (8 ms) required, 40 (5 ms) desired

NxDS0, no signalling, N = 2..4 32 (4 ms)

NxDS0, no signalling, N > 4 8 (1 ms)

NxDS0 with DS1 CAS 24 (3 ms)

NxDS0 with E1 CAS 16 (2 ms)

Values 1: no signalling and N value is 1

2: no signalling and N value is 2 to 4

3: no signalling and N value is greater than 4

4: ds1 cas

5: e1 cas

Default 1: no signalling and N value is 1

Access read-write

pcmEncoding

Type of PCM Encoding.

Values 1: mulaw

2: alaw

Default 1: mu law

Access read-write

Notes Each Encoding Data Unit (EDU) is an octet-aligned concatenation of one or more frames of an audio algorithm, entailing a specific format of the bits. Every audio packet and SDU contains an integral number of EDUs.

periodFirstOrder

This attribute specifies the packet period selection interval in milliseconds.

Values Integer32 (10..30)

Default 10

Access read-write

periodSecondOrder

This attribute specifies the packet period selection interval in milliseconds.

Values Integer32 (10..30)

Default 10

Access read-write

periodThirdOrder

This attribute specifies the packet period selection interval in milliseconds.

Values Integer32 (10..30)

Default 10

Access read-write

periodFourthOrder

This attribute specifies the packet period selection interval in milliseconds.

Values Integer32 (10..30)

Default 10

Access read-write

pfcEnable

Enable or disable Protocol Field Compression (PFC) negotiation.

Values 1: enabled

2: disabled

Default 2: disabled

Access read-write

phone

Phone number of the customer.

Access read-write

physAddress

The interface's address at its protocol sub-layer. For example, for an 802.x interface, this object normally contains a MAC address. The interface's media-specific MIB must define the bit and byte ordering and the format of

the value of this object. For interfaces which do not have such an address (e.g., a serial line), this object should contain an octet string of zero length.

Values 0 to 16

Access init-only

physicalIndex

This variable uniquely identifies the interface to which the Ethernet/MAU described by this entry is connected.

Values 0 to 2147483647

Default 0

Access init-only

piggyBackEvents

Enables or disables RTP piggyback events.

Values 1 (enabled), 2 (disabled)

Default 2 (disabled)

Access read-write

pingFailMax

This object contain the maximum number of times that the ping of the nexthop of an active route in a fallback route pair must fail, in a row, before this route is demoted to fallback and the other route in the fallback pair becomes the active route.

Values 0 to 2147483647

Default 0

Access read-write

pingInterval

This object defines the interval, in milliseconds, of successive pings of the nexthop of an active fallback route pair. The minimum value is 500 milliseconds, and must be specified in increments of 100 milliseconds.

Values 0 to 4294967295 (in milliseconds)

Default 0

Access read-create

plsMaxAllocSize

Indicates a size of unassigned allocation that should be attached to the end of ranging window.

Values Integer

Default 120

Access read-write

point

Values 0 to 32

Default 0D0.N 0D0.W

Minimum 0

Maximum 32

Access read-only

pointToPoint

Specifies whether the interface point-to-point.

Values 1: True

2: False

Default 2: False

Access read-write

polygonPoints

Values 0 to 255

Access read-only

portAlignmentReq

A request to block or unblock all the user ports associated with the current IG by port type.

Values 1: noRequest

2: blockAll

Block all user ports associated with this IG.

3: blockIsdn

Block all of the isdnBa and isdnPra user ports associated with this IG.

4: blockPstn

Block all of the PSTN user ports associated with this IG.

5: unblockAll

Unblock all user ports associated with this IG.

6: unblockIsdn

Unblock all of the isdnBa and isdnPra user ports associated with this IG.

7: unblockPstn

Unblock all the pstn user ports associated with this IG.

Default 1: noRequest

Access read-create

portAlignmentStatus

Indicates the status result for user port block/unblock request by port type. The values of this attribute indicate both type of request and its success or fail status.

Values 1: blockAllSucceeded

2: blockAllFailed

3: blockIsdnSucceeded

4: blockIsdnFailed

5: blockPstnSucceeded

6: blockPstnFailed

7: unblockAllSucceeded

8: unblockAllFailed

9: unblockIsdnSucceeded

10: unblockIsdnFailed

11: unblockPstnSucceeded

12: unblockPstnFailed

Access read-only

Notes The value is invalid when v52IgPortAlignmentRequest is noRequest.

portDesc

Port description that is stored only in DZMS database. This value will be reset to emptyString after reading the device in DZMS.

Values A string of maximum of 32 characters.

Default empty string

Access read-write

portEnd (DynOmciCpePortFwd)

Highest value port number for the range. This can be equal to port-start if there is only one port.

Values 0..65535

Default 0

Access read-write

portFwdListProfile (DynOmciCpeIfVlan)

The address of the port fwd list profile associated with this interface. It is only supported on a WAN interface. The value 0 indicates no associated cpe-port-fwding profile.

Values 0..2147483647

Default 0

Access read-write

portId (DynOmciCpeLdpRem)

The string value used to identify the port component associated with the remote system.

Access read-create

portId (AAL2 ELCP)

The port ID to identify an IAD port on the CPE side. PortId in combination with PortType uniquely identifies a port on the IAD side for this voice connection.

Values 0 to 32767

Default 0

Access init-only

portId (DynOmciCpeVoipServer)

This attribute specifies the TCP/UDP port number of the VoIP protocol. The default value -1 select the default port number for the VoIP protocol. It is 2944 for H.248 and 5060 for SIP. Type is integer 32.

Values -1 to 32767

Default -1

Access Read-Write

portIndex

The port number of the physical port on the card or device. The numbering should start at 1 for the first port on the device and proceed sequentially. This object has been replaced by zhonePortIndex.

Values 0 to 2147483647

Default 1

Access init-only

portNumber

The port number that port access control applies to.

Values 1 to 1023

Access read-create

portNumber (for DynOmciCpeEthSubscriber, DynOmciCpePweSubscriber, DynOmciCpeRfSubscriber, DynOmciCpeVoipSubscriber object)

This represents the CPE Subscriber Port Numbers for Dynamic OMCI.

Values Unsigned32 (1..2147483647)

Access read-only

portStart (DynOmciCpePortFwd)

Lowest value port number for the range.

Values 0 to 65535

Default 0

Access read-write

portType

The port's hardware type.

Values 1: other
2: rs232
3: rs422
4: rs423
5: v35
6: x21
Default 1: other
Access read-only

portType (AAL2 ELCP)

The port type for the CPE (IAD) port for the current connection.

Values 1: pots
2: isdnDChannel
3: isdnBChannel
Default 1: pots
Access read-write

portType (PAT)

Specifies whether the connection is TCP or UDP.

Values 1: TCP
2: UDP
Default 1: TCP
Access read-write

portType (DynOmciCpeEthSubscriber)

This attribute specifies the Ethernet interface wiring.

Values 1: dce
2: dte
Default 1: dce
Access read-write

potsLbrRelayStatus

Indicates whether a particular Adsl Pots Bypass Relay is in a normal or closed loop condition. By setting this parameter, the manager can switch the relays

in order to bypass the Low Pass Filter for POTS closed loop testing (closed) or force a relay back to normal operation mode before the zhoneAdslPotsBypassRelayMaxDuration timer expires.

This object is only valid for Adsl Pots Splitter cards containing the Pots LBF Relays.

Values 1: normal
2: closed
Default 1: normal
Access read-write

powerFeed

This attribute controls whether power is provided to an external equipment over the Ethernet interface.

Values 1: Enabled
2: Disabled
Default 2: Disabled
Access read-write

powerlevel

Olt power level. For the Broadlight 3000, values will be rounded down to one of 0, 3, or 6.

Values Unsigned Integer
Default 0
Access read-write

powerScale

The zhoneSdslConfigPowerScale is used to adjust transmit power in small increments to compensate for minor differences on power between units. It is recommended that value should remain in the range of 20992 (-3 db) to 32767 (+0.8 db). This value is calculated using the following formula $\text{PowerScale} = (10^{(\text{Db} / 20)}) * 29952$. Where Db is in the range of -3 to 0.8.

A value of 29952 corresponds to 0 db. The value should not be limited to the range 20992 to 32767. In some circumstances a value outside of this range maybe needed.

Values 1 to 844162056
Default 29952
Access read-write

precedence

Precedence is the attribute of McastControlList object. Whenever a McastControlList is added, you must provide an unique combination of controlId and precedence for successful operation. Hence, within a McastControlLists object, all McastControlList elements must be provided with unique precedence during their addition.

Note that since precedence attribute of McastControlList is a read-create attribute as per MIB, it can not be modified using OSS. If you have to modify it, the corresponding McastControlList object should be deleted first, and then added.

Values 1 to 2147483647

Default 1

Access read-create

prefCommName

Values 0 to 255

Default ZhonePrivate

Access sync-only

preferredApIndex

Index for the preferred audio encoding format within the Audio Profile. It identifies an audio profile row of the selected audio profile at the AAL2 VCL.

Values 1 to 512

Default 1

Access read-write

Notes The Index must be an agreed operating parameter between the transmitter and receiver for both directions.

preassignedEqD

Combined with max rt propagation delay and max onu response time determines max logical PON distance.

Values Integer

Default 0

Access read-write

prevStats

Blob of previous statistics information returned. Format is:

PhysPort,I1,A1,I2,A2,...In,An

where

PhysPort - physical port ID value

Ii = attribute index (0-15, ascii-coded decimal integer) within ME according to G984.4 standard.

Ai = attribute value (counter, ascii-coded decimal integer)

n = number of attributes returned = number requested.

Maximum number of attributes per ME is 16.

Values Octet String (0..256)

Access read-only

priLineId

The id entered in the line profile table for the profile of the primary line. The default value of 0 indicates no primary line is configured. When this field is set, the matching lineProfile entry will also be updated. Note that the matching entry (ifIndex) must be adminned down before this field can be modified. Required by row creation.

Values 0 to 2147483647

Default 0

Access read-write

primaryClockSource

The ifIndex of the interface from which clock can be derived. Only used for OC3, DS1, and DS3 interfaces. This interface will be used to drive the 8 khz backplane signal. The interface must be set to loop or local timing in order for it to be considered as the primary clock source. It must also be administratively turned on and be in an operational state. Only one interface in the system can be the primary clock source at any given time.

Values 0 to 2147483647

Default 0

Access read-write

primaryDns

This attribute specifies the primary DNS IP address. If this value is 0.0.0.0, no primary SIP DNS is defined.

Values IP address
Default 0.0.0.0
Access read-write

primaryServer

This attribute contains the name (IP address or resolved name) of the primary MGC or SIP proxy server that controls the signaling messages.

Values IP address
Default 0.0.0.0
Access read-write

primaryServerIP

The IP address of the primary Network Time Protocol (NTP) server.

NTP, described in RFC 2030, synchronizes devices on the network with a timing source. When NTP is enabled, setting the system time with the **setdatetime** command is not allowed.

The device will get the time from the NTP server every hour. If the system clock has drifted more than 9 seconds from the source clock, the system time will be reset.

Values A valid IP address
 A value of 0.0.0.0 disables NTP.
Default 0.0.0.0
Access read-write

priNameServer

IP Address of the primary domain name server that the client should use for DNS resolution.

Default 0.0.0.0
Access read-write

priority

1 is the lowest priority. When conflicting rules match, the one with the highest priority is used.

Values 1 to 2147483647
Default 1
Access read-create

priWeight

The preference for this line to be active. A value of 1 is low preference, and 10 is high. Default is 5 to match the secondary line so the default is that both lines are equal and whichever becomes active first remains the active line until an error condition occurs. Value of 0 is reserved.

Values 0 to 10

Default 0

Access read-write

privatelp (DynOmciCpePortFwd)

The port IP Address with which to send the traffic.

Default 0.0.0.0

Access read-write

privatePort (DynOmciCpePortFwd)

The port number with which to send the traffic.

Values 0 to 65535

Default 0

Access read-write

processorType

A string is returned that describes the processor type used on the unit or card.

Default 0

Access init-only

profileIndex

The index of OMCI profile records. This attribute will be auto-created or selected by DZMS server. This is created by using next available profileIndex of OMCI ME profiles.

Values NUMBER(0..2147483647)

Access read-only

profileIndex (for DynOmciCpePweSubscriber)

This attribute points to the associated PWE profile. 1 indicates a default profile is used.

Values Unsigned32 (1..2147483647)

Default 1

Access read-write

profileName (for DynOmciCpeIpServer, DynOmciCpeVoipServer, DynOmciCpeVoipFeatures, DynOmciCpeVoipMedia, DynOmciCpePwe, DynOmciCpeVideo, DynOmciCpeVideoAcl, DynOmciCpeTrafficMgmt, DynOmciCpeStaticRouteList, DynOmciCpeSystemCommon, DynOmciCpeWlan, DynOmciCpeWlanAdv, DynOmciCpeDhcpServer, DynOmciCpePortFwdList, DynOmciCpeDnsHostList)

Unified Service Provisioning OMCI CPE profiles for the dynamic bridging. Including Dynamic OMCI CPE profiles and Residential Gateway feature related OMCI CPE profiles. This field is only writable upon creation and must be unique within its table. The default value of profileName for DynOmciCpeDnsHostList is zhone.

Values STRING (SIZE (0..32))

Access read-create

profileName (for DynOmciCpeCfgRule)

The profile name of the rule.

Values STRING (SIZE (0..32))

Access read-create

projectionName

Values 0 to 32

Access read-only

protocol

Enables or disables Challenge Handshake Authentication Protocol (CHAP).

Values enabled

disabled

Default disabled

Access read-write

protocol (VoipServer)

The signalling protocol used for VOIP. Choices are SIP (Session Initiation Protocol) and MGCP (Media Gateway Control Protocol).

Values 1: sip
2: mgcp
3: megaco
Default 1: sip
Access read-create

protocol (DynOmciCpePortFwd)

Indicate which protocols to monitor for the port numbers.

Values 1: none
2: tcp
3: udp
4: tcpudp
5: icmp
6: icmpv4
Default 1: none
Access read-write

protocolEmulation

Indicates whether the device is acting as Network-side or CPE with respect to this DS1. For example, if the system is configured with a PBX on the network side of the device, then the device will act as a CPE with respect to the DS1. If the device is on the network side of a PBX, then this leaf should be set to network, as the PBX would be acting as a CPE.

Values 1: network
2: cpe
Default 1: network
Access read-write

protocolSpec

This variable specifies which variation of the v5.2 protocol is used by this V5.2 IG.

Values 1: edition1

The V5.2 specification defined in ETSI ETS 300 347-1 ed.1 (1994-09).

2: edition2

The V5.2 specification defined in ETSI EN 300 347-1 V2.2.2 (1999-12).

Default 1: edition2

Access read-write

providerId

Unique service provider ID. This value is used throughout the management system to link objects with customers or service providers.

Values 0 to 2147483647

Default 0

Access read-write

provVariantReq

A request by the user regarding the provisioning variant.

Values 1: noRequest

The default value. Setting the variable to **noRequest** has no effect.

2: verifywithle

A request to verify if the LE is ready for reprovisioning.

3: requestfromle

A request to get the LE Variant ID and Interface ID.

4: switchletolocal

A request to the LE side to change its provisioning variant to be the same as **local-prov-variant**.

Default 3: requestfromle

Access read-write

provVariantReqStatus

Status result for v52IgProvVariantRequest.

Values 1: verifyWithLeSucceeded
 2: verifyWithLeFailed
 3: requestFromLeSucceeded
 4: requestFromLeFailed
 5: switchToLocalSucceeded
 6: switchToLocalFailed

Access read-only

Notes When v52IgProvVariantRequest has a value of noRequest, the value for this variable is invalid.

psdShape

This field is used to configure the VDSL2 PSD Shape on Vectoring enabled blades. Vectoring expects all ports to have the same or compatible PSD shapes. When set to a PSD shape this field updates the PSD shape for all ports on the blade and sets the vectoring applicaton PSD shape.

Values 0 to 255

Default 255

Access read-only

pstnLayer3StartAddr

The start address for PSTN users. When PSTN users are added, they must have an address greater or equal to this one. This value must match the value on the switch.

Values 0 to 32768, inclusive

Default 0

Access read-write

ptypePayload

This attribute specifies the RTP payload type for the payload channel in the TDM to PSN direction. Assignable PTYPES lie in the dynamic range 96..127.

Values Integer32 (96..127)

Default 96

Access read-write

ptypeSignalling

This attribute specifies the RTP payload type direction for the signaling channel in the TDM to PSN. Assignable PTYPES lie in the dynamic range 96..127. If signalling is not transported in its own channel, the value should be set to 0. So the valid values used are 0, 96-127.

Values Integer32 (0..127)

Default 0

Access read-write

publicAddr (PAT)

The IP address that is acting as the port forwarding agent, this should be a public internet address.

Access read-write

publicPort (PAT)

The port number on the forwarding agent that is to be used.

Values 1 to 49151

Default 1

Access read-write

pulseInterDigitDuration

Outgoing Pulse dialing inter-digit timing in milliseconds. The default value of this object is 500 milliseconds.

Values 100 to 1000

Default 500

Access read-write

pulseRate

Outgoing pulse dialing rate in pulses per second (pps). The default value of this object is 10 pps.

Values 10 to 20

Default 10

Access read-write

pweIndex

This represents the CPE PWE index value.

Values Unsigned32 (1..2147483647)

Access read-create

Q

qualityProtocol

The link Quality Protocol used.

Values lqr

Default lqr

Access read-write

Notes The only valid quality protocol is LQR.

queryOrder

Kind of resolver query for this routing domain.

Values 1: hostsFirst

Searches the local hosts table first, then the list of nameservers.

2: dnsFirst

Searches the list of nameservers first, then the local hosts table

3: dnsOnly

searches only the list of nameservers

Default 1: hostsFirst

Access read-write

R

radiusAuthIndex

This object contains the index of the Radius Server profile to be used if RADIUS authentication is desired.

Values 0 to 2147483647

Access Read-Write

radiusKey (DynOmciCpeWlanSubscriber)

Remote Authentication Dial In User Server (RADIUS) authentication key.
This field cannot contain a SPACE and is returned as a string of asterisks.

Values String (1..13)

Access Read-Write

radiusPort (DynOmciCpeWlan)

UDP port to use for accessing the Remote Authentication Dial In User Server (RADIUS).

Values 0 to 99999

Default 1812

Access Read-Write

radiusServerIp(DynOmciCpeWlan)

IP address of the Remote Authentication Dial In User Server (RADIUS) used for 802.1x authentication.

Default 0.0.0.0

Access Read-Write

range1End

Last IP address of the first address pool in this subnet. If either the start or end range has a value of 0 then the entire address pool is ignored.

Default 0.0.0.0

Access read-write

range1Start

Starting IP address of the first address pool in this subnet. If either the start or end range has a value of 0 then the entire address pool is ignored.

Default 0.0.0.0

Access read-write

range2End

Last IP address of the second address pool in this subnet. If either the start or end range has a value of 0 then the entire address pool is ignored.

Default 0.0.0.0

Access read-write

range2Start

Starting IP address of the second address pool in this subnet. If either the start or end range has a value of 0 then the entire address pool is ignored.

Default 0.0.0.0

Access read-write

range3End

Last IP address of the third address pool in this subnet. If either the start or end range has a value of 0 then the entire address pool is ignored.

Default 0.0.0.0

Access read-write

range3Start

Starting IP address of the third address pool in this subnet. If either the start or end range has a value of 0 then the entire address pool is ignored.

Default 0.0.0.0

Access read-write

range4End

Last IP address of the fourth address pool in this subnet. If either the start or end range has a value of 0 then the entire address pool is ignored.

Default 0.0.0.0

Access read-write

range4Start

Starting IP address of the fourth address pool in this subnet. If either the start or end range has a value of 0 then the entire address pool is ignored.

Default 0.0.0.0

Access read-write

rasTimer

Reassembly timer, in milliseconds. The timer assures that the reassembly of an SSSAR-SDU does not exceed a specified time. When the time expires, the

I366.1 receive state machine goes into the abort state, looks for the end of the frame, and goes back to idle.

Values 0 to 2147486547

Default 0

Access read-write

Notes If no such timer supervision is required, set this value to 0 to disable the timer.

rate

This attribute sets the Ethernet port rate.

Values 1: auto

2: tenMbps

3: oneHundredMbps

4: oneThousandMbps

Default 1: auto

Access read-write

rate54G (DynOmciCpeWlanAdv)

The rate when the radio is operating in 802.11g mode. This object only applies when the Multiple Input Multiple Output (MIMO) 802.11m Rate is set to use 54g.

Values 1: auto

2: rate1m

3: rate2m

4: rate5dot5m

5: rate6m

6: rate9m

7: rate11m

8: rate12m

9: rate18m

10: rate24m

11: rate36m

12: rate48m

13: rate54m

Default 1: auto

Access read-write

rbitRxPolicy

This attribute defines the action toward the Nx64 TDM interface when remote failure is indicated on packets received from the PSN.

Values 1: nothing
2: alarm
3: idlechannel

Default 1: nothing

Access read-write

rbitTxClearPolicy

This attribute defines the number of consecutive valid packets that causes the transmitted R bit to be cleared in the TDM to PSN direction, removing the remote failure indication to the far end. The default value 0 selects the ONT internal policy.

Values Integer32 (0..255)

Default 0

Access read-write

rbitTxSetPolicy

This attribute defines the number of consecutive lost packets that causes the transmitted R bit to be set in the TDM to PSN direction, indicating lost packets to the far end. The default value 0 selects the ONT internal policy.

Values Integer32 (0..255)

Default 0

Access read-write

rdIndex

The Routing Domain (RD) table index.

Values 0 to 2147483647

Default 1

Access init-only

reasmMaxSize

Size of the largest IP datagram that this entity can reassemble from incoming IP fragments received on this interface.

Values 0 to 65535

Default 65535

Access read-write

reboot

Set this object to 'reboot' to reboot an ONU. The reboot command is sent to the ONU over the OMCI. Successful transmission of the reboot command to the ONU does not guarantee the ONU will reboot. The ONU may be in a fatal lockup which prevents the processing of the reboot command.

Values 0: none

1: reboot

Default 0

Access read-write

receiveAuthEnable

Enable or disable incoming authentication.

Values 1: enabled

2: disabled

Default 1: enabled

Access read-write

Notes This parameter provides a method to negotiate the use of a specific protocol for authentication.

receiveAuthProtocol

The preferred protocol used for incoming authentication.

Values 1: pap

2: chap

Default 1: pap

Access read-write

Notes If **receive-auth-enable** is enabled, the authentication protocol will be negotiated with the peer.

receiveTlp

Receive Transmission Level Point (RTLTP) settings control the amount gain or loss added to the incoming signal after it is decoded to analog. to increase the signal level set the RTLTP setting to higher values.

Values 1: fxsRtlpN9db
 2: fxsRtlpN8db
 3: fxsRtlpN7db
 4: fxsRtlpN6db
 5: fxsRtlpN5db
 6: fxsRtlpN4db
 7: fxsRtlpN3db
 8: fxsRtlpN2db
 9: fxsRtlpN1db
 10: fxsRtlp0db
 11: fxsRtlp1db
 12: fxsRtlp2db
 13: fxsRtlp3db
Default 10: fxsRtlp0db
Access read-write

Region_Id

Values an integer, 0 to 2147483647
Access read-only

regionalFocusX

Values -2147483647 to 2147483647
Default 15557
Access read-only

regionalFocusY

Values -2147483647 to 2147483647
Default 9318
Access read-only

regionalZoomFactor

Values 0 to 2147483647

Default 0.861998

Access read-only

register

For user agent clients, this value is inserted into the Expires header. For registrar servers, if a received request contained an Expires header, then the value in the header takes precedence.

Values 0 to 4294967295 (in seconds)

Access read-create

regOrLeaseUser

What kind of lease the user carries.

Values 1: regular

2: permanentLease

semiPermanentLease

Default 1: regular

Access read-write

relayId

A numeric identifier which identifies the relay.

Default 0

Access init-only

releaseTimer

This attribute contains a release timer defined in seconds. The value 0 specifies that the ONT is to use its internal default. The default value of this 45 attribute is 10 seconds.

Values Integer32(0..255)

Default 10

Access read-write

remIndex

The Cpe LLDP remote index.

Values 1 to 2147483647

Default 1

Access read-create

remoteInterfaceId

24-bit LE side V5.2 Interface identifier provisioned. This is read only can only be updated indirectly by setting **provVariantReq** to **requestFromLe**.

Values 0 to 16777215

Access read-only

remoteLoop

Which DS1 or E1 on the DS3 or E3 will be indicated in the remote DS1 loopback request. A value of 0 means no DS1s will be looped. A value of 29 means all DS1s or E1s will be looped.

Values 0 to 29

Default 0

Access read-write

remoteProvVariant

LE side provisioning variant used to differentiate variations in configuration. This is read only can only be updated indirectly by setting **v52IgProvVariantRequest** to **requestFromLe**.

Values 0 to 127, inclusive

Default 0

repeaterId

This value identifies if the STU is being used as a repeater.

Values 1: **repeaterEnable**

2: **repeaterDisable**

3: **noChangeRepeater**

Default 2: **repeaterDisable**

Access read-create

reserveEnd

The default number of IP addresses at the end of a subnet's IP address space that are reserved by the DHCP server. To override this default a specific subnet rule must be created for each subnet that needs to be handled differently.

Values 0 to 255

Default 5

Access read-write

reserveStart

The default number of IP addresses at the start of a subnet's IP address space that are reserved by the DHCP server. To override this default a specific subnet rule must be created for each subnet that needs to be handled differently. Default value is 5

Values 0 to 255

Default 5

Access read-write

reserveState

Reserves a port for provisioning.

Values 1: free

Port can be provisioned.

2: reserved

Port is reserved for provisioning.

3: provisioned

Port is provisioned.

Default free

Notes Applies to DSL ports only.

restartTimer

Number of seconds before retransmission of an Link Control Protocol (LCP) packet.

Values 1 to 999

Default 3

Access read-write

resync

OMCI ONT resync operation.

Values 0 and 1

Access read-write

retryInterval(DynOmciCpePppoe)

Time in seconds before retrying connection.

Values 1..2147483647

Default 3

Access read-write

retVal

DZS Enhanced Status return value, used to return LLDP error codes (LLDP disabled, no devices attached, etc) from the linecard back to the uplink.

Access read-create

revertiveClockSource

Indicates whether revertive behavior of the system clock is desired. If set to **true (1)**, the system will automatically revert to the higher weighted clock when it becomes available again. Used by systems that have the Clock Manager enabled.

Values 1: true

2: false

Default 1: true

Access read-write

rgMode (DynOmciCpeConnection)

Use to specify the RG service for a CPE. When creating or deleting a CPE connection, if the CPE does not support RG service then this object should not be provided. If a CPE does support RG service then this object must be provided. When reading this object, the value returned will be 'notApplicable'. This data is not preserved in the CPE connection profile.

Values	1: routed
	2: brouted
	3: rpppoe
	4: bpppoe
	5: bridged
	6: notApplicable
Default	6: notApplicable
Access	read-write

rgMode (DynOmciCpelfVlan)

Specifies the VLAN connection type of the interface.

Values	1: routed
	2: brouted
	3: rpppoe
	4: bpppoe
	5: bridged
	6: notApplicable
Default	6: notApplicable
Access	read-create

ringBack

The ring back is requested if this variable is set to on.

Values	1: on
	2: off
Default	2: off
Minimum	1
Maximum	2
Access	read-write

ringCadence

Ringling cadence to use in test.

Values 1: ringCadenceR0
 2: ringCadenceR1
 3: ringCadenceR2
 4: ringCadenceR3
 5: ringCadenceR4
 6: ringCadenceR5
 7: ringCadenceR6
 8: ringCadenceR7
 9: ringCadenceCommon
 10: ringCadenceSplash

Access read-create

ringFrequency

The ring frequency to be used in the FXS interface.

Values 1: ringFrequency25
 The ring frequency is 25 Hertz.

2: ringFrequency50
 The ring frequency is 50 Hertz.

3: ringFrequency20
 The ring frequency is 20 Hertz.

4: ringFrequency30
 The ring frequency is 30 Hertz.

Default 3: ringFrequency20

Access read-write

ringSource

Ring voltage source selection (internal or external).

Values 1: internal
 Select internal ring generator (located within MTAC) as source for ring voltage to line cards.

2: external
 Select external ring generator (input connector located on MTAC) as source for ring voltage to line cards.

Default 1: internal

Access read-write

ringTimer

Time in seconds to continue ringing.

Values 0 to 900 (in seconds)

Access read-create

robustNess

This attribute allows tuning for possible packet loss in the network.

The default value 0 causes the ONT to follow the IETF recommendation to copy the robustness value from query messages originating further upstream.

Values Integer32 (0..255)

Default 0

Access read-write

rogueOnuDetect

This attribute controls detection of rogue ONU on fiber.

Values 1: detection disabled

2: detect via measurement from rssi process

3: detect via indication from background process

4: normally off, but switch to rssi mode if rogue is suspected

Default 1: detection disabled

Access read-write

rohTimer

This attribute defines the time in seconds for the receiver off hook condition before ROH tone is applied. The value 0 disables ROH timing. The default value is 15 seconds.

Values Integer32 (0..255)

Default 15

Access read-write

role

RTCP packet interval, in milliseconds. Range is 2500 to 10000 ms. This object's value is only pertinent if RTCP has been enabled.

Values 0 to 2147483647

Default 5000

Access read-create

rowStatus

Used to create and/or remove rows from this table.

Values 1 to 6

Default 4

Access read-write

rstpEnable

Indicates if the RSTP shall be enabled on the bridge.

Values 1: true

2: false

Default 2: false

Access read-create

rtcpEnabled

Indicates whether the Real-Time Control Protocol is enabled for the device.

Values 1: true

2: false

Default 2: false

Access read-create

rtcpPktInterval

Values A valid traffic descriptor.

Default 0

Access read-write

rtpDscp (DynOmciCpeVoipServer)

This parameter indicates DSCP value for cpe voip rtp traffic.

Values Valid code values:

<0-63> Differentiated services codepoint value

af11	Mark packets with AF11 dscp (001010)
af12	Mark packets with AF12 dscp (001100)
af13	Mark packets with AF13 dscp (001110)
af21	Mark packets with AF21 dscp (010010)
af22	Mark packets with AF22 dscp (010100)
af23	Mark packets with AF23 dscp (010110)
af31	Mark packets with AF31 dscp (011010)
af32	Mark packets with AF32 dscp (011100)
af33	Mark packets with AF33 dscp (011110)
af41	Mark packets with AF41 dscp (100010)
af42	Mark packets with AF42 dscp (100100)
af43	Mark packets with AF43 dscp (100110)
cs1	Mark packets with CS1(precedence 1) dscp (001000)
cs2	Mark packets with CS2(precedence 2) dscp (010000)
cs3	Mark packets with CS3(precedence 3) dscp (011000)
cs4	Mark packets with CS4(precedence 4) dscp (100000)
cs5	Mark packets with CS5(precedence 5) dscp (101000)
cs6	Mark packets with CS6(precedence 6) dscp (110000)
cs7	Mark packets with CS7(precedence 7) dscp (111000)
default	Mark packets with default dscp (000000)
ef	Mark packets with EF dscp (101110)

Default 0**Access** read-write

rtpTimeStampMode

This attribute determines the mode in which RTP timestamps are generated in the TDM to PSN direction.

Values 1:unknown

2:absolute

3:differential

Default 1:unknown**Access** read-write

rtsThreshold (DynOmciCpeWlanAdv)

The packet size of a request-to-send (RTS) transmission. A low threshold implies RTS packets are sent more frequently, thus requiring more bandwidth but ensuring packet transmission on a busy network.

Values 0..2347**Default** 2347**Access** read-write

rxACCMMap

The Asynchronous Control Character (ACC) map that the local PPP entity requires for use on its receive side. In effect, this is the ACC map that is required in order to ensure that the local modem will successfully receive all characters.

Access read-write

Notes Changing this parameter will take effect when the link is next restarted.

rxGain

This attribute specifies a gain value for the received signal in the form of a 2s complement number. Valid values are -120 (-12.0 dB) to 60 (+6.0 dB).

Values -12..6

Default 0

Access read-write

rxTrafficDesc

The traffic descriptor used for the receive direction.

Values A valid traffic descriptor.

Default 0

Access read-write

rxTrafficDescrIndex

The value of this object identifies the row in the ATM Traffic Descriptor Table which applies to the receive direction of this VCL.

Values 0 to 2147483647

Default 0

Access read-write

S

scrambleEnabled

The status of the DS3 Scramble mode.

Values 1: true
Scramble mode is enabled

2: false
Scramble mode is disabled.

Default 1: true

Access read-write

scrambler

Description.

Values 1: Enabled

2: Disabled

Default 1: Enabled

Access read-write

sdpEnable

Enable or disable Self-Describing Padding (SDP) negotiation.

Values 1: enabled

2: disabled

Default 2: disabled

Access read-write

secLineId

The name as entered in the line profile table for the profile of the secondary line. The default value of 0 indicates that no secondary line is configured. When this field is set, the matching lineProfile entry will also be updated. Note that the matching entry (ifIndex) must be admined down before this field can be modified.

Values 0 to 2147483647

Default 0

Access read-write

secNameServer

The IP address of the secondary domain name server that the client should use for DNS resolution.

Default 0.0.0.0

Access read-write

secondaryDns

This attribute specifies the secondary DNS IP address. If this value is 0.0.0.0, no secondary SIP DNS is defined.

Values IP address
Default 0.0.0.0
Access read-write

secondaryPoint

Values 0 to 32
Default 0 0
Access read-only

secondaryServer

This attribute contains the name (IP address or resolved name) of the secondary or backup MGC that controls the signaling messages.

Values IP address
Default 0.0.0.0
Access read-write

secondaryServerIP

The IP address of the secondary NTP server. If the primary NTP server is not available, the device will use the secondary server.

Values A valid IP address
 A value of 0.0.0.0 disables NTP.
Default 0.0.0.0
Access read-write

secondNameServer

The IP address of the secondary name server for this routing domain. This nameserver is queried if the first nameserver cannot resolve the query. If a secondary name server has not been defined, this field is set to 0.0.0.0

Default 0.0.0.0
Access read-write

secureFwd (DynOmciCpelpServer)

When this mode is enabled, packets are not flooded to all ports. Instead, all packets are forwarded to the port that is designated as the uplink port. In this mode, users are prevented from directly communicating with each other, and broadcast frames are discarded.

Values 1: Enabled

2: Disabled

Default 2: Disabled

Access read-write

secWeight

The preference for this line to be active. A value of 1 is low preference, and 10 is high. Default is 5 to match the primary line so that the default is that both lines are equal and whichever line becomes active first remains the active line until an error condition occurs. Value of 0 is reserved.

Values 0 to 10

Default 0

Access read-write

seltCable

This object configures the selt algorithm with the type of cable being tested. For real-world testing with actual wire, this should be set to 'real'. In a lab environment, this can be set to match the type of loop simulator being used. The values supported depend on which chip is being used to perform the selt measurement. Real cable is always supported. The Conexant G24 also supports simulated cable using a Consultronics/Spirent DLS90 or DLS400 line simulator.

Other chips may support other options.

Values 1: real

2: simulatedWithDls90

3: simulatedWithDls400

Default 1: real

Access read-write

seltCmd

This object corresponds to the list of test operations that can be performed. Reading this object will always return the value 'noOp (1)'.

If SELT test is already running on an ADSL interface, the only possible test command is 'stop (1)'.

The clearResults(4) command clears test results by removing an entry in both the zhoneAdslSeltLcTable and the zhoneAdslSeltLnfTable. After the execution of this command, the appropriate value for zhoneAdslSeltStatus would be resultsCleared(5).

Values **1: noOp**
used when reading the object

2: start
start test

3: stop
stop an initiated test

4: clearResults
clears the results of that test

Default **1: noOp**

Access read-write

seltDuration

This object provides the maximum time in seconds that the SELT test measurement can take. If the test runs longer than this, it will be aborted, and the results (if any) will be removed.

Values **0 to 4294967295 (in seconds)**

Default **600**

Access read-write

seltGaugeExpected

This object configures the expected wire gauge of the loop. The values supported depend on which chip is being used to perform the selt measurement. For example, the Conexant G24 must be configured as either awg24 or awg26 (or the metric equivalents .50mm or .40mm respectively).

The values are:

unknown (1),
AWG--
awg26 (26),
awg24 (24),
awg22 (22),

awg19 (19),

Metric--

metric32 (1032), -- (0.32mm)

metric40 (1040), -- (0.40mm)

metric50 (1050), -- (0.50mm)

metric63 (1063), -- (0.63mm)

metric90 (1090), -- (0.90mm)

Metric/Japan--

metricJapan32 (2032), -- (0.32mm)

metricJapan40 (2040), -- (0.40mm)

metricJapan50 (2050), -- (0.50mm)

metricJapan65 (2065), -- (0.65mm)

metricJapan90 (2090) -- (0.90mm)

Values 1: unknown

19: awg19

22: awg22

24: awg24

26: awg26

1032: metric32

1040: metric40

1050: metric50

1063: metric63

1090: metric90

2032: metricJapan32

2040: metricJapan40

2050: metricJapan50

2065: metricJapan65

2090: metricJapan90

Default 26: awg26

Access read-write

sendCallProceedingTone

Indicates whether on call originations destined to the PSTN, the device should play a call proceeding tone while waiting for ring-back tone from the switch.

Values 1: true
2: false
Default 2: false
Access read-create

sendCode

What type of code is being sent across the DS3/E3 interface by the device. (Optional for E3 interfaces.) Setting this variable causes the interface to begin sending the code requested.

Values 1: **dsx3SendDS1LoopCode**
Sending looped or normal data
2: **dsx3SendLineCode**
Sending a request for a line loopback
3: **dsx3SendPayloadCode**
Sending a request for a payload loopback
4: **dsx3SendResetCode**
Sending a loopback deactivation request
5: **dsx3SendDS1LoopCode**
Requesting to loopback a particular DS1/E1 within a DS3/E3 frame
6: **dsx3SendDS1LoopCode**
Sending a test pattern.
Default 1: **dsx3SendDS1LoopCode**
Access read-write

sendAuthEnable

Enable or disable outgoing authentication.

Values 1: enabled
2: disabled
Default 2: disabled
Access read-write

sendAuthIdentity

The identity (PAP peer ID or CHAP name) used with **send-auth-protocol** to look up the outgoing authentication information.

Default empty string

Access read-write

sendAuthProtocol

the protocol used within **send-auth-identity** to look up the outgoing authentication information.

Example: 'ZNTS' for Zhone.

Values 1: pap

2: chap

Default 1: pap

Access read-write

serialNoVendorId

This is a string of 4 ascii characters representing the vendor ID portion of the onu serial number.

Example: 'ZNTS' for Zhone.

Default ZNTS

Access read-write

serialNoVendorSpecific

This is a string of maximum of 10 ascii decimal digits representing the vendor-specific portion of the onu serial number.

Default 0

Access read-write

serialNoVendorSpecificHex

This is a string of maximum of 9 hexa-decimal digits representing the vendor-specific portion of the onu serial number.

Default 0

Access read-write

serialNum

The vendor-specific serial number as advertised by the remote endpoint.

Access read-create

serialNumber

ONU serial number.

Values 1 to 32

Access read-write

serverIdType

Identifies the voip server type

Values 1: longboard

- 2: asterisk
- 3: sipexpressrouter
- 4: metaswitch
- 5: sylanro
- 6: broadsoft
- 7: ubiquity
- 8: generalbandwidth
- 9: tekelec-T6000
- 10: generic
- 11: sonus
- 12: siemens
- 13: tekelec-T9000
- 14: lucent-telica
- 15: nortel-cs2000
- 16: nuera
- 17: lucent-imerge
- 18: coppercom
- 19: newcross
- 20: cisco

Default 10: generic

Access read-create

serverIndex

This represents the unique CPE Ip server index value. This is a mandatory attribute for addition operation.

Values Unsigned32 (1..2147483647)

Access read-create

serverIndex (DynOmciCpeDhcpServer)

This represents the CPE DHCP server index value.

Values Unsigned32 (1..2147483647)

Access read-create

ServerIndex(for DynOmciCpeVoipServer)

This represents the unique CPE Voip Server Index value. This is a mandatory attribute for addition operation.

Values Unsigned32 (1..2147483647)

Access read-create

serverIndex (for DynOmciCpeSipDialPlan)

This represents the SIP dial plan server index value.

Values Unsigned32 (1..2147483647)

Access read-write

serverIndex (for DynOmciCpelp)

This attribute points to the cpe-ip-server profile associated with this host IP. When cpe-ip profile is missing, use the default cpe-ip-server profile with DHCP enabled.

Values Unsigned32 (1..2147483647)

Default 1

Access read-write

serverListProfile (DynOmciCpeDhcpServer)

Index of the dhcp-conditional-server-list profile associated with this entry, or 0 if none.

Values 0..2147483647

Default 0

Access read-write

serverProfileIndex

This attribute points to the associated cpe-voip-server profile.

Values Unsigned32 (1..2147483647)

Default 1

Access read-write

serviceCategory

The ATM service category. DZS devices support currently support the following classes of traffic:

- Unspecified bit rate (UBR) for data. Supported on the BAN, MALC, Z-Edge.
- Constant bit rate (CBR) for voice. Supported on the MALC, Sechtor 100, Z-Edge.
- Real time variable bit rate (rt-VBR) for voice or data, depending on the application. Supported on the BAN, MALC, Sechtor 100.
- Non-real time variable bit rate (nrt-VBR) for voice or data, depending on the application. Supported on the MALC.

serviceCategory

The ATM service category.

Values 2: CBR

Constant bit rate (CBR) for voice. Supported on the MALC, Sechtor 100, Z-Edge.

3: RTVBR

Real time variable bit rate (rt-VBR) for voice or data, depending on the application. Supported on the BAN, MALC, Sechtor 100.

4: NRTVBR

Non-real time variable bit rate (nrt-VBR) for voice or data, depending on the application. Supported on the MALC.

6: UBR

Unspecified bit rate (UBR) for data. Supported on the BAN, MALC, Z-Edge.

Default 6: UBR

Access read-write

servicePathKey

Values 0 to 2147483647
Default 0
Access read-write

serviceTemplate

The ifIndex of the virtual ONU supplying the service template for this rule.
Access read-create

serviceType

Values 0 to 2147483647
Default 0
Access read-write

serviceType (for DynOmciCpePwe object)

This attribute specifies the basic service type, either a transparent bit pipe or an encapsulation that recognizes the underlying structure of the payload.
Values 1: unstructured
2: octetalignedunstruct
3: structured
Default 1: unstructured
Access read-write

serviceType (for DynOmciCpelp object)

This represents the CPE Ip service types.
Values 1: voip
2: pwe
3: wifi
Access read-only

sessionExpiration

The time at which an element (proxy, UAC, UAS) will consider the SIP session timed out, if no successful session refresh transaction occurs beforehand.

Values 90 to 2147483647

Default 180

Access read-create

sessionMinSE

Because of the processing load of mid-dialog requests, all elements (proxy, UAC, UAS) can have configured minimum value for the session interval that they are willing to accept.

Values 90 to 2147483647

Default 180

Access read-create

sessionTimer

Support session timers.

Values 1: on

Session timers are supported, and SIP messages will have a 'Supported: timer' header.

2: off

Session timers are NOT supported, and SIP messages will NOT have a 'Supported: timer' header.

Default 2: off

Access read-create

sesThreshold

Number of lost, malformed or otherwise unusable packets expected in the PSN to TDM direction within a one-second interval that causes a severely errored second to be counted. Stray packets do not count toward a severely errored second, nor do packets whose L bit is set at the far end. The default value is 3.

Values Integer32 (0..65535)

Default 3

Access read-write

shared

TRUE if the GEM Port which uses this traffic descriptor shares and allocation ID with another GEM port. FALSE otherwise. When used more than once in the same ONU, the GEM ports are assigned the same alloc-id value.

Values 1: true

2: false

Default 2: false

Access read-write

shelfIndex

The global shelf index. This index can be used by tables that index by shelf and slot eliminating the need for shelf and slot columns in the table.

Values 1 to 255

Default 1

Access init-only

sigCode

This attribute specifies the POTS-side signaling.

Values 1: loopstart

2: groundstart

3: loopreversebattery

4: coinfirst

5: dialtonefirst

6: multiparty

Default 1: loopstart

Access read-write

signaling

This attribute specifies how CAS signalling is carried.

Values 1: none

2: inbandcas

3: outofbandcas

Default 1: none

Access read-write

signalingDscp (DynOmciCpeVoipServer)

This parameter indicates DSCP value for cpe voip signaling traffic.

Values Valid code values:

<0-63>	Differentiated services codepoint value
af11	Mark packets with AF11 dscp (001010)
af12	Mark packets with AF12 dscp (001100)
af13	Mark packets with AF13 dscp (001110)
af21	Mark packets with AF21 dscp (010010)
af22	Mark packets with AF22 dscp (010100)
af23	Mark packets with AF23 dscp (010110)
af31	Mark packets with AF31 dscp (011010)
af32	Mark packets with AF32 dscp (011100)
af33	Mark packets with AF33 dscp (011110)
af41	Mark packets with AF41 dscp (100010)
af42	Mark packets with AF42 dscp (100100)
af43	Mark packets with AF43 dscp (100110)
cs1	Mark packets with CS1(precedence 1) dscp (001000)
cs2	Mark packets with CS2(precedence 2) dscp (010000)
cs3	Mark packets with CS3(precedence 3) dscp (011000)
cs4	Mark packets with CS4(precedence 4) dscp (100000)
cs5	Mark packets with CS5(precedence 5) dscp (101000)
cs6	Mark packets with CS6(precedence 6) dscp (110000)
cs7	Mark packets with CS7(precedence 7) dscp (111000)
default	Mark packets with default dscp (000000)
ef	Mark packets with EF dscp (101110)

Default 0

Access read-write

signalMode

Signaling mode of the interface. Default is set to **robbedBit** for T1 platforms and **messageOriented** for E1 platforms.

Values 1: none

Indicates that no bits are reserved for signaling on this channel.

2: robbedBit

Indicates that DS1 Robbed Bit Signaling is in use. (Not supported for E1 lineTypes.)

3: bitOriented

Indicates that E1 Channel Associated Signaling is in use. (Only supported for E1 lineTypes.)

4: messageOriented

Indicates that Common Channel Signaling is in use either on channel 16 of an E1 link or channel 24 of a DS1.

5: other

Not available on BAN.

Default 2: robbedBit (for T1)

4: messageOriented (for E1)

Access read-write

signalType (E&M)

The signaling type of E&M tie trunk interface.

Values 1: winkStart

The calling side seizes the line by going off-hook on its E lead but waits for a short off-hook “wink” indication on its M lead from the called side before sending address information as DTMF digits.

2: immediateDial

The calling side seizes the line by going off-hook on its E lead and sends address information as DTMF digits.

3: delayDial

The calling side seizes the line by going off-hook on its E lead. After a timing interval, the calling side looks at the supervision from the called side. If the supervision is on-hook, the calling side starts sending information as DTMF digits; otherwise, the calling side waits until the called side goes on-hook and then starts sending address information.

Default 1: winkStart

Access read-write

signalType (Gr303crv)

Specifies the signaling method currently used by this GR-303 CRV.

Values 2: loopstart

3: groundstart

4: loopreversebattery

5: electronicbusinessset

Default 2: loopstart

Access read-write.

sigProtocol

This attribute specifies the VoIP signaling protocol to use.

Values 1 (sip), 2 (sipplar), 3 (h248), 4(mgcp)

Default 3 (h248)

Access read-write

sipDomain

This attribute contains the host or domain part of the SIP address of record for users connected to this ONT.

Values IPAddress

Default 0.0.0.0

Access read-write

sipRegRetryTime (DynOmciCpeVoipServer)

Sip Reg Retry Time specifies the SIP registration retry time in seconds. The default value is 60 seconds.

Values 0..2147483647

Default 60

Access read-write

sipRegExpiryTime

Sip Reg Expiration Time specifies the SIP registration expiration time in seconds. If its value is 0, the SIP agent does not add an expiration time to the registration requests and does not perform re-registration. The default value is 3600 seconds.

Values Unsigned32 (0..2147483647)

Default 3600

Access read-write

sipRegistrar

Sip Registrar points to a network address ME that contains the name (IP 25 address or resolved name) of the registrar server for SIP signaling messages.

Values IPAddress

Default 0.0.0.0

Access read-write

sipRegStartTime

Sip ReReg Head Start Time specifies the time in seconds prior to timeout that causes the SIP agent to start the re-registration process.

Values Unsigned32 (0..2147483647)
Default 360
Access read-write

slanId

The secure LAN ID for SLAN Tag Insertion.

Values Integer32(0 to 4095)
Access read-create

slanId (DynOmciCpelfVlan)

The secure LAN ID for SLAN Tag Insertion.

Values Integer32(0 to 4096)
Default 0
Access read-create

slanTpId

The value of the tag protocol identifier that is being used for QinQ packets. (0x8100 is default.).

Values 1. 33024 (0x8100)
 2. 34984 (0x88A8)
 3. 37120 (0x9100)
 4. 37376 (0x9200)
Default 33024
Access read-write

slanXlatId

If a packet's SLAN ID value matches this object's value then the packet's SLAN ID will be translated to the value of bridgeIfStagId. A zero value disables translation. This attribute is shown as Original SLAN ID in DZMS client.

Values 0 to 4095
Default 0
Access read-create

slanXlateId (DynOmciCpeConnection)

The Secure LAN Id for SLAN Tag Insertion.

Values Integer32 (0..4096)

Default 0

Access read-write

slanXlateId (DynOmciCpelfVlan)

The Secure LAN Id for SLAN Tag Insertion.

Values Integer32 (0..4096)

Default 0

Access read-create

slanXlateCos (DynOmciCpeConnection, DynOmciCpelfVlan)

The Class-Of-Service (COS) value used for SLAN Tag Insertion.

Values Integer32 (0..8)

Default 0

Access read-write

slanXlateTpId (DynOmciCpeConnection, DynOmciCpelfVlan)

The value of the tag protocol identifier that is being used for QinQ packets.

Values 33024 (0x8100)

34984 (0x88A8)

37120 (0x9100)

37376 (0x9200)

Default 33024 (0x8100)

Access read-write

slot

This field is used to configure which slot will use the VDSL2 PSD Shape configured in the **psdShape** field for the given index.

Access Read-only

slotIndex

The global slot index. This index can be used by tables that index by shelf and slot eliminating the need to include shelf and slot columns in the table.

Values 1 to 17

Default 1

Access init-only

slotsRequired

The slots required.

Values 1 or 2

Default 1

Access read-only

slotsUsed

Values 0 to 18

Default 0

Minimum 0

Maximum 18

Access read-only

softSwitch

Soft switch identifies the SIP gateway softswitch vendor. The format is four ASCII coded alphabetic characters [A..Z] as defined in ANSI T1.220. A 30 value of four null characters indicates no particular vendor.

Values STRING (SIZE (0..32))

Access read-write

softwareRev

The vendor-specific software revision string as advertised by the remote endpoint.

Access read-create

spectralPlan

The VDSL spectral plan.

Values	1: plan-998-1 Low frequency band, US standard. Optimal performance. 2: plan-997-1 Low frequency band, European standard. Reduced performance. 3: asym-vdsl-1 Not currently supported. 4: asym-vdsl-2 Not currently supported. 5: asym-vdsl-3 Not currently supported.
Default	1
Access	read-write

srcAddr

Source IP address, the IP address that accept packets from.

Access	read-create
---------------	-------------

srcAddrHigh

The inclusive upper bound for the source IP address range. See the filterStatementEntry description for details.

Default	0.0.0.0
Access	read-create

srcAddrLow

The inclusive lower bound for the source IP address range. See the filterStatementEntry description for details.

Default	0.0.0.0
Access	read-create

srcIP

This is the source IP address. Default value is 0.0.0.0, it specifies that the source IP address is to be ignored.

Values	IPAddress
Default	0.0.0.0

Access read-write

srcPortHigh

The inclusive upper bound for the transport layer source port range. See the filterStatementEntry description for details.

Values 0 to 65535

Default 0

Access read-create

srcPortLow

The inclusive lower bound for the transport layer source port range. See the filterStatementEntry description for details.

Values 0 to 65535

Default 0

Access read-create

specDescription

Textual description of the filter specification. This should briefly describe the nature of the filter defined by the associated filter statements.

Values An octet string of maximum of 255 characters

Access read-create

specIndex

The index that identifies an entry in the filterSpecTable. The fltGlobalIndexNext object is used to determine the next value of this object.

Values 0 to 2147483647

Default 0

Access init-only

sscsType

The SSCS type that applies to the CID.

Values 1: i3661

2: i3662

Default 2: i3662

Access read-write

Notes Channels that carry media streams (POTS or ISDN B-channels) use the SSCS defined in I.366.2. Channels that carry control and management plane traffic (CCS, ELCP, ISDN D-channels and LES-EOC) use the SSCS defined in I.366.1.

ssid (DynOmciCpeWlanSubscriber)

The SSID assigned to the wireless LAN interface.

Values 1..32

Access read-write

ssrsPayload

This attribute specifies the RTP synchronization source for the payload channel in the TDM to PSN direction.

Values Integer32 (0.. 2147483647)

Default 0

Access read-write

ssrsSignalling

This attribute specifies the RTP synchronization source for the signaling channel in the TDM to PSN direction. If signalling is not transported in its own channel, this value should be set to 0.

Values Integer32 (0.. 2147483647)

Default 0

Access read-write

startAddress (DynOmciCpeDhcpServer)

First address in the pool to be assigned by the DHCP server.

Values IP address

Default 0.0.0.0

Access read-write

stdAdminStatus

The desired operational status of this PW.

Values 1 (up), 2(down), 2 (testing)

Default 2

Access read-create

stdDescription

A textual string containing information about the PW. If there is no description this object contains a zero length string.

Values Char (1..255)

Default empty string

Access read-create

stdInboundLabel

The PW label used in the inbound direction (i.e. packets received from the PSN). It may be set manually if pwOwner is 'manual', otherwise setting done automatically. For MPLS, MPLS over IP or MPLS over GRE PSN, it represents the 20 bits of PW tag, for L2TP it represents the 32 bits Session ID and for IP PSN it represents the source UDP port number. If the label is not yet known (signaling in process), the object SHOULD return a value of 0xFFFFFFFF.

Values 56251..60100

Access read-create

stdOutboundLabel

The PW label used in the outbound direction (i.e. toward the PSN). It might be set manually if pwOwner is 'manual', otherwise setting done automatically. For MPLS, MPLS over IP or MPLS over GRE PSN, it represents the 20 bits of PW tag, for L2TP it represents the 32 bits Session ID and for IP PSN it represents the destination UDP port number. If the label is not yet known (signaling in process), the object SHOULD return a value of 0xFFFFFFFF.

Values 2142, 56251..60100

Access read-create

stdPeerIPAddress

This object contains the value of the peer node address of the PW/PE maintenance protocol entity

Access read-create

stagCOS

The s-tag COS used for s-tag insertion.

Values 0 to 7

Default 0

Access init only

stagId

The s-tag value to put on packets that are received on this interface.

Values 0 to 4095

Default 0

Access init only

stagTPID

This value of the tag protocol identifier that is being used for Q-in-Q packets (0x8100 is the default).

Values 33024 | 34984 | 37120 | 37376

Default 33024

Access init only

standard

This variable determines the SHDSL standards

Values 1: annex-a

2: annex-b

Default 2: annex-b

Access read-create

startMulticastAddress

The starting address for the multicast map.

Values A valid IP address

Default 0.0.0.0

Access read-write

startVci

The starting VCI for the multicast map.

Values 32 to 65535

Default 32

Access read-write

startupCheckLinkId

Indicates whether or not a Link ID check will be performed when the V5.2 interface is starting up.

Values 1: true

2: false

Default 2: false

Access read-write

startupMargin

The value of this variable is used to negotiate the bit rate during startup. The amount of margin is specified in dB and its value ranges from 0 to 15. Current Implementation supports only 0dB.

Values 0 to 15

Default 6

Access read-create

staticProfileIndex (DynOmciCpeSystemCommon)

Address of the static-route-list profile associated with this CPE.

Values 0..2147483647

Default 0

Access read-write

statusText

Blob of status information returned. Format is:

PhysPort,attrId1,attrName1,attrNumValue1,attrStrValue1...
attrIdn,attrNamen,attrNumValuen,attrStrValuenwhere

PhysPort = OMCI port ID value. For example, a value of 1025 indicates the Ethernet port 1 in the ONT. 1025 is 0x401 in hex format. 0x400 is the UNI type and 1 is the UNI #. For CIG UNI type, 0x400 and 0x500 are Ethernet, 0x600 is POTS, and 0x700 is RF video, Wifi is not an OMCI managed UNI port.

attrIdn = index number (0-15) of the attribute within the the ME (ascii decimal)

attrNamen = name of the attribute (returned by OCS)

attrNumValuen = value of the attribute (ascii decimal)

attrStrValuen = interpreted value ('up', 'down', etc)

n = number of attributes returned = number requested.

Maximum number of attributes per ME is 16.

For example, if you got a statusText value as :

1025,4,AdminState,0,up,5,OperState,1,inactive,6,ConfigInd,0,no connection.

This is a variable number of 4 object groupings, and here are the explanation:

- 1025 indicates the Ethernet port 1 in the ONT
- For group (4,AdminState,0,up), 4 is the ME attribute ID which translates to “AdminState” has a value of “0” which translates to “up”.
- For group (5,OperState,1,inactive), 5 is the ME attribute ID which translates to “OperState” has a value of “1” which translates to “inactive”.
- For group (6,ConfigInd,0,no connection), 6 is the ME attribute ID which translates to “ConfigInd” has a value of “0” which translates to “no connection”.

Values Octet String (0..512)

Access read-only

statusTrapEnable

This field describes the enable status of the Status Trap alarm. If this field is true (1) then the Status Trap alarm is enabled. If this field is false (2) then the Status Trap alarm is disabled.

Values 1: true

2: false

Default 1: true

Access read-write

startupUnblockUserPorts

Indicates whether or not user ports will be unblocked when the V5.2 interface is starting up.

Values 1: true

2: false

Default 2: false

Access read-write

starts

Time in GMT that the lease was assigned.

Values 0 to 4294967295

Default 0

Access read-write

state

The state.

Values 0 to 32

Access read-write

statEnabled

Enables or disables collecting statistics for the OID specified in the statInstance.

Values 1: false

2: true

Default 2: true

Access read-write

statOid

The SNMP Object Identifier (OID) to collect as part of the bulk statistics feature.

Values A valid SNMP OID

Access read-write

statInstance

The SNMP OID instance to collect as part of the bulk statistics feature.

Values A valid SNMP instance

Access read-write

status (PPPoA)

Indicates whether the authentication is valid or invalid.

Values 1: valid

2: invalid

Default 1: valid

Access read-write

Notes Setting this parameter to **invalid** causes the information for that record to become invalidated. Calls using the authentication information from an invalidated record will not be accepted or initiated.

statusTrapEnable

This field describes the enable status of the Status Trap alarm. If this field is true (1) then the Status Trap alarm is enabled. If this field is false (2) then the Status Trap alarm is disabled.

Values 1: true

2: false

Default 1: true

Access read-write

stbServerAddress

The IP address of the set-top box head-end server.

Values 4 to 4

Default 0.0.0.0

Access read-write

stickyAddr

Specifies that the DHCP server attempts to hand out the same IP address to the same host, whenever possible, based on hardware address.

Values 1: disable

2: enable

Default 1: enable

stmtIndex

The table index that identifies a filter statement. These indices should be sparse to allow for insertion into the list.

Values 0 to 2147483647

Default 0

Access init-only

stopBits

The port's number of stop bits.

Values 1: one
2: two
3: oneAndHalf
4: dynamic
Default 1: one
Access read-write

striplInsert

If true, enable VLAN Tagging.

Values 1: true
2: false
Default 2: false
Access read-create

subMode

This attribute specifies whether the Ethernet interface is bridged or derived from an IP router function.

Values 1: bridged
2: routed
Default 1: bridged
Access read-write

subnetGroup

The DHCP subnet group ID that indicates the address pool grouping to be used for the CPE subnet. This value is needed to support point-to-multipoint configurations.

Values 0 to 2147483647
Default 0
Access read-write

subPortIndex

This entry provides flexibility to define more than one interface on a port. The most common use of subport will be to handle structured TDM lines allowing for example an interface number to be assign to each DS1 in a DS3. Subport may also be used to handle other situations where a physical port contains more than one interface. This object has replaced by zhoneSubPortIndex.

Values 1 to 2147483647

Default 1

Access init-only

SubRegion_Group_1

Values an integer, 0 to 2147483647

Access read-only

SubRegion_Group_2

Values an integer, 0 to 2147483647

Access read-only

SubRegion_Group_3

Values an integer, 0 to 2147483647

Access read-only

SubRegion_Id_1

Values an integer, 0 to 2147483647

Access read-only

SubRegion_Id_2

Values an integer, 0 to 2147483647

Access read-only

SubRegion_Id_3

Values an integer, 0 to 2147483647

Access read-only

subscriberKey

Values 0 to 4294967295

Default 0

Access read-only

subscriberIndex(DynOmciCpeWlanSubscriber)

This represents the CPE WLAN subscriber index value.

Values 1 to 2147483647

Default 1

Access read-create

supervisoryDisconnect

Specifies whether a supervisory disconnect signal (CPC) will be presented to the interface when the connected party at the interface hangs up.

Values 1: true

2: false

Default 1: true

Access read-write

supportPassword (DynOmciCpeSystemCommon)

Log-in password for user account support.

Values String (0..16)

Access read-write

suppressionFirstOrder

This attribute specifies whether silence suppression is on or off.

Values 1: enabled

2: disabled

Default 2: disabled

Access read-write

suppressionSecondOrder

This attribute specifies whether silence suppression is on or off.

Values 1: enabled
2: disabled
Default 2: disabled
Access read-write

suppressionThirdOrder

This attribute specifies whether silence suppression is on or off.

Values 1: enabled
2: disabled
Default 2: disabled
Access read-write

suppressionFourthOrder

This attribute specifies whether silence suppression is on or off.

Values 1: enabled
2: disabled
Default 2: disabled
Access read-write

swAdmin

How the software upgrade is administered. Some options are only valid if warm upgrade is supported.

Values 1: **loadUpgradeSw**
The unit or card requests a load of the upgrade software. If an upgrade version of software is not available or the server is unable to respond to the load request, the card will set the cardSwStaus to upgradeloadFailed (warm upgrade only - this is ignored if warm upgrade is not supported).

2: **upgradeNow**
The unit or card upgrades to the already loaded upgrade software if warm upgrade is supported or reboot and load the upgrade software if warm upgrade is not supported. If the upgrade software has not been loaded or there is no grade software load available, the unit or card will set the cardSwStatus to upgradeFailed. When this value is set, the zhoneSysCardSwUpgradeEnable field is set to true in the genSystem MIB associated with this card.

3: upgradeOnReset

The unit or card will upgrade to the already loaded software the next time it is initialized. If the upgrade software is not loaded the card will set the cardSwStatus to upgradeFailed following reset.

4: reloadCurrRev

The unit or card will re-initialize and load the previous version of code prior to the upgrade, which is either the specific version or default version of the software, depending upon what has been provisioned. When this value is set, the zhoneSysCardSwUpgradeEnable field is set to false.

Default 4: reloadCurrRev

Access read-write

switchVendor

The switch vendor for this V5.2 IG.

Values 2: lucent

3: nortel

4: alcatel

5: ericsson

6: nokia

7: siemens

8: samsung

Default 5: ericsson

Access read-write

swPreviousVersion

The previous software version on the unit or card.

Values 0 to 32

Access read-only

swRunningVersion

The software version that is currently running on this card. This object complies to the DZS software revision syntax.

Values 0 to 32

Access read-only

symmetry

Symmetry of the IMA group.

Values 1 to 3

Default 1

Access read-create

syncBytesClkDivisor

Synchronization bytes clock divisor.

Values 0 to 65535

Default 1

Access read-write

sysCardSpecificSwFileName

The file or code image name of the specific software version that should be loaded by the unit or card for operation. If a failure occurs when trying to boot an upgrade version, the system will automatically revert to a good software version which is either the `zhoneSysCardSpecificSwFileName` if it is not NULL or the default for this card type if it is NULL. The operation will vary if the unit is a BAN card or a standalone device.

If the unit is a BAN, the card will load the indicated module and set `cardSwStatus` as appropriate. If the `cardSpecificFileName` is set to NULL then the card will load the image indicated by the `zhoneSoftwareLoadTable` in the system mib.

If the unit (card) is a Z-Edge, Sechtor 100A, or similar device, the system will first check if there is a local copy of the code image indicated. If there is, the card will load from the local copy. If the local copy is not the same revision as `zhoneSysCardSpecificSwFileName` the card will load this file.

Values 0 to 64

Access read-write

sysCardSwEnable

Whether or not the card has been enabled to be loaded. If this value is false, no software will be loaded, otherwise the logic flow defined in the table description will be used to determine which software to load.

Values 1 or 2

Default 1

Access read-write

sysCardSwType

The card type for the card in this shelf/slot. When first created, this value is unknown. When the card requests a load, the value in this field is compared to the requesting card type. If the value is unknown and a load can be serviced (the card is software enabled, and software is specified in either the upgrade, specific, or default for this type fields), the value is updated to the requesting card type. Otherwise, the type in this field is compared to the requesting card type.

If it matches, a load based on the other parameters is attempted. If the card types do not match, the load request is rejected and the initial request causes a trap to be generated. This processing is provided to ensure a card is not accidentally installed which attempts to load or start provisioning for a non-matching card type. The user may switch to a different card type than currently executing by setting this value to the new card type after the old card has been removed.

Values	1: unknownType 3: infoServices 5: voiceASP 6: t3Tdm 7: t3Atm 8: atmTrnkCr 9: ethernet 10: hdsl2 11: adsl 12: e3Atm 13: oc3AtmSm 14: oc3AtmMm 15: atmTrnk 16: atmTrnkAal25 17: atmTrnkT1e1 1001: sechotr100 2001: zedge64 3001: zplex10 5001: malcUplinkNt1 5002: malcPots 5003: malcMtac 5004: malcAdsl 5005: malcAdslPots 5006: maclGshdsl 5007: malcGshdslPots 5015: AdslAC6MalcCard
Default	10: hdsl2
Access	read-write

sysCardSwUpgradeEnable

This field indicates whether the zhoneSysCardUpgradeSwFileName should be used as the software image when the card requests a software load. This value can either be set directly or may be set indirectly through changes to the cardSwAdmin field in the card MIB.

Values 1 or 2

Default 2

Access read-write

sysCardUpgradeSwFileName

The file name of the software version that should be loaded by the unit or card into the upgrade region of the unit or card memory or storage. This file takes precedence over the current software file name and is used to attempt to upgrade the software on the system or card. If a user determines that the software upgrade version is acceptable, this value should be set in the CurrSwFileName.

Values 0 to 64

Access read-write

sysCfgCountryRegion

The country or region in which this system operates. This variable is used to adjust system settings for different functional modules. For example, both DSP tones and TDM compounding are country-specific settings that can be determined by this variable.

Values	1: Unknown 2: Argentina 3: Australia 4: Belgium 5: China 6: Costa Rica 7: Finland 8: France 9: Germany 10: Hong Kong 11: Italy 12: Japan 13: Korea 14: Mexico 15: Netherlands 16: New Zealand 17: Singapore 18: Spain 19: Sweden 20: Switzerland 21: UK 22: US 116: ireland 224: unitedarabemirates
Default	9: Germany (for S100E1 devices) 22: US (for other device types)
Access	read-only

sysContact

Textual identification of the contact person for this device, together with information on how to contact this person.

Access read-write

sysDescription

Textual description of the entity. This value should include the full name and version identification of the system's hardware type, software operating-system, and networking software.

Access sync-only

sysLocation

Textual description of the physical location of this device (for example, Telephone closet, 3rd floor).

Access read-write

sysName

The string value used to identify the system name of the remote system.

Access read-create

sysObjectID

The vendor's authoritative identification of the network management subsystem contained in the entity. This value is allocated within the SMI enterprises subtree (1.3.6.1.4.1) and provides an easy and unambiguous means for determining what kind of box is being manage.

Access read-only

sysRebootAdmin

Indicates which cards to reboot.

Values **1: System Reboot**

Reboot all the cards.

2: Reboot By Card Type

Reboot all cards of a particular card type (specified by sysRebootCardType).

Default 0

Access read-write

sysRebootCardType

Numerical representation of the unit/card type to be rebooted.

Values

- 1: unknownType
- 3: infoServices
- 5: voiceASP
- 6: t3Tdm
- 7: t3Atm
- 8: atmTrnkCr
- 9: ethernet
- 10: hdsl2
- 11: adsl
- 12: e3Atm
- 13: oc3AtmSm
- 14: oc3AtmMm
- 15: atmTrnk
- 16: atmTrnkAal25
- 17: atmTrnkT1e1
- 1001: sechotr100
- 2001: zedge64
- 3001: zplex10
- 5001: malcUplinkNt1
- 5002: malcPots
- 5003: malcMtac
- 5004: malcAdsl
- 5005: malcAdslPots
- 5006: maclGshdsl
- 5007: malcGshdslPots

Default 0

Access read-write

sysSwAdmin

System software administration.

Values

- 1: downLoadSw
Download the image from the Primary load Server. If that fails, try to download from secondary load server. This value is only allowed if the zhoneSysSwStatus field is NOT downLoadSwInProgress.

2: abortDownloadSw

Abort an in-progress download. This value is only allowed if the `zhoneSysSwStatus` is `downloadSwInProgress`.

3: delete

deletes the filename specified in `zhoneSysSwLocalFilePath`. This is used by the manager to manage the amount of storage space used by the system for sw load images. A delete will fail if the file is specified in either a default software entry, specific software entry, or an upgrade software entry. This is only supported on devices having a file system.

Default 0

Access read-write

sysSwLocalFilePath

The path on the system for the image to be stored.

Values 0 to 64

Access read-write

sysSwLogin

The login to be used on the remote system when downloading the image. This value is maintained throughout (i.e. persisted) and can be used for additional/future download requests. This login is used for both CLI 'update-image' commands and snmp/zms requests using this mib. The login is defaulted to NULL the first time the system is booted and is maintained across sw download requests and reboots. That is, the value is not cleared upon completion and can be reused by the manager. When read, the last value used is returned.

Values 0 to 32

Access read-write

sysSwPassword

The password to be used on the remote system when downloading the image. This value is maintained throughout (i.e. persisted) and can be used for additional/future download requests. This password is used for both CLI 'update-image' commands and snmp/zms requests using this mib. The password is defaulted to NULL the first time the system is booted and is maintained across sw download requests and reboots. That is, the value is not cleared upon completion and can be reused by the manager. When read, a NULL value is always returned.

Values 0 to 32

Access read-write

sysSwPriLoadServer

The IP address of the primary software load server. If the value of the `zhoneSysSwPriLoadServer` is set to 0.0.0.0 then no primary load server is available. If the system is unable to load the software file from the Primary Load Server, the system should attempt to load the software from the Secondary Load Server (`zhoneSysSecLoadServer`). This value must be set on each download request.

Default 0.0.0.0

Access read-write

sysSwRemoteFilePath

The path on the remote system for the image to be loaded.

Values 0 to 64

Access read-write

sysSwSecLoadServer

The IP address of the secondary or back-up software load server, if there is no secondary server this should be set to 0.0.0.0

Default 0.0.0.0

Access read-write

T

t200

The maximum layer two wait time for an acknowledgment of a transmitted frame. Units are in milliseconds, in increments of 50.

Values 100 to 1000, inclusive.

Default 1000

Access read-write

t203

Maximum layer two idle time before connection verification is necessary. Units are in seconds, in increments of ten.

Values 10 to 300, inclusive.

Default 10

Access read-only

TableBasedFilter

none : Dynamic IP Filtering on bridge based on DHCP, is disabled.

BIT 1(mac) : MAC address based Filtering based on DHCP, is enabled.

BIT 2(ip) : IP address based dynamic IP Filtering based on DHCP, is enabled.

Values 0: mac

1: ip

Default none**Access** read-create

targetWorseCaseSnrMgn

targetWorstCaseSnrMgn is a value used to compensate for the worst possible noise conditions in a bundle when link rates are negotiated. Where a value of zero would consider the worst noise condition and would train at a much lower rate to compensate for potential noise in the bundle later. Where a value higher than zero would consider a little less noise in the bundle and would train at a slightly higher rate than with the zero value.

Desired EFMcu port SNR Margin to be achieved on all PMEs assigned to the port, during initialization if efmCuTargetWorstCaseMode is TRUE. (The SNR margin is the difference between the desired SNR and the actual SNR).

Note that 802.3ah recommends using default Target SNR Margin of 5dB for 2BASE-TL ports and 6dB for 10PASS-TS ports in order to achieve mean Bit Error Rate (BER) of 10^{-7} at the PMA service interface. This object is read-write for the -O subtype EFMcu ports (2BaseTL-O/10PassTS-O) and not available for the -R subtypes.

Changing of the Target SNR Margin MUST be performed when the link is Down. Attempts to change this object MUST be rejected with the error inconsistentValue, if the link is Up or Initializing.

Note that current SNR Margin of the PMEs comprising the EFMcu port is represented by efmCuPmeSnrMgn. This object MUST be maintained in a persistent manner.

Values -10..21**Default** 1**Access** read-write

targetCurrCondMode

Enabling this mode only (disabling targetWorseCaseMode) will allow for the consideration of noise conditions that are currently occurring in the bundle at the time that data rates are being negotiated.

Values 1: True

2: False

Default 2: False

Access read-write

targetCurrCondSnrMgn

targetCurrCondSnrMgn is a value used to compensate for the current noise conditions in the bundle when link rates are being negotiated. Where a value of zero would give a lower train rate and a higher value would give a higher train rate. This mode can be affected by noise that is introduced in the bundle after train rates have been determined and can cause a loss of link and subsequent rates to be negotiated lower next time if the new noise is still persisting.

Values -10..21

Default 6

Access read-write

targetUni

Which unis from the template should be applied by this rule.

This consists of a uni type followed by one or more uni numbers, separated by commas, or with ranges indicated by '-'. The word 'all' may be used to indicate all ports of that type.

Example: eth 1-3.

In this example, apply services on the provisioned ethernet port on the service template to ethernet ports 1, 2, and 3 of the matched CPE.

If this field is blank, then the entire service template is applied to the CPE.

Access read-create

targetWorseCaseMode

Enabling this mode only (disabling targetCurrCondMode) will allow for the worst noise conditions in a bundle to be considered when negotiating data rates.

Values 1: True
2: False
Default 2: False
Access read-write

tdmJitterBufferDepth

The size of this buffer SHOULD be locally configured to allow accommodation to the PSN-specific packet delay variation. If configured to a value not supported by the implementation, the agent MUST return an error code 'jtrBfrDepth' in 'pwTDMConfigError '

NOTE: jitter buffers are a limited resource to be managed. The actual size should be at least twice as big as the value of pwTDMCfgJtrBfrDepth.

Values : 1.. 170000
Default 12500
Access read-create

tdmPayloadSize

The value of this object indicates the PayLoad Size (in bytes) to be defined during the PW setUp. Upon TX, implementation must be capable of carrying that amount of bytes. Upon RX, when the LEN field is set to 0, the payload of packet MUST assume this size, and if the actual packet size is inconsistent with this length, the packet MUST be considered to be malformed.

Values : 192..250
Default 192
Access read-create

tdmPktFillPattern

Filler byte pattern played out on the TDM interface if pwTDMCfgPktReplacePolicy was set to filler(3). Object MAY be changed only when the related PW is defined as not active.

Default value is 255, it indicates play All Ones, equal to AIS indications.

Values 0-255
Default 255
Access read-create

tdmPktReorder

If set True: as CE bound packets are queued in the jitter buffer, out of order packets are re-ordered. The maximum sequence number differential (i.e., the range in which re-sequencing can occur) is dependant on the depth of the jitter buffer. See pwTDMCfgJtrBfrDepth.

NOTE: Some implementations may not support this feature. The agent should then reject a SET request for true.

Values 1(True)
2(False)
Default 2
Access read-create

tdmPktReplacePolicy

This parameter determines the value to be played when CE bound packets have over/underflow the jitter buffer, or are missing for any reason. This byte pattern is sent(played)on the TDM line. Selecting implementationSpecific(2) implies agent specific algorithm. Selecting filler(3) requires setting of pwTDMCfgPktFiller.

Values 1 (All Ones)
2(Implementation Specific)
3 (Filler)
User defined
Default 1
Access read-create

testId

This object is only valid for Mtac Cards which support the new Legerity Suite of tests. This object identifies which test is to be run, or can be set to abort the current test.

Test results are sent to the zhoneMetallicTestResultsTable.

There are 3 Legerity software packages: Basic Test (VCP-BT), Advanced Test (VCP-AT), and Advanced Test Plus (VCP-ATP).

We currently support Basic Test package.

The following lists which tests are supported in each software package.

Calibration VCP-ATP

Foreign DC Voltage Test VCP-BT, VCP-AT, VCP-ATP

Foreign AC VoltageTest VCP-BT, VCP-AT, VCP-ATP
DC Loop Resistance Test VCP-BT, VCP-AT, VCP-ATP
Three-Element Insulation Resistance Test VCP-BT, VCP-AT, VCP-ATP
Five-Element Insulation Resistance Test VCP-ATP
Three-Element Capacitance Test VCP-BT, VCP-AT
Receiver Off-Hook Test VCP-BT, VCP-AT, VCP-ATP
Distance to Open Test VCP-BT, VCP-AT
Foreign AC Currents Test VCP-BT, VCP-AT
Ringer Equivalency Number Test VCP-BT, VCP-AT, VCP-ATP
DTMF and Pulse Digit Measurement Test VCP-BT, VCP-AT
Noise Measurement Test VCP-BT, VCP-AT, VCP-ATP
Signal to Noise Ratio Test VCP-ATP
Arbitrary Single Tone Measurement Test VCP-ATP
Tone Generation Test VCP-BT, VCP-AT
Trans-Hybrid Loss Test VCP-BT, VCP-AT, VCP-ATP
Draw and Break Dial Tone Test VCP-BT, VCP-AT
Inward Current Test VCP-ATP
DC Feed Self-Test VCP-BT, VCP-AT, VCP-ATP
On/Off Hook Self-Test VCP-BT, VCP-AT, VCP-ATP
Ringing Self-Test VCP-BT, VCP-AT, VCP-ATP
Ringing Monitor Test VCP-BT, VCP-AT, VCP-ATP
Metering Self-Test VCP-BT, VCP-AT
Transmission Self-Test VCP-BT, VCP-AT, VCP-ATP
Dialing Sef Test VCP-BT, VCP-AT
Howler Test VCP-BT, VCP-AT
Fuse Test VCP-ATP
Read Loop and Battery Conditions VCP-BT, VCP-AT, VCP-ATP

Values	1: none 2: runAllTests 3: abortTest 4: calibration 5: foreignDCVoltage 6: foreignACVoltage 7: dcLoopResistance 8: threeElementInsulationResistance 9: fiveElementInsulationResistance 10: threeElementCapacitance 11: receiverOffHook 12: distanceToOpen 13: foreignACCurrents 14: ringerEquivalencyNumber 15: dtmfAndPulseDigitMeasurement 16: noiseMeasurement 17: signalToNoiseRatio 18: arbitrarySignalToneMeasurement 19: toneGeneration 20: transHybridLoss 21: drawAndBreakDialTone 22: inwardCurrent 23: dcFeedSelf, 24: onAndOffHookSelfTest 25: ringingSelfTest 26: ringingMonitor 27: meteringSelfTest 28: transmissionSelfTest 29: dialingSelfTest 30: howlerTest 31: fuseTest 32: readLoopAndBatteryConditions
Default	1: none
Access	read-write

testLinkIfIndex

This object is used to designate an interface as the test link for use in the Test Pattern Procedure. The distinguished value of zero specifies that the implementation may choose the Test Link, in which case, the implementation may also choose the value of 'imaGroupTestPattern'. The value zero may also be used if no link has yet been added to the group. Note that this value is NOT the same as the Tx LID value, but instead either identifies the ifIndex value of the test link to be used by the Test Pattern Procedure (i.e., the link whose LID value is inserted in the Tx LID field of the transmitted ICP cells), or identifies that the implementation may choose the test link (if the value is zero).

Values 0 to 2147483647

Default 0

Access read-create

testMode

This field is used to set the metallic test mode for a given line. The mtac test mode may be changed only if the zoneMetallicInterfaceIndex is set, Otherwise it defaults to mtacModeNone, and can be changed again by setting InterfaceIndex to non-zero values.

Values 1: **mtacModeBridge**

All traffic on the given line is bridged across the mtac lines.

2: **mtacModeLookIn**

All inbound traffic on the given line originates exclusively to the mtac lines.

3: **mtacModeLookOut**

All outbound traffic on the given line originates exclusively to the mtac lines.

4: **mtacModeNone**

No mtac test is in progress.

Default 4: **mtacModeNone**

Access read-write

testPattern

The value of this object is used to specify the Tx Test Pattern in an IMA group loopback operation. A value in the range 0 to 255 designates a specific pattern. The distinguished value of -1 specifies that the implementation may choose the value. In this case, the implementation may also choose the value of 'imaGroupTestLinkIfIndex'. It is recommended that the specific value 255 not be used for testing, since by (R-137) the IMA interface is required to transmit 0xFF (i.e., 255) when the incoming test command is inactive or the test link is not detected, and thus it cannot be established for certain whether

255 was received due to an actual loopback operation or due to the normal operation of an IMA that is not performing (or else cannot perform) the test pattern procedure.

Values -1 to 255
Default -1
Access read-create

testProcStatus

This object is used to enable or disable the Test Pattern Procedure, and to note whether at least one link failed the test. The test is started by setting operating(2) status. If any link should fail the test, the IMA will set the status to linkFail(3). The linkFail(3) state will persist until either the disabled(1) state is set or until no instance of imaLinkTestProcStatus has the value linkFail(3). Only the values disabled(1) and operating(2) may be written. Writing the operating(2) value will not cause clearing of the linkFail(3) state.

Values 1: disabled
 2: operating
 3: linkfail
Default 1: disabled
Access read-create

text (OMCI ME profile)

This list of ME attributes that are retrieved from OMCI ME profile file. This is a large string and consists of all attribute names separated with colons.

Values STRING(SIZE(0..65535 characters))
Access read-only

thermoALowerThold

Specifies the lower threshold for thermometer A in degrees Celsius.

Values -10 to 5
Default -5
Access read-write

thermoAUpperThold

Specifies the upper threshold for thermometer A in degrees Celsius.

Values 45 to 80
Default 65

Access read-write

thermoBLowerThold

Specifies the lower threshold for thermometer B in degrees Celsius.

Values -10 to 5

Default -5

Access read-write

thermoBUpperThold

Specifies the upper threshold for thermometer B in degrees Celsius.

Values 45 to 80

Default 60

Access read-write

thirdNameServer

The IP address of the third name server for this routing domain. This nameserver is queried if the first nameserver and the secondary nameserver cannot resolve the query. If a third name server has not been defined, this field is set to 0.0.0.0

Default 0.0.0.0

Access read-write

threshold15MinLofs

The number of Loss of Framing seconds on a DSL interface within any given 15 minutes performance data collection period that causes the SNMP agent to send a trap. A value of 0 (zero) will disable the trap.

Values 0 to 900

Default 0

Access read-write

threshold15MinLols

The number of Loss of Link seconds on a DSL interface within any given 15 minutes performance data collection period that causes the SNMP agent to send a trap. A value of 0 (zero) will disable the trap.

Values 0 to 900

Default 0

Values	0: ts0	
		1: ts1
	2: ts2	
	3: ts3	
	4: ts4	
	5: ts5	
	6: ts6	
	7: ts7	
	8: ts8	
	9: ts9	
	10: ts10	
	11: ts11	
	12: ts12	
	13: ts13	
	14: ts14	
	15: ts15	
	16: ts16	
	17: ts17	
	18: ts18	
	19: ts19	
	20: ts20	
	21: ts21	
	22: ts22	
	23: ts23	
	24: ts24	
	25: ts25	
	26: ts26	
	27: ts27	
	28: ts28	
	29: ts29	
	30: ts30	
	31: ts31	

Access read-write

tmProfileIndex

This object points to the associated Traffic Management profile. If there is no traffic control for the UNI, set it to 0.

Values Unsigned32 (1..2147483647)

Default 0

Access read-create

tosCOS

Specifies the maximum number of unique multicast (video) streams that this interface has subscribed to.

Values 0 to 7

Default 0

Access init-only

tosOption

Specifies the IP TOS Class-of-Service bit setting on packets sent out this interface. The setting is stored in the tosCOS field.

Values 1: Disable (COS value is left untouched.)

2: Originate (Load COS bits only if we originate the packet. COS for packets routed through this interface are not affected.)

3: All (Load COS bits on all packets)

Default 1: Disable

Access init-only

totalMemory

The amount of physical memory contained by the unit or card.

Values 0 to 2147483647

Default 0

Access init-only

tpIndex (DynOmciCpelfVlan, DynOmciCpeLldpRem)

TpIndex identifies the Index of termination point associated with this MAC bridge port.

Values Integer32 (0..255)

Access read-create

tpType(DynOmciCpelfVlan, DynOmciCpeConnection, DynOmciCpeLldpRem)

TpType identifies the type of termination point associated with this MAC bridge port.

Values	physicalPathTerminationPointEthernetUNI(1), interworkingVCCTerminationPoint(2), mapperServiceProfile(3), ipHostConfigData(4), gemInterworkingTerminationPoint(5), multicastGEMInterworkingTerminationPoint(6), physicalPathTerminationPointxDSLUNIPartOne(7), physicalPathTerminationPointVDSLUNI(8), ethernetFlowTerminationPoint(9), physicalPathTerminationPointUNI(10), physicalPathTerminationPointRFvideoUNI(11), physicalPathTerminationPointIPTVUNI(12), h248VoiceTerminationPoint(13), sipVoiceTerminationPoint(14), sipPlarVoiceTerminationPoint(15), pweTerminationPoint(16), virtualEthernetInterfacePoint(17), dot11TerminationPoint(18), fiberLanTerminationPoint(19), tr69TerminationPoint(20), mgmtTerminationPoint(21), mgcpVoiceTerminationPoint(22)
Access	read-create

tr69Inform (DynOmciCpeSystemCommon)

Enable or Disable the generation of Inform messages to the ACS (Auto Configuration Server).

Values	1: enabled 2: disabled
Default	1. enabled

Access read-write

trafficClass

Whether traffic is Constant Bit Rate or Unspecified Bit Rate.

Values 1: cbr

2: ubr

Default 2: ubr

Access read-write

trafficDescrIndex

Any integer. DZMS overwrites the value with a unique value obtained from the device.

Values 1 to 2147483647

Default 2

Access init-only

trafficDescrParam1

The peak cell rate (PCR) of the interface. Once the peak cell rate (PCR) has been set, traffic policing ensures that the PCR limitations are enforced. The system will discard cells of incoming (ingress) traffic that exceed the PCR.

- Notes**
- Refer to the documentation that came with your device for information about supported traffic parameters.
 - The **trafficDescrParam1** and **trafficDescrParam2** parameters must be set to one of the configured rates in the **atm-vcl-param** profile.
 - Refer to the following specifications for more information about PCR:
 - ATM Forum, ATM User-Network Interface, Version 3.0 (UNI 3.0) Specification, 1994.
 - ATM Forum, ATM User-Network Interface, Version 3.1 (UNI 3.1) Specification, November 1994.

trafficDescrParam1

The peak cell rate, in cells/second, for the combined high- and low-priority traffic (the CLP=0+1 aggregate cell stream).

Values 0 to 2147483647

Default 100

Access read-write

trafficDescrParam2

Sustained cell rate (SCR) of the interface. Refer to the documentation that came with your device for information about supported traffic parameters.

Notes The **trafficDescrParam1** and **trafficDescrParam2** must be set to one of the configured rates in the **atm-vcl-param** profile.

trafficDescrParam3

Maximum burst size (MBS) of the interface. Refer to the documentation that came with your device for information about supported traffic parameters.

trafficDescrParam4

Cell delay variation tolerance (CDVT) of the interface. Refer to the documentation that came with your device for information about supported traffic parameters.

trafficDescrParam5

Currently not supported.

trafficDescrType

Traffic descriptor type. Note that not all products support all traffic types. Refer to the documentation that came with your device for information about supported traffic categories.

- Values**
- 2: atmNoClpNoScr**
(OID 1.3.6.1.2.1.37.1.1.2)
No CLP and no Sustained Cell Rate.
 - 4: atmClpTaggingNoScr**
(OID 1.3.6.1.2.1.37.1.1.4)
CLP with tagging and no Sustained Cell Rate.
 - 6: atmClpNoTaggingScr**
(OID 1.3.6.1.2.1.37.1.1.6)
CLP with no tagging and Sustained Cell Rate.
 - 7: atmClpTaggingScr**
(OID 1.3.6.1.2.1.37.1.1.7)
CLP with tagging and Sustained Cell Rate.
 - 9: atmClpTransparentNoScr**
(OID 1.3.6.1.2.1.37.1.1.9)
CLP transparent and no Sustained Cell Rate.

10: atmClpTransparentScr**(OID 1.3.6.1.2.1.37.1.1.10)**

CLP transparent and Sustained Cell Rate.

12: atmNoClpNoScrCdv**(OID 1.3.6.1.2.1.37.1.1.12)**

No CLP and no Sustained Cell Rate.

14: atmClpNoTaggingScrCdv**(OID 1.3.6.1.2.1.37.1.1.14)**

CLP with no tagging, Sustained Cell Rate.

15: atmClpTaggingScrCdv**(OID 1.3.6.1.2.1.37.1.1.15)**

CLP with tagging, Sustained Cell Rate.

trafficProfileTemplate

Specify the name of the GPON Traffic Profile (GTP) template. If trafficProfileIdx is included as part of OSS XML/ OSS request and it is >0, trafficProfileTemplate is not specified or specified as 0 as part of OSS XML/ OSS GEM port modify request, DZMS will check for the existence of matching GPON Traffic Profile on device and auto-create the GTP if it does not exist already.

Values A string of 60 characters**Access** write-only**transmitClockSource**

The transmit clock source.

Values 1: loopTiming

The recovered receive clock is used as the transmit clock

2: localTiming

A local clock source is used or that an external clock is attached to the box containing the interface

3: throughTiming

Transmit clock is derived from the recovered receive clock of another DS3 interface.

Default 1: loopTiming**Access** read-write**transmitPowerBackoffMode**

Determines if the transmit power backoff defined in SHDSL standard is used. The default value is backoffDisable

Values 1: backoffDisable
 2: backoffEnable
 3: noChangeBackoff

Default 2: backoffEnable

Access read-create

transmitTlp

In case of FXO, it specifies FXO attenuation and default value is 0dB. In case of FXS this setting controls the amount of gain or loss added to a voice signal from CPE before it is encoded to digital PCM. To increase the signal level, reduce the TTLTP setting to lower value. The default is fxsTtlp0db.

Values 1: fxsTtlp9db
 2: fxsTtlp8db
 3: fxsTtlp7db
 4: fxsTtlp6db
 5: fxsTtlp5db
 6: fxsTtlp4db
 7: fxsTtlp3db
 8: fxsTtlp2db
 9: fxsTtlp1db
 10: fxsTtlp0db
 11: fxsTtlpN1db
 12: fxsTtlpN2db
 13: fxsTtlpN3db

Default 10: fxsTtlp0db

Access read-write

transport

The underlying transport for the Pseudowire service.

Values 1: ethernet
 2: udp
 3: mpls

Default 2: udp

Access read-write

trapEnable (ULC)

Allows SNMP users to disable or enable the sending of line level traps from the device to the NMS.

Values 1: enabled
2: disabled
Default 1
Access read-write

trunkConditioning

The trunk conditioning setting specifies whether the FXS/FXO port should send the idle pattern to the network upon declaration of a Carrier Group Alarm (CGA) on the WAN link to which the port is assigned. In most cases leave the trunk conditioning setting at its default setting (off).

Values 1: off
2: idle
3: busy
Default 1: off
Access read-write

trunkRXT

Refers to the trunk side receiving ATM Traffic Descriptor template handle id.

trunkStartVci

Refers to the trunk side ATM VCL's VCI. During ATM Vc Cross Connect creation, if a card slot is chosen on the line side then the "trunkStartVci" value is used as a starting VCI value and is incremented by 1 for every VCL that is created on the trunk side for every line side port's VCL.

trunkTXTD

Refers to the trunk side transmitting ATM Traffic Descriptor template handle id.

trunkVpi

Refers to the trunk side ATM VCL's VPI.

txACCMMap

The Asynchronous Control Character (ACC) map that the local PPP entity requires for use on its transmit side. In effect, this is the ACC map that is required in order to ensure that all characters can be successfully transmitted through the local modem.

Access read-write

Notes Changing this parameter will take effect when the link is next restarted.

txDiscardNonMatchingVpi

Discard non-matching VPI.

Values 1: true

2: false

Default 2: false

Access read-write

txFrameLength

The frame length to be used by the IMA group in the transmit direction. Can only be set when the IMA group is startup.

Values 32 to 256

Default 128

Access read-create

txGain

This attribute specifies a gain value for the transmitted signal in the form of a 2s complement number. Valid values are -120 (-12.0 dB) to 60 (+6.0 dB).

Values -12..6

Default 0

Access read-write

txImald

The IMA ID currently in use by the near-end IMA function.

Values 0 to 255

Default 0

Access init-only

txPath

This attribute allows setting the POTS UNI either to full-time on-hook transmission or part-time on-hook transmission.

Values 1:fulltimeonhookxmit
2: parttimeonhookxmit
Default 1:fulltimeonhookxmit
Access read-write

txPower (DynOmciCpeWlanAdv)

The percentage of total power that should be used for data transmissions.

Values 1..100
Default 100
Access read-write

txSyncBytes

Enable transmission of sync bytes for ONUs to get their clock from a network source.

Values 1: txSyncBytesEnable
2: txSyncBytesDisable
Default 2: txSyncBytesDisable
Access read-write

txTrafficDescrIndex

The value of this object identifies the row in the ATM Traffic Descriptor Table which applies to the transmit direction of this VCL.

Values 0 to 2147483647
Default 0
Access read-write

type

The object type of the object as stored in the database.

Values 1 to 32
Access read-only
Notes This is a database-specific object and should not be modified.

Type (VDSL system config)

Values 0 to 32

Access read-only

U

unbalancedFrames

unicastAging

Multicast Aging period in tenths of seconds for the specified VLAN. After the specified aging period is met, the VLAN address is deleted from the routing table. A value of 0 indicates that aging is turned off. Maximum value is 2147483647 tenths of seconds, which equates to approximately 2485.51 days.

Values 0 to 2147483647

Access read-create

unitConversionFactor

The unit conversion factor.

Values 0 to 2147483647

Default 0

Access read-only

unnumberedIndex

The index used to obtain the local-side unnumbered address.

Access read-write

updateLogicalNames

A boolean variable to indicate if the logical names could be updated or not.

Access read-create

updateTime

The time in seconds between transmissions of RIP updates on all interfaces within a RD. The default is 30 seconds, which follows RFC 1058 and subsequent RIP RFCs.

Values 0 to 4294967294

Default 30

Access read-write

upDisableBandEndTone <1,2,3,4> upDisableBandStartTone<1,2,3,4>

The four downstream fields downDisableBandStartTone(1-4) and downDisableBandEndTone(1-4) configure bands in which VDSL vectoring is disabled or not applied at the CO side in the downstream direction. The four upstream fields upDisableBandStartTone(1-4) and upDisableBandEndTone(1-4) configure bands in which VDSL vectoring is disabled or not applied at the CPE side for the upstream direction. The values are VDSL tone indexes. The range for each field is 0 - 4096. The default for all vectoring disable tone fields is 0 which disabled the band.

NOTE:

1) The disable band end tone index must be greater than the corresponding disable band start tone index.

e.g. downDisableBandEndTone1 > downDisableBandStartTone1

upDisableBandEndTone1 > upDisableBandStartTone1

2) disable band start tone index for a subsequent band must be greater than the preceding disable band's disable band end tone index.

e.g. downDisableBandStartTone2 > downDisableBandEndTone1

upDisableBandStartTone2 > upDisableBandEndTone1

3) Only tone bands which follow the first two rules will be applied. If any tone band is zero, or is not starting at a higher index, it and all subsequent tone bands are ignored.

Values 0-4096

Default 0

Access read-create

upstreamPvcKey

Values 0 to 4294967294

Default 0

Access read-write

upstreamPvcType

Values 1 to 32

Access read-write

upstreamSir

This attribute specifies the upstream sustained information rate. Unit is kbps.

Values Integer32 (0..1310720)

Maximum 0

Access read-write

upstreamPir

This attribute specifies the upstream peak information rate. Unit is kbps.

Values Integer32 (0..1310720)

Maximum 0

Access read-write

upstreamPriority

Upstream Priority, currently the only used value range is 0-7.

Values Integer32 (0..65535)

Maximum 0

Access read-write

upstreamWeight

Upstream Weight for weighted round robin scheduling, variable size is 1 byte, default 0.

Values Integer32 (0..255)

Maximum 0

Access read-write

usageControl

UPC is the process of monitoring and controlling the ATM traffic by enforcing the traffic parameters. By using this parameter, you can disable on a per-traffic descriptor basis. This is typically only done during debugging.

Values 1: true

Enforcement of traffic descriptor parameters is enabled.

2: false

Enforcement of traffic descriptor parameters is disabled.

Default 1: true

Notes The **usage-parameter-control** parameter must be set to **true** if the ATM service category is CBR.

usBerInterval

Interval for upstream bit error rate calculation.

Values Integer

Default 5000

Access read-write

usBerLimit

Maximum allowable bit error rate prior to forward error correction for VDSL interfaces.

Values 1: 10^{-3}

2: 10^{-4}

3: 10^{-5}

4: 10^{-6}

Access read-write

userAuthMode

This object determines how users access to the device is authenticated. A value of local indicates that user should be authenticated using locally provisioned user-profile profiles. A value of radius indicates that only RADIUS authentication should be used. A value of radiusThenLocal indicates that the user should be authenticated first by using local user-profile profiles and if that fails, by authentication with a remote RADIUS server.

Values 1: local
 2: radius
 3: radiusThenLocal
 4: radiusThenCraft
Default 1: local
Access Read-Write

userName

This attribute contains the SIP user identification used for authentication.

Values STRING (SIZE (0..25))
Access read-write

userName (DynOmciCpePppoe)

The login user name to be used for PPP authentication

Values STRING (SIZE (1..64))
Access read-write

userPassword

The password is used by the subscriber in conjunction with the subUserLogonID to gain access to features or services provided by the service provider. The password object is only valid if the subUserLogonId is used. The default value is a NULL string.

Values 0 to 32
Access read-write

userPassword (DynOmciCpeSystemCommon)

Log-in password for user account.

Values String (0..16)
Access read-write

userPortId

The user port identifier for this connection.

Values 0-4799 if the userType is POTS (1)
 0-1599 if the userType is ISDN (2)
Access read-create

userPortType

The type of V5.2 user port.

Values 1: unknown
 2: POTS
 3: ISDN basic access
 4: ISDN primary rate access
Default 2: POTS
Access read-only

userType

The type of user port being defined.

Values 1: pots
 2: isdnDChannel
 3: isdnBChannel
Default 1: pots
Access read-create

useTDTmpl

Specifies whether the multicast map uses an ATM traffic descriptor template.

Values 1: true
 2: false
Default 1: true
Access read-write

usIgmpRate

This attribute limits the maximum rate of upstream IGMP traffic.

Traffic in excess of this limit is silently discarded. The attribute value is specified in messages/second. The default value 0 imposes no rate limit on this traffic.

Values Integer32 (0..2147483647)
Default 0
Access read-write

usMaxConstant

Restrict upstream constellations to use those that do not exceed this maximum.

Values 1: QAM-4
2: QAM-8
3: QAM-16
4: QAM-32
5: QAM-64
6: QAM-128
7: QAM-256
Default 6: QAM-128
Access read-write

usMaxRate

The target downstream line rate (in Mbps) that the interface will attempt to achieve. If this rate cannot be achieved, the highest possible rate will be obtained. The actual downstream rate may differ from this value, depending on line conditions. Applies to VDSL interfaces only.

Values 0 to 2147483647
Default 70875
Access read-write

usMinMargin

Equals the minimum upstream signal-to-noise ratio margin measured at quadrature phase shift keying (QPSK). This is used to compute the maximum constellation size. Applies to VDSL interfaces only.

Values 0 to 255 (in 1/8 of a decibel)
Default 24 (3.0 dB)
Access read-write

usPboEnabled

Upstream power back-off enable for the VTU-R. Applies to VDSL interfaces only.

Values 1: upstream-power-backoff-enable
 2: upstream-power-backoff-disable
Default 1
Access read-write

usRsFrame

Specifies the desired final value for upstream Reed-Solomon code word size and interleaving depth. This determines the final upstream Reed-Solomon parameters that are used. Applies to VDSL interfaces only.

Values 1: rs-1-1-1
 2: rs-255-239-1
 3: rs-255-239-16
 4: rs-142-126-29
 5: rs-255-239-143
 6: rs-193-177-191
Default 5: rs-255-239-143
Access read-write

usSymbolRate

Select symbol rates to use by the rate adaptive algorithm. Applies to VDSL interfaces only.

Values 1 ds-sym-rate-sel-bit-0
 2: ds-sym-rate-sel-bit-1
 4: ds-sym-rate-sel-bit-2
 8: ds-sym-rate-sel-bit-3
 16: ds-sym-rate-sel-bit-4
 32: ds-sym-rate-sel-bit-5
 64: ds-sym-rate-sel-bit-6
 128: ds-sym-rate-sel-bit-7
Default 128: ds-sym-rate-sel-bit-7
Access read-write

utopiaDiscard

Discard Utopia cells with HEC errors or arriving on unknown VPI.

Values 1: true
2: false
Default 2: false
Access read-write

V

value (GponOnuGenProfileVariable/ GponOnuSpecProfileVariable)

The value to assign the GponOnuGenProfileVariable or GponOnuSpecProfileVariable.

Values STING size (0...64 characters), the value has to be less than 64 characters.
Default An empty string
Access read-write

vccAalType

The type of AAL used on this VCC. An instance of this object only exists when the local VCL endpoint is also the VCC endpoint, and AAL is in use. The AAL type includes AAL1, AAL2, AAL3/4, and AAL5.

Values 1: aal1
2: aal34
3: aal5
4: other
5: unknown
6: aal2

Default 3: aal5

Access read-write

Notes The other (4) may be user-defined AAL type. The unknown type indicates that the AAL type cannot be determined.

vcCrossConnAdminStatus

Cross connect administrative status.

Values 1: Up
Allows the cross connect to pass packets.
2: Down
Disables the cross connect from passing packets.

Default 2: Down

Access read-write

vcCrossConnHighIfIndex

The ifIndex value of the high ATM interface for this VP cross-connect (the ATM interface whose ifIndex value is numerically higher than the other ATM interface in the cross connect).

Values 1 to 2147483647

Default 1

Access init-only

vcCrossConnHighVci

VCI value at the high ATM interface.

Values 1 to 2147483647

Default 1

Access init-only

vcCrossConnHighVpi

VPI value at the high ATM interface.

Values 1 to 2147483647

Default 1

Access init-only

vcCrossConnIndex

A unique value to identify this VC cross-connect. For each VCL associated with this cross-connect, the agent reports this cross-connect index value in the atmVclCrossConnectIdentifier attribute of the corresponding atmVclTable entries.

Values 1 to 2147483647

Default 1

Access init-only

vcCrossConnLowIfIndex

The ifIndex value of the low ATM interface for this VP cross-connect (the ATM interface whose ifIndex value is numerically lower than the other ATM interface in the cross connect).

Values 1 to 2147483647

Default 1

Access init-only

vcCrossConnLowVci

VCI value at the low ATM interface.

Values 1 to 2147483647

Default 1

Access init-only

vcCrossConnLowVpi

VPI value at the low ATM interface.

Values 1 to 2147483647

Default 1

Access init-only

vci

If this IP interface is on an ATM VC, the ATM VCI is specified here. If the IP interface is not an ATM VC, this field is ignored. The maximum VCI value cannot exceed the value allowable by atmInterfaceMaxVciBits defined in ATM-MIB.

Values 0 to 65535

Default 0

Access read-write

vci (bridgelogicals)

If the bridge interface is on an ATM VC, the ATM VCI is specified here. The range is 32...65,535. The maximum VCI value cannot exceed the value allowable by atmInterfaceMaxVciBits defined in ATM-MIB. If the bridge interface is not an ATM VC, but for example, an Ethernet / GPON interface, this field should be set to 0.

Default 32

Access read-create

vciRange

The VCI range used for creating bulk video cross connects.

Values 0 to 64
Default null
Access read-write

vclAdminStatus

Administrative status of the VCL. This object is instantiated only for a VCL that terminates a VCC (one that is not cross-connected to other VCLs).

Values **1: up**
 Allows the interface to pass packets.
2: down
 Disables the interface from passing packets. When a managed system initializes, all interfaces start in the down state.
Default **2: down**
Access read-write

vclCastType

The connection topology type.

Values **1: Point-to-Point**
Default **1: Point-to-point**
Access read-write

vclConnKind

The use of call control.

Values **1: Application**
Default **1: Application**
Access read-write

vclCrossConnId

Cross Connection between two VCLs. The value is indexed by the ATM interface port number and VPI/VCI values of the two ends of that cross connection.

Values 0 to 2147483647
Default 0
Access init-only

vclRate

This parameter is used to describe the minimum guaranteed rate for a VCL which is bound to this traffic descriptor for BAN ATM Trunk cards. It is applicable only for the rt-VBR service category.

Access read-write

Notes The **vclRate** parameter is only used on BAN release 1.5.2. In other releases, it does not apply.

vdslMode

Type of VDSL equipment.

Values **1: vtu-o**
central office equipment
2: vtu-r
customer premise equipment

Default **1: vtu-o**

Access read-write

vendorClientId

Values **4 to 255**

Access read-write

vendorMatchLength

Length of the vendor class identifier substring to match. A value of -1 indicates that the entire string starting at the match offset value should be matched.

Values **-1 to 255**

Default **-1**

Access read-write

vendorMatchOffset

Offset into the client passed vendor class identifier string where the string match should start.

Default **0**

Access read-write

vendorMatchString

Vendor class identifier match string that is used to determine which clients should be placed in this group. The default value for this object is an empty octet string.

Access read-write

version1

The version number of the filter specification. This is used to flag any changes in the statements that comprise a filter. Each time a modification occurs to an object in a filter spec (including the list of filter statements of the same fltSpecIndex in filterStatementTable), the value of this object, and fltSpecVersion2 must be incremented.

Values 0 to 4294967295

Default 0

Access read-create

version2

The version number of the filter specification. The value of this object must be equal to fltSpecVersion1, otherwise the filter spec is inconsistent.

Values 0 to 4294967295

Default 0

Access read-create

videoCrossConnAdminStatus

The desired administrative status of this bi-directional VC cross-connect.

Values 1 to 2

Default 2

Access read-write

videoCrossConnAdminStatus

The desired administrative status of this bi-directional VC cross-connect.

Values 1 to 2

Default 2

Access read-write

videoCrossConnHighIfIndex

The ifIndex value for the ATM interface for this VC cross-connect. The term high implies that this ATM interface has the numerically higher ifIndex value than the other ATM interface identified in the same atmVcCrossConnectEntry.

Values 1 to 2147483647

Default 1

Access init-only

videoCrossConnHighVci

The VCI value at the ATM interface associated with the VC cross-connect that is identified by atmVcCrossConnectHighIfIndex.

Values 1 to 2147483647

Default 1

Access init-only

videoCrossConnHighVpi

The VPI value at the ATM interface associated with the VC cross-connect that is identified by atmVcCrossConnectHighIfIndex.

Values 1 to 2147483647

Default 1

Access init-only

videoCrossConnIndex

A unique value to identify this VC cross-connect.

Values 1 to 2147483647

Default 1

Access init-only

videoCrossConnLowIfIndex

The ifIndex value of the ATM interface for this VC cross-connect. The term low implies that this ATM interface has the numerically lower ifIndex value than the other ATM interface identified in the same atmVcCrossConnectEntry.

Values 1 to 2147483647

Default 1

Access init-only

videoCrossConnLowVci

The VCI value at the ATM interface associated with this VC cross-connect that is identified by atmVcCrossConnectLowIfIndex.

Values 1 to 2147483647

Default 1

Access init-only

videoCrossConnLowVpi

The VPI value at the ATM interface associated with the VC cross-connect that is identified by atmVcCrossConnectLowIfIndex.

Values 1 to 2147483647

Default 1

Access init-only

videoCrossConnTemplateId

The template ID of the video cross connect.

Values 0 to 2147483647

Default 0

Access read-write

videoDomain

The video domain for the IP video server.

Values 0 to 65535

Default 0

Access read-write

videoGport

This is the downstream only multicast GEM port for IPTV content streams.

Values Integer32 (0..4095)

Default 0

Access read-write

videoIndex

This represents the CPE Video Index. This is a mandatory attribute for addition operation.

Values Integer32 (1..2147483647)

Access read-create

videoProfileIndex

This object points to the associated video profile.

Values Unsigned32 (0..2147483647)

Default 0

Access read-write

videoSystemConfigIndex

A unique value to identify this video system profile entry.

Values 1 to 65535

Default 1

Access read-write

vlan

This represents the CPE Vlan Id. This is a mandatory attribute for addition operation.

Values Integer32 (0..4096)

Access read-create

vlanCOS

The COS value loaded into the VLAN header. Applicable only if the vlan ID is non-zero.

Values 0 to 7

Default 0

Access init-only

vlanCOS (for DynOmciCpeConnection)

The Class-Of-Service (COS) value used for VLAN Tag Insertion.

Values Integer32 (0..8)
Default 0
Access read-write

vlanId

The Virtual LAN ID.

Values 1 to 4095
Default 1
Access init-only

vlanId (bridgelogicals)

The virtual LAN Id for VLAN Tag Insertion.

Values 0...214,783,647
Default 0
Access read-create

vlanId (IPlogicals)

The virtual LAN Id for the IP interface If the VLAN ID specified is non-zero, all traffic destined to this interface should have the VLAN ID specified.

Values 0 to 4095
Default 0
Access init only

vlanId (DynOmciCpeConnection)

The Virtual LAN Id for VLAN Tag Insertion.

Values Integer32(0 to 4095)
Access Read-Create

vlanIP (DynOmciCpelfVlan)

The address used for IP host services. Default is 0.0.0.0.

Values Integer32(0 to 4096)
Access Read-Create

vlanTpId

The value of the tag protocol identifier that is being used for QinQ packets. (0x8100 is default.).

Values 33024
34984
37120
37376
Default 33024
Access Read-Write

vlanRouting (DynOmciCpeSystemCommon)

If enabled is selected, routing between VLANs is allowed.

Enabled: Route table lookups ignore the VLAN ID of the ingress and egress ports. If there is a match, the packet is routed out the interface specified in the Route table, regardless of which VLAN it is a member of. Cross VLAN Routing enabled is the default behavior.

Disabled: Packets will be forwarded to the configured Default Route for the VLAN that they arrived on unless there is a Route Table match within that same VLAN. Routing of packet across VLANs is prevented, providing traffic isolation.

Values 1: enabled
2: disabled
Default 2: disabled
Access Read-Write

vlanXlatId

If a packet's VLAN ID value matches this object's value then the packet's VLAN ID will be translated to the value of bridgeIfVlanId. A zero value disables translation. This attribute is shown as Original VLAN ID in DZMS client.

Values 0 to 4095
Default 0
Access Read-Create

vlanXlatId (DynOmciCpeConnection)

The Virtual LAN Id for VLAN Tag Insertion.

Values Integer32 (0..4096)

Default 0

Access Read-Write

vlanXlateId (DynOmciCpelfVlan)

The Virtual LAN Id for VLAN Tag Insertion.

Values Integer32 (0..4096)

Default 0

Access Read-Create

vlanXlateCos (DynOmciCpeConnection, DynOmciCpelfVlan)

The Class-Of-Service (COS) value used for VLAN Tag Insertion.

Values Integer32 (0..8)

Default 0

Access Read-Write

vlanXlateTpld (for DynOmciCpeConnection object)

The value of the tag protocol identifier that is being used for QinQ packets. (0x8100 is default.).

Values 1. 33024 (0x8100)

2. 34984 (0x88A8)

3. 37120 (0x9100)

4. 37376 (0x9200)

Default 1. 33024 (0x8100)

Access Read-Write

voicebandwidthcheck

Enables or disables ATM bandwidth checking for voice calls. The system will reject a call if there is not enough bandwidth. On a MALC or a Z-Edge, if a call is rejected, the device sends a fast busy to the caller. On the BAN or the Sechtor 100A, if a call is rejected, the switch sends a busy signal to the caller.



Note: Bandwidth calculations are enabled or disabled for the entire system.

The bandwidth check is based on the configuration of the transmit and receive ATM traffic descriptors for the ATM VC/CID used by the call. The system uses the following parameters to calculate the required bandwidth:

- rt-VBR traffic: SCR
- CBR traffic: PCR
- All other traffic types: calls are always accepted, there are no voice quality guarantees.

The system will reject calls if:

$[\text{Receive or transmit cell rate} - (180 \times \text{number of active calls on the VC})] < 180$

(The bandwidth calculations assume that approximately 180 AAL2 cells per second are used for each voice call.)

Values 1: true

2: false

Default 2: false

vpi

The VPI value for a VPL or VCL. If this IP interface is on an ATM VC, the ATM VPI is specified here. If the IP interface is not an ATM VC, this field is ignored. The value VPI=0 is only allowed for a VCL. For ATM UNIs supporting VPCs the VPI value ranges from 0 to 255. The VPI value 0 is supported for ATM UNIs conforming to the ATM Forum UNI 4.0 Annex 8 (Virtual UNIs) specification. For ATM UNIs supporting VCCs the VPI value ranges from 0 to 255. For ATM NNIs the VPI value ranges from 0 to 4095. The maximum VPI value cannot exceed the value allowable by atmInterfaceMaxVpiBits defined in ATM-MIB.

Values 0 to 4095

Default 0

Access read-write

vpi (bridgelogicals)

If the bridge interface is on an ATM VC, the ATM VPI is specified here. If the bridge interface is not an ATM VC, this field is ignored. The VPI value is for a VPL or VCL. The value VPI=0 is only allowed for a VCL. For ATM UNIs supporting VPCs, the VPI value ranges from 0 to 255. The VPI to the ATM Forum UNI 4.0 Annex 8 (Virtual UNIs) specification. For ATM UNIs supporting VCCs, the VPI value ranges from 0 to 255. For ATM NNIs, the VPI value ranges from 0 to 4095. The maximum VPI value cannot exceed the value allowable by atmInterfaceMaxVpiBits defined in ATM-MIB. If the bridge interface is not an ATM VC, but for example, an Ethernet / GPON interface, this field should be set to 0.

Default 0
Access read-create

vpi (IP video)

The VPI for the ATM interface carrying the IP video stream.

Values 0 to 4095
Default 0
Access read-write

vpiState

Indicates whether a new vpi requires a card reboot

Values 1: clear
 2: VPI updating
 3: reboot required

W

wdsMacListProfileIndex (DynOmciCpeWlanSubscriber)

The address of cpe-wds-mac-ctrl-list profile associated with this CPE.

Values 0 to 2147483647
Default 0
Access read-write

weight

A weight given to this card in determining whether this card should become Active on system boot.

Values 1 to 5
Default 2
Access read-create

wepEncryption (DynOmciCpeWlan)

WEP encryption mode.

Values 1: enabled

2: disabled

Default 2: disabled

Access read-write

wepStrength (DynOmciCpeWlan)

WEP encryption strength.

Values 1: bit64

2: bit128

Default 2: bit128

Access read-write

wettingCurrent

Specifies whether wetting current is disabled or enabled(standard).

Values 1: disabled

2: enabled

Access read-create

wlanIndex (DynOmciCpeWlan, DynOmciCpeWlanAdv)

This represents the CPE WLAN index value.

Values 1..2147483647

Default 1

Access read-create

wlanAdvProfileIndex (DynOmciCpeWlanSubscriber)

The address of cpe-wlan-advanced profile associated with this CPE.

Values 0..2147483647

Default 1

Access read-write

wlanProfileIndex (DynOmciCpeWlanSubscriber)

The address of cpe-wlan profile associated with this CPE.

Values 0..2147483647

Default 1

Access read-write

wmm (DynOmciCpeWlanAdv)

Enable or disable WiFi Protected Setup (WPS) security method. If WPS is enabled, the network authentication method, the data encryption, and network key should also be configured in order to authenticate to this wireless network. It is available for WPA-PSK, WPA2-PSK, Mixed WPA2/WPA-PSK and Open Network Authentication methods.

Values 1: enabled

2: disabled

Default 2: enabled

Access read-write

wmmAdvertise (DynOmciCpeWlan)

WMM(Wireless Mult Mode) provides a subset of the IEEE 802.11e QoS standard, which adds prioritization to wireless to optimize their performance. When multiple concurrent applications are on the wireless network each application may have different latency and throughput needs.

WMM provides for this optimization, however WMM may provide slower.

Values 1: enabled

2: disabled

Default 2: disabled

Access read-write

wmmApsd (DynOmciCpeWlanAdv)

Enable or disable the Automatic Power Save Delivery (APSD) power management method. This feature is useful for bi-directional applications, such as VoIp phones WMM provides for this optimization, however WMM may provide slower.

Values 1: enabled

2: disabled

Default 1: enabled

Access read-write

wmmNoAck (DynOmciCpeWlanAdv)

Enable or disable the suppression of acknowledgements for frames that do not require a QOS Acknowledgement. This avoids the unnecessary transmission of acknowledgements for highly time-critical data.

Values 1: enabled
2: disabled
Default 1: enabled
Access read-write

wireMode

This variable is used to disable or enable four-wire-framing.

Values 1: four-wire-disable
four-wire-disable
2: four-wire-enable-byte-interleave
four-wire-enable-byte-interleave
3: four-wire-enable-bit-interleave
four-wire-enable-bit-interleave
4: four-wire-enable-non-interleave
four-wire-enable-non-interleave
Default 1: four-wire-disable
Access read-create

wpa2PreAuthentication (DynOmciCpeWlan)

Enable or disable WPA2 pre-authentication.

Values 1: enabled
2: disabled
Default 2: disabled
Access read-write

wpaEncryption (DynOmciCpeWlan)

WPA encryption mode

Values 1: aes
2: tkipaes
Default 1: aes
Access read-write

wpaGroupRekeyInterval (DynOmciCpeWlan)

WPA Group Rekey Interval.

Values 0..2147483647

Default 0

Access read-write

wps (DynOmciCpeWlanAdv)

Enable or disable WiFi Protected Setup (WPS) security method. If WPS is enabled, the network authentication method, the data encryption, and network key should also be configured in order to authenticate to this wireless network. It is available for WPA-PSK, WPA2-PSK, Mixed WPA2/WPA-PSK and Open Network Authentication methods.

Values 1: enabled

2: disabled

Default 2: disabled

Access read-write

wpsAddClientMethod (DynOmciCpeWlanAdv)

A client can be added via three different methods.

Values 1: pushbutton

2: stapin

3: appin

Default 1: pushbutton

Access read-write

wpsApMode (DynOmciCpeWlanAdv)

If the provider is using an external registrar for security, select configured. The PIN for AP mode is specified by the registrar. Provide this PIN to the client. Issue Config AP to begin the registration process with the client.

Values 1: configured

2: unconfigured

Default 1: configured

Access read-write

X

xpressTechnology (DynOmciCpeWlanAdv)

Enable or disable the XPress(TM) Technology.

Values 1: enabled
2: disabled
Default 2: disabled
Access read-write

Z

zip

The zip code.

Values 0 to 32
Access read-write

zmsBlockCli

If TRUE, CLI only allows **show** commands and cannot **modify/add** any data records. This is used by DZMS to prevent race conditions and to allow only single access to system resources especially during problem analysis times.

Values 1: true
2: false
Default 2: false
Access read-write

zmsConnectionStatus

The status of the connection between this device and the DZS Management System, if one exists. This value can either be set directly by DZMS or by the device internally when a SNMP request is processed from the DZMS. For devices that are being proxied by another entity, since they are not being polled by DZMS, will set this state according to the communication channel state between this device and the connected entity.

Values 1: active
2: inactive
Default 2: inactive

Access read-write

zmsExists

Identifies whether or not this device is being managed by the DZS Management System. This is only configurable by the management system and NOT by the CLI as it indicates existence of the manager. Different operations are performed based on this setting (no configuration synchronization processing is performed if there is no DZS Management System).

Values 1: true

2: false

Default 2: false

Access read-write

zmsIPAddress

This is the DZMS IP address for which configuration synchronization messages and any other special handling is performed. This can be set either via CLI or DZMS.

Default 0.0.0.0

Access read-write

zoomFactor

Values 0 to 2147483647

Default 1

Access read-only

4 PERFORMANCE REQUESTS REFERENCE

This chapter describes the performance request attributes supported for the objects listed in the performance request tables.

A

aal2CpsPerfBadCPSLength

The number of octets expected for a CPS-Packet overlapping into this CPS-PDU does not match the information contained in the STF; if the value of the OSF is less than 47, processing starts at the octet pointed to by the OSF.

aal2CpsPerfBadPayloadLength

The OSF of the STF contains a value 48 or greater; the complete CPS-PDU is discarded.

aal2CpsPerfHEC

The Header Error Control (HEC) Code of a CPS-Packet header indicates transmission errors in the CPS-Packet header; the unprocessed information in the CPS-PDU is discarded.

aal2CpsPerfIllegalCID

The CID value in the received CPS-Packet header is not associated with a SAP.

aal2CpsPerfIllegalUUI

The UUI field in the received CPS-Packet header contains a value (28 or 29) that is reserved for future standardization.

aal2CpsPerfPayloadTooLong

The length of the received CPS-Packet Payload (CPS-SDU) exceeds the maximum length indicated in **Max_SDU_Deliver_Length**.

aal2CpsPerfRessError

A partial CPS-Packet has been received previously and must be discarded due to errors detected before the reassembly of the CPS-Packet could be completed.

aal2CpsPerfSTFBadSeq

The sequence number of the STF is wrong; if the OSF is less than 47, processing starts at the octet pointed to by the OSF, otherwise the complete CPS-PDU is discarded.

aal2CpsPerfSTFParity

The parity of the STF indicates transmission errors; the complete CPS-PDU was discarded.

aal2CpsPerfTransError

The Header Error Control (HEC) Code of a CPS-Packet header that was overlapping a CPS-PDU boundary indicates transmission errors in the CPS-Packet header; if the value of the OSF is less than 47, processing starts at the octet pointed to by the OSF.

aal2ElcplgLapvReceived

Number of LAPV frames received by this interface group.

aal2ElcplgLapvSent

Number of LAPV frames sent from this interface group.

aal2ElcplgOperStatus

The operational status of an ELCP interface group.

aal2ElcplgOperStatusChangeCount

Number of times the aal2ElcplgOperStatus has changed.

aal2Sscsl3661BadLength

The value of the Length field in the SSTED-PDU does not match the length of the received SSTED-PDU.

aal2Sscsl3661CRC

The value of the CRC field is not equal to the CRC calculated over the received information.

aal2Sscsl3661MsgTooLong

The maximum permissible size for a reassembled SSSAR-SDU (**Max_SDU_Length**) has been exceeded.

aal2Sscsl3661MsgTooShort

An SSTED-PDU of length 8 or less has been received.

aal2Sscsl3661RasTimerExpired

The reassembly timer RAS_Timer has expired.

aal2Sscsl3662IllegalUUI

The UUI field in the received CPS-Packet header contains a value in the reserved ranges (16..22, 28..30). See I.366.2, Table 12.

aal2Sscsl3662ProfileError

Invalid UUI codepoint and UUI length columns. Does not match any Audio Profile entry, based on the VCL entry's audio profile index.

aal2Sscsl3662Type3CRC

A Type 3 packet using CRC-10 validation with a checksum error will increase this counter.

aal2VclCellsReceived

Total cells received.

aal2VclCellsSent

Total cells sent.

aal2VcICircuitModeData

The service provided is the transfer of circuit mode data. The circuit mode data service is an emulation of the $N \times 64$ kbit/s ($N = 1, 2, \dots, 30$) unrestricted, 8 kHz structured circuit mode service of Recommendations I.231.1 and I.231.10. The value $N = 31$ is also allowed. The value of N does not change during the lifetime of a call.

Values 1: enabled
2: disabled

aal2VcIDTMFDialedDigits

DTMF Dialed Digits are multifrequency audio tones typically used for inter-register signalling of addresses during call setup or for end-to-end device control during an established call. Depending on the system, codes are defined for the digits 0-9 of a telephone keypad and other auxiliary signals. The service provided is the transfer of dialed digits, namely, the transfer of five descriptors: digit type, digit character, start time, end time, and power level of the dual frequency pulse.

Values 1: enabled
2: disabled

aal2VcIFaxDemoRemo

Facsimile Demodulation/Remodulation is the process of detecting facsimile traffic, extracting digital information from the incoming analogue modulated signal, transporting this across a trunk in packet formats, and reproducing the facsimile control and image information by remodulation at the other end. The service provided is the transfer of demodulated facsimile image information and control information, from the demodulating User to the remodulating User. Valid values are enabled and disabled.

Values 1: enabled
2: disabled

aal2VcIMfR1DialedDigits

MF-R1 Dialed Digits. See description for atmVccAal2DTMFDialedDigits. Valid values are enabled and disabled.

Values 1: enabled
2: disabled

aal2VcIMfR2DialedDigits

MF-R2 Dialed Digits. Valid values are enabled and disabled.

Values 1: enabled

2: disabled

aal2VciStatsTimeElapsed

The number of seconds that have elapsed since the beginning of the current collection period. If, for some reason, such as an adjustment in the system's time-of-day clock, the current interval exceeds the maximum value, the agent will return the maximum value.

aal5VccCrcErrors

The number of AAL5 CPCS PDUs received with CRC-32 errors on this AAL5 VCC at the interface associated with an AAL5 entity.

aal5VccOverSizedSDUs

The number of AAL5 CPCS PDUs discarded on this AAL5 VCC at the interface associated with an AAL5 entity because the AAL5 SDUs were too large.

aal5VccSarTimeOuts

The number of partially reassembled AAL5 CPCS PDUs which were discarded on this AAL5 VCC at the interface associated with an AAL5 entity because they were not fully reassembled within the required time period. If the reassembly timer is not supported, then this object contains a zero value.

aal5VccVci

The VCI value of the AAL5 VCC at the interface identified by the ifIndex.

aal5VccVpi

The VPI value of the AAL5 VCC at the interface identified by the ifIndex.

actAtp

Actual Aggregate Transmit Power from the ATU (ATU-R on downstream direction and ATU-C on upstream direction), at the instant of measurement. It ranges from -310 to 310 units of 0.1 dB (Physical values are -31 to 31 dBm). A value of all 1's indicates the measurement is out of range to be represented. This object reflects the value of the parameter following the most recent DELT performed on the associated line. Once the DELT process is over, the parameter no longer changes until the row is deleted or a new DELT process is initiated.

actChans

Number of active multicast channels.

actHosts

Number of active multicast hosts.

age

The number of seconds since this route was last updated or otherwise determined to be correct. Note that no semantics of 'too old' can be implied except through knowledge of the routing protocol by which the route was learned.

allocatedAssuredBw

The Total Available DBA Allocated Assured bandwidth per OLT.

Access read-only

allocatedBestEffortBw

The Total Available DBA Allocated Best Effort bandwidth per OLT.

Access read-only

allocatedCbrBw

The Total Available DBA Allocated CBR bandwidth per OLT.

Access read-only

allocatedCompensatedCbrBw

The Total Available DBA Compensated CBR bandwidth per OLT.

Access read-only

allocatedNonAssuredBw

The Total Available DBA Allocated Non Assured bandwidth per OLT.

Access read-only

allocatedUbrBw

The Total Available DBA Allocated UBR bandwidth per OLT.

Access read-only

attainableRate

Maximum Attainable Data Rate. The maximum net data rate currently attainable by the ATU-C transmitter and ATU-R receiver (when referring to downstream direction) or by the ATU-R transmitter and ATU-C receiver (when referring to upstream direction). Value is coded in bits/s. This object reflects the value of the parameter following the most recent DELT performed on the associated line. Once the DELT process is over, the parameter no longer changes until the row is deleted or a new DELT process is initiated.

atucActualTxMode

This field provides actual standard used for the connection based on outcome of negotiation with ATU-R.

Values	unknownMode(1) unknown mode or no connection made T1413Mode(2) connection using T1.413 standard alctl14Mode(3) connection with Alcatel 14 CPE alctlMode(4) connection with Alcatel CPE. adiMode(5) connection with Analog Device CPE gdmrMode(6) connection using G.dmr standard gliteMode(7) connection using G.lite standard gads1PlusMode(8) connection using ADSL+ standard gspanPlusMode(9) connection with GlobeSpan+ CPE adsl2PlusMode(10) connection using ADSL2+ standard adsl2Mode(11) connection using ADSL2 standard readsl2Mode(12) connection using Reach Extended ADSL2 standard uawgMode(13) connection with UAWG (Universal ADSL Working Group) type CPE
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centLiteMode(14)

connection with Centillium G.Lite CPE

centFullMode(15)

connection with Centillium CPE

tiAnnBMode(16)

connection with TI AnnexB CPE

alctAnnBMode(17)

connection with Alcatel AnnexB CPE

reachDSLMode(18)

connection with Paradyne ReachDSL CPE

adsl2PlusAnnexMMode(19)

connection using ADSL2+ and Annex-M standards

adsl2AnnexMMode(20)

connection using ADSL2 and Annex-M standards

atucChanBackgroundErrBlks

Background errored blocks at near end.

atucChanCorrectedBlks

Count of all blocks received with errors that were corrected since agent reset. These blocks are passed on as good data.

atucChanCrcBlockLength

Indicates the length of the channel data-block on which the CRC operates. Refer to Line Code Specific MIBs, [11] and [12] for more information.

atucChanCurr15MinCorrectedBlks

Count of all blocks received with errors that were corrected on this channel within the current 15 minute interval.

atucChanCurr15MinRxBlks

Count of all encoded blocks received on this channel within the current 15 minute interval.

atucChanCurr15MinTxBlks

Count of all encoded blocks transmitted on this channel within the current 15 minute interval.

atucChanCurr15MinTimeElapsed

Total elapsed seconds in this interval.

atucChanCurr15MinUncorrectedBlks

Count of all blocks received with uncorrectable errors on this channel within the current 15 minute interval.

atucChanCurr1DayCorrectedBlks

Count of all blocks received with errors that were corrected on this channel during the current day as measured by `adslAtucChanPerfCurr1DayTimeElapsed`.

atucChanCurr1DayRxBlks

Count of all encoded blocks received on this channel during the current day as measured by `adslAtucChanPerfCurr1DayTimeElapsed`.

atucChanCurr1DayTimeElapsed

Number of seconds that have elapsed since the beginning of the current 1-day interval.

atucChanCurr1DayTxBlks

Count of all encoded blocks transmitted on this channel during the current day as measured by `adslAtucChanPerfCurr1DayTimeElapsed`.

atucChanCurr1DayUncorrectBlks

Count of all blocks received with uncorrectable errors on this channel during the current day as measured by `adslAtucChanPerfCurr1DayTimeElapsed`.

atucChanCurrTxRate

Actual transmit rate on this channel.

atucChanErrFast

CRC errors on fast buffer at near end.

atucChanErrInterleave

FEC corrections on interleaved buffer at near end.

atucChanFastRetrainFailures

Fast retrains failures.

atucChanFastRetrains

Fast retrains.

atucChanFECCorrectedFast

FEC corrections on fast buffer at near end.

atucChanFECCorrectedInterleave

FEC corrections on interleaved buffer at near end.

atucChanInterleaveDelay

Interleave Delay for this channel. Interleave delay applies only to the interleave channel and defines the mapping (relative spacing) between subsequent input bytes at the interleaver input and their placement in the bit stream at the interleaver output. Larger numbers provide greater separation between consecutive input bytes in the output bit stream allowing for improved impulse noise immunity at the expense of payload latency. In the case where the ifType is Fast (125), use noSuchObject.

atucChanIntervCorrectedBlks

Count of all blocks received with errors that were corrected on this channel during this interval.

atucChanIntervRxBlks

Count of all encoded blocks received on this channel during this interval.

atucChanIntervTxBlks

Count of all encoded blocks transmitted on this channel during this interval.

atucChanIntervUncorrectBlks

Count of all blocks received with uncorrectable errors on this channel during this interval.

atucChanIntervValidData

This variable indicates if the data for this interval is valid.

atucChanInvalidIntervals

The number of intervals in the range from 0 to the value of `adslAtucChanPerfValidIntervals` for which no data is available. This object will typically be zero except in cases where the data for some intervals are not available (e.g., in proxy situations).

atucChanNonSES

Non SES blocks received at the near end.

atucChanPerfBE

Near-end idle cell bit error count. Supported for ADSL2/ADSL2+ only.

atucChanPerfCD

Near-end delineated total cell count. Supported for ADSL2/ADSL2+ only.

atucChanPerfCU

Near-end user total cell count. Supported for ADSL2/ADSL2+ only.

atucChanPrevTxRate

The rate at the time of the `lastatucRateChangeTrap` event. It is also set at initialization to prevent a trap being sent. Rate changes less than `adslAtucThresh(*)RateDown` or less than `adslAtucThresh(*)RateUp` will not cause a trap or cause this object to change.

atucChanRxBlks

Count of all encoded blocks received on this channel since agent reset.

atucChanTxBlks

Count of all encoded blocks transmitted on this channel since agent reset.

atucChanUncorrectBlks

Count of all blocks received with uncorrectable errors since agent reset.

atucChanValidIntervals

The number of previous 15-minute intervals in the interval table for which data was collected. Given that $\langle n \rangle$ is the maximum # of intervals supported. The value will be $\langle n \rangle$ unless the measurement was (re-)started within the last $(\langle n \rangle * 15)$ minutes, in which case the value will be the number of complete 15 minute intervals for which the agent has at least some data. In certain cases (e.g., in the case where the agent is a proxy) it is possible that some intervals are unavailable. In this case, this interval is the maximum interval number for which data is available.

atucCurr1DayTimeElapsed

Number of seconds that have elapsed since the beginning of the current 1-day interval.

atucCurr15MinEss

The number of Errored Seconds in the current 15-minute interval. The errored second parameter is a count of one-second intervals containing one or more CRC anomalies, or one or more los or sef defects.

atucCurr15MinInits

The number of the line initialization attempts in the current 15-minute interval. Includes both successful and failed attempts.

atucCurr15MinLofs

The number of seconds in the current 15-minute interval when there was Loss of Framing.

atucCurr15MinLols

The number of seconds in the current 15-minute interval when there was Loss of Link.

atucCurr15MinLoss

The number of seconds in the current 15-minute interval when there was Loss of Signal.

atucCurr15MinLprs

The number of seconds in the current 15-minute interval when there was Loss of Power.

atucCurr15MinTimeElapsed

Total elapsed seconds in this interval.

atucCurr1DayEss

The number of Errored Seconds during the current day as measured by adslAtucPerfCurr1DayTimeElapsed. The errored second parameter is a count of one-second intervals containing one or more crc anomalies, or one or more los or sef defects.

atucCurr1DayInits

The number of the line initialization attempts in the day as measured by adslAtucPerfCurr1DayTimeElapsed. Includes both successful and failed attempts.

atucCurr1DayLofs

The number of seconds when there was Loss of Framing during the current day as measured by adslAtucPerfCurr1DayTimeElapsed.

atucCurr1DayLols

The number of the number of seconds when there was Loss of Link during the current day as measured by adslAtucPerfCurr1DayTimeElapsed.

atucCurr1DayLoss

The number of seconds when there was Loss of Signal during the current day as measured by adslAtucPerfCurr1DayTimeElapsed.

atucCurr1DayLprs

The number of seconds when there was Loss of Power during the current day as measured by `adslAtucPerfCurr1DayTimeElapsed`.

atucCurr1DayTimeElapsed

Number of seconds that have elapsed since the beginning of the current 1-day interval.

atucCurrAtn

Measured difference in the total power transmitted by the peer ATU and the total power received by this ATU.

atucCurrAttainableRate

Indicates the maximum currently attainable data rate by the ATU. This value will be equal or greater than the current line rate.

atucCurrOutputPwr

Measured total output power transmitted by this ATU. This is the measurement that was reported during the last activation sequence.

atucCurrSnrMgn

Noise Margin as seen by this ATU with respect to its received signal in tenth dB.

atucCurrStatus

Indicates current state of the ATUC line. This is a combination of possible conditions.

Values 0: noDefect

No defects on the line.

1: lossOfFraming

ATUC failure due to not receiving valid frame.

2: lossOfSignal

ATUC failure due to not receiving signal.

3: lossOfPower

ATUC failure due to loss of power. Note that the agent may still function.

4: lossOfSignalQuality

Loss of Signal Quality is declared when the Noise Margin falls below the Minimum Noise Margin, or the bit-error-rate exceeds 10^{-7} .

5: lossOfLink

ATUC failure due to inability to link with ATUR.

6: dataInitFailure

ATUC failure during initialization due to bit errors corrupting startup exchange data.

7: configInitFailure

ATUC failure during initialization due to peer ATU not able to support requested configuration.

8: protocolInitFailure

TUC failure during initialization due to incompatible protocol used by the peer ATU.

9: noPeerAtuPresent

ATUC failure during initialization due to no activation sequence detected from peer ATU. This is intended to supplement ifOperStatus.

atucDisconnects

Number of disconnects.

atucElapseds

Elapsed seconds at the near end.

atucElapseds

Elapsed seconds at the near end.

atucEss

Count of the number of Errored Seconds since agent reset. The errored second parameter is a count of one-second intervals containing one or more crc anomalies, or one or more los or sef defects.

atucFECs

Seconds with one or more FECs at the near end.

atucHiBER

Seconds declared as hi BER by the near end.

atucInits

Count of the line initialization attempts since agent reset. Includes both successful and failed attempts.

atucIntervInits

The number of line initialization attempts during the interval. Includes both successful and failed attempts.

atucIntervLofs

The number of seconds in the interval when there was Loss of Framing.

atucIntervLols

The number of seconds in the interval when there was Loss of Link.

atucIntervLoss

The number of seconds in the interval when there was Loss of Signal.

atucIntervLprs

The number of seconds in the interval when there was Loss of Power.

atucIntervNum

Performance Data Interval number one is the most recent previous interval; interval 96 is 24 hours ago. Intervals 2..96 are optional.

atucIntervValidData

This variable indicates if the data for this interval is valid.

atucInvSerialNumber

The vendor specific string that identifies the vendor equipment.

atucInvVendorID

The vendor ID code is a copy of the binary vendor identification field defined by the PHY[10] and expressed as readable characters.

atucInvVersionNumber

The vendor specific version number sent by this ATU as part of the initialization messages. It is a copy of the binary version number field defined by the PHY[10] and expressed as readable characters.

atucInvalidIntervals

The number of intervals in the range from 0 to the value of `adslAtucPerfValidIntervals` for which no data is available. This object will typically be zero except in cases where the data for some intervals are not available (e.g., in proxy situations).

atucLofs

Count of the number of Loss of Framing failures since agent reset.

atucLols

Count of the number of Loss of Link failures since agent reset.

atucLoss

Loss of signal seconds at the near end.

atucLprs

Count of the number of Loss of Power failures since agent reset.

atucPerfCurr1DayFecsL

Number of seconds with one or more near end FECs in the last 24 hours.

atucPerfCurr15MinFecsL

Number of seconds with one or more near end FECs in the current 15 minute interval.

atucPerfFecsL

Seconds with one or more FECs at the near end.

atucPerfLossL

Number of near end loss of signal failures.

atucPerfPrev1DayFecsL

Number of seconds with one or more near end FECs in the previous 24 hours.

atucPerfSuccessfulConnects

Successful connects.

atucPrev1DayEss

The number of Errored Seconds within the most recent previous 1-day period. The errored second parameter is a count of one-second intervals containing one or more crc anomalies, or one or more los or sef defects.

atucPrev1DayInits

The number of the line initialization attempts in the most recent previous 1-day period. Includes both successful and failed attempts.

atucPrev1DayLofs

The number of seconds in the interval when there was Loss of Framing within the most recent previous 1-day period.

atucPrev1DayLols

The number of seconds in the interval when there was Loss of Link within the most recent previous 1-day period.

atucPrev1DayLoss

The number of seconds in the interval when there was Loss of Signal within the most recent previous 1-day period.

atucPrev1DayLprs

The number of seconds in the interval when there was Loss of Power within the most recent previous 1-day period.

atucPrev1DayMoniSecs

The amount of time in the previous 1-day interval over which the performance monitoring information is actually counted. This value will be the same as the interval duration except in a situation where performance monitoring data could not be collected for any reason.

atucSESSs

Severely errored seconds at the near end.

atucUASs

Unavailable seconds at the near end.

atucValidIntervals

The number of previous 15-minute intervals in the interval table for which data was collected. Given that <n> is the maximum # of intervals supported. The value will be <n> unless the measurement was (re-)started within the last (<n>*15) minutes, in which case the value will be the number of complete 15-minute intervals for which the agent has at least some data. In certain cases (e.g., in the case where the agent is a proxy) it is possible that some intervals are unavailable. In this case, this interval is the maximum interval number for which data is available.

aturChanBackgroundErrBlks

Background errored blocks at far end.

aturChanCrcBlockLength

Indicates the length of the channel data-block on which the CRC operates. Refer to Line Code Specific MIBs, [11] and [12] for more information.

aturChanCorrectedBlks

Count of all blocks received with errors that were corrected since agent reset. These blocks are passed on as good data.

aturChanCurr15MinCorrectedBlks

Count of all blocks received with errors that were corrected on this channel within the current 15 minute interval.

aturChanCurr15MinUncorrectedBlks

Count of all encoded blocks transmitted on this channel within the current 15 minute interval.

aturChanCurr15MinRxBlks

Count of all encoded blocks received on this channel within the current 15 minute interval.

aturChanCurr15MinTimeElapsed

Total elapsed seconds in this interval. A full interval is 900 seconds.

aturChanCurr15MinTxBlks

Count of all encoded blocks transmitted on this channel within the current 15 minute interval.

aturChanCurr1DayCorrectedBlks

Count of all blocks received with errors that were corrected on this channel during the current day as measured by `adslAturChanPerfCurr1DayTimeElapsed`.

aturChanCurr1DayRxBlks

Count of all encoded blocks received on this channel during the current day as measured by `adslAturChanPerfCurr1DayTimeElapsed`.

aturChanCurr1DayTimeElapsed

Number of seconds that have elapsed since the beginning of the current 1-day interval.

aturChanCurr1DayTxBlks

Count of all encoded blocks transmitted on this channel during the current day as measured by `adslAturChanPerfCurr1DayTimeElapsed`.

aturChanCurr1DayUncorrectBlks

Count of all blocks received with uncorrectable errors on this channel during the current day as measured by `adslAturChanPerfCurr1DayTimeElapsed`.

aturChanCurrTxRate

Actual transmit rate on this channel.

aturChanCurrErrFast

CRC errors on fast buffer at far end.

aturChanErrInterleave

CRC errors on interleaved buffer at far end.

aturChanFECCorrectedFast

FEC corrections on fast buffer at far end.

aturChanFECCorrectedInterleave

FEC corrections on interleaved buffer at far end.

aturChanInterleaveDelay

Interleave Delay for this channel. Interleave delay applies only to the interleave channel and defines the mapping (relative spacing) between subsequent input bytes at the interleaver input and their placement in the bit stream at the interleaver output. Larger numbers provide greater separation between consecutive input bytes in the output bit stream allowing for improved impulse noise immunity at the expense of payload latency. In the case where the ifType is Fast(125), use noSuchObject.

aturChanIntervCorrectedBlks

Count of all blocks received with errors that were corrected on this channel during this interval.

aturChanIntervRxBBlks

Count of all encoded blocks received on this channel during this interval.

aturChanIntervTxBlks

Count of all encoded blocks transmitted on this channel during this interval.

aturChanIntervUncorrectBlks

Count of all blocks received with uncorrectable errors on this channel during this interval.

aturChanIntervValidData

This variable indicates if the data for this interval is valid.

aturChanInvalidIntervals

The number of intervals in the range from 0 to the value of `adslAturChanPerfValidIntervals` for which no data is available. This object will typically be zero except in cases where the data for some intervals are not available (e.g., in proxy situations).

aturChanNonSES

Non SES blocks received at the far end.

aturChanPerfBE

Far-end idle cell bit error count. Supported for ADSL2/ADSL2+ only.

aturChanPerfCD

Far-end delineated total cell count. Supported for ADSL2/ADSL2+ only.

aturChanPerfCU

Far-end user total cell count. Supported for ADSL2/ADSL2+ only.

aturChanPrev1DayCorrectedBlks

Count of all blocks received with errors that were corrected on this channel within the most recent previous 1-day period.

aturChanPrev1DayMoniSecs

The amount of time in the previous 1-day interval over which the performance monitoring information is actually counted. This value will be the same as the interval duration except in a situation where performance monitoring data could not be collected for any reason.

aturChanPrev1DayRxBlks

Count of all encoded blocks received on this channel within the most recent previous 1-day period.

aturChanPrev1DayTxBlks

Count of all encoded blocks transmitted on this channel within the most recent previous 1-day period.

aturChanPrev1DayUncorrectBlks

Count of all blocks received with uncorrectable errors on this channel within the most recent previous 1-day period.

aturChanPrevTxRate

The rate at the time of the last `adslAturRateChangeTrap` event. It is also set at initialization to prevent a trap being sent. Rate changes less than `adslAturThresh(*)RateDown` or less than `adslAturThresh(*)RateUp` will not cause a trap or cause this object to change. (*) == Fast or Interleave. See `AdslLineAlarmConfProfileEntry`.

aturChanRxBlks

Count of all encoded blocks received on this channel within the most recent previous 1-day period.

aturChanTxBlks

Count of all encoded blocks transmitted on this channel within the most recent previous 1-day period.

aturChanUncorrectBlks

Count of all blocks received with uncorrectable errors since agent reset.

aturChanValidIntervals

The number of previous 15-minute intervals in the interval table for which data was collected. Given that `<n>` is the maximum # of intervals supported. The value will be `<n>` unless the measurement was (re-)started within the last `(<n>*15)` minutes, in which case the value will be the number of complete 15 minute intervals for which the agent has at least some data. In certain cases (e.g., in the case where the agent is a proxy) it is possible that some intervals are unavailable. In this case, this interval is the maximum interval number for which data is available.

aturCurr15MinEss

The number of Errored Seconds in the current 15-minute interval. The errored second parameter is a count of one-second intervals containing one or more crc anomalies, or one or more los or sef defects.

aturCurr15MinLofs

The number of seconds in the current 15-minute interval when there was Loss of Framing.

aturCurr15MinLoss

The number of seconds in the current 15-minute interval when there was Loss of Signal.

aturCurr15MinLprs

The number of seconds in the current 15-minute interval when there was Loss of Power.

aturCurr15MinTimeElapsed

Total elapsed seconds in this interval.

aturCurr1DayEss

The number of Errored Seconds during the current day as measured by adslAturPerfCurr1DayTimeElapsed. The errored second parameter is a count of one-second intervals containing one or more crc anomalies, or one or more los or sef defects.

aturCurr1DayLofs

The number of seconds when there was Loss of Framing during the current day as measured by adslAturPerfCurr1DayTimeElapsed.

aturCurr1DayLoss

The number of the number of seconds when there was Loss of Signal during the current day as measured by adslAturPerfCurr1DayTimeElapsed.

aturCurr1DayLprs

The number of the number of seconds when there was Loss of Power during the current day as measured by adslAturPerfCurr1DayTimeElapsed.

aturCurr1DayTimeElapsed

Number of seconds that have elapsed since the beginning of the current 1-day interval.

aturCurrAtn

Measured difference in the total power transmitted by the peer ATU and the total power received by this ATU.

aturCurrAttainableRate

Indicates the maximum currently attainable data rate by the ATU. This value will be equal or greater than the current line rate.

aturCurrOutputPwr

Measured total output power transmitted by this ATU. This is the measurement that was reported during the last activation sequence.

aturCurrSnrMgn

Noise Margin as seen by this ATU with respect to its received signal in tenth dB.

aturCurrStatus

Indicates current state of the ATUR line. This is a combination of possible conditions. Due to the isolation of the ATUR when line problems occur, many state conditions like loss of power, loss of quality signal, and initialization errors, can not be determined. While trouble shooting ATUR, also use object, adslAtucCurrStatus.

Values 0: noDefect

There no defects on the line.

1: lossOfFraming

ATUR failure due to not receiving valid frame.

2: lossOfSignal

ATUR failure due to not receiving signal.

3: lossOfPower

ATUR failure due to loss of power.

4: lossOfSignalQuality

Loss of Signal Quality is declared when the Noise Margin falls below the Minimum Noise Margin, or the bit-error-rate exceeds 10^{-7} . This is intended to supplement ifOperStatus.

aturElapseds

Elapsed seconds at the far end.

aturEss

Count of the number of Errored Seconds since agent reset. The errored second parameter is a count of one-second intervals containing one or more crc anomalies, or one or more los or sef defects.

aturFECs

Seconds with one or more FECs at the far end.

aturHiBER

Seconds declared as hi BER by the far end.

aturIntervEss

The number of Errored Seconds in the interval. The errored second parameter is a count of one-second intervals containing one or more crc anomalies, or one or more los or sef defects.

aturIntervLofs

The number of seconds in the interval when there was Loss of Framing.

aturIntervLoss

The number of seconds in the interval when there was Loss of Signal.

aturIntervLprs

The number of seconds in the interval when there was Loss of Power.

aturIntervNum

Performance Data Interval number one is the most recent previous interval; interval 96 is 24 hours ago. Intervals 2..96 are optional.

aturIntervValidData

This variable indicates if the data for this interval is valid.

aturInvSerialNumber

The vendor specific string that identifies the vendor equipment.

aturInvVendorID

The vendor ID code is a copy of the binary vendor identification field defined by the PHY[10] and expressed as readable characters.

aturInvVersionNumber

The vendor specific version number sent by this ATU as part of the initialization messages. It is a copy of the binary version number field defined by the PHY[10] and expressed as readable characters.

aturInvalidIntervals

The number of intervals in the range from 0 to the value of `adslAturPerfValidIntervals` for which no data is available. This object will typically be zero except in cases where the data for some intervals are not available (e.g., in proxy situations).

aturLofs

Loss of frame seconds at the far end.

aturLoss

Loss of signal seconds at the far end.

aturLprs

Count of the number of Loss of Power failures since agent reset.

aturPerfCurr1DayFecsl

Number of seconds with one or more far end FECs in the last 24 hours.

aturPerfCurr15MinFecsL

Number of seconds with one or more far end FECs in the current 15 minute interval.

aturPerfFecsL

Seconds with one or more FECs at the far end.

aturPerfLossL

Number of far end loss of signal failures.

aturPerfLprs

Loss of power seconds at the far end (dying gasp).

aturPerfPrev1DayFecsL

Number of seconds with one or more far end FECs in the previous 24 hours.

aturPrev1DayEss

The number of Errored Seconds within the most recent previous 1-day period. The errored second parameter is a count of one-second intervals containing one or more crc anomalies, or one or more los or sef defects.

aturPrev1DayLofs

The number of seconds in the interval when there was Loss of Framing within the most recent previous 1-day period.

aturPrev1DayLoss

The number of seconds in the interval when there was Loss of Signal within the most recent previous 1-day period.

aturPrev1DayLprs

The number of seconds in the interval when there was Loss of Power within the most recent previous 1-day period.

aturPrev1DayMoniSecs

The amount of time in the previous 1-day interval over which the performance monitoring information is actually counted. This value will be the same as the interval duration except in a situation where performance monitoring data could not be collected for any reason.

aturSESSs

Severely errored seconds at the far end.

aturUASs

Unavailable seconds at the far end.

aturValidIntervals

The number of previous 15-minute intervals in the interval table for which data was collected. Given that <n> is the maximum # of intervals supported. The value will be <n> unless the measurement was (re-)started within the last (<n>*15) minutes, in which case the value will be the number of complete 15-minute intervals for which the agent has at least some data. In certain cases (e.g., in the case where the agent is a proxy) it is possible that some intervals are unavailable. In this case, this interval is the maximum interval number for which data is available.

availAllocIds

Total number of available alloc-Ids for this OLT.

Default 0

Access read-only

availDballocIds

Total number of available DBA alloc-Ids for this OLT.

Default 0

Access read-only

availMemory

The amount of physical memory that is unallocated and not in use by the unit/card.

avgTimeForLease

The average time to obtain a lease.

B

badAddr

The number of packets received with an incorrect address field. This counter is a component of the ifInErrors variable that is associated with the interface that represents this PPP link.

BadCtrls

The number of packets received on this link with an incorrect control field. This counter is a component of the ifInErrors variable that is associated with the interface that represents this PPP Link.

BadFCS

The number of received packets that have been discarded due to having an incorrect FCS. This counter is a component of the ifInErrors variable that is associated with the interface that represents this PPP link.

blockStatus

The V5.2 user port operational status.

- Values**
- 1: unblocked**
 - 2: localBlocked**
 - 3: remoteBlocked**
 - 4: bothBlocked**

Location V52UserPort

blockStatus

The link block operational status of the V5.2 link. The link blocking procedure is asymmetrical. The AN requests a link block, the LE decides whether or not to block the link, the LE releases the connection, then the LE re-establishes semi-permanent and preconnected connections.

Values 1: unblocked
 2: local
 3: remote
 4: both
Location V52Link

bridgeStatsBcastRcvd

Broadcast packets received.

bridgeStatsBcastSent

bridgeStatsBcastSent.

bridgeStatsErrorSent

bridgeStatsErrorSent.

bridgeStatsMcastRcvd

Multicast packets received.

bridgeStatsMcastSent

Multicast packets sent.

bridgeStatsUcastRcvd

Unicast packets received.

bridgeStatsUcastSent

Unicast packets sent.

C

cardActiveCalls

The number of currently active calls on the card.

CardAdminResets

The number of administration resets to this card.

CardCLEICode

CLEI (Common Language Equipment Identifier) code of the shelf assembly as stored in the EEPROM of the unit if defined otherwise the value will be set to NULL by the shelf monitor card.

CardEepromVersion

The EEPROM version for the card in this slot

CardFaultResets

The number of fault resets to this card.

CardFeatureBits

A string of 8 octets indicating the setting of the feature bits.

cardIncomingCallsBlocked

The number of blocked incoming calls on the card.

cardIncomingCallsCompleted

The number of completed outgoing calls on the card.

cardIndex

The slot where this Interface Group and GR-303 stack reside. A value of 0 indicates the Interface Group is not currently installed on a specific card by the GR-303 protocol stack.

Values 0 to 17

cardLineType

The card line type for this slot. This is only valid if the slot is provisioned.

cardMemStatus

Memory status of the card sent with memory trap.

- Values**
- 1: ramMemOK**
Less than 90% of RAM is used.
 - 2: ramMemLow**
More than 90% of RAM is used.
 - 3: flashMemOK**
Enough flash for maximum database.
 - 4: flashMemLow**
Not enough flash for maximum database.
 - 5: flashMemOut**
No more flash memory, data no longer persistent.

cardOutgoingCallsBlocked

The number of blocked outgoing calls on the card.

cardOutgoingCallsCompleted

The number of completed outgoing calls on the card.

CardSerialNumber

The serial number for the card in this slot.

CardState

The card state for the card.

- Values**
- 0: cardStateNone**
 - 1: cardStateBooting**
Card is booting from EEPROM.
 - 2: cardStatePost**
Card is performing POST tests.
 - 3: cardStateLoading**
Card is loading code from code loader.
 - 4: cardStateConfiguring**
Card is configuring.
 - 5: cardStateRunning**
Card is running.
 - 6: cardStateResetHold**
Card is blocked from booting.

7: cardStateDumping

8: cardStateFault

Card is in fault state.

9: cardStateReset

Card is resetting.

10: cardNotPresent

Card is not present in slot.

11: cardNotProvisioned

Card in slot is not provisioned.

cardSubType

The card sub type for this slot.

CardType

Type of DZS card.

Values

- 1: unknownType**
- 2: interShelf**
- 3: infoServices**
- 5: voiceASP**
- 6: t3Tdm**
- 7: t3Atm**
- 8: atmTrnkCr**
- 9: ethernet**
- 10: hdsl2**
- 11: adsl**
- 13: oc3Atm**
- 15: atmTrnk**
- 16: atmTrnkAal25**
- 17: atmTrnkT1e1**
- 18: tdmT1e1**
- 1001: sechtor100**
- 2001: zedge64**
- 5001: malcUplinkNt1**
- 5002: malcPots**
- 5003: malcMtac**
- 5004: malcAdsl**
- 5005: malcAdslPots**
- 5006: malcGshdsl(5006)**
- 5007: malcGshdslPots**
- 5008: malcPots900**
- 5009: malcDs3**
- 5010: malcAdslAc5**
- 5011: malcOc3**
- 5012: malcMtacFc**
- 5013: malcAc5AnxB**
- 5014: malcT1E1Tdm**
- 5015: malcAdslAc6**
- 5016: malcAdslPotsAc6**
- 5017: malcAdslSplAc6**

5018: malcAdslAnxBAc6

5019: malcAdslAnxBSpltAc6

5020: malcAdslC

5021: malcAdslPotsC

5022: malcAdslSpltC

cardVersion

The card version for the card in this slot.

cellsReceived

Total cells received.

cellsSent

Total cells sent.

chanCrcBlockLength

Indicates the length of the channel data-block on which the CRC operates.

chanCurrTxFastFec

Actual Forward Error Correction (FEC) redundancy related overhead for a fast channel. This parameter is not applicable to an interleaved (slow) channel.

For interleaved channels, a value of zero shall be returned..

chanCurrTxRate

Actual transmit data rate on this channel. Note: 1 kbps = 1000 bps.

chanCurrTxSlowBurstProtect

Actual level of impulse noise (burst) protection for an interleaved (slow) channel. This parameter is not applicable to fast channels. For fast channels, a value of zero shall be returned.

chanInterleaveDelay

Interleave Delay for this channel.

Interleave delay applies only to the interleave (slow) channel and defines the mapping (relative spacing) between subsequent input bytes at the interleaver input and their placement in the bit stream at the interleaver output. Larger numbers provide greater separation between consecutive input bytes in the output bit stream allowing for improved impulse noise immunity at the expense of payload latency.

In the case where the ifType is fast(125), return a value of zero.

cidrDest

The destination IP address of this route. This object may not take a Multicast (Class D) address value. Any assignment (implicit or otherwise) of an instance of this object to a value x must be rejected if the bitwise logical-AND of x with the value of the corresponding instance of the ipCidrRouteMask object is not equal to x.

cidrInfo

A reference to MIB definitions specific to the particular routing protocol which is responsible for this route, as determined by the value specified in the route's ipCidrRouteProto value. If this information is not present, its value should be set to the OBJECT IDENTIFIER {0 0}, which is a syntactically valid object identifier, and any implementation conforming to ASN.1 and the Basic Encoding Rules must be able to generate and recognize this value.

cidrMask

Indicate the mask to be logical-ANDed with the destination address before being compared to the value in the ipCidrRouteDest field. For those systems that do not support arbitrary subnet masks, an agent constructs the value of the ipCidrRouteMask by reference to the IP Address Class. Any assignment (implicit or otherwise) of an instance of this object to a value x must be rejected if the bitwise logical-AND of x with the value of the corresponding instance of the ipCidrRouteDest object is not equal to ipCidrRouteDest.

cidrNextHop

The address of the next system en route on remote routes. Otherwise, the value is 0.0.0.0.

cidrRouteNumber

The number of current ipCidrRouteTable entries in this routing domain that are not invalid.

circuitModeData

Circuit Mode Data. Refer to the corresponding object in the VCL table. If this value is not set using the Command Line Interface (CLI) or the DZS Management System (DZMS), the agent will use the AAL2 VCL entry default.

Values **1: enabled**
2: disabled

clientErrors

The number of errors that the DHCP client has encountered.

collCount

The number of per-frame media collisions for which a particular collision histogram cell represents the frequency on a particular interface.

collFrequencies

A count of individual MAC frames for which the transmission (successful or not) on a particular interface occurs after the frame has experienced exactly the number of collisions in the associated dot3CollCount object. For example, a frame which is transmitted on interface 77 after experiencing exactly 4 collisions would be indicated by incrementing only dot3CollFrequencies.77.4. No other instance of dot3CollFrequencies would be incremented in this example. This counter does not increment when the interface is operating in full duplex mode. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

controlInUnknownOpCodes

A count of MAC Control frames received on this interface that contain an opcode that is not supported by this device. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

cur15MinCrcAnom

The number of CRC anomalies since the start of the current measurement period.

cur15MinES

The number of errored-seconds since the start of the current measurement period.

cur15MinInits

The number of line initialization attempts in the current 15-minute interval. Includes both successful and failed attempts.

cur15MinLofs

The number of Loss of Framing failures in the current 15-minute interval.

cur15MinLols

The number of Loss of Link failures in the current 15-minute interval.

cur15MinLoss

The number of Loss of Signal failures in the current 15-minute interval.

cur15MinLOSW

The number of loss-of-sync-word seconds since the start of the current measurement period.

cur15MinLofs

The number of Loss of Framing failures in the current 15 minute interval.

cur15MinLols

The number of Loss of Link failures in the current 15 minute interval.

cur15MinLoss

The number of Loss of Signal failures in the current 15 minute interval.

cur15MinSES

The number of severely-errored-seconds since the start of the current measurement period.

cur15MinTimeElapsed

The number of seconds elapsed since the start of the current measurement period.

cur15MinUAS

The number of unavailable seconds since the start of the current measurement period.

currActiveCalls

The number of calls currently in use. **CurrActiveCalls** is used in conjunction with **MaxActiveCalls**, the maximum number of calls available, to provide a metric for describing current utilization. This does not imply that a call has completed, but that the DS0 timeslot has been allocated for a call by the GR-303 interface. The DS1 frame is composed of 24 consecutively numbered (1-24) timeslots (channels or DS0s). The maximum number of currently active calls is the maximum number of DS1s assigned to the IG times the maximum number of channels or DS0s per DS1 less 2 or 4 channels for primary only EOC/TCM and primary/secondary EOC/TCM = $(\{1..28\} * 24 - \{2,4\}) = \{22..670\}$ 28*24 equals maximum number of DS0s on a DS3, with 0 signaling channels. If you have one 28 T1 IG with no redundancy you can have 672-2 signaling DS0's, or 670, or 670 DS0s available for active calls. If you have a single T1 IG, you have only 24-2 signaling DS0s, or 22, DS0s available for active calls.

currant

Measured difference in the total power transmitted by the peer Vtu and the total power received by this Vtu. The effective range is 0 to +63.75 dB.

currAttainableRate

Indicates the maximum currently attainable data rate in steps of 1000 bits/second by the Vtu. This value will be equal to or greater than vds1PhysCurrLineRate. Note that for SCM, the minimum and maximum data rates are equal.

currBadAmiViolation

The current number of frames that have an incorrect number of AMI violations.

currCallsBlocked

The number of calls that were blocked due to no available time-slot.

currentBESs

The current number of Bursty Errored Seconds.

currFeNumFailures

The current number of Far End failures.

currLineRate

Indicates the current data rate in steps of 1000 bits/second by the Vtu. This value will be less than or equal to vdslPhysCurrAttainableRate.

currIsdnCalls

The number of ISDN calls completed.

currentCCVs

The current number of C-bit Coding Violations.

currentCESs

The number of C-bit Errored Seconds.

currentCSEs

The number of C-bit Severely Errored Seconds.

currentCSSs

The current number of Controlled Slip Seconds.

currentDMs

The current number of degraded minutes.

currentESs

The number of Errored Seconds.

currentGr303ProtocolErrors

The number of GR-303 protocol errors.

currentInBoundCalls

The number of inbound calls completed.

currentIndex

The index value that uniquely identifies the DS1 interface to which this entry is applicable. The interface identified by a particular value of this index is the same interface as identified by the same value as a zhLineIndex object instance.

currentLCVs

The number of Line Code Violations (LCVs).

currentLESS

The number of Line Errored Seconds (LESS).

currLogicalChannelId

The value of this variable indicates the current logical c-channel identifier of the physical c-channel. This is useful when a physical c-channel that was provisioned as an active channel fails - if the physical c-channel was a member of a V5.2 protection group the logical c-channel identifier will be assigned to a standby physical c-channel. This variable is only valid when read in conjunction with v52CChannel status.

currentOutBoundCalls

The number of outbound calls completed.

currentOutBoundCallsBlocked

The number of outbound calls that were blocked due to no available timeslot.

currentOutEOCLapdRcvd

The number of LAPD frames received on the Embedded Operations Channel (EOC).

currentOutEOCLapdSent

The number of LAPD frames sent on the Embedded Operations Channel (EOC).

currentOutEOCRcvdErrs

The number of LAPD frames received in error on the Embedded Operations Channel (EOC).

currentOutTMCLapdRcvd

The number of LAPD frames received on the Timeslot Management Channel (TMC).

currentOutTMCLapdRcvdErrs

The number of LAPD frames received in error on the Timeslot Management Channel (TMC).

currentOutTMCLapdSent

The number of LAPD frames sent on the Timeslot Management Channel (TMC).

currentPCVs

The number of Path Coding Violations.

currentPESs

The counter associated with the number of P-bit Errored Seconds.

currentPSEs

The counter associated with the number of P-bit Severely Errored Seconds.

currentSEFSs

The number of Severely Errored Framing Seconds.

currentSEs

The number of Severely Errored Seconds.

currentUASs

The number of Unavailable Seconds.

curlpAddr

The IP address currently assigned to the subscriber. If more than one address is assigned then this value should indicate the most recently assigned IP address assigned to the subscriber. The full list of addresses assigned to the subscriber can be found in the DHCP tables.

currActiveCalls

The number of calls currently active. Not supported.

currErrorSeconds

The current number of ISDN activation seconds.

currFsyncSeconds

The current number of frame synch loss seconds.

currInboundCalls

The number of inbound calls completed.

currLapdRcvd

The number LAPD frames received on all c-paths for this interface group.

currLapdRcvdErrs

The number LAPD frames received in error on all c-paths for this interface group.

currLapdSent

The number of LAPD frames sent on all c-paths for this interface group.

currOutboundCalls

The number of outbound calls completed.

currOutboundCallsBlocked

The number of outbound calls that were blocked due to no available timeslot.

currOutputPwr

Measured total output power transmitted by this VTU. This is the measurement that was reported during the last activation sequence.

currPstnInboundCalls

The number of PSTN inbound calls completed.

currPstnOutboundCalls

The number of PSTN outbound calls completed.

currSnrMargin

Noise Margin as seen by this Vtu with respect to its received signal in 0.25dB. The effective range is -31.75 to +31.75 dB.

currStatus

Indicates current state of the Vtu line. This is a bit-map of possible conditions. The various bit positions are:

0 => noDefect => There are no defects on the line.

1 => lossOfFraming => Vtu failure due to not receiving a valid frame.

2 => lossOfSignal => Vtu failure due to not receiving signal.

3 => lossOfPower => Vtu failure due to loss of power.

4 => lossOfSignalQuality => Loss of Signal Quality is declared when the Noise Margin falls below the Minimum Noise Margin, or the bit-error-rate exceeds 10^{-7} .

5 => lossOfLink => Vtu failure due to inability to link with peer Vtu. Set whenever the transceiver is in the 'Warm Start' state.

6 => dataInitFailure => Vtu failure during initialization due to bit errors corrupting startup exchange data.

7 => configInitFailure => Vtu failure during initialization due to peer Vtu not able to support requested configuration.

8 => protocolInitFailure Vtu failure during initialization due to incompatible protocol used by the peer Vtu.

9 => noPeerVtuPresent => Vtu failure during initialization due to no activation sequence detected from peer Vtu. This is intended to supplement ifOperStatus.

currTimeElapsed

The number of seconds that have elapsed that these counts represent during the current 15-minute period.

currUnbalancedFrame

The current count of unbalanced frames.

currV52ProtocolErrors

The number of V5.2 Protocol errors.

D

deltDevice

This object provides a way to query which device (chip) is providing the DELT measurement on this interface.

Values **other (1)**
a device not in the list below

conexantG24 (2)
Conexant G24 chip

deltFailReason

The status of the recent occasion the loop diagnostics mode forced (LDSF) was issued for the associated line.

Values **none (1)**
The default value in case LDSF was never requested for the associated line.

success(2)
The recent command completed successfully.

inProgress(3)
The Loop Diagnostics process is in progress.

unsupported (4)
The NE (Network Element) or the line card doesn't support LDSF.

cannotRun (5)

The NE cannot initiate the command, due to a non specific reason.

aborted (6)

The Loop Diagnostics process aborted.

failed (7)

The Loop Diagnostics process failed.

illegalMode (8)

The NE cannot initiate the command, due to the specific mode of the relevant line.

adminUp (9)

The NE cannot initiate the command, as the relevant line is administratively 'Up'.

tableFull (10)

The NE cannot initiate the command, due to reaching the maximum number of rows in the results table.

noResources (11)

The NE cannot initiate the command, due to lack of internal memory resources.

deltTimeCompleted

This object provides a date and time stamp corresponding to the last completion of a test where valid results were obtained. If no test results are available, perhaps in the situation where no test has ever been started, this object shall return a value that has all the fields set to zero.

downAttainLineRate

This is the MAX line rate that can be supported on the line in the downstream direction.

downLineRate

The DSL downstream (co->cpe) line rate on this interface.

dsConst

Downstream constellation size in use.

dsCorrOctets

Number of errored octets corrected by Reed-Solomon in the downstream channel.

dsLineRate

Measured downstream line rate.

dspVersion

The version of the DSP/Framer VLSI in the transceiver.

dsRemoteBip8Errors

Downstream Reed-Solomon code word and depth.

dsRsFinal

Downstream Reed-Solomon code word and depth.

dsSnr

Measured downstream EQ output SNR (in dB times 8).

dsUncorrOctets

Number of uncorrectable octets in the downstream channel.

dtmfDialedDigits

DTMF Dialed Digits. Refer to the corresponding object in the VCL table. Valid values are enabled and disabled. If this value is not set by CLI or DZMS, the agent will use the AAL2 VCL entry default.

Values 1: enabled
2: disabled

E

eocPrimarySvcState

The conditions that are currently present at the entity that can affect the service state.

Values 1: active
 2: standby
 3: outOfService

eocSecondarySvcState

The conditions that are currently present at the entity that can affect the service state.

Values 1: active
 2: standby
 3: outOfService

errorRx

IGMP packet is bad received.

F

farEndCurrentBESSs

The number of Far End Bursty Errored Seconds.

farEndCurrentCCVs

The counter associated with the number of Far End C-bit Coding Violations reported using the far end block error count.

farEndCurrentCESs

The counter associated with the number of Far End C-bit Errored Seconds.

farEndCurrentCSESSs

The counter associated with the number of Far End C-bit Severely Errored Seconds.

farEndCurrentCSSs

The number of Far End Controlled Slip Seconds.

farEndCurrentDMs

The number of Far End Degraded Minutes.

farEndCurrentESs

The number of Far End Errored Seconds.

farEndCurrentIndex

The index value that uniquely identifies the DS1 interface to which this entry is applicable. The interface identified by a particular value of this index is the same as the interface identified by the same value of zhLineIndex.

farEndCurrentInvalidIntervals

The number of intervals in the range from 0 to dsx3FarEndValidIntervals for which no data is available. This object will be zero except in cases where the data for some intervals are not available, for example, in proxy situations.

farEndCurrentLESS

The number of Far End Line Errored Seconds.

farEndCurrentPCVs

The number of Far End Path Coding Violations.

farEndCurrentSEFSs

The number of Far End Severely Errored Framing Seconds.

farEndCurrentSESSs

The number of Far End Severely Errored Seconds.

farEndCurrentUASs

The number of Unavailable Seconds.

farEndIntervalBESSs

The number of Far End Bursty Errored Seconds.

farEndIntervalCCVs

The counter associated with the number of Far End C-bit Coding Violations reported using the far end block error count.

farEndIntervalCESs

The counter associated with the number of Far End C-bit Errored Seconds encountered by a DS3 interface in one of the previous 96 individual 15-minute intervals. In the case where the agent is a proxy and data is not available, noSuchInstance will be returned.

farEndIntervalCSEs

The counter associated with the number of Far End C-bit Severely Errored Seconds.

farEndIntervalCSSs

The number of Far End Controlled Slip Seconds.

farEndIntervalDMs

The number of Far End Degraded Minutes.

farEndIntervalESs

The number of Far End Errored Seconds.

farEndIntervalIndex

The index value that uniquely identifies the DS1 interface to which this entry is applicable. The interface identified by a particular value of this index is identical to the interface identified by the same value of zhLineIndex.

farEndIntervalLESs

The number of Far End Line Errored Seconds.

farEndIntervalNumber

A number between one and 96, where one is the most recently completed 15-minute interval and 96 is the 15-minutes interval completed 23 hours and 45 minutes prior to interval 1.

farEndIntervalPCVs

The number of Far End Path Coding Violations.

farEndIntervalSEFSs

The number of Far End Severely Errored Framing Seconds.

farEndIntervalSESSs

The number of Far End Severely Errored Seconds.

farEndIntervalUASs

The number of Unavailable Seconds.

farEndIntervalValidData

This variable indicates if the data for this interval is valid.

farEndInvalidIntervals

The number of intervals in the range from 0 to zhFarEndValidIntervals for which no data is available. This object will be zero except in cases where the data for some intervals are not available, for example, proxy situations.

farEndTimeElapsed

The number of seconds that have elapsed since the beginning of the far end current error measurement period. If, for some reason the current interval exceeds the maximum value, the agent will return the maximum value.

farEndTotalBESSs

The number of Bursty Errored Seconds (BESSs) encountered by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

farEndTotalCCVs

The counter associated with the number of Far End C-bit Coding Violations reported using the far end block error count encountered by a DS3 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

farEndTotalCESs

The counter associated with the number of Far End C-bit Errored Seconds encountered by a DS3 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

farEndTotalCSEsSs

The counter associated with the number of Far End C-bit Severely Errored Seconds encountered by a DS3 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

farEndTotalCSSs

The number of Far End Controlled Slip Seconds encountered by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

farEndTotalDMs

The number of Degraded Minutes (DMs) encountered by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

farEndTotalESs

The number of Far End Errored Seconds encountered by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

farEndTotalIndex

The index value which uniquely identifies the DS1 interface to which this entry is applicable. The interface identified by a particular value of this index is identical to the interface identified by the same value of zhLineIndex.

farEndTotalLESSs

The number of Far End Line Errored Seconds encountered by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

farEndTotalPCVs

The number of Far End Path Coding Violations reported using the far end block error count encountered by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

farEndTotalSEFSs

The number of Far End Severely Errored Framing Seconds encountered by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

farEndTotalSESs

The number of Far End Severely Errored Seconds encountered by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

farEndTotalUASs

The number of Unavailable Seconds encountered by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

farEndValidIntervals

The number of previous far end intervals for which data was collected. The value will be 96 unless the interface was brought online within the last 24 hours, in which case the value will be the number of complete 15-minute far end intervals since the interface has been online.

faxDemoRemo

Fax Demodulation/Remodulation. Refer to the corresponding object in the VCL table. If this value is not set by CLI or DZMS, the agent will use the AAL2 VCL entry default.

Values **1: enabled**
 2: disabled

feLineCurrCV

The counter associated with the number of Far End Coding Violations reported through the far end block error count that is detected by a SONET/SDH medium/section/line interface during the current 15-minute interval

feLineCurrES

The counter associated with the number of Far End Errored Seconds detected by a SONET/SDH interface during the current 15-minute interval.

feLineCurrSES

The counter associated with the number of Far End Severely Error Seconds detected by a SONET/SDH medium/section/line interface during the current 15-minute interval.

feLineCurrUAS

The counter associated with the number of Far End Unavailable Seconds detected by a SONET/SDH medium/section/line interface during the current 15-minute interval

feLineIntervCV

The counter associated with the number of Far End Coding Violations reported using the far end block error count detected by a SONET/SDH line interface in a particular 15-minute interval during the last 24 hours.

feLineIntervES

The counter associated with the number of Far End Errored Seconds detected by a SONET/SDH line interface during a particular 15-minute interval during the last 24 hours.

feLineIntervNum

A number between one and 96 that identifies the interval for which the statistics are available. The interval identified by the number one is the 15-minute interval completed most recently, and interval number N is the interval that immediately proceeds the interval identified as N-1.

feLineIntervSES

The counter associated with the number of Far End Severely Errored Seconds detected by a SONET/SDH line interface during a particular 15-minute interval during the last 24 hours.

feLineIntervUAS

The counter associated with the number of Far End Unavailable Seconds encountered by a SONET/SDH line interface during a particular 15-minute interval during the last 24 hours

feLineIntervValid

Indicates if the data for this interval is valid.

fePathCurrCV

The counter associated with the number of Far End Coding Violations reported using the far end block error count detected by a SONET/SDH path interface during the current 15-minute interval.

fePathCurrES

The counter associated with the number of Far End Errored Seconds detected by a SONET/SDH interface during the current 15-minute interval.

fePathCurrSES

The counter associated with the number of Far End Severely Errored Seconds encountered by a SONET/SDH path interface during the current 15-minute interval.

fePathCurrUAS

The counter associated with the number of Far End Unavailable Seconds detected by a SONET/SDH path interface during the current 15-minute interval.

fePathIntervCV

The counter associated with the number of Far End Coding Violations reported using the far end block error count detected by a SONET/SDH path interface during a particular 15-minute interval during the last 24 hours.

fePathIntervES

The counter associated with the number of FarEnd Errored Seconds detected by a SONET/SDH path interface during a particular 15-minute interval during the last 24 hours.

fePathIntervNum

A number between one and 96 that identified the interval for which the statistics are available. The interval one is the most recently completed 15-minute interval, and the interval N is the interval immediately proceeding the interval called N-1.

fePathIntervSES

The counter associated with the number of Far End Severely Errored Seconds detected by a SONET/SDH path interface during a particular 15-minute interval during the last 24 hours.

fePathIntervUAS

The counter associated with the number of Far End Unavailable Seconds detected by a SONET/SDH path interface during a particular 15-minute interval during the last 24 hours.

fePathIntervvalid

Indicates if the data for this interval is valid.

fifoOverflowCount

Number of cells dropped because of first in, first out (FIFO) overflow.

firmwareVersion

The firmware release number in ASCII.

fixBitStatus

Indicates that the fix bit feature is enabled. This value applies only to the Central Office (CO) side and should be ignored for the customer (CPE) side

Values **1: fix-bit-enable**
 2: fix-bit-disable

forwDatagrams

The number of input datagrams for which this entity was not the final IP destination, as a result of which an attempt was made to find a route to forward them to that final destination. In entities which do not act as IP routers, this counter will include only those packets which were Source-Routed using this entity when the Source-Route option processing was successful.

fragCreates

The number of IP datagram fragments that have been generated as a result of fragmentation at this entity.

fragFails

The number of IP datagrams that have been discarded because they needed to be fragmented at this entity but could not be, usually because the Don't Fragment flag was set.

fragOks

The number of IP datagrams that have been successfully fragmented at this entity.

framingErrs

The total number of characters with a framing error, input from the port since system reinitialization and while the port state was up or test.

G

genQueryRx

general IGMP queries received.

genQueryTx

general IGMP queries transmitted.

H

h2LlastChange

The value of sysUpTime at the time this VP cross-connect entered its current operational in the high to low direction.

hecCorrectedCount

Number of corrected HEC cells.

hecUncorrectedCount

Number of uncorrected dropped HEC cells.

h2LoperStatus

The operational status of the VP cross-connect in one direction; (that is, from the high to low direction).

HighIfIndex

The ifIndex value of the ATM interface for this VP cross-connect. The term high implies that this ATM interface has the numerically higher ifIndex value than the other ATM interface identified in the same atmVpCrossConnectEntry.

HighThreshold

The upper temperature when an alarm will be sent.

HighVpi

The VPI value at the ATM interface associated with the VP cross-connect that is identified by atmVpCrossConnectHighIfIndex.

I

ifAdminStatus

The desired state of the interface.

- Values**
- 1: up**
Ready to pass packets.
 - 2: down**
When a managed system initializes, all interfaces start with **ifAdminStatus** in the **down** state.
 - 3: testing**
In some test mode. No operational packets can be passed.

ifDescr

A textual string containing information about the interface. This string should include the name of the manufacturer, the product name and the version of the interface hardware/software.

ifDs3PlcpAlarmState

Indicates if there is an alarm present for the DS3 PLCP. The value `receivedFarEndAlarm` means that the DS3 PLCP has received an incoming Yellow Signal, the value `incomingLOF` means that the DS3 PLCP has declared a loss of frame (LOF) failure condition, and the value `noAlarm` means that there are no alarms present. Transition from the failure to the no alarm state occurs when no defects (LOF) are received for more than 10 seconds.

Values

- 1: noAlarm**
- 2: receivedFarEndAlarm**
- 3: incomingLOF**

ifDs3PlcpSEFSs

The number of DS3 PLCP Severely Errored Framing Seconds (SEFS). Each SEFS represents a one-second interval which contains one or more SEF events.

ifDs3PlcpUASs

The counter associated with the number of Unavailable Seconds encountered by the PLCP.

ifFeHECEvents

The number of times Near-end Header Error Control (HEC) violations occur.

ifFeNCDEvents

The number of times the Near-end No Cell Delineation (NCD) events occur.

ifFeOCDEvents

The number of times Far-end Out of Cell Delineation (OCD) events occur.

ifHCInBits

The total number of bits received on the interface, including framing characters. This object is a 64-bit version of `ifInBits`.

ifHCInBroadcastPkts

The number of packets, delivered by this sub-layer to a higher (sub-)layer, which were addressed to a broadcast address at this sub-layer.

ifHCInMulticastPkts

The number of packets, delivered by this sub-layer to a higher (sub-)layer, which were addressed to a multicast address at this sub-layer. For a MAC layer protocol, this includes both Group and Functional addresses.

ifHCInOctets

The total number of octets received on the interface, including framing characters. This object is a 64-bit version of ifInOctets.

ifHCInUcastPkts

The number of packets, delivered by this sub-layer to a higher (sub-)layer, which were not addressed to a multicast or broadcast address at this sub-layer. This object is a 64-bit version of ifInUcastPkts.

ifHCOutBits

The total number of bits transmitted out of the interface, including framing characters. This object is a 64-bit version of ifOutBits.

ifHCOutBroadcastPkts

The total number of packets that higher-level protocols requested be transmitted, and which were addressed to a broadcast address at this sub-layer, including those that were discarded or not sent. This object is a 64-bit version of ifOutBroadcastPkts.

ifHCOutMulticastPkts

The total number of packets that higher-level protocols requested be transmitted, and which were addressed to a multicast address at this sub-layer, including those that were discarded or not sent. For a MAC layer protocol, this includes both Group and Functional addresses. This object is a 64-bit version of ifOutMulticastPkts.

ifHCOutOctets

The total number of octets transmitted out of the interface, including framing characters. This object is a 64-bit version of ifOutOctets.

ifHECEvents

The number of times Near-end Header Error Control (HEC) violations occur.

ifHCOutUcastPkts

The total number of packets that higher-level protocols requested be transmitted, and which were not addressed to a multicast or broadcast address at this sub-layer, including those that were discarded or not sent. This object is a 64-bit version of ifOutUcastPkts.

ifHighSpeed

An estimate of the interface's current bandwidth in units of 1,000,000 bits per second. If this object reports a value of `n' then the speed of the interface is somewhere in the range of `n-500,000' to `n+499,999'. For interfaces which do not vary in bandwidth or for those where no accurate estimation can be made, this object should contain the nominal bandwidth. For a sub-layer which has no concept of bandwidth, this object should be zero.

ifInBits

The total number of bits received on the interface, including framing characters. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

ifInBroadcastPkts

The number of packets, delivered by this sub-layer to a higher (sub-)layer, which were addressed to a broadcast address at this sub-layer.

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

ifIndex (IpCidrRoute)

The ifIndex value which identifies the local interface through which the next hop of this route should be reached. A value of 0 means there is no next hop interface.

ifIndex

A unique value, greater than zero, for each interface. It is recommended that values are assigned contiguously starting from 1. The value for each interface

sublayer must remain constant at least from one reinitialization of the entity's network management system to the next reinitialization.

iflnIndex

The value of rs232PortIndex for the port to which this entry belongs.

Location RS232InSignal, RS232OutSignal

iflnDiscards

The number of inbound packets that were chosen to be discarded even though no errors had been detected to prevent their being deliverable to a higher-layer protocol. One reason for discarding a packet could be to free up buffer space. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

iflnErrors

For packet-oriented interfaces, the number of inbound packets that contained errors preventing them from being deliverable to a higher-layer protocol. For character-oriented or fixed-length interfaces, the number of inbound transmission units that contained errors preventing them from being deliverable to a higher-layer protocol.

iflnOctets

The total number of octets received on the interface, including framing characters. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

iflnUcastPkts

The number of packets, delivered by this sub-layer to a higher (sub-)layer, which were not addressed to a multicast or broadcast address at this sub-layer.

iflnUnknownProtos

For packet-oriented interfaces, the number of packets received via the interface which were discarded because of an unknown or unsupported protocol. For character-oriented or fixed-length interfaces that support protocol multiplexing the number of transmission units received via the interface which were discarded because of an unknown or unsupported protocol. For any interface that does not support protocol multiplexing, this counter will always be 0.

ifInErrors

The number of inbound packets on a packet oriented interface that contained errors preventing them from being deliverable to a higher-layer protocol. For character-oriented or fixed-length interfaces, the number of inbound transmission units that contained errors preventing them from being deliverable to a higher-layer protocol. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of `ifCounterDiscontinuityTime`.

ifInNUcastPkts

The number of packets, delivered by this sublayer to a higher (sub-)layer, which were addressed to a multicast or broadcast address at this sublayer. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of `ifCounterDiscontinuityTime`. This object is deprecated in favour of `ifInMulticastPkts` and `ifInBroadcastPkts`.

ifInUcastPkts

The number of packets, delivered by this sublayer to a higher layer, that were not addressed to a multicast or broadcast address at this sublayer. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of `ifCounterDiscontinuityTime`.

ifInUnknownProtos

For packet-oriented interfaces, the number of packets received using the interface which were discarded because of an unknown or unsupported protocol. For character-oriented or fixed-length interfaces that support protocol multiplexing the number of transmission units received using the interface which were discarded because of an unknown or unsupported protocol. For any interface that does not support protocol multiplexing, this counter will always be 0. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of `ifCounterDiscontinuityTime`.

ifLastChange

The value of `sysUpTime` at the time the interface entered its current operational state. If the current state was entered prior to the last reinitialization of the local network management subsystem, then this object contains a zero value.

ifMtu

The size of the largest packet which can be sent/received on the interface, specified in octets. For interfaces that are used for transmitting network datagrams, this is the size of the largest network datagram that can be sent on the interface.

ifNCDEvents

The number of times the Near-end No Cell Delineation (NCD) events occur.

ifOCDEvents

The number of times the Out of Cell Delineation (OCD) events occur. If seven consecutive ATM cells have Header Error Control (HEC) violations, an OCD event occurs. A high number of OCD events may indicate a problem with the TC Sublayer.

ifOperStatus

The operational status of the interface.

Values 1: up

Ready to pass packets.

2: down

3: testing

In test mode.

4: unknown

Status cannot be determined.

5: dormant

6: notPresent

Some component is missing.

7: lowerLayerDown

Interface is down due to the state of lower layer interface(s).

ifOutBits

The total number of bits transmitted out of the interface, including framing characters. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

ifOutDiscards

The number of outbound packets that were chosen to be discarded even though no errors had been detected to prevent their being transmitted. One reason for discarding a packet could be to free up buffer space. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of `ifCounterDiscontinuityTime`.

ifOutErrors

The number of outbound packets on packet oriented interface that could not be transmitted because of errors. For character-oriented or fixed-length interfaces, the number of outbound transmission units that could not be transmitted because of errors. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of `ifCounterDiscontinuityTime`.

ifOutNUcastPkts

The total number of packets that higher-level protocols requested be transmitted, and which were addressed to a multicast or broadcast address at this sublayer, including those that were discarded or not sent. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of `ifCounterDiscontinuityTime`. This object is deprecated in favour of `ifOutMulticastPkts` and `ifOutBroadcastPkts`.

ifOutOctets

The total number of octets transmitted out of the interface, including framing characters. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of `ifCounterDiscontinuityTime`.

ifOutUcastPkts

The total number of packets that higher-level protocols requested be transmitted, and which were not addressed to a multicast or broadcast address at this sublayer, including those that were discarded or not sent. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of `ifCounterDiscontinuityTime`.

InPauseFrame

The count of MAC Control frames received on this interface with an opcode indicating the PAUSE operation.

ifPhysAddress

The interface address at its protocol sublayer. For example, for an 802.x interface, this object normally contains a MAC address. The interface's media-specific MIB must define the bit and byte ordering and the format of the value of this object. For interfaces which do not have such an address (e.g., a serial line), this object should contain an octet string of zero length.

ifSpeed

An estimate of the interface current bandwidth in bits per second. For interfaces which do not vary in bandwidth or for those where no accurate estimation can be made, this object should contain the nominal bandwidth. If the bandwidth of the interface is greater than the maximum value reportable by this object then this object should report its maximum value (4,294,967,295) and ifHighSpeed must be used to report the interface's speed. For a sublayer which has no concept of bandwidth, this object should be zero.

ifTCArmState

Whether there is an alarm present for the TC Sublayer.

Values 1: noAlarm

The TC Sublayer is currently not in the LCD defect maintenance state.

2: lcdFailure

The TC Sublayer is currently in the Loss of Cell Delineation (LCD) defect maintenance state.

igCurrentEOCLapdRcvdErrs (Gr303lg)

The current number of LAPD frames received in error on the Embedded Operations Channel (EOC).

igTotalEOCLapdRcvdErrs (Gr303lg)

The total number of LAPD frames received in error on the Embedded Operations Channel (EOC).

inAddrErrors

The number of input datagrams that were discarded because the IP address in their IP header destination field was not a valid address to be received at this entity. This count includes invalid addresses (0.0.0.0) and addresses of unsupported Classes (Class E). Entities that are not IP routers and therefore do not forward datagrams, use this counter because it includes datagrams that were discarded because the destination address was not a local address.

inDelivers

The total number of input datagrams successfully delivered to IP user-protocols (including ICMP).

Index

An index into the temperature table. It is index1. Number of temperature sensors with no missing indices.

inDiscards

The number of input IP datagrams for which no problems were encountered to prevent their continued processing, but that were discarded for lack of buffer space. This counter does not include any datagrams discarded while awaiting reassembly.

InGoodOctets

The LQR InGoodOctets counter for this link. Refer to section 2.2, Counters, of RFC1333.

inHdrErrors

The number of input datagrams discarded due to errors in their IP headers, including bad checksums, version number mismatch, other format errors, time-to-live exceeded, and errors discovered in processing their IP options.

InLQRs

The value of the InLQRs counter on the local node for the link identified by ifIndex.

inReceives

The total number of input datagrams received from interfaces, including those received in error.

inSigNumber

The number of input signals for the port in the input signal table (rs232PortInSigTable). The table contains entries only for those signals the software can detect and that are useful to observe.

interclsdnCalls

The number of ISDN calls completed.

intervalBESs

The number of Bursty Errored Seconds.

intervalCCVs

The number of C-bit Coding Violations.

intervalCESs

The number of C-bit Errored Seconds.

intervalCSEs

The number of C-bit Severely Errored Seconds.

intervalCSSs

The number of Controlled Slip Seconds.

intervalDMs

The number of Degraded Minutes.

intervalESs

The number of Errored Seconds.

intervalGr303ProtocolErrors

The number of GR-303 protocol errors.

intervalInBoundCalls

The number of inbound calls completed.

intervalIndex

The index value which uniquely identifies the DS1 interface to which this entry is applicable. The interface identified by a particular value of this index is the same interface as identified by the same value as a zhLineIndex object instance.

intervalLCVs

The number of Line Code Violations.

intervalLEsS

The number of Line Errored Seconds.

intervalNum

A number between one and 96, where one is the most recently completed 15-minute interval and 96 is the 15-minutes interval completed 23 hours and 45 minutes prior to interval 1.

intervalNumber

A number between one and 96, where one is the most recently completed 15-minute interval and 96 is the 15-minutes interval completed 23 hours and 45 minutes prior to interval 1.

intervalOutBoundCalls

The number of outbound calls completed.

intervalOutBoundCallsBlocked

The number of outbound calls that were blocked due to no available timeslot.

intervalOutEOCLapdErrs

The number of LAPD frames received in error on the Embedded Operations Channel (EOC) in this interval.

intervalOutEOCLapdRcvd

The number of LAPD frames received on the Embedded Operations Channel (EOC) in this interval.

intervalOutEOCLapdSent

The number of LAPD frames sent on the Embedded Operations Channel (EOC) in this interval.

intervalOutTMCLapdRcvd

The number LAPD frames received in error on the Time-slot Management Channel (TMC) in this interval.

intervalOutTMCLapdRcvdErrs

The number of LAPD frames received in error on the Embedded Operations Channel (EOC) in this interval.

intervalOutTMCLapdSent

The number of LAPD frames sent on the Time-slot Management Channel (TMC) in this interval.

intervalPCVs

The counter associated with the number of P-bit Coding Violations.

intervalPESs

The counter associated with the number of P-bit Errored Seconds.

intervalPSESs

The counter associated with the number of P-bit Severely Errored Seconds.

intervalSEFSs

The number of Severely Errored Framing Seconds.

intervalSESs

The number of Severely Errored Seconds.

intervalPCVs

The number of Path Coding Violations.

intervalUASs

The number of Unavailable Seconds. This object may decrease if the occurrence of unavailable seconds occurs across an interval boundary.

intervalValidData

This variable indicates if the data for this interval is valid.

intervCallsBlocked

The number of calls that were blocked due to no available time-slot.

intervInboundCalls

The number of inbound calls completed.

intervLapdRcvd

The number of LAPD frames received on all c-paths for this interface group.

intervLapdRcvdErrs

The number of LAPD frames received in error on all c-paths for this interface group.

intervLapdSent

The number of LAPD frames sent on all c-paths for this interface group.

intervOutboundCalls

The number of outbound calls completed.

intervOutboundCallsBlocked

The number of outbound calls that were blocked due to no available timeslot.

intervPstnInboundCalls

The number of PSTN inbound calls completed.

intervPstnOutboundCalls

The number of PSTN outbound calls completed.

intervV52ProtocolErrors

The number of V5.2 protocol errors.

invalidIntervals

The number of intervals in the range from 0 to zwValidIntervals for which no data is available. This object will be zero except in cases where the data for some intervals are not available, for example, in proxy situations.

ipAddrInUse

This value indicates the number of IP addresses that are currently being used by the subscriber. This value cannot exceed subDataMaxAddrAllowed.

ipIfOperStatus

The current IP operational status of the subscriber's data interface, this status reflects the ifOperStatus of the interface MIB using the identical syntax.

- Values**
- 1: up**
Ready to pass packets.
 - 2: down**
The admin status is down or there is a fault that prevents the interface from passing packets testing.
 - 3: testing**
The interface has been placed in a testing state by the administrator.
 - 4: unknown**
The status can not be determined for some reason.
 - 5: dormant**
Not defined in the interface MIB.
 - 6: notPresent**
Some component is missing.
 - 7: lowerLayerDown**
Down due to state of lower-layer interface(s).

L

l2HlastChang

The value of sysUpTime at the time this VP cross-connect entered its current operational state in the low to high direction.

I2HoperStatus

The operational status of the VP cross-connect in one direction; (that is, from the low to high direction).

I2pACCM

The current value of the ACC Map used for sending packets from the local PPP entity to the remote PPP entity. The value of this object is meaningful only when the link has reached the open state (ifOperStatus is up).

I2rACCComp

Indicates whether the local PPP entity will use Address and Control Compression when transmitting packets to the remote PPP entity. The value of this object is meaningful only when the link has reached the open state (ifOperStatus is up).

I2rCompProtocol

The IP compression protocol that the local PPP link uses when sending packets to the remote PPP link. The value of this object is meaningful only when the link has reached the open state (pppIpOperStatus is opened).

Values 1: none

2: vj-tcp

I2rProtoComp

Indicates whether the local PPP entity will use Protocol Compression when transmitting packets to the remote PPP entity. The value of this object is meaningful only when the link has reached the open state (ifOperStatus is up).

Values 1: enabled

2: disabled

lastCacRc

This object contains the result code of the last GPON CAC request for the OLT.

Values

- 1 : ok
- 2 : noReq
- 3 : badParms
- 4 : cfgLookup
- 5 : genError
- 6 : maxOnuGemPorts
- 7 : maxOltAllocIds
- 8 : maxOltDbalAllocIds
- 9 : maxBw
- 10 : maxCbrBw
- 11 : trafficClassChg
- 12 : dbaStatusChg

Access read-only

lastChange

The value of sysUpTime at the time this CID entered its current operational state, indicated by aal2CidOperStatus.

lastRecvLqrPkt

This object contains the most recently received LQR packet. The format of the packet is as described in the LQM Protocol specification. All fields of the packet, including the save fields, are stored in this object.

The LQR packet is stored in network byte order. The LAP-B and PPP headers are not stored in this object; the first four octets of this variable contain the Magic-Number field, the second four octets contain the LastOutLQRs field and so on. The last four octets of this object contain the SaveInOctets field of the LQR packet.

leavesRx

IGMP leaves received.

leavesTx

IGMP leaves transmitted.

lineAlarmStatus

Indicates current state of the DSL line and this is only applicable to HDSL2 and SHDSL interface types. `adslAtucCurrStatus` defined in RFC 2662 should be used to query line state for ADSL line types.

This is a bit-map of possible conditions. The various bit positions are:

Values 0: noDefect

There no defects on the line

4: snrMarginAlarm

Declared when the Noise Margin falls below the Minimum Noise Margin, or the bit-error-rate exceeds 10^{-7} .

6: loswFailureAlarm

Indicates a forward LOSW alarm.

7: configInitFailure

Endpoint failure during initialization due to paired endpoint not able to support requested configuration.

8: protocolInitFailure

Endpoint failure during initialization due to incompatible protocol used by the paired endpoint.

9: noNeighborPresent

Endpoint failure during initialization due to activation sequence detected from paired endpoint.

10: loopbackActive

A loopback is currently active at this Segment Endpoint.

lineBipCount

Number of Receive Line Overhead Processor (RLOP) BIP-8 errors. The RLOP is responsible for line-level alarms and for monitoring performance.

lineCapabilities

The DSL types supported on this interface. This is a bit-map of possible types. This variable can be used to determine possible values for `zhoneDslLineType`.

Values 0: hdsl2

1: shdsl

2: sdsl

3: adsl

lineCoding

Specifies the ADSL coding type used on this line.

lineCurrCV

The counter associated with the number of Coding Violations detected by a SONET/SDH line during the current 15-minute interval.

lineCurrES

The counter associated with the number of Errored Seconds detected by a SONET/SDH line during the current 15-minute interval.

lineCurrSES

The counter associated with the number of Severely Errored Seconds detected by a SONET/SDH line during the current 15-minute interval

lineCurStatus

Indicated the status of the interface. It is a bit map represented as a sum, so it can represent multiple defects simultaneously. sonetLineNoDefect should be set only if no other flag is set.

Values 1: sonetLineNoDefect

2: sonetLineAIS

4: sonetLineRDI

lineCurrUAS

The counter associated with the number of Unavailable Seconds detected by a SONET/SDH line during the current 15-minute interval.

lineFebeCount

Number of RLOP far-end block errors (FEBE).

lineIntervCV

The counter associated with the number of Coding Violations detected by a SONET/SDH line during a particular 15-minute interval during the last 24 hours.

lineIntervES

The counter associated with the number of Errored Seconds encountered by a SONET/SDH line during a particular 15-minute interval during the last 24 hours.

lineIntervNum

A number between one and 96 that identifies the interval for which the statistics are available. The interval one is the most recently completed 15-minute interval, and the interval N is the interval immediately proceeding the interval called N-1.

lineIntervSES

The counter associated with the number of Unavailable Seconds detected by a SONET/SDH line in a particular 15-minute interval during the last 24 hours.

lineIntervUAS

The counter associated with the number of Unavailable Seconds detected by a SONET/SDH line in a particular 15-minute interval during the last 24 hours.

lineIntervValid

Indicates if the data for this interval is valid.

lineState

Line status for a VDSL channel.

- Values**
- 1: idle**
 - 2: training**
 - 3: active**
 - 4: loopback**

lineStatus (E1ATMPPhysical, E3UnchannelizedPhysical)

Indicates the Line Status of the interface. It contains loopback, failure, received alarm and transmitted alarms information. If the zhloopbackState bit is set, the loopback in effect can be determined from the loopbackConfig object.

- Values**
- 1: noAlarm**
 - 2: rcvFarEndLOF Far end LOF**
 - 4: xmtFarEndLOF**
Near end sending LOF Indication
 - 8: rcvAIS**
Far end sending AIS
 - 16: xmtAIS**
Near end sending AIS

- 32: lossOfFrame**
Near end loss of frame
- 64: lossOfSignal**
Near end loss of signal
- 128: loopbackState**
Near end is looped.
- 256: ts16AIS**
E1 TS16 AIS
- 512: rcvFarEndLOMF**
Far End Sending TS16 LOMF
- 1024: xmtFarEndLOMF**
Near End Sending TS16 LOMF
- 2048: rcvTestCode**
Near End detects a test code
- 4096: otherFailure**
any line status not defined
- 8192: unavailSigState**
Near End in Unavailable Signal State
- 16384: netEquipOOS**
Carrier Equipment Out of Service

lineStatus (DSLPhysical)

The DSL modem status. This field indicates the status of the HDSL2/SDSL Transceiver.

- Values**
- 1: down**
The transceiver is in idle mode.
 - 2: downloading**
The transceiver is downloading the boot code.
 - 3: activated**
The transceiver is activated
 - 4: training**
The transceiver is in synchronization phase.
 - 5: up**
The transceiver is in data mode.
 - 6: test**
The transceiver is in test mode.

lineStatusLastChange

The value of MIB II sysUpTime object at the time this DS1/DS3/E3 entered its current line status state. If the current state was entered prior to the last reinitialization of the proxy-agent, then this object has a zero value.

lineType

Defines the type of ADSL physical line entity that exists, by defining whether and how the line is channelized. If the line is channelized, the value will be other than noChannel(1). This object defines which channel type(s) are supported. In the case that the line is channelized, the manager can use the ifStackTable to determine the ifIndex for the associated channel(s).

Values

- 1: noChannel**
- 2: fastOnly**
- 3: interleavedOnly**
- 4: fastOrInterleaved**
- 5: fastAndInterleaved**

lineUpTime

Shows how long this link has been UP in DD:HH:MM format.

InAtten

When referring to the downstream direction, it is the measured difference in the total power transmitted by the ATU-C and the total power received by the ATU-R over all sub-carriers during diagnostics mode. When referring to the upstream direction, it is the measured difference in the total power transmitted by the ATU-R and the total power received by the ATU-C over all sub-carriers during diagnostics mode. It ranges from 0 to 1270 units of 0.1 dB (Physical values are 0 to 127 dB). A special value of 0x7FFFFFFF (2147483647) indicates the line attenuation is out of range to be represented. A special value of 0x7FFFFFFE (2147483646) indicates the line This object reflects the value of the parameter following the most recent DELT performed on the associated line. Once the DELT process is over, the parameter no longer changes until the row is deleted or a new DELT process is initiated.

localMaxSlotID

The Max-Slot-Id parameter that the local node has advertised and that is in use on the link. If VJ-TCP header compression is not in use on the link then the value of this object is 0. The value of this object is meaningful only when the link has reached the open state (pppIpOperStatus is opened).

LocalMRU

The current value of the MRU for the local PPP Entity. This value is the MRU that the remote entity is using when sending packets to the local PPP entity. The value of this object is meaningful only when the link has reached the open state (ifOperStatus is up).

LocalPeriod

The LQR reporting period, in hundredths of a second that is in effect for the local PPP entity.

Refer to section 2.5, Configuration Option Format, of RFC1333.

Location

The location of the this temperature sensors.

locdCount

Loss of cell delineation.

log

An array of up to 512 real $H(f)$ logarithmic representation values in dB for the respective transmission direction. It is designed to support up to 512 (downstream) sub-carriers. The number of utilized values on downstream direction depends on NSCs, and on upstream direction it depends on NSCs. This value is referred here as NSC. Each array entry represents the real $Hlog(f = i \cdot Df)$ value for a particular sub-carrier index i , ($0 \leq i < NSC$). The real $Hlog(f)$ value is represented as $(6 - m(i)/10)$, with $m(i)$ in the range 0 to 1022. A special value $m=1023$ indicates that no measurement could be done for the subcarrier because it is out of the passband or that the attenuation is out of range to be represented. This parameter is applicable in loop diagnostic procedure and initialization. Each value in this array is 16 bits wide and is stored in big endian format.

loopbackStatus

Represents the current state of the loopback on the DS1 interface. It contains information about loopbacks established by a local manager and remotely from the far end. This status is combination of loopbackConfig, and sendCode options as this status represents the local as well as far loopbacks.

Values 1: noLoopback
2: nearEndLineLoopback
4: nearEndOtherLoopback
8: nearEndLocalLoopback
16: farEndPayloadLoopback
32: farEndLineLoopback

Location DS1

loopbackStatus (DS1ATM, DS3)

Represents the current state of the loopback on the DS3 interface. It contains information about loopbacks established by a manager and remotely from the far end. The dsx3LoopbackStatus is a bit map represented as a sum, therefore is can represent multiple loopbacks simultaneously.

Values 1: dsx3NoLoopback
2: dsx3NearEndPayloadLoopback
4: dsx3NearEndLineLoopback
dsx3NearEndOtherLoopback
16: dsx3NearEndInwardoopback
32: dsx3FarEndPayloadLoopback
64: dsx3FarEndLineLoopback

Location DS3

LowIfIndex

The ifIndex value of the ATM interface for this VP cross-connect. The term low implies that this ATM interface has the numerically lower ifIndex value than the other ATM interface identified in the same atmVpCrossConnectEntry.

LowThreshold

The lower temperature that an alarm will be sent.

LowVpi

The VPI value at the ATM interface associated with the VP cross-connect that is identified by atmVpCrossConnectLowIfIndex

M

mauAutoNegSupported

Indicates whether or not auto-negotiation is supported.

mauDefaultType

Physical interface type.

Values

- 1: mauOther**
- 5: mau10BaseT**
- 10: mau10BaseTHD**
- 11: mau10BaseTFD**
- 15: mau100BaseTXHD**
- 16: mau100BaseTXFD**

mauFalseCarriers

A count of the number of false carrier events during IDLE in 100BASE-X and 1000BASE-X links. For all other MAU or interface types, this counter will always indicate zero. This counter does not increment at the symbol rate. It can increment after a valid carrier completion at a maximum rate of once per 100 ms for 100BASE-X and once per 10us for 1000BASE-X until the next CarrierEvent. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

mauJabberState

The value other(1) is returned if the jabber state is not 2, 3, or 4. unknown(2) is returned when the MAU true state is unknown; when it is being initialized. If the MAU is not jabbering the agent returns noJabber(3). This is the normal state. If the MAU is in jabber state the agent returns the jabbering(4) value.

Values

- 1: other**
- 2: unknown**
- 3: noJabber**
- 4: jabbering**

mauJabberStateEnters

A count of the number of times that zhoneMauJabberState for this MAU or interface instance enters the state jabbering(4). This counter will always

indicate zero for MAUs of type dot1MauTypeAUI and those of speeds above 10Mbps. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

mauMediaAvailable

If the MAU or interface is a link or fiber type (FOIRL, 10BASE-T, 10BASE-F) then this is equivalent to the link test fail state/low light function

For an AUI MAU this indicates whether or not loopback is detected on the DI circuit. other(1) is returned if the mediaAvailable state is not one of 2 through 11, and unknown(2) is returned when the MAU true state is unknown; for example, when it is being initialized. At power-up or following a reset, the value of this attribute will be unknown for AUI and 10BASE-FP MAUs. For these MAUs loopback will be tested on each transmission during which no collision is detected. If DI is receiving input when DO returns to IDL after a transmission and there has been no collision during the transmission then loopback will be detected. The value of this attribute will only change during non-collided transmissions for AUI and 10BASE-FP MAUs. For 100Mbps and 1000Mbps MAUs, the enumerations match the states within the respective link integrity states. Any MAU which implements management of auto-negotiation will map remote fault indication to remote fault.

The values remoteJabber(7), remoteLinkLoss(8), and remoteTest(9) SHOULD be used instead of remoteFault(5) where the reason for remote fault is identified in the remote signaling protocol.

The value notAvailable(4) takes precedence over the value remoteFault(5). Any MAU that implements management of clause 37 Auto-Negotiation will map the received RF1 and RF2 bit values for Offline to offline(10), Link Failure to remoteFault(5) and Auto-Negotiation Error to autoNegError(11).

Values 1: other

2: unknown

3: available

The link, light, or loopback is normal.

4: notAvailable

Link loss, low light, or no loopback.

5: remoteFault

A fault has been detected at the remote end of the link. This value applies to 10BASE-FB, 100BASE-T4 Far End Fault Indication and non-specified remote faults from a system running auto-negotiation.

6: invalidSignal

Indicates that an invalid signal has been received from the other end of the link. InvalidSignal (6) applies only to MAUs of type 10BASE-FB. Where an IEEE Std 802.3u-1995 clause 22 MII is present, a logic one in the remote fault bit (reference section 22.2.4.2.8 of that document) maps to

the value remoteFault(5), and a logic zero in the link status bit (reference section 22.2.4.2.10 of that document) maps to the value notAvailable(4).

7: remoteJabber

8: remoteLinkLoss

9: remoteTest

10: offline

11: autoNegError

mauOperStatus

The current operational state of the MAU. The value other(1) is returned if the MAU or interface is in a state other than one of the states 2 through 6.

A value of unknown(2) is returned when the true state of the MAU is unknown; for example, when it is being initialized. A value of operational(3) means the MAU is fully functional, operates, and passes signals to its attached DTE. A value of standby(4) forces DI and CI to idle and the media transmitter or interface to idle or fault, if supported. Standby(4) mode only applies to link type MAUs. The state of zhoneMauMediaAvailable is unaffected.

A MAU in shutdown(5) state assumes the same condition on DI, CI, and the media transmitter as though it were powered down or not connected. The MAU may return an other(1) value for the zhoneMauJabberState and zhoneMauMediaAvailable objects when it is in this state. For an AUI, this state will remove power from the AUI. Setting this variable to the value reset(6) resets the MAU in the same manner as a power-off, power-on cycle of at least one-half second would. The agent is not required to return the value reset (6).

- Values**
- 1: other**
 - 2: unknown**
 - 3: operational**
 - 4: standby**
 - 5: shutdown**
 - 6: reset**

mauTypeListBits

A value that uniquely identifies the set of possible IEEE 802.3 MAU types. If auto-negotiation is present on this MAU, this object corresponds to ifMauAutoNegCapability. This MAU may be capable of operating as a MAU type that is beyond the scope of this MIB. This is indicated by returning the bit value bOther in addition to any bit values for capabilities that are listed.

Values	0: bOther
	1: bAUI
	2: b10base5
	3: bFoir1
	4: b10base2
	5: b10baseT
	6: b10baseFP
	7: b10baseFB
	8: b10baseFL
	9: b10broad36
	10: b10baseTHD
	11: b10baseTFD
	12: b10baseFLHD
	13: b10baseFLFD
	14: b100baseT4
	15: b100baseTXHD
	16: b100baseTXF
	17: b100baseFXHD
	18: b100baseFXFD
	19: b100baseT2HD
	20: b100baseT2FD
	21: b1000baseXHD
	22: b1000baseXFD
	23: b1000baseLXHD
	24: b1000baseLXFD
	25: b1000baseSXHD
	26: b1000baseSXFD
	27: b1000baseCXHD
	28: b1000baseCXFD
	29: b1000baseTHD
	30: b1000baseTFD

maxActiveCalls

maxAllocIds

Maximum number of alloc-Ids supported by this OLT.

Default 0

Access read-only

maxConfiguredCalls

The maximum number of active calls, equal to the maximum number of links (up to 16) assigned to the IG times the maximum number of time slots per E1 link (31) less the total number of provisioned c-channels on these links.

maxDbAllocIds

Maximum number of DBA alloc-Ids supported by this OLT.

Default 0

Access read-only

metric1

The primary routing metric for this route. The semantics of this metric are determined by the routing-protocol specified in the route ipCidrRouteProto value. If this metric is not used, its value should be set to -1.

metric2

An alternate routing metric for this route. The semantics of this metric are determined by the routing-protocol specified in the route ipCidrRouteProto value. If this metric is not used, its value should be set to -1.

metric3

An alternate routing metric for this route. The semantics of this metric are determined by the routing-protocol specified in the route ipCidrRouteProto value. If this metric is not used, its value should be set to -1.

metric4

An alternate routing metric for this route. The semantics of this metric are determined by the routing-protocol specified in the route ipCidrRouteProto value. If this metric is not used, its value should be set to -1.

metric5

An alternate routing metric for this route. The semantics of this metric are determined by the routing-protocol specified in the route `ipCidrRouteProto` value. If this metric is not used, its value should be set to -1.

mfR1DialedDigits

MF-R1 Dialed Digits. Refer to the corresponding object in the VCL table. If this value is not set by CLI or DZMS, the agent will use the AAL2 VCL entry default.

Values **1: enabled**
 2: disabled

mfR2DialedDigits

MF-R2 Dialed Digits. Refer to the corresponding object in the VCL table. If this value is not set by CLI or DZMS, the agent will use the AAL2 VCL entry default.

Values **1: enabled**
 2: disabled

N

negCapRecvBits

A value that uniquely identifies the set of capabilities received from the remote auto-negotiation entity. Note that interfaces that support this MIB may be attached to remote auto-negotiation entities which have capabilities beyond the scope of this MIB. This is indicated by returning the bit value `bOther(0)` in addition to any bit values for capabilities listed.

Values	0: bOther
	1: b10baseT
	2: b10baseTFD
	3: b100baseT4
	4: b100baseTX
	5: b100baseTXFD
	6: b100baseT2
	7: b100baseT2FD
	8: bFdxPause
	9: bFdxAPause
	10: bFdxSPause
	11: bFdxBPause
	12: b1000baseX
	13: b1000baseXFD
	14: b1000baseT
	15: b1000baseTFD

negCapabilityBits

A value that uniquely identifies the set of capabilities of the local auto-negotiation entity. Note that interfaces that support this MIB may have capabilities that extend beyond the scope of this MIB. Note that the local auto-negotiation entity may support some capabilities beyond the scope of this MIB. This is indicated by returning the bit value bOther in addition to any bit values for capabilities that are listed.

Values	0: bOther
	1: b10baseT
	2: b10baseTFD
	3: b100baseT4
	4: b100baseTX
	5: b100baseTXFD
	6: b100baseT2
	7: b100baseT2FD
	8: bFdxPause
	9: bFdxAPause
	10: bFdxSPause
	11: bFdxBPause
	12: b1000baseX
	13: b1000baseXFD
	14: b1000baseT
	15: b1000baseTFD

negConfig

A value indicating the current status of the auto-negotiation process. The enumeration parallelDetectFail(5) maps to a failure in parallel detection as defined in 28.2.3.1 of [IEEE 802.3 Std].

Values	1: other
	2: configuring
	3: complete
	4: disabled
	5: parallelDetectFail

negRemoteFaultRecv

A value that identifies any fault indications received from the far end of a link by the local auto-negotiation entity for 1000Mbps MAUs.

Values 1: noError
 2: offline
 3: linkFailure
 4: autoNegError

negRemoteSignaling

A value indicating whether the remote end of the link is using auto-negotiation signaling. It takes the value detected(1) if and only if, during the previous link negotiation, FLP Bursts were received.

Values 1: detected
 2: notdetected

nextHopAS

The Autonomous System Number of the Next Hop. The semantics of this object are determined by the routing-protocol specified in the route ipCidrRouteProto value. When this object is unknown or not relevant its value should be set to zero.

number

A number between one and 96, where one is the most recently completed 15-minute interval and 96 is the 15-minutes interval completed 23 hours and 45 minutes prior to interval 1. If a 6, 12, or 18 hour period is selected, the max interval number will be 24, 48, or 72.

O

onuAlarmDACT

Instructs the ONU to deactivate itself.

onuAlarmDIS

Disabled ONU alarm. The device stays in this state even after power off.

onuAlarmERR

Block error detection alarm The downstream received BIP8 is compared with the calculated BIP8 on the received stream. The number of differing bits is accumulated as errors.

onuAlarmFRML

Loss of downstream frame alarm, which is activated when the frame bit is 0 for three consecutive frames.

onuAlarmLCD

Loss of cell delineation alarm, which is activated when seven consecutive ATM cells have an invalid HEC.

onuAlarmLOS

Loss of signal alarm, which is activated when there is no valid optical signal.

onuAlarmMEM

Message error message alarm, which is activated when the ONU receives an unknown message.

onuAlarmMIS

Link mismatching alarm, which is activated when the ONU detects that the received PST and transmitted PST are different.

onuAlarmOAML

PLOAM cell loss alarm, which is activated when three consecutive PLOAM headers are wrong.

onuAlarmPEE

Physical equipment error alarm, which is activated when the ONU receives a PEE message.

onuAlarmSD

Signal Degraded alarm, which is activated when the downstream BER greater than or equal to $10E^{-5}$.

onuAlarmSUF

Startup Failure alarm, which is activated when the ranging of this ONU has failed (see ranging protocol for exact condition).

onuAlarmTF

Transmitter Failure alarm, which is activated when the ONU transmitter is declared in failure, when there is no nominal photocurrent, or when the drive currents go beyond the maximum specification.

onuLoopback

ONU loopback status.

Values 1: **onuNone**
 2: **onuInward**
 3: **onuLine**

onuOperStatus

ONU operational status.

Values 1: **onuStateDown**
 2: **onuStateUp**

oofCount

Near end is out of frame. False indicates that the line is up and in frame.

operStatus (ATM VPL)

The operational status of the VP cross-connect in one direction (that is, from the high to low direction).

operStatus (card)

Indicates the operational status of the unit/card.

Values 1: **unknown**
 The card is in an unknown state

2: **operating**
 Software has been loaded and is operating, the unit/card is accepting requests from other system elements or if a standalone unit it is processing requests form the WAN or local interfaces.

3: **nonoperating**
 The unit/card is not operating due to internal errors or it has been disabled by the management system.

4
 The unit/card is in the maintenance state.

operStatus (GR-303)

The line operational status is the composition (logical OR'ing) of definitions from GR-303-CORE. Some of them are service affecting, some of them do not affect the ability of service.

Values 0: up

The line is operational.

1: fault

A fault condition (FLT).

2: manualOos

The object is manually taken OOS for maintenance purpose

3: removedFromServiceBySwitch

The entity has been removed from service by the switch.

4: unEquipped

The default state when new CRV entry is added

5: notConnected

Line is not connected

Location GR-303

operStatus (CID)

The current operational status of the CID.

Values 1: up

2: down

3: unknown

Location ATMCid

operStatus (V5.2)

The V5.2 interface operational status. Indicates whether it is fully inoperable and unable to provide service to the user, or it is fully operable and available for use. Valid values are operable, inoperable, standby, degradedService and the transitional states of inoperableIsInProgress and inoperableOosInProgress. Depending on the V52 configuration, the transition between inService and outOfService can take quite some time for an interface group. The transitional states give operator a more accurate indication of a V52 IG behavior.

Values 1: operable
 2: inoperable
 3: standby
 4: degradedService
 5: inoperableIsInProgress
 6: inoperableOosInProgress

Location v52Ig

operStatusCause

The cause of the interface operational status.

Values 1: localDisable
 2: localDisable
 3: remoteOutOfService
 4: protocolFailure
 protocolFailure

opticRssi

RSSI power of the SFP per ONU. The value is in tenths of dB.

Values Integer (-600 .. 200)

Access read-only

opticStatus

When the End Of Life (EOL) Alarm bit is set an alarm will be raised. Alarm is set at about 150% of original current. Alarm will be level Major. It will be cleared when the SFP is replaced, or when the Alarm bit is cleared. Warning indication will be provided when user asks for status. Warning is set when EOL is at about 130% original current.

Values 1: ok
 2: warning
 4: alarm
 8: ok

Access read-only

opticTemperature

Internally measured Transceiver Temperature of the OLT in Celsius.

Values Integer (-128..128)

Access read-only

opticTxBiasCurrent

Measured Tx Bias current per OLT in milli Amperes.

Values Integer (0..131)

Access read-only

opticTxPower

Measured Tx Output Power of the OLT in dB. The measured value will be in tenths of dB.

Values Integer (-600..200)

Access read-only

opticVoltage

Internally measured Transceiver Supply Voltage of the OLT in Volts. The measured value will be in the order of hundredths of Volts.

Values Integer (0..655)

Access read-only

outDiscards

The number of output IP datagrams for which no problem was encountered to prevent their transmission to their destination, but which were discarded for lack of buffer space. Note that this counter would include datagrams counted in ipForwDatagrams if any such packets met the (discretionary) discard criteria.

outNoRoutes

The number of IP datagrams discarded because no route could be found to transmit them to their destination. Note that this counter includes any packets counted in ipForwDatagrams which meet this no-route criterion and that this includes any datagrams which a host cannot route because all of its default routers are down.

outPauseFrame

The count of MAC Control frames transmitted on this interface with an opcode indicating the PAUSE operation.

outLQRs

The value of the OutLQRs counter on the local node for the link identified by `ifIndex`.

Refer to section 2.2, Counters, of RFC1333.

outRequests

The total number of IP datagrams that local IP user- protocols (including ICMP) supplied to IP in requests for transmission. This counter does not include any datagrams counted in `ipForwDatagrams`.

outSigNumber

The number of output signals for the port in the output signal table (`rs232PortOutSigTable`). The table contains entries only for those signals the software can assert and that are useful to observe.

overrunErrs

The total number of characters with an overrun error that were input from the port since system reinitialization and while the port state was up or test.

P

p2lACCM

The ACC Map used by the remote PPP entity when transmitting packets to the local PPP entity. The value of this object is meaningful only when the link has reached the open state (`ifOperStatus` is up).

parityErrs

The total number of characters with a parity error that were input from the port since system reinitialization and while the port state was up or test.

pathBipCount

Number of Receive Path Overhead Processor (RPOP) BIP-8 errors. The RSOP interprets pointers and extracts path overhead and the synchronous payload envelope. It is also responsible for path-level alarms and for monitoring performance.

pathCurrCV

The counter associated with the number of Coding Violations detected by a SONET/SDH path during the current 15-minute interval.

pathCurrES

The counter associated with the number of Errored Seconds encountered by a SONET/SDH path during the current 15-minute interval.

pathCurrSES

The counter associated with the number of Severely Errored Seconds encountered by a SONET/SDH path during the current 15-minute interval.

pathCurrStatus

Indicates the state of the interface. It is a bit map represented by a sum, and can represent multiple defects simultaneously. sonetPathNoDefect (1) should be set only if there are no other flags set.

Values

- 1: sonetPathNoDefect**
- 2: sonetPathSTSLOP**
- 4: sonetPathSTSAIS**
- 8: sonetathSTSRDI**
- 16: sonetPathUnequipped**
- 32: sonetPathSignalLabelMismatch**

pathCurrUAS

The counter associated with the number of Unavailable Seconds encountered by a path during the current 15-minute interval.

pathFebeCount

Number of RPOP far-end block errors (FEBE).

pathIntervCV

The counter associated with the number of Coding Violations detected by a SONET/SDH path during a particular 15-minute interval during the last 24 hours.

pathIntervES

The counter associated with the number of Errored Seconds detected by a SONET/SDH path during a particular 15-minute interval during the past 24 hours.

pathIntervNum

A number between one and 96 that identifies the interval for which statistics are available. The interval one is the most recently completed 15-minute interval, and the interval N is the interval immediately proceeding N-1.

pathIntervSES

The counter associated with the number of Severely Errored Seconds detected by a SONET/SDH path during a particular 15-minute interval during the last 24 hours.

pathIntervUAS

The counter associated with the number of Unavailable Seconds detected by a path during a particular 15-minute interval during the past 24 hours.

pathIntervValid

Indicates if the data for this interval is valid.

pauseOperMode

Reflects the PAUSE mode currently in use on this interface, as determined by either (1) the result of the auto-negotiation function or (2) if auto-negotiation is not enabled or is not implemented for the active MAU attached to this interface, by the value of dot3PauseAdminMode. Interfaces operating at 100 Mbps or less will never return enabledXmit(2) or enabledRcv(3). Interfaces operating in half-duplex mode will always return disabled(1). Interfaces on which auto-negotiation is enabled but not yet completed should return the value disabled(1).

Values 1: disabled
2: enabledXmit
3: enabledRcv
4: enabledXmitAndRcv

peakMemoryUsage

The maximum physical memory that has been allocated at any time by the unit/card.

pEocTmcIfIndex

Systems that support more than one DS1 require protection switching for the Embedded Operations Channel (EOC) and a Time-slot Management Channel (TMC).

This object refers to the ifIndex for the DS1 line supporting the active EOC and TMC path protection switching. The value for this object is equal to the value of ifIndex from the Interfaces table of MIB II (RFC 1213).

Per-call timeslot assignment is accomplished over a 64 Kbps TMC that carries messages between the LDS and RDT. These messages are used to make and break timeslot assignments between line units and DS0s on a per-call basis.

A separate DS0 data link, called the Embedded Operations Channel (EOC), is used to transmit operations messages between RDT and LDS/OSs. The initial protection channel for the EOC is channel number 12 of a DS1 and channel number 24 on the same DS1 is reserved for TMC protection.

GR-303-CORE states that protection switching for DS1 lines is optional. GR-303-CORE R5-263 [1835] requires that an RDT and an IDT shall support EOC and TMC/CSC path protection switching on systems supporting more than one DS1. R5-264 [505] states that a RDT and an IDT shall receive from a remote managing system and process retrieval of time slots for the second EOC (the initial standby) and the second TMC or CSC. R5-266 [507] states that an RDT and an IDT shall receive from a remote managing system and process requests to assign and to retrieve the DS1 line for the standby EOC and the standby TMC/CSC. The value will be zero if the Interface Group is not installed on a specific card.

peerCardIndex

The slot where the Interface Group peer (or standby) resides. Peer shelf and peer slot define where this Interface Group peer and GR-303 stack resides. Range of valid values is 0-17. A peer slot value of 0 represents a no backup configuration.

peerShelfIndex

The shelf where this Redundancy Group peer (or standby) resides. Peer shelf and peer slot define where this peer Interface Group and GR-303 stack resides. Range of valid values is 0-255. A peer shelf value of 0 represents a no backup configuration.

peerSlot

The slot where this Interface Group's peer (or standby) resides. Peer shelf and peer slot define where this Interface Group peer and V5.2 stack reside. Range of valid values is 0-17. A peer slot value of 0 represents a no backup configuration.

peerStatus

The status of the peer (or standby) Interface Group

Values **1: noStandbyConfigured**
2: configuredAndAvailable
3: configuredAndUnavailable

PktTooLongs

The number of received packets that have been discarded because their length exceeded the MRU. This counter is a component of the ifInErrors variable that is associated with the interface that represents this PPP Link. Note that packets that are longer than the MRU but which are successfully received and processed are not included in this count.

portAlignmentStatus

The status result for a user port block/unblock request by port type. The values of this variable indicate both type of request and its success or fail status. The value is invalid when v52IgPortAlignmentRequest is noRequest.

Values

- 1: blockAllSucceeded**
- 2: blockAllFailed**
- 3: blockIsdnSucceeded**
- 4: blockIsdnFailed**
- 5: blockPstnSucceeded**
- 6: blockPstnFailed**
- 7: unblockAllSucceeded**
- 8: unblockAllFailed**
- 9: unblockIsdnSucceeded**
- 10: unblockIsdnFailed**
- 11: unblockPstnSucceeded**
- 12: unblockPstnFailed**

portStatus

Summary of the status of all physical ports on the unit/card. Each port is identified by it's ifTable index and the status is indicated as up|down|unknown.

postResults

A results string generated by the unit/card that displays the results of the POST (power-on-self-test). The first item in the text string indicates the overall results of the test, Pass or Fail. The reminder of the string enumerates the results of each test which will vary by unit/card type or, the string may only enumerate failed tests.

prevBadAmiViolation

The number of frames with the incorrect number of AMI violations during the previous 15-minutes.

prevErrorSeconds

The number of ISDN activation error seconds during the previous 15-minute interval.

prevFsyncSeconds

The number of frame synch error seconds during the previous 15-minute interval.

prevUnbalancedFrame

The number of unbalanced frames during the previous 15-minute interval.

processorIdle

The total time the CPU has spent executing the idle loop. Note:
 $\text{unitProcessorUsage} / (\text{unitProcessorUsage} + \text{unitProcessorIdle}) = \% \text{ processor utilization}$

processorType

A string is returned that describes the processor type used on the unit/card.

processorUsage

The total time the CPU has spent executing processes.

protocol

The routing mechanism that the route used to learn. The inclusion of values for gateway routing protocols is not intended to imply that hosts should support those protocols.

- Values**
- 1: other**
not specified
 - 2: local**
 - 3: netmgmt**
 - 4: icmp**
result of ICMP Redirect
 - 8: rip**
Berkeley RIP or RIP-II IS-IS
 - 9: isis**
Dual IS-IS
 - 13: ospf**
Open Shortest Path First
 - 14: bgp**
Border Gateway Protocol
 - 15: idpr**
InterDomain Policy Routing

provVariantReqStatus

The status result for a v52IgProvVariantRequest. When v52IgProvVariantRequest has a value of noRequest, the value for this variable is invalid.

Values 1: verifyWithLeSucceeded

LE is ready for reprovision with the verifyWithLe request.

2: verifyWithLeFailed

LE is not ready for reprovision with the verifyWithLe request.

3: requestFromLeSucceeded

Received Variant ID and Interface ID from LE with the requestFromLe request.

4: requestFromLeFailed

Did not receive Variant ID and Interface ID from LE with the requestFromLe request.

5: switchToLocalSucceeded

LE side has changed its own provisioning variant to be the same as v52IgLocalProvVariant with the switchToLocal request.

6: switchToLocalFailed

LE side failed to change its own provisioning variant to be the same as v52IgLocalProvVariant with the switchToLocal request.

Q

qln

An array of up to 512 real Quiet Line Noise values in dBm/Hz for the respective transmission direction. It is designed for up to 512 (downstream) sub-carriers. The number of utilized values on downstream direction depends on NSCs, and on upstream direction it depends on NSCs. This value is referred here as NSC. Each array entry represents the $QLN(f = i \cdot Df)$ value for a particular sub-carrier index i , ($0 \leq i < NSC$). The $QLN(f)$ is represented as $(-23 - n(i)/2)$, with $n(i)$ in the range 0 to 254. A special value $n(i)=255$ indicates that no measurement could be done for the subcarrier because it is out of the passband or that the noise PSD is out of range to be represented. This parameter is applicable in loop diagnostic procedure and initialization. Each value in this array is 8 bits wide.

Quality

The current quality of the link as declared by the local PPP entity's Link-Quality Management modules. No effort is made to define good or bad, nor the policy used to determine it. The not-determined value indicates that the entity does not actually evaluate the link's quality.

This value is used to differentiate the determined to be good case from the no determination made and presumed to be good case.

Values 1: good
2: bad
3: not determined

queries

The number of responses sent to RIP queries from other systems.

R

r2lACCComp

Indicates whether the remote PPP entity will use Address and Control Compression when transmitting packets to the local PPP entity. The value of this object is meaningful only when the link has reached the open state (ifOperStatus is up).

Values 1: enabled
2: disabled

r2lCompProtocol

The IP compression protocol that the remote PPP-IP entity uses when sending packets to the local PPP-IP entity. The value of this object is meaningful only when the link has reached the open state (pppIpOperStatus is opened).

Values 1: none
2: vj-tcp

r2lProtoComp

Indicates whether the remote PPP entity will use Protocol Compression when transmitting packets to the local PPP entity. The value of this object is meaningful only when the link has reached the open state (ifOperStatus is up).

Values 1: enabled
2: disabled

reasmFails

The number of failures detected by the IP reassembly algorithm (for whatever reason). Note that this is not necessarily a count of discarded IP fragments

since some algorithms (notably the algorithm in RFC 815) can lose track of the number of fragments by combining them as they are received.

reasmOks

The number of IP datagrams successfully reassembled.

reasmReqds

The number of IP fragments received that needed to be reassembled at this entity.

reasmTimeout

The maximum number of seconds that received fragments are held while they are awaiting reassembly at this entity.

rebind

Seconds until DHCP rebind occurs.

remoteInterfaceId

The 24-bit LE side V5.2 Interface identifier provisioned. This is read-only and can only be updated indirectly by setting v52IgProvVariantRequest to requestFromLe.

Values 1 to 16777215

remoteMaxSlotID

The Max-Slot-Id parameter that the remote node has advertised and that is in use on the link. If VJ-TCP header compression is not in use on the link then the value of this object shall be 0. The value of this object is meaningful only when the link has reached the open state (pppIpOperStatus is opened).

RemoteMRU

The current value of the MRU for the remote PPP Entity. This value is the MRU that the local entity is using when sending packets to the remote PPP entity. The value of this object is meaningful only when the link has reached the open state (ifOperStatus is up).

RemotePeriod

The LQR reporting period, in hundredths of a second, that is in effect for the remote PPP entity.

remoteProvVariant

LE side provisioning variant used to differentiate variations in configuration. This is read only and can only be updated indirectly by setting `v52IgProvVariantRequest` to `requestFromLe`.

Values 0 to 127

Default 0

renew

Seconds until DHCP renewal occurs.

rdIndex

The Routing Domain (RD) table index.

routingDiscards

The number of routing entries which were chosen to be discarded even though they are valid. One possible reason for discarding such an entry could be to free up buffer space for other routing entries.

routeChanges

The number of route changes made to the IP Route Database by RIP. This does not include the refresh of a route age.

routeType

The type of route. Note that `local(3)` refers to a route for which the next hop is the final destination; `remote(4)` refers to a route for which the next hop is not the final destination. Routes which do not result in traffic forwarding or rejection should not be displayed even if the implementation keeps them stored internally. `reject(2)` refers to a route which, if matched, discards the message as unreachable. This is used in some protocols as a means of correctly aggregating routes.

Values 1: other
2: reject
3: local
4: remote

rxCellCount

Receive ATM Cell Processor (RACP) receive cell count.

RxFCSSize

The size of the Frame Check Sequence (FCS) in bits that the remote node will generate when sending packets to the local node. The value of this object is meaningful only when the link has reached the open state (ifOperStatus is up).

S

sectionBipCount

Number of Receive Section Overhead Processor (RSOP) bit-interleaved parity (BIP)-8 errors. The RSOP synchronizes and descrambles frames and provides section-level alarms and performance monitoring.

sectionCurrCV

The counter associated with the number of Coding Violations detected by a SONET/SDH section during the current 15-minute interval.

sectionCurrES

The counter associated with the number of Errored Seconds encountered by a SONET/SDH Section in the current 15 minute interval.

sectionCurrSES

The counter associated with the number of Severely Errored Seconds encountered by a SONET/SDH Section in the current 15 minute interval.

sectionIntervES

The counter associated with the number of Errored Seconds detected by a SONET/SDH section during the current 15-minute interval.

sectionCurrSEFS

The counter associated with the number of Severely Errored Framing Seconds detected by a SONET/SDH section in the current 15-minute interval.

sectionIntervCV

The counter associated with the number of Coding Violations detected by a SONET/SDH section during a particular 15-minute interval during the last 24 hours.

sectionIntervES

The counter associated with the number of Errored Seconds detected by a SONET/SDH section during a particular 15-minute interval during the last 24 hours.

sectionIntervNum

A number between one and 96 that identifies the interval for which the statistics are available. Interval one is the most recently completed 15-minute interval, and interval N is the interval immediately proceeding N-1.

sectionIntervSEFS

The counter associated with the number of Severely Errored Framing Seconds detected by a SONET/SDH section during a particular 15-minute interval during the last 24 hours.

sectionIntervSES

The counter associated with the number of Severely Errored Seconds detected by a SONET/SDH section during a particular 15-minute interval during the last 24 hours.

sectionIntervValid

Indicates if the data for this interval is valid.

sectionStatus

Indicates the status of the interface. It is a bit map represented as a sum, and can represent multiple defects simultaneously. The value sonetSectionNoDefect should be set only if there are no other flags set.

Values 1: sonetSectionNoDefect
2: sonetSectionLOS
3: sonetSectionLOF

segmentLength

Indicates an estimate of the segment length being reported.

segmentGauge

Indicates the estimated Gauge of the segment being reported. The gauge reported back is based on the zhoneAdslSeltUnits configuration parameter passed when the test is started. The values are:

Values unknown (1),
AWG--
awg26 (26),
awg24 (24),
awg22 (22),
awg19 (19),

Metric--
metric32 (1032), -- (0.32mm)
metric40 (1040), -- (0.40mm)
metric50 (1050), -- (0.50mm)
metric63 (1063), -- (0.63mm)
metric90 (1090), -- (0.90mm)

Metric/Japan--
metricJapan32 (2032), -- (0.32mm)
metricJapan40 (2040), -- (0.40mm)
metricJapan50 (2050), -- (0.50mm)
metricJapan65 (2065), -- (0.65mm)
metricJapan90 (2090) -- (0.90mm)

segmentType

Indicates the segment type being reported.

Values **1 : notPresent**
 2 : inline
 3 : bridgeTap

seltBridgeTapsSupported

This object provides a way to query whether or not the measurement device (zhoneAdslSeltDevice) supports SELT on loops that have bridge taps. If this object returns false, and SELT is run on a loop with bridge taps, the SELT results will be unreliable.

seltDevice

This object provides a way to query which device (chip) is providing the SELT measurement on this interface.

Values **other (1)**
 a device not in the list below
 conexantG24 (2)
 Conexant G24 chip

seltLnflnbandNoise

This object provides the inband noise of all of the sub-carriers on this line. It is an array of 16-bit signed integers stored in big-endian format. For example, string position 0 is the most-significant octet of the first noise value, and string position 1 is the least-significant byte of the first noise value.

Each value is a signed 16-bit 2's complement number scaled as least-significant bit = 1/10 dBm/Hz. e.g. an integer value of 35 represents 3.5 dBm/Hz. If the measurement is invalid it will read as full-scale negative (0x8000).

seltStatus

This object provides status about any SELT test running or ever run (since last reset) on an ADSL interface. This object can be read by NMS managers to determine the status of a test; that is:

Values **inProgress (1)**
 If the test is currently running.
 stoppedInProgress (2)
 If a request to stop the test is still in progress.
 complete (3)
 If the test has been run at least once since system reset and has completed.
 notStarted (4)
 If the test has not been run on this interface since last system reset.

resultsCleared (5)

If the entries in the test results table, Lc and Lnf, were removed as the result of the execution of the clearResults(4) zhoneAdslSeltCmd.

aborted (6) .

If the test was aborted by the user before it could complete. In this case, the Lc and Lnf tables will have been removed.

timedOut (7)

If the test was aborted automatically because it did not complete before the timeout specified by zhoneAdslSeltDuration. In this case, the Lc and Lnf tables will have been removed.

unsupported (8)

The device does not support SELT.

seltTimeCompleted

This object provides a date and time stamp corresponding to the last completion of a test where valid results were obtained. If no test results are available, perhaps in the situation where no test has ever been started, this object shall return a value that has all the fields set to zero.

seltTimeLeft

This object provides the approximate time left in seconds for the test running on this ADSL interface. This object will return 0 if no test is actively running on the ADSL interface specified.

sEocTmcIfIndex

This object refers to the ifIndex for the DS1 line supporting the standby EOC and TMC path protection switching. The value for this object is equal to the value of ifIndex from the Interfaces table of MIB II (RFC 1213). The value will be zero if the Interface Group is not installed on a specific card or no secondary DS1 is provisioned.

serialNumber

Name of the DHCP offering server.

serverName

The vendor specific string that identifies the vendor equipment.

shelf (UlcConfig)

The shelf to which a ULC port belongs. It also indicates the ULC port type (ISDN, POTS, or EBS).

Values 1 to 255

shelf (V5.2)

The shelf where this interface and V5.2 stack resides. Zero (0) is an invalid value.

Values 1 to 255

Notes This is a database-specific object and should not be modified.

shelfIndex

The shelf where this Interface Group and GR-303 stack reside. A value of 0 indicates the Interface Group is not currently installed on a specific card by the GR-303 protocol stack.

Values 0 to 255

Notes This is a database-specific object and should not be modified.

sigAtten

When referring to the downstream direction, it is the measured difference in the total power transmitted by the ATU-C and the total power received by the ATU-R over all sub carriers during Showtime after the diagnostics mode. When referring to the upstream direction, it is the measured difference in the total power transmitted by the ATU-R and the total power received by the ATU-C over all sub carriers during Showtime after the diagnostics mode. It ranges from 0 to 1270 units of 0.1 dB (Physical values are 0 to 127 dB). A special value of 0x7FFFFFFF (2147483647) indicates the signal attenuation is out of range to be represented. A special value of 0x7FFFFFFE (2147483646) indicates the signal attenuation measurement is unavailable. This object reflects the value of the parameter following the most recent DELT performed on the associated line. Once the DELT process is over, the parameter no longer changes until the row is deleted or a new DELT process is initiated.

sigChanges

The number of times the signal has changed from on to off or from off to on.

sigId

Identification of a hardware signal.

Values	1: rts Request to Send
	2: cts Clear to Send
	3: dsr Data Set Ready
	4: dtr Data Terminal Ready
	5: ri Ring Indicator
	6: dcd Received Line Signal Detector
	7: sq Signal Quality Detector
	8: srs Data Signaling Rate Selector
	9: srts Secondary Request to Send
	10: scts Secondary Clear to Send
	11: sdcd Secondary Received Line Signal Detector

signalQuality

Used to compute the signal-to-noise ratio. To determine the signal-to-noise ratio, take the returned high and low bytes to form a 16 bit integer, I. Using this information apply the following formula $\text{Received S/N(dB)} = 58.4 - 10 \log_{10}(I)$

sigState

The current signal state.

Values	1: none
	2: on
	3: off

slot

The shelf where this Interface and V5.2 stack resides. Zero (0) is an invalid value.

Values 1 to 17

snr

The SNR Margin per sub-carrier, expressing the ratio between the received signal power and received noise power per subscriber. It is an array of 512 octets, designed for supporting up to 512 (downstream) sub-carriers. The number of utilized octets on downstream direction depends on NSCds, and on upstream direction it depends on NSCus. This value is referred here as NSC. Octet i ($0 \leq i < \text{NSC}$) is set to a value in the range 0 to 254 to indicate that the respective downstream or upstream sub-carrier has SNR of: $(-32 + \text{zhoneAdsl2SCStatusSnr}(i)/2)$ in dB i.e., -32 to 95dB). In other words, the least-significant bit of each octet is 0.5 dB, and there is a -32 dB bias. The special value 255 means that no measurement could be done for the subcarrier, probably because it is out of the PSD mask passband or that the noise PSD is out of range to be represented. Each value in this array is 8 bits wide.

snrMargin

SNR Margin is the maximum increase in dB of the noise power received at the ATU (ATU-R on downstream direction and ATU-C on upstream direction), such that the BER requirements are met for all bearer channels received at the ATU. It ranges from 640 to 630 units of 0.1 dB (Physical values are -64 to 63 dB).

A special value of 0x7FFFFFFF (2147483647) indicates the SNR Margin is out of range to be represented. A special value of 0x7FFFFFFE (2147483646) indicates the SNR Margin measurement is currently unavailable. This object reflects the value of the parameter following the most recent DELT performed on the associated line. Once the DELT process is over, the parameter no longer changes until the row is deleted or a new DELT process is initiated.

Speed

Fan speed of the fan in RPM

specQueryRx

Specific IGMP queries received.

specQueryTx

specQueryTx.

startProgress

Defines the current detailed startup state of the transceiver.

Values 16: startup-handshake-in-progress

24: startup-training-in-progress

26: startup-framer-sync-in-progress

statementIndex

IpFilterSpec

statsAlignmentErrors

A count of frames received on a particular interface that are not an integral number of octets in length and do not pass the FCS check.

The count represented by an instance of this object is incremented when the alignmentError status is returned by the MAC service to the LLC (or other MAC user). Received frames for which multiple error conditions obtain are, according to the conventions of IEEE 802.3 Layer Management, counted exclusively according to the error status presented to the LLC.

This counter does not increment for 8-bit wide group encoding schemes.

Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

statsCarrierSenseErrors

The number of times that the carrier sense condition was lost or never asserted when attempting to transmit a frame on a particular interface.

The count represented by an instance of this object is incremented at most once per transmission attempt, even if the carrier sense condition fluctuates during a transmission attempt.

This counter does not increment when the interface is operating in full-duplex mode. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

statsDeferredTransmission

A count of frames for which the first transmission attempt on a particular interface is delayed because the medium is busy. The count represented by an instance of this object does not include frames involved in collisions.

This counter does not increment when the interface is operating in full-duplex mode. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of `ifCounterDiscontinuityTime`.

statsDuplexStatus

The current mode of operation of the MAC entity. A value of unknown indicates that the current duplex mode could not be determined. Management control of the duplex mode is accomplished through the MAU MIB. When an interface does not support autonegotiation, or when autonegotiation is not enabled, the duplex mode is controlled using `ifMauDefaultType`. When autonegotiation is supported and enabled, duplex mode is controlled using `ifMauAutoNegAdvertisedBits`. In either case, the currently operating duplex mode is reflected both in this object and in `ifMauType`. Note that this object provides redundant information with `ifMauType`, allowing a management application to determine the duplex status of an interface without having to know every possible value of `ifMauType`.

Values

- 1: unknown**
- 2: halfDuplex**
- 3: fullDuplex**

statsExcessiveCollisions

A count of frames for which transmission on a particular interface fails due to excessive collisions. This counter does not increment when the interface is operating in full-duplex mode. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of `ifCounterDiscontinuityTime`.

statsFCSErrors

A count of frames received on a particular interface that are an integral number of octets in length but do not pass the FCS check. This count does not include frames received with frame-too-long or frame-too-short error. The count represented by an instance of this object is incremented when the `frameCheckError` status is returned by the MAC service to the LLC (or other MAC user). Received frames for which multiple error conditions obtain are, according to the conventions of IEEE 802.3 Layer Management, counted exclusively according to the error status presented to the LLC.

Note: Coding errors detected by the physical layer for speeds above 10 Mbps will cause the frame to fail the FCS check. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of `ifCounterDiscontinuityTime`.

statsFrameTooLongs

A count of frames received on a particular interface that exceed the maximum permitted frame size. The count represented by an instance of this object is incremented when the frameTooLong status is returned by the MAC service to the LLC (or other MAC user). Received frames for which multiple error conditions obtain are, according to the conventions of IEEE 802.3 Layer Management, counted exclusively according to the error status presented to the LLC. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

statsIndex

An index value that uniquely identifies an interface to an ethernet-like medium. The interface identified by a particular value of this index is the same interface as identified by the same value of ifIndex.

statsInternalMacReceiveErrors

A count of frames for which reception on a particular interface fails due to an internal MAC sublayer receive error. A frame is only counted by an instance of this object if it is not counted by the corresponding instance of either the dot3StatsFrameTooLongs object, the dot3StatsAlignmentErrors object, or the dot3StatsFCSErrors object.

The precise meaning of the count represented by an instance of this object is implementation-specific. An instance of this object may represent a count of receive errors on a particular interface that are not otherwise counted. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

statsInternalMacTransmitErrors

A count of frames for which transmission on a particular interface fails due to an internal MAC sublayer transmit error. A frame is only counted by an instance of this object if it is not counted by the corresponding instance of either the dot3StatsLateCollisions object, the dot3StatsExcessiveCollisions object, or the dot3StatsCarrierSenseErrors object. The precise meaning of the count represented by an instance of this object is implementation-specific. In particular, an instance of this object may represent a count of transmission errors on a particular interface that are not otherwise counted. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

statsInvalidIntervals

The number of intervals in the range from 0 to `igStatsValidIntervals` for which no data is available. This object will be zero except in cases where the data for some intervals are not available in proxy situations.

statsLateCollisions

The number of times that a collision is detected on a particular interface later than one `slotTime` into the transmission of a packet. A late collision included in a count represented by an instance of this object is also considered a generic collision for purposes of other collision-related statistics. This counter does not increment when the interface is operating in full-duplex mode. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of `ifCounterDiscontinuityTime`.

statsMultipleCollisionFrames

A count of successfully transmitted frames on a particular interface for which transmission is inhibited by more than one collision. A frame that is counted by an instance of this object is also counted by the corresponding instance of either the `ifOutUcastPkts`, `ifOutMulticastPkts`, or `ifOutBroadcastPkts`, and is not counted by the corresponding instance of the `dot3StatsSingleCollisionFrames` object.

This counter does not increment when the interface is operating in full-duplex mode. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of `ifCounterDiscontinuityTime`.

statsSQETestErrors

A count of times that the SQE TEST ERROR message is generated by the PLS sublayer for a particular interface. The SQE TEST ERROR is set in accordance with the rules for verification of the SQE detection mechanism in the PLS Carrier Sense Function as described in IEEE Std. 802.3, 1998 Edition, section 7.2.4.6.

This counter does not increment on interfaces operating at speeds greater than 10 Mbps, or on interfaces operating in full-duplex mode. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of `ifCounterDiscontinuityTime`.

statsSingleCollisionFrames

A count of successfully transmitted frames on a particular interface for which transmission is inhibited by exactly one collision. A frame that is counted by an instance of this object is also counted by the corresponding instance of either the ifOutUcastPkts, ifOutMulticastPkts, or ifOutBroadcastPkts, and is not counted by the corresponding instance of the dot3StatsMultipleCollisionFrames object.

This counter does not increment when the interface is operating in full-duplex mode. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

statsStatus

Indicates if stats are enabled for this user for debugging purposes. If the status are enabled the stats can be examined using the IpRec MIB stats table. The stats associated with this subscriber are indicated by the subIpIfIndex.

Values 1: enabled

2: disabled

statsSymbolErrors

For an interface operating at 100 Mbps, the number of times there was an invalid data symbol when a valid carrier was present. For an interface operating in half-duplex mode at 1000 Mbps, the number of times the receiving media is non-idle (a carrier event) for a period of time equal to or greater than slotTime, and during which there was at least one occurrence of an event that causes the PHY to indicate Data reception error or carrier extend error on the GMII.

For an interface operating in full-duplex mode at 1000 Mbps, the number of times the receiving media is non-idle (a carrier event) for a period of time equal to or greater than minFrameSize, and during which there was at least one occurrence of an event that causes the PHY to indicate Data reception error on the GMII.

The count represented by an instance of this object is incremented at most once per carrier event, even if multiple symbol errors occur during the carrier event. This count does not increment if a collision is present. Discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime.

statsTimeElapsed

The number of seconds that have elapsed since the beginning of the current collection period. If, for some reason, such as an adjustment in the system's time-of-day clock, the current interval exceeds the maximum value, the agent will return the maximum value.

statsValidIntervals

The number of previous intervals for which data was collected. The value will typically be 24, 48, 72, or 96 unless the interface group was brought online within the last 6, 12, 18, or 24 hours, in which case the value will be the number of complete 15-minute intervals since the interface has been online. In the case where the agent is an SNMP transport bridging proxy or SNMP translating proxy, it is possible that some intervals are unavailable. In this case, this interval is the maximum interval number for which data is available.

status

V5.2 c-channel status.

Values **1: active**
 2: standby
 3: failed

subEPActiveCalls

The number of currently active calls at the subscriber voice endpoint.

subEPIncomingCallsBlocked

The number of blocked incoming calls at the subscriber voice endpoint.

subEPIncomingCallsCompleted

The number of completed incoming calls at the subscriber voice endpoint.

subEPOutgoingCallsBlocked

The number of blocked outgoing calls at the subscriber voice endpoint.

subEPOutgoingCallsCompleted

The number of completed outgoing calls at the subscriber voice endpoint.

subVoiceEndPointIndex

This index is equal to the end-point indices defined for the subscriber voice endpoint tables in the zhoneSubscriberRecords. These endpoints include AAL2, GR-303, V5.2, POTS, etc. For all the endpoint types, the endpoint index is unique to the system.

swRunningVers

The software version that is currently running on this card. This object complies to the DZS software revision syntax.

swStatus

The upgrade status.

Values **1: upgradeLoadSuccessful**

Following the setting of cardSwAdmin to loadUpgradeSw this status indicates the load to memory or storage was successful.

2: loadingSw

The unit/card is loading upgrade software from the internal software load service (warm upgrade only).

3: upgrading

The unit/card is in the process of switching to the upgrade software (warm upgrade only).

4: upgradeLoadFailed

The upgrade version software load was not able to complete due to an error or the unit has determined the downloaded software to be invalid.

5: upgradeFailed

The upgrade was not able to complete due a load error, invalid version loaded or upgrade software was loaded and unable to initialize. The unit/card has reverted to the 'current' version of software.

6: upgradeImageNotAvail

The upgrade image is not available on this system.

7: pendingUpgradeOnReset

The unit/card needs a reboot to upgrade to the already loaded software.

8: upgradeNotRequested

The upgrade image is available on the system but the cardSwAdmin is not set to perform upgrade.

swUpgradeVers

If warm upgrade is supported, this is the version of software that is loaded on the card that will be switched to with a warm upgrade. This value is set by the

card software after a loadUpgradeSoftware command has completed successfully.

sysCardSwSpecificVers

The software version of the file name specified in zhoneSysCardSpecificSwFileName. If the revision is not known or the file name is NULL, this should return an empty string.

sysCardSwUpgradeVers

The software version specified in the zhoneSysCardUpgrade FileName variable in this table that will be loaded on the next card reboot. This object complies to the DZS software revision syntax.

sysSwStatus

The system software status.

- Values**
- 1: downloadSwUnknown**
No state known. This is the value if no sw download had been requested since the system was initialized or a delete command is in progress.
 - 2: downloadSwInProgress**
Download currently in progress.
 - 3: downloadSwAborted**
abort command processed. Sw not loaded.
 - 4: downloadSwFailed**
FTP/TFTP request failed to both the primary and secondary server.
 - 5: downloadSwSuccessful**
Software successfully downloaded from either the primary or secondary server.
 - 6: deleteFileNotFound**
The file specified in zhoneSysSwLocalFilePath cannot be found on the device.
 - 7: deleteFileInUse**
The file specified in zhoneSysSwLocalFilePath is currently specified in a sw configuration record.
 - 8: deleteSuccessful**
The file was successfully deleted.

sysSwUpgradeStatus

Describes status of software upgrade.

Values	1: not-started
	2: standby-upgrade-started
	3: standby-upgrade-completed
	4: standby-upgrade-aborted
	5: swact-started
	6: swact-complete
	7: swact-aborted
Default	1: not-started

systemActiveCalls

The number of currently active calls in the system.

systemConfigurationNumCards

The total number of cards in the system (physically). The number per shelf is defined in the shelf mib.

systemConfigurationNumShelves

The number of shelves physically in the system

systemIncomingCallsBlocked

The number of blocked incoming calls in the system.

systemIncomingCallsCompleted

The number of completed incoming calls in the system.

systemOutgoingCallsBlocked

The number of blocked outgoing calls in the system.

systemOutgoingCallsCompleted

The number of completed outgoing calls in the system.

systemUpTime

The time (in hundredths of a second) since the network management portion of the system was last reinitialized.

T

tftp

Tftp address for the current interface.

threshold

IpfilterSpec

timeElapsed

The number of seconds that have elapsed since the beginning of the current collection period. If, for some reason, such as an adjustment in the system's time-of-day clock, the current interval exceeds the maximum value, the agent will return the maximum value.

tipRingStatus

Indicates whether the tip and ring points from the local unit match the tip and ring points of the remote.

Values 1: normal
2: reverse

tmcPrimarySvcState

Represents conditions that are currently present at the given entity and can affect the service state.

Values 1: active
2: standby
3: outOfService

tmcSecondarySvcState

Represents conditions that are currently present at the given entity and can affect the service state.

Values 1: active
2: standby
3: outOfService

tmcState

Time-slot Management Channel (TMC) state. Indicates when the line is off-hook. A line in a permanent signal condition initiates call clearing, during which time the RDT disconnects the RDT line unit from the DS0 (timeslot) assigned to the call, and continues to monitor the line. The RDT will report the return to on-hook status.

Values 1: inService
2: permanentSignal
Default inService

totalAvailableBw

The Total Available DBA Bandwidth per OLT.

Access read-only

totalAvailableCompensatedCbrBw

The Total Available DBA Compensated CBR bandwidth per OLT.

Access read-only

totalBESs

The number of Bursty Errored Seconds (BESs) encountered by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

totalCallsBlocked

The number of calls that were blocked due to no available time-slot.

totalCCVs

The number of C-bit Coding Violations encountered by a DS3 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

totalCESs

The number of C-bit Errored Seconds encountered by a DS3 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

totalCrcAnom

The number of the line CRC anomalies.

totalCSEs

The number of C-bit Severely Errored Seconds encountered by a DS3 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

totalCSSs

The number of Controlled Slip Seconds encountered by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

totalDiscardedCellsRx

Total number of discarded cells received on this VPL.

totalDiscardedCellsTx

Total number of discarded cells transmitted on this VPL.

totalDMs

The number of Degraded Minutes (DMs) encountered by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

totalErrorSeconds

The total number of ISDN activation seconds.

totalES

The number of the line errored seconds since agent reset.

totalESs

The sum of Errored Seconds encountered by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

totalFabricCellsRx

Total cells received across the ATM fabric interface for a given VC, including cells discarded due to parity errors.

TotalFabricCellsTx

Total cells transmitted across the ATM fabric interface for a given VC, including cells discarded due to parity errors.

totalFinalCLP0CellsRx

Total CLP0 Cells received by fabric. This field monitors egress total cells transmitted CLP0.

totalFinalCLP0CellsTx

Total CLP0 cells received by fabric. This field monitors egress total cells transmitted CLP0.

totalFinalCLP1CellsRx

Total CLP1 cells received by fabric. This field is setup to monitor egress total cells transmitted CLP1.

totalFinalCLP1CellsTx

Total CLP1 Cells transmitted. This field monitors egress total cells transmitted CLP1.

totalFsyncSeconds

The total number of frame synch error seconds.

totalGemPorts

Total number GEM Ports currently on this OLT.

Access read-only

totalGr303ProtocolErrors

The number of GR-303 protocol errors.

totalInBoundCalls

The number of inbound calls completed.

totalIsdnCalls

The number of ISDN calls completed.

totalIndex

The index value that uniquely identifies the DS1 interface to which this entry is applicable. The interface identified by a particular value of this index is the same interface as identified by the same value as a zhLineIndex object instance.

DS3/E3: The index value which uniquely identifies the DS3/E3 interface to which this entry is applicable. The interface identified by a particular value of this index is the same interface as identified by the same value an dsx3LineIndex object instance.

TotalInitialCellsRx

Total cells received across the ingress interface for a given VC, and not discarded by policing prior to thresholding (includes CPU insertions but excludes ATM switch fabric generated ingress OAM insertions).

TotalInitialCellsTx

Total cells received across the ingress interface for a given VC, and not discarded by policing prior to thresholding (includes CPU insertions but excludes Atm switch fabric generated ingress OAM insertions).

totalInits

The number of line initialization attempts since agent reset. Includes both successful and failed attempts.

totalLapdRcvd

The number of LAPD frames received on all c-paths for this interface group.

totalLapdRcvdErrs

The number of LAPD frames received in error on all c-paths for this interface group.

totalLapdSent

The number of LAPD frames sent on all c-paths for this interface group.

totalLCVs

The number of Line Code Violations (LCVs) detected by a DS1 interface in the current 15-minute interval. Invalid 15-minute intervals count as 0.

totalLESSs

The number of Line Errored Seconds encountered by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

totalLofs

The number of the number of Loss of Framing failures since agent reset.

totalLols

The number of the number of Loss of Link failures since agent reset.

totalLoss

The number of the number of Loss of Signal failures since agent reset.

totalLOSW

The number of line loss-of-sync-word seconds since agent reset.

totalOutBoundCalls

The number of outbound calls completed.

totalOutBoundCallsBlocked

The number of outbound calls that were blocked due to no available timeslot.

totalOutEOCLapdRcvd

The total number of LAPD frames received on the Embedded Operations Channel (EOC).

totalOutEOCLapdSent

The total number of LAPD frames sent on the Embedded Operations Channel (EOC).

totalOutEOCRcvdErrs

The total number of Embedded Operations Channel (EOC) frames received in error.

totalOutTMCLapdRcvd

The total number of LAPD frames received in error on the Embedded Operations Channel (EOC) in this interval.

totalOutTMCLapdRcvdErrs

The total number LAPD frames received in error on the Time-slot Management Channel (TMC).

totalOutTMCLapdSent

The number of LAPD frames sent on the Time-slot Management Channel (TMC) in this interval.

totalPCVs

The number of Path Coding Violations encountered by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

totalPeriodsElapsed

The number of 15-minute time periods that are used in the performance total table. This value is generally 96, indicating a valid 24-hour period, except when the agent has been restarted.

totalPESs

The counter associated with the number of P-bit Errored Seconds, encountered by a DS3 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

totalPstnInboundCalls

The number of PSTN inbound calls completed.

totalPstnOutboundCalls

The number of PSTN outbound calls completed.

totalSEFSs

The number of Severely Errored Framing Seconds detected by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

totalSES

The number of line severely errored seconds since agent reset.

totalSESSs

The number of Severely Errored Seconds detected by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

totalUAS

The number of line unavailable seconds since agent reset.

totalUASs

The number of Unavailable Seconds detected by a DS1 interface in the previous 24 hour interval. Invalid 15-minute intervals count as 0.

totalUnbalancedFrame

The total number of unbalanced frames.

totalV52ProtocolErrors

The number of V5.2 Protocol errors.

transmitClockSource

The source of the Transmit Clock.

loopTiming: the recovered receive clock is used as the transmit clock. Only set if this DS1 has zhoneClockSourceEligibility equal to eligible and the state of that DS1 is up and the DS1 resource provider has decided that this DS1 is to provide the transmit clock for all the DS1s. Only one DS1 will have this set per platform.

localTiming: a local clock source is used or when an external clock is attached to the box containing the interface.

throughTiming: recovered receive clock from another interface is used as the transmit clock. If a DS1 other than this one reflects loopTiming, this DS1 will use that DS1 as its transmit clock.

Values **1: loopTiming**
 2: localTiming
 3: throughTiming

transmitPower

Nominal transmit power

txCellCount

Transmit ATM Cell Processor (TACP) transmit cell count.

TxFCSSize

The size of the Frame Check Sequence (FCS) in bits that the local node will generate when sending packets to the remote node. The value of this object is meaningful only when the link has reached the open state (ifOperStatus is up).

U

unknownProtos

The number of locally-addressed datagrams received successfully but discarded because of an unknown or unsupported protocol.

unknownRx

Unknown IGMP packets received.

upAttainLineRate

This is the MAX line rate that can be supported on the line in the upstream direction.

upLineRate

The DSL upstream (Customer Premise Equipment to Central Office) line rate on this interface.

upTime

The amount of time since this unit/card was last initialized (warm or full reset). The units are in 100ths of a second.

usConst

Upstream constellation size in use.

usCorrOctets

Number of errored octets corrected by Reed-Solomon in the upstream channel.

usedAllocIds

Total number of alloc-Ids in use for this OLT.

Default 0
Access read-only

usedDballocIds

Total number of DBA alloc-Ids in use for this OLT.

Default 0
Access read-only

usLineRate

Measured upstream line rate.

usRsFinal

Upstream Reed-Solomon code word and depth.

usSnr

Measured upstream EQ output SNR (in dB times 8).

usUncorrOctets

Number of uncorrectable octets in the upstream channel.

V

v2ReportsRx

Version 2 reports received (JOINs).

v2ReportsTx

Version 2 reports sent.

validIntervals

The number of previous near end intervals for which data was collected. The value will be 96 unless the interface was brought on line within the last 24 hours, in which case the value will be the number of complete 15-minute near end intervals since the interface has been online. In the case where the agent is a proxy, it is possible that some intervals are unavailable. In this case, the interval is the maximum interval number for which data is available.

A SONET/SDH interface must be capable of supporting at least n intervals. The minimum value of n is four. The default for n is 32. The maximum value of n is 96. The value will be <n>, unless the measuring was started or restarted within the last <n>*15-minutes, in which case the value will be the number of complete 15-minute intervals for which the agent has some data.

vendorClassId

Vendor Class ID.

vendorID

The vendor ID code is a copy of the binary vendor identification field expressed as readable characters in hexadecimal notation.

vendorSpecificPointer

OID instance in vendor-specific MIB. The Instance may be used to determine shelf/slot/port of the ATUC interface in a DSLAM.

versionNumber

The vendor specific version number sent by this Vtu as part of the initialization messages. It is a copy of the binary version number field expressed as readable characters in hexadecimal notation.

videoCrossConnH2LLastChange

The value of sysUpTime at the time this VC cross-connect entered its current operational state in high to low direction.

videoCrossConnH2LOperStatus

The current operational status of the VC cross-connect in one direction (for example, from the high to low direction).

videoCrossConnL2HLastChange

The value of sysUpTime at the time this VC cross-connect entered its current operational state in low to high direction.

videoCrossConnL2HOperStatus

The current operational status of the VC cross-connect in one direction (for example, from the low to high direction).

Z

All the zhoneEtherStats have values 0 to 4294967294, and access method read-only.

zhoneDsIPerfCRCAnomalies

The number of CRC anomalies since the start of the current measurement period.

zhoneDsIPerfES

The number of errored-seconds since the start of the current measurement period.

zhoneDsIPerfInits

Count of the line initialization attempts in the current 15 minute interval. Includes both successful and failed attempts.

zhoneDsIPerfLOSWS

The number of loss-of-sync-word seconds since the start of the current measurement period.

zhoneDslPerfLofs

Count of the number of Loss of Framing failures in the current 15 minute interval.

zhoneDslPerfLols

Count of the number of Loss of Link failures in the current 15 minute interval.

zhoneDslPerfLoss

Count of the number of Loss of Signal failures in the current 15 minute interval.

zhoneDslPerfSES

The number of severely-errored-seconds since the start of the current measurement period.

zhoneDslPerfTimeElapsed

The number of seconds elapsed since the start of the current measurement period.

zhoneDslPerfTotalCRCAnomalies

Count of the line CRC anomalies since agent reset.

zhoneDslPerfTotalES

Count of the line errored seconds since agent reset.

zhoneDslPerfTotalInits

Count of the line initialization attempts since agent reset. Includes both successful and failed attempts.

zhoneDslPerfTotalLOSWS

Count of the line loss-of-sync-word seconds since agent reset.

zhoneDslPerfTotalLofs

Count of the number of Loss of Framing failures since agent reset.

zhoneDslPerfTotalLols

Count of the number of Loss of Link failures since agent reset.

zhoneDslPerfTotalLoss

Count of the number of Loss of Signal failures since agent reset.

zhoneDslPerfTotalSES

Count of the line severely errored seconds since agent reset.

zhoneDslPerfTotalUAS

Count of the line unavailable seconds since agent reset.

zhoneDslPerfUAS

The number of unavailable seconds since the start of the current measurement period.

zhoneEtherStatsAverageRxThroughput

Average receive throughput in bits per second since last query.

zhoneEtherStatsAverageTxThroughput

Average transmit throughput in bits per second since last query.

zhoneEtherStatsBroadcastPkts

The total number of good packets transmitted and received that were directed to the broadcast address.

Note that this does not include multicast packets.

zhoneEtherStatsCollisions

The best estimate of the total number of collisions on this Ethernet segment.

The value returned will depend on the location of the RMON probe. Section 8.2.1.3 (10BASE-5) and section 10.3.1.3 (10BASE-2) of IEEE standard 802.3 states that a station must detect a collision, in the receive mode, if three or more stations are transmitting simultaneously. A repeater port must detect a collision when two or more stations are transmitting simultaneously. Thus a probe placed on a repeater port could record more collisions than a probe connected to a station on the same segment would.

Probe location plays a much smaller role when considering 10BASE-T. 14.2.1.4 (10BASE-T) of IEEE standard 802.3 defines a collision as the simultaneous presence of signals on the DO and RD circuits (transmitting and receiving at the same time). A 10BASE-T station can only detect collisions when it is transmitting. Thus probes placed on a station and a repeater, should report the same number of collisions.

Note also that an RMON probe inside a repeater should ideally report collisions between the repeater and one or more other hosts (transmit collisions as defined by IEEE 802.3k) plus receiver collisions observed on any coax segments to which the repeater is connected.

zhoneEtherStatsCRCAAlignErrors

The total number of packets received that had a length (excluding framing bits, but including FCS octets) of between 64 and 1518 octets, inclusive, but had either a bad Frame Check Sequence (FCS) with an integral number of octets (FCS Error) or a bad FCS with a non-integral number of octets (Alignment Error).

zhoneEtherStatsDropEvents

The total number of events in which packets were dropped due to lack of resources.

Note that this number is not necessarily the number of packets dropped; it is just the number of times this condition has been detected.

zhoneEtherStatsDroppedFrames

The total number of frames that were received and therefore not accounted for in the Total Dropped Events, but that the system chose not to count this frame (for whatever reason).

Most often, this event occurs when the system is out of some resources and decides to shed load from this collection.

This count does not include packets that were not counted because they had MAC-layer errors.

Note that, unlike the Total Dropped Events counter, this number is the exact number of frames dropped.

zhoneEtherStatsFragments

The total number of packets received that were less than 64 octets in length (excluding framing bits but including FCS octets) and had either a bad Frame Check Sequence (FCS) with an integral number of octets (FCS Error) or a bad FCS with a non-integral number of octets (Alignment Error).

Note that it is entirely normal for Total Fragments to increment. This is because it counts both runs (which are normal occurrences due to collisions) and noise hits.

zhoneEtherStatsIPMCBridgedPkts

Total number of bridged multicast packets.

zhoneEtherStatsIPMCRoutedPkts

Total number of routed multicast packets.

zhoneEtherStatsIPMCTxDroppedPkts

Transmitted multicast bytes.

zhoneEtherStatsIPMCRxDroppedPkts

Total number of received multicast packets dropped.

zhoneEtherStatsJabbers

The total number of packets received that were longer than 1518 octets (excluding framing bits, but including FCS octets), and had either a bad Frame Check Sequence (FCS) with an integral number of octets (FCS Error) or a bad FCS with a non-integral number of octets (Alignment Error).

Note that this definition of jabber is different than the definition in IEEE-802.3 section 8.2.1.5 (10BASE5) and section 10.3.1.4 (10BASE2). These documents define jabber as the condition where any packet exceeds 20 ms. The allowed range to detect jabber is between 20 ms and 150 ms

zhoneEtherStatsMcastRxBytes

Received multicast bytes.

zhoneEtherStatsMcastRxDroppedBytes

Received multicast bytes which were dropped.

zhoneEtherStatsMcastTxBytes

Transmitted multicast bytes.

zhoneEtherStatsMulticastPkts

The total number of good packets transmitted and received that were directed to a multicast address.

Note that this number does not include packets directed to the broadcast address.

zhoneEtherStatsOctets

The total number of octets of data (including those in bad packets) transmitted and received on the network (excluding framing bits but including FCS octets).

This object can be used as a reasonable estimate of 10-Megabit Ethernet utilization. If greater precision is desired, the Total packets and Total Octets can be sampled before and after a common interval. The differences in the sampled values are packets and octets respectively, and the number of seconds in the interval is Interval.

These values are used to calculate the Utilization as follows:

Utilization=

Packets * (9.6 + 6.4) + (Octets *.8)

Interval * 10,000

The result of this equation is the value Utilization which is the percent utilization of the Ethernet segment on a scale of 0 to 100 percent.

zhoneEtherStatsOversizePkts

The total number of packets transmitted and received that were longer than 1518 octets (excluding framing bits, but including FCS octets) and were otherwise well formed.

zhoneEtherStatsOversizeRxPkts

The total number of packets received that were longer than 1518 octets (excluding framing bits, but including FCS octets) and were otherwise well formed.

zhoneEtherStatsOversizeTxPkts

The total number of packets transmitted that were longer than 1518 octets (excluding framing bits, but including FCS octets) and were otherwise well formed.

zhoneEtherStatsPkts

The total number of packets (including bad packets, broadcast packets, and multicast packets) transmitted and received.

zhoneEtherStatsPkts64Octets

The total number of packets (including bad packets) transmitted and received that were 0 to 64 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsPkts65to127Octets

The total number of packets (including bad packets) transmitted and received that were between 65 and 127 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsPkts128to255Octets

The total number of packets (including bad packets) transmitted and received that were between 128 and 255 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsPkts256to511Octets

The total number of packets (including bad packets) transmitted and received that were between 256 and 511 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsPkts512to1023Octets

The total number of packets (including bad packets) transmitted and received that were between 512 and 1023 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsPkts1024to1518Octets

The total number of packets (including bad packets) transmitted and received that were between 1024 and 1518 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsPkts1519to2047Octets

The total number of packets (including bad packets) transmitted and received that were between 1519 and 2047 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsPkts2048to4095Octets

The total number of packets (including bad packets) transmitted and received that were between 2048 and 4095 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsPkts4095to9216Octets

The total number of packets (including bad packets) transmitted and received that were between 4095 and 9216 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsReceivedPkts64Octets

The total number of packets (including bad packets) received that were between 0 and 64 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsReceivedPkts65to127Octets

The total number of packets (including bad packets) received that were between 65 and 127 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsReceivedPkts128to255Octets

The total number of packets (including bad packets) received that were between 128 and 255 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsReceivedPkts256to511Octets

The total number of packets (including bad packets) received that were between 256 and 511 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsReceivedPkts512to1023Octets

The total number of packets (including bad packets) received that were between 512 and 1023 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsReceivedPkts1024to1518Octets

The total number of packets (including bad packets) received that were between 1024 and 1518 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsReceivedPkts1519to2047Octets

The total number of packets (including bad packets) received that were between 1519 and 2047 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsReceivedPkts2048to4095Octets

The total number of packets (including bad packets) received that were between 2048 and 4095 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsReceivedPkts4095to9216Octets

The total number of packets (including bad packets) received that were between 4095 and 9216 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsRxBandwidthOccupancy

Percentage of bandwidth currently being utilized for receiving traffic.

zhoneEtherStatsRxPkts

The total number of packets (including bad packets, broadcast packets, and multicast packets) received.

zhoneEtherStatsRXNoErrors

Total number of packets received. without errors.

zhoneEtherStatsTransmittedPkts64Octets

The total number of packets (including bad packets) transmitted and received that were 0 to 64 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsTransmittedPkts65to127Octets

The total number of packets (including bad packets) transmitted and received that were between 65 and 127 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsTransmittedPkts128to255Octets

The total number of packets (including bad packets) transmitted that were between 128 and 255 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsTransmittedPkts256to511Octets

The total number of packets (including bad packets) transmitted that were between 256 and 511 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsTransmittedPkts512to1023Octets

The total number of packets (including bad packets) transmitted and received that were between 512 and 1023 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsTransmittedPkts1024to1518Octets

The total number of packets (including bad packets) transmitted that were between 1024 and 1518 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsTransmittedPkts1519to2047Octets

The total number of packets (including bad packets) transmitted that were between 1519 and 2047 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsTransmittedPkts2048to4095Octets

The total number of packets (including bad packets) transmitted that were between 2048 and 4095 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsTransmittedPkts4095to9216Octets

The total number of packets (including bad packets) transmitted and received that were between 4095 and 9216 octets in length inclusive (excluding framing bits but including FCS octets).

zhoneEtherStatsTxBandwidthOccupancy

Percentage of bandwidth currently being utilized for transmitting traffic.

zhoneEtherStatsTXNoErrors

Total number of packets transmitted without errors.

zhoneEtherStatsTxPkts

The total number of packets (including bad packets, broadcast packets, and multicast packets) transmitted.

zhoneEtherStatsUndersizePkts

The total number of packets received that were less than 64 octets long (excluding framing bits, but including FCS octets) and were otherwise well formed.

zhoneGponOltStatisticsDownstreamDiscardedFrames

The number of downstream packets discarded due to CRC errors, MAC lookup miss, congestion, etc.

zhoneGponOltStatisticsDownstreamGemFrames

Total number of GEM frames sent in downstream direction.

zhoneGponOltStatisticsDownstreamOmciFrames

Total number of OMCI frames sent in downstream direction.

zhoneGponOltStatisticsDownstreamPloamFrames

Total number of PLOAM frames sent in downstream direction.

zhoneGponOltStatisticsDownstreamPonCpuPkts

The number of downstream packets generated by the CPU (MIPS).

zhoneGponOltStatisticsDownstreamPonValidEthernetPkts

Total number of valid Ethernet packets sent in downstream direction.

zhoneGponOltStatisticsDownstreamTmCrcPkts

The number of dropped downstream packets due to CRC errors.

zhoneGponOltStatisticsDownstreamTmDroppedCpuPkts

The number of dropped downstream packets originated by the CPU (MIPS).

zhoneGponOltStatisticsDownstreamTmMacLookupMiss

The number of downstream MAC lookup miss events.

zhoneGponOltStatisticsDownstreamTmPktsDroppedGemPidNotEnabled

The number of downstream packets dropped because the GEM port ID was not configured correctly.

zhoneGponOltStatisticsDownstreamTmPktsForwardedFromHmToPon

The number of downstream packets forwarded from the header modification stage to the PON.

zhoneGponOltStatisticsDownstreamTmQNDroppedPkts (N=0 to7)

The number of downstream packets dropped by egress priority queue [0 to 7] due to congestion. Queue 0 is the highest priority; queue 7 is the lowest priority. Packets in the lowest priority queues are dropped first.

When the PON link is not active, this counter will not increment.

zhoneGponOltStatisticsDownstreamTmQNValidPkts (N=0 to7)

The number of downstream packets forwarded by egress priority queue [0 to 7] to the PON. Queue 0 is the highest priority; queue 7 is the lowest priority. Packets in the lowest priority queues are dropped first.

When the PON link is not active, this counter will not increment.

zhoneGponOltStatisticsDownstreamTmValidPkts

Total number of valid packets that were sent in downstream direction. When the PON link is not active, this counter will not increment.

zhoneGponOltStatisticsDownstreamTxBytes

Total number of bytes transmitted sent in downstream direction.

zhoneGponOltStatisticsDownstreamValidGemFrames

Total number of valid GEM frames sent in downstream direction.

zhoneGponOltStatisticsUpstreamDiscardedFrames

Total number of discarded GEM frames sent in upstream direction.

zhoneGponOltStatisticsUpstreamDroppedPktsInactivePorts

Total number of packets sent from the ONT to MxK that were dropped because the GEM port ID was not configured.

zhoneGponOltStatisticsUpstreamDroppedPloamsFifoFull

Total number of PLOAMs sent from the ONT to MxK that were dropped because the FIFO buffer was full.

zhoneGponOltStatisticsUpstreamGemFrames

The number of GEM frames sent in the upstream direction.

zhoneGponOltStatisticsUpstreamIdlePloamFrames

Total number of idle PLOAM frames sent in upstream direction.

zhoneGponOltStatisticsUpstreamOmciFrames

The number of OMCI frames sent in the upstream direction.

zhoneGponOltStatisticsUpstreamPloamFrames

Total number of PLOAM frames sent in the upstream direction. This includes:

- Total number of PLOAM messages, including idle PLOAMs.
- Total number of valid PLOAM messages (not including idle PLOAMs)

- Total number of PLOAM messages dropped due to CRC errors.

zhoneGponOltStatisticsUpstreamPonErrorPloams

Total number of PON error PLOAM messages sent in upstream direction.

zhoneGponOltStatisticsUpstreamPonInvalidPkts

The number of upstream errored packets.

zhoneGponOltStatisticsUpstreamPonValidNotIdlePloams

Total number of valid non-idle PLOAM messages sent in upstream direction.

zhoneGponOltStatisticsUpstreamPonValidPkts

Total number of valid PON packets sent in upstream direction.

zhoneGponOltStatisticsUpstreamTmDroppedCpuPkts

The number of dropped upstream packets sent by the MIPS to SGMII.

zhoneGponOltStatisticsUpstreamTmDroppedPktsCrcError

The number of upstream packets that were dropped because of CRC errors.

zhoneGponOltStatisticsUpstreamTmDroppedPktsSecurity

Total number of upstream packets that were dropped because they didn't pass the security rules.

zhoneGponOltStatisticsUpstreamTmLearnFailures

MAC address learning failures from traffic sent in upstream direction that were due to a full FIFO buffer.

zhoneGponOltStatisticsUpstreamTmQNDroppedPkts (N= 0 to 7)

Number of upstream packets dropped by egress priority queue [0 to 7] due to congestion. Queue 0 is the highest priority; queue 7 is the lowest priority. Packets in the lowest priority queues are dropped first.

When the PON link is not active, this counter will not increment.

zhoneGponOltStatisticsUpstreamTmQNValidPkts (N= 0 to 7)

Number of upstream packets forwarded by egress priority queue [0 to 7] to the MxK. Queue 0 is the highest priority; queue 7 is the lowest priority. Packets in the lowest priority queues are dropped first.

When the PON link is not active, this counter will not increment. When the PON link is not active, this counter will not increment.

zhoneGponOltStatisticsUpstreamValidGemFrames

The number of valid GEM frames sent in upstream direction.

zhoneVdslPerfMonDsCorrOctets

Number of errored octets corrected by Reed- Solomon in the downstream channel.

zhoneVdslPerfMonDsUncorrOctets

Number of uncorrectable octets in the downstream channel.

zhoneVdslPerfMonUsCorrOctets

Number of errored octets corrected by Reed-Solomon in the upstream channel.

zhoneVdslPerfMonUsUncorrOctets

Number of uncorrectable octets in the upstream channel.

zhoneVdslPhysSide

This is a mandatory attribute to retrieve the VDSL status information. Specifying 1 will retrieve the CO side information and 2 will retrieve the CPE side information.

zhoneVdslPhyrVtucCorrectedRetransmittedCodewords

VTUC Retransmitted corrected Codewords.

zhoneVdslPhyrVtucRetransmittedCodewords

VTUC Retransmitted uncorrectable Codewords.

zhoneVdslPhyrVtucUncorrectableRetransmittedCodewords

VTUC Retransmitted uncorrectable Codewords.

zhoneVdslPhyrVturCorrectedRetransmittedCodewords

VTUR Retransmitted corrected Codewords.

zhoneVdslPhyrVturRetransmittedCodewords

VTUR Retransmitted Codewords.

zhoneVdslPhyrVturUncorrectableRetransmittedCodewords

VTUR Retransmitted uncorrectable Codewords.

zmsBlockCli

Indicates whether the CLI is blocked or unblocked. If the CLI is blocked, the device cannot be provisioned using the CLI. Provisioning using the CLI is usually enabled, except when the device is being managed by DZMS.

Parent: PSBANDevice

Values **1: blocked**
 2: unblocked

Access read-only

